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SERIES

Calculus

for

IIT-JEE

**and other engineering
entrance examinations**

$$\int_a^b f(x)dx = F(b) - F(a)$$

$$u(t) = \frac{d}{dt} v(t) = \frac{dv}{dt} = \dot{v}$$

$$\frac{d^2}{dt^2} x(t) = \frac{d^2 x}{dt^2} = \ddot{x}$$

$$a(t) = \frac{d}{dt} v(t) = \frac{dv}{dt} = \dot{v}$$



SANNU RAHI



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Calculus for IIT-JEE

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chapter
I
Function

1.1 CALCULUS

Calculus is that branch of mathematics in which we study the relation between variables, its properties and application.

1.2 FUNCTION

Function is that branch of calculus in which we study the geometrical figures and its properties obtained by relation between variables.

1.2.1 Types of Function

There are two types of function:

- (a) Single-valued function
- (b) Multi-valued function

Single-valued Function

Graphically if a straight line parallel to y-axis intersect the curve of function at only one point then function is called single valued function (Figs. 1.1 & 1.2).

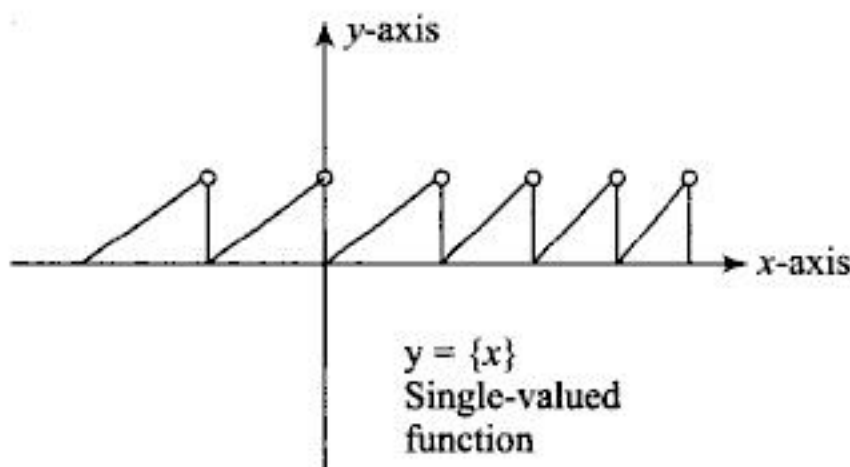


Fig. 1.1

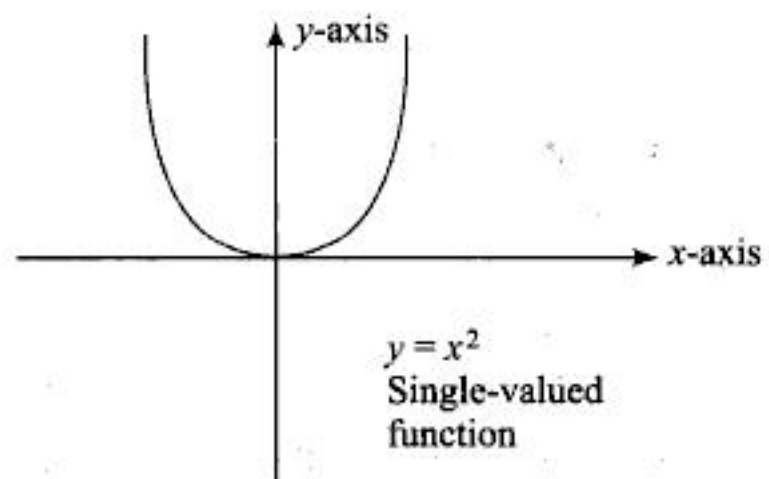


Fig. 1.2

Mathematically, if one value of independent variable of any function represents one value of their dependent variable at a time then the function is called single-valued function.

e.g. $f(x) = \sin\{\log x\} + 1$

Here, put $x = 1$, $y = f(1) = \sin\{\log 1\} + 1 = 1$

Hence, given function is a single-valued function.

Multi-valued Function

Graphically, if a straight line parallel to y-axis intersects the curve of function at more than one point then the function is called multi-valued function.

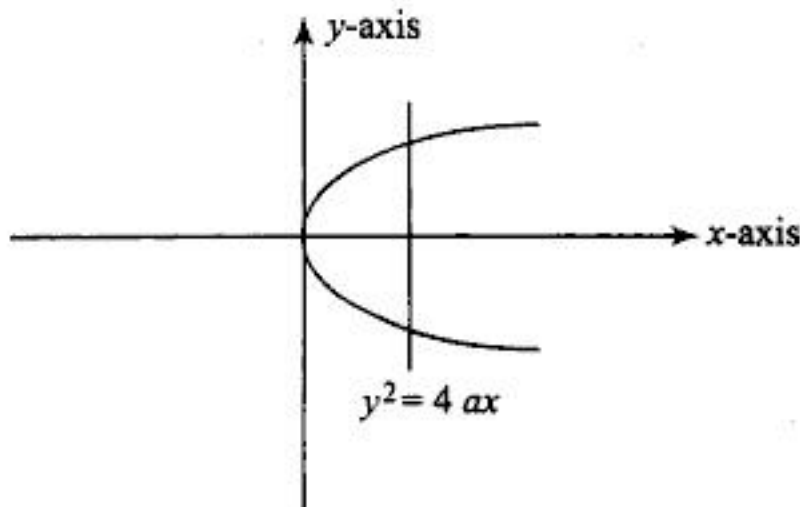


Fig. 1.3

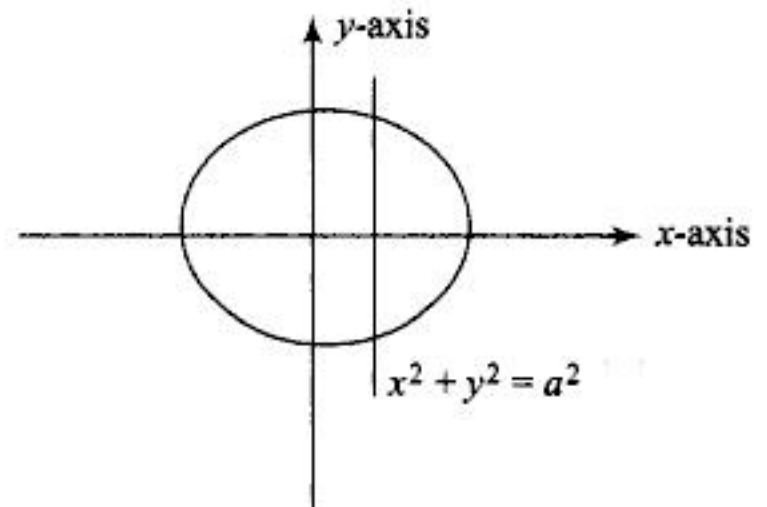


Fig. 1.4

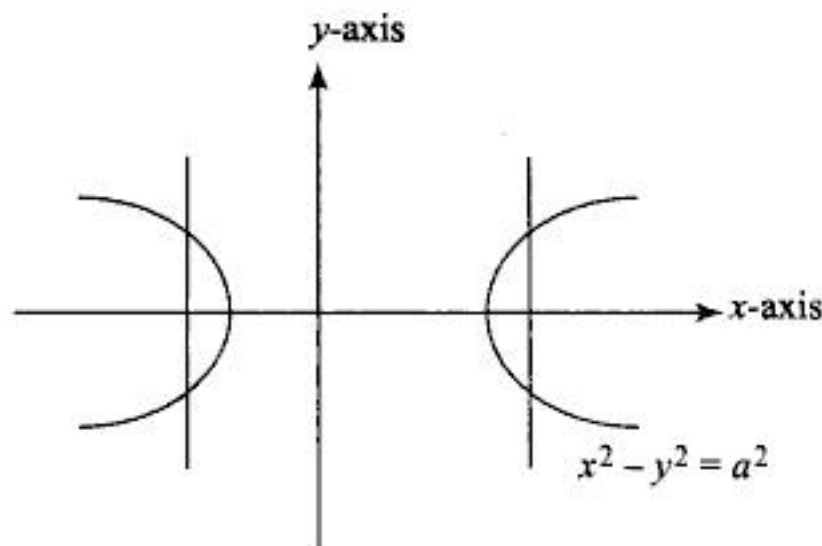


Fig. 1.5

Mathematically, if one value of an independent variable of any function represents more than one value of the dependent variable at a time then the function is called multi-valued function.

Generally, multi-valued function has undefined form.

e.g. $y^2 + x^2 = 4$

Put $x = 1$, $y^2 + 1^2 = 4 \Rightarrow y = \pm\sqrt{3}$

Hence, given function is a multi-valued function.

1.2.2 Representation of Function

A function can be represented by the following forms:-

- Arrow form
- Order pair form

- (c) Tabular form
- (d) Verbal description form
- (e) Standard equation or formula form
- (f) Graphical form

Arrow Form

If A and B are two non-empty sets and f is a relation which associates each element of set A with unique element of set B , then f is called function from A to B . It is denoted as $f: A \rightarrow B$

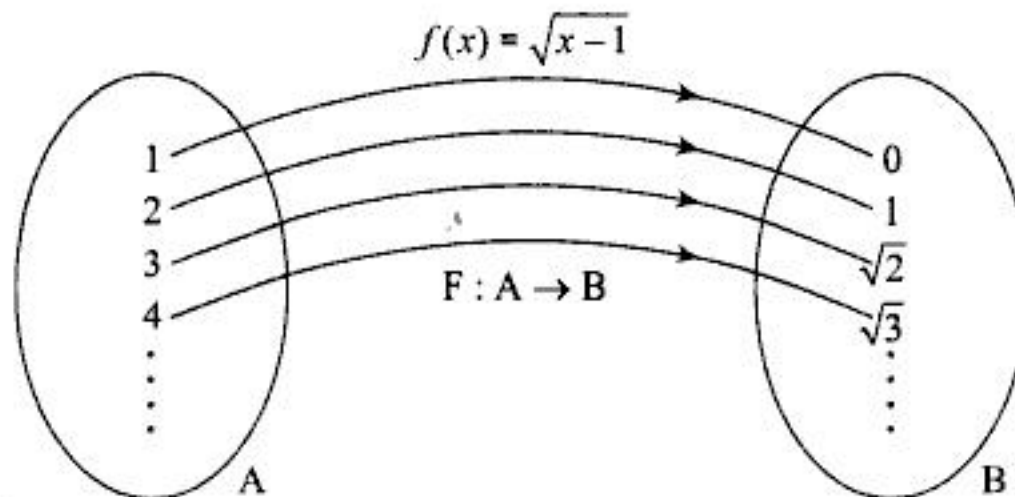


Fig. 1.6

For $f: A \rightarrow B$ in Fig. 1.6:

- (i) Set A is called domain of function $f(x)$.
- (ii) Set B is called co-domain or range of function.
- (iii) If co-domain of a function is equal to its range then the function is called onto function but if the range of function is a subset of its co-domain then the function is called into function.
- (iv) If A and B are two non-empty sets having m and n numbers of elements respectively, then number of functions $f: A \rightarrow B$ is (n^m)
e.g. If $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c\}$ then number of functions or mappings $f: A \rightarrow B$ is $(n^m) = 3^4 = 81$
- (v) For every $x \in A$, there exist $y \in B$, such that, $y = f(x)$ i.e. each element of set A has image in B but there may be elements in B which are not image of any element of A .
- (vi) Same element of A cannot be associated to distinct elements of B i.e. image of each element of A is unique but distinct element of A may be associated to the same element of B . i.e. more than one element of A may have same image in set B .

Order Pair Form

Form of function

$$f(x, y) = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

is called function in order pair form having domain

$$x \in \{x_1, x_2, x_3, \dots, x_n\} \text{ and range, } y \in \{y_1, y_2, \dots, y_n\}$$

Example 1.1 If $f: N \rightarrow R$ where $f(n) = a_n = \frac{n}{(2n+1)^2}$ then write the function in ordered pair form.

Solution Given $a_n = \frac{n}{(2n+1)^2}$

Putting $n = 1, 2, 3, 4, \dots$ successively, we get

$$a_1 = \frac{1}{(2 \cdot 1 + 1)^2} = \frac{1}{3^2} = \frac{1}{9}$$

$$a_2 = \frac{2}{(2 \cdot 2 + 1)^2} = \frac{2}{5^2} = \frac{2}{25}$$

$$a_3 = \frac{3}{(2 \cdot 3 + 1)^2} = \frac{3}{7^2} = \frac{3}{49}$$

$$a_4 = \frac{4}{(2 \cdot 4 + 1)^2} = \frac{4}{9^2} = \frac{4}{81}$$

Hence, order pair form of given function is $\left\{ \left(1, \frac{1}{9}\right), \left(2, \frac{2}{25}\right), \left(3, \frac{3}{49}\right), \dots \right\}$

Tabular Form

If set of values of independent and dependent variables of any function are given in table due to this form of function can be evaluate called function in tabular form (Table 1.1).

Table 1.1

e.g.

x	1	10	10^2	10^3	...
y	0	1	2	3	...

From table, form of function is

$$\begin{array}{ccc}
 y = \log_{10} x & & \\
 \downarrow & & \downarrow \\
 \text{Dependent variable} & & \text{Independent variable}
 \end{array}$$

Here, domain, $x \in \{1, 10, 10^2, 10^3, \dots\}$ and range, $y \in \{0, 1, 2, 3, \dots\}$

Verbal Description Form

Form of function in which independent variable and dependent variable are defined in verbal form then function is called verbal description form.

e.g.

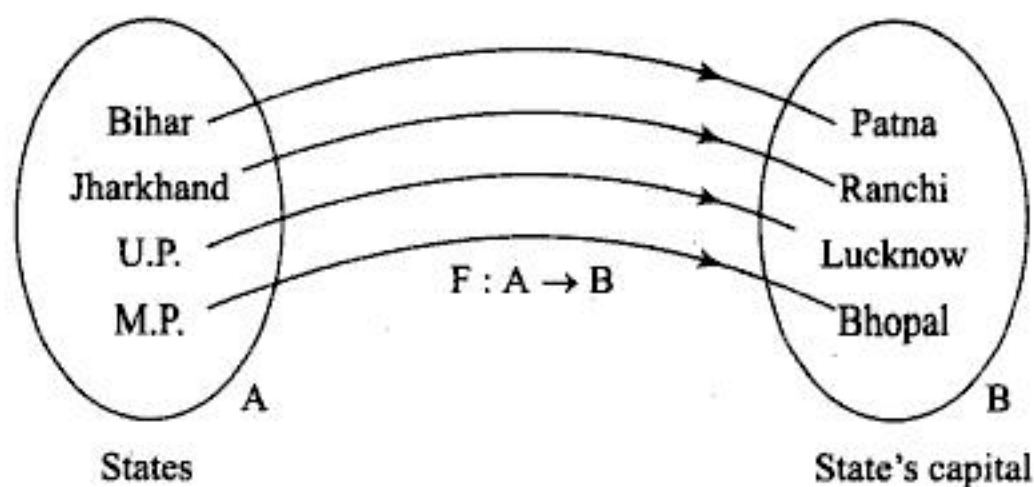


Fig. 1.7

Standard Equation of Formula Form

Any standard formula or equation in two variables represent function.

e.g. Area of a circle = πr^2

$$\Rightarrow y = \pi r^2$$

$\downarrow$ $\downarrow$
 Dependent variable Independent variable

e.g. $2x - 3y = 5$

$$\Rightarrow y = \frac{2}{3}x - \frac{5}{3}$$

$\downarrow$ $\downarrow$
 Dependent variable Independent variable

Graphical Form

Any specified curve represent single valued function or multivalued function x -axis represents domain of function and y -axis represent co-domain or range of function.

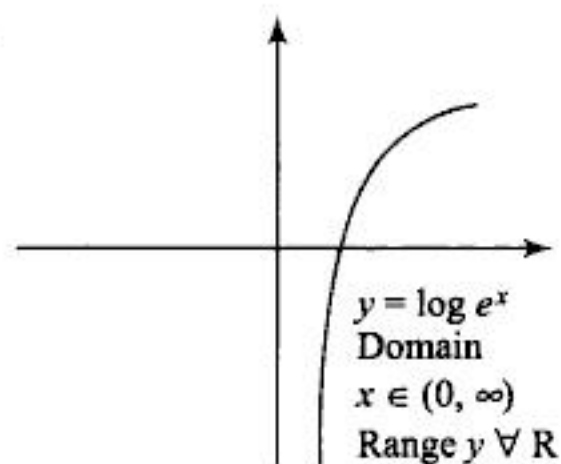


Fig. 1.8

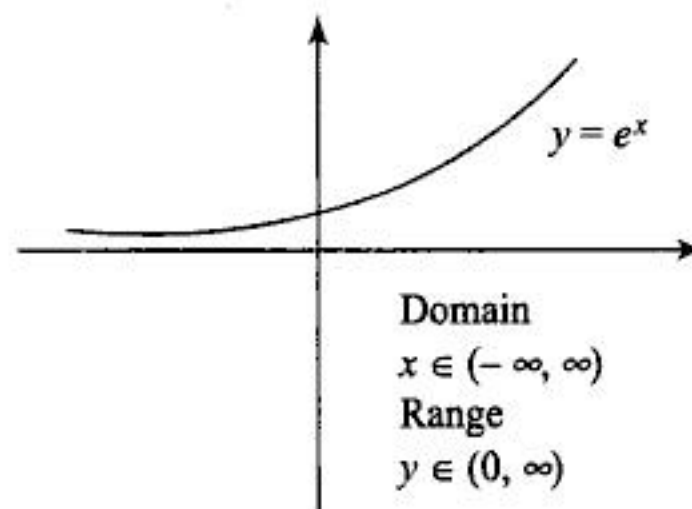


Fig. 1.9

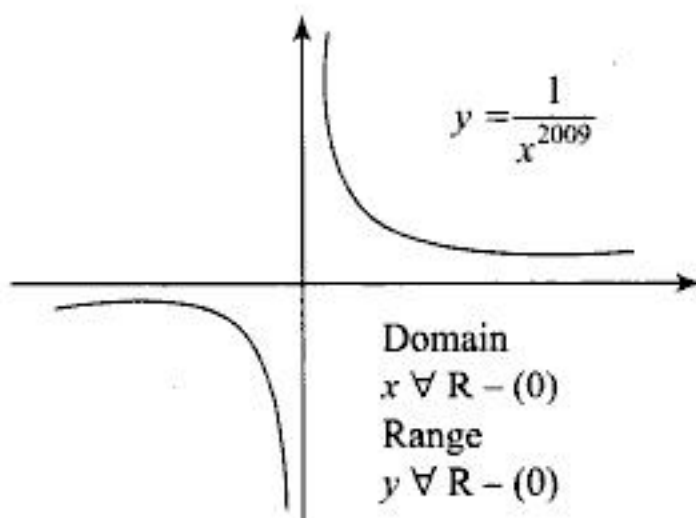


Fig. 1.10

1.3 MORE TYPES OF FUNCTIONS

1. Algebraic Functions

A function in which dependent and independent variables are held in simple form with algebraic sign is called algebraic function.

Types of Algebraic Function

- (a) Polynomial function
- (b) Constant function
- (c) Identity function
- (d) Rational function
- (e) Irrational function
- (f) Modulus function
- (g) Signum function
- (h) Greatest integral function
- (i) Least integral function
- (j) Fractional part of function
- (k) Geometrical function

2. Transcendental Functions

A function in which independent variable is defined with mathematical term is called transcendental function.

Types of Transcendental Function

- (a) Trigonometric function
- (b) Inverse trigonometric function
- (c) Logarithmic function
- (d) Exponential function

3. Multi Functions

A function which is neither algebraic nor transcendental is called multi function.

Types of Multi Function

- (a) Composite function
- (b) Periodic function
- (c) Even function
- (d) Odd function
- (e) Even extensive function
- (f) Odd extensive function
- (g) Implicit function
- (h) Explicit function
- (i) Monotonic function
- (j) Increasing function
- (k) Strictly Increasing function
- (l) Decreasing function
- (m) Strictly decreasing function
- (n) One-One function or Injective function
- (o) Many-one function
- (p) Onto function or surjective function
- (q) Into function
- (r) One-One onto function or Bijective function
- (s) Partial function
- (t) Identical function
- (u) Inverse function
- (v) Bounded function

1.3.1 Algebraic Functions

Polynomial Function

A function of form $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n$, where $a_n \neq 0$ and power of x is a whole number is called polynomial function.

Some Basic Polynomial Functions

- (i) $y = f(x) = x^2$
Domain, $x \in (-\infty, \infty)$
Range, $y \in [0, \infty)$
Graph is symmetric about y -axis
- (ii) $y = f(x) = x^3$
Domain, $x \forall R$
Range, $y \forall R$
Graph is symmetric about origin or opposite Quardants

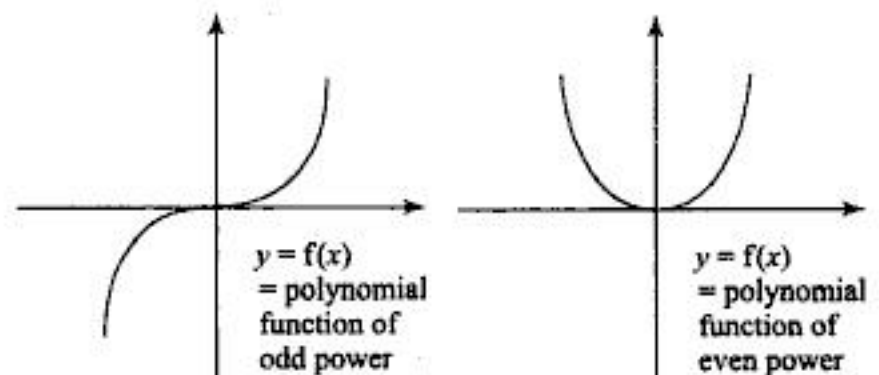


Fig. 1.11

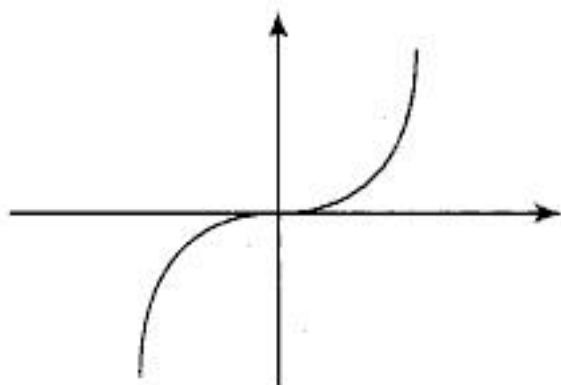


Fig. 1.12

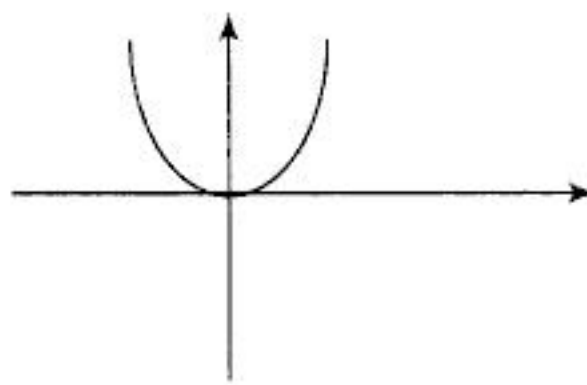


Fig. 1.13

(iii) $y = f(x) = x^4$

Domain $x \forall \mathbb{R}$

Range, $y \in [0, \infty)$

Graph is symmetric about y -axis

(iv) $y = f(x) = x^{2n}, n \in \mathbb{I}^+$

Domain $x \forall \mathbb{R}$

Range, $y \in [0, \infty)$

Graph is symmetric about y -axis

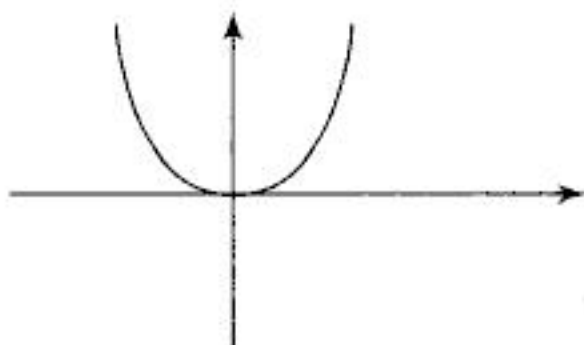


Fig. 1.14

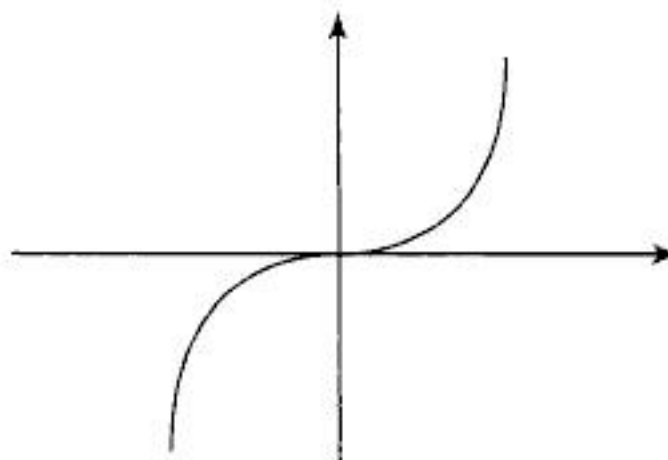


Fig. 1.15

(v) $y = f(x) = x^{2n-1}, n \in \mathbb{I}$

Domain, $x \forall \mathbb{R}$

Range, $y \forall \mathbb{R}$

Graph is symmetric about origin or opposite quadrants

Tips: (i) Graph of function $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n$ where $a_n \neq 0$ is

- { symmetrical about y -axis, when n is even integer
- { symmetrical about origin or opposite quadrant, when n is odd integer

(ii) If $y = f(x)$ is any function then its graph is

- { symmetrical about y -axis, when $f(-x) = f(x)$
- { symmetrical about origin or opposite quadrant, $f(-x) = -f(x)$

Constant Function

A function in which the value of independent variable changes but the value of dependent variable does not change then the function is called a constant function

Domain, $x \forall R$

Range, $y \in \{k\}$

Graph is symmetric about y-axis

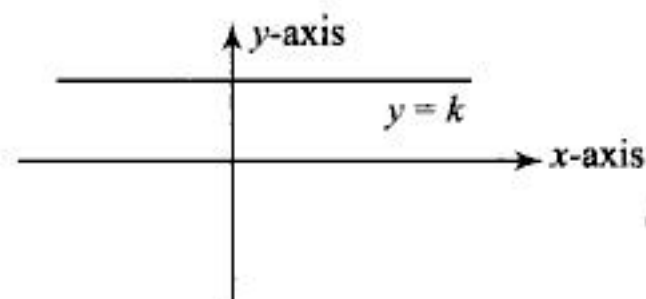


Fig. 1.16

Tips: If the first order derivative of function is equal to zero, then function must be constant function.

Tips: Graph of constant function must be parallel to x-axis.

Example 1.2 If $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ where $f''(x) = -f(x)$ and $g(x) = f'(x)$ and given $F(5) = 5$ then $F(10)$

is

(a) 5

(b) 10

(c) 0

(d) 15

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Solution Given

$$F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$$

$$\Rightarrow F'(x) = f\left(\frac{x}{2}\right) \cdot f'\left(\frac{x}{2}\right) + g\left(\frac{x}{2}\right) \cdot g'\left(\frac{x}{2}\right)$$

$$= f\left(\frac{x}{2}\right) \cdot f'\left(\frac{x}{2}\right) - f'\left(\frac{x}{2}\right) \cdot f''\left(\frac{x}{2}\right)$$

$$\{\because g(x) = f'(x) \Rightarrow g'(x) = f''(x) = -f(x)\}$$

$$\Rightarrow F'(x) = f\left(\frac{x}{2}\right) \cdot f'\left(\frac{x}{2}\right) - f'\left(\frac{x}{2}\right) \cdot f\left(\frac{x}{2}\right)$$

$$\Rightarrow F'(x) = 0$$

$\Rightarrow F(x)$ is constant function

$$\therefore F(x) = F(5) = F(10) = 5$$

Identity Function

A function of form, $y = f(x) = x$ is called Identity function.

Graph

$$y = f(x) = x$$

Domain, $x \forall R$

Range, $y \forall R$

The graph is symmetrical about origin or, opposite quadrants

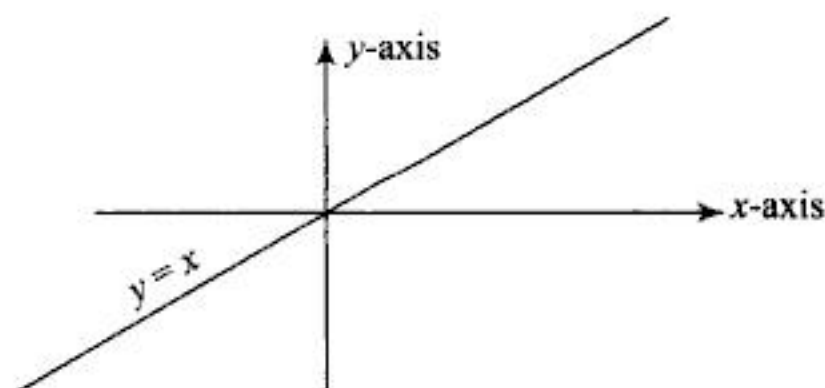


Fig. 1.17

Rational Function

If $f(x)$ and $g(x)$ are any two polynomial functions then a function of the form $y = \frac{f(x)}{g(x)}$ where $g(x) \neq 0$ is called rational function.

Some Basic Rational Functions

(i) $y = f(x) = \frac{1}{x}$

Domain, $x \in \mathbb{R} - \{0\}$

Range, $y \in \mathbb{R} - \{0\}$

The graph is symmetric about origin or opposite quadrant.

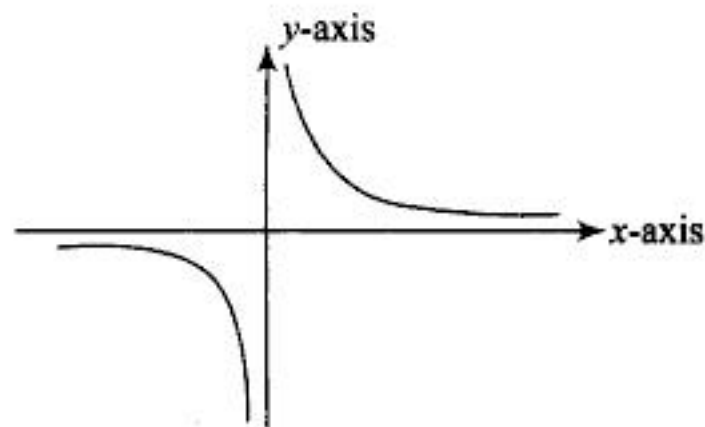


Fig. 1.18

(ii) $y = f(x) = \frac{1}{x^2}$

Domain, $x \in \mathbb{R} - \{0\}$

Range, $y \in (0, \infty)$

The graph is symmetrical about y-axis.

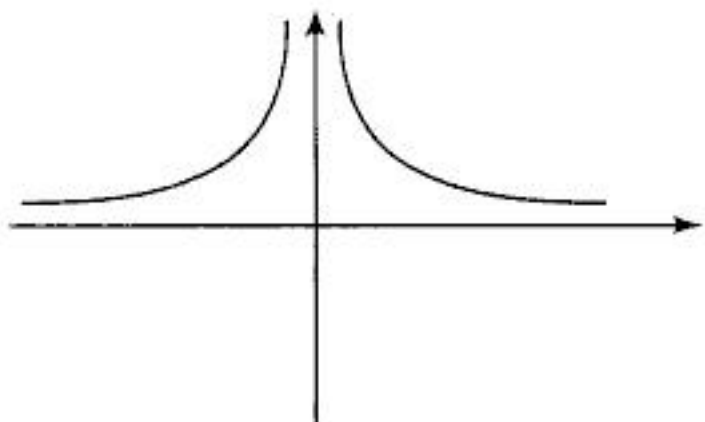


Fig. 1.19

(iii) $y = f(x) = \frac{1}{x^{2n-1}}, n \in \mathbb{I}^+$

Domain, $x \in \mathbb{R} - \{0\}$

Range, $y \in \mathbb{R} - \{0\}$

The graph is symmetric about origin or opposite quadrants

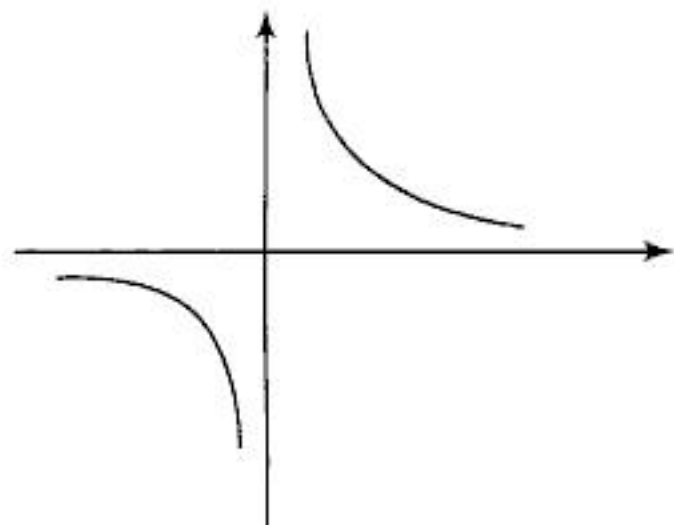


Fig. 1.20

$$(iv) \ y = \frac{1}{x^{2n}}, n \in I^+$$

Domain, $x \in R - \{0\}$

Range, $y \in (0, \infty)$

And graph is symmetrical about y-axis.

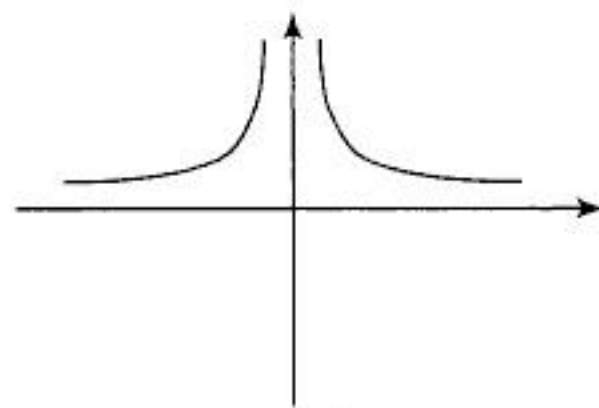


Fig. 1.21

$$(v) \ y = -\frac{1}{x^{2n}}, n \in I^+$$

Domain, $x \in R - \{0\}$

Range, $y \in (-\infty, 0)$

And graph is symmetrical about y-axis.

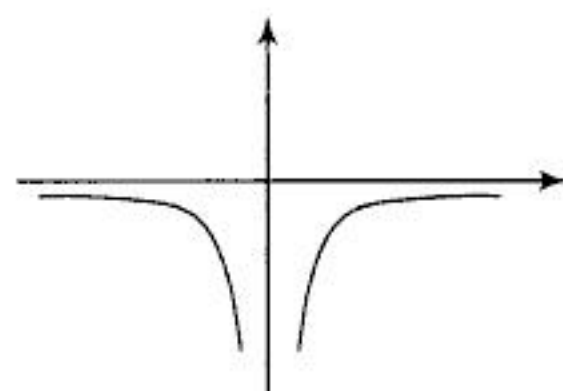


Fig. 1.22

$$(vi) \ y = -\frac{1}{(x^{2n-1})}, n \in I^+$$

Domain $x \in R - \{0\}$

Range $y \in R - \{0\}$

And graph is symmetric about origin or opposite quadrant.

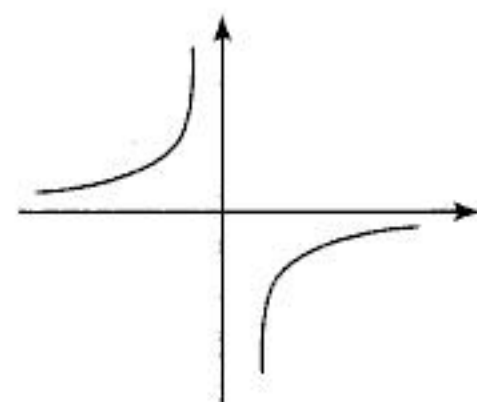


Fig. 1.23

Irrational Functions

A function of the form in which an independent variable is held under a radical sign.

Tips:

$$(a) \ f(\sqrt[2k]{x}) = \text{defined for } x \geq 0$$

$$(b) \ f(\sqrt[2k+1]{x}) = \text{defined } x \in R$$

Some basic irrational functions

$$(i) \ y = f(x) = \sqrt{x}$$

Domain, $x \in [0, \infty)$

Range, $y \in [0, \infty)$

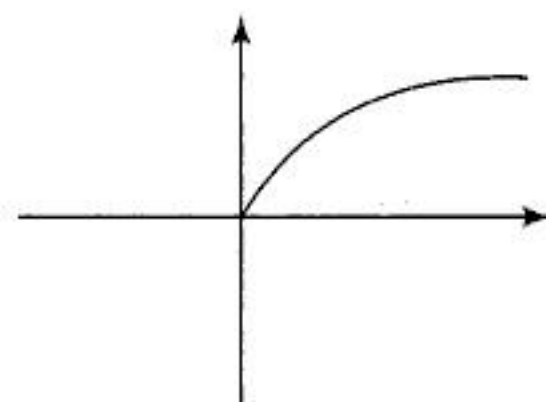


Fig. 1.24

$$(ii) y = f(x) = \sqrt[3]{x}$$

Domain, $x \forall R$

Range, $y \forall R$

And graph is
symmetrical about
origin or opposite quadrants

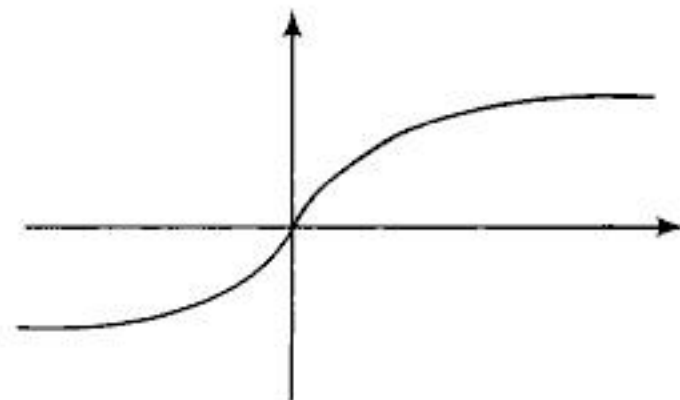


Fig. 1.25

$$(iii) y = f(x) = \sqrt[2n]{x}, n \in I^+$$

Domain

$$x \in [0, \infty)$$

Range, $y \in [0, \infty)$

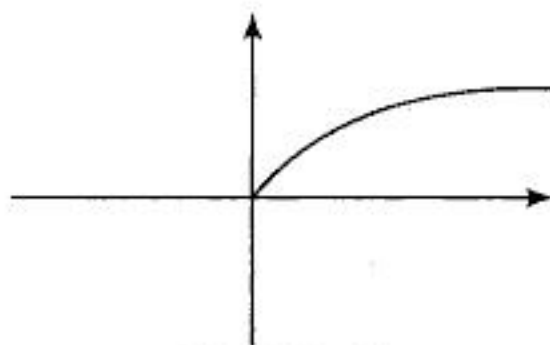


Fig. 1.26

$$(iv) y = f(x) = \sqrt[2n+1]{x}, n \in I^+$$

Domain, $x \forall R$

Range, $y \forall R$

And graph is symmetric
about origin or opposite quadrants.

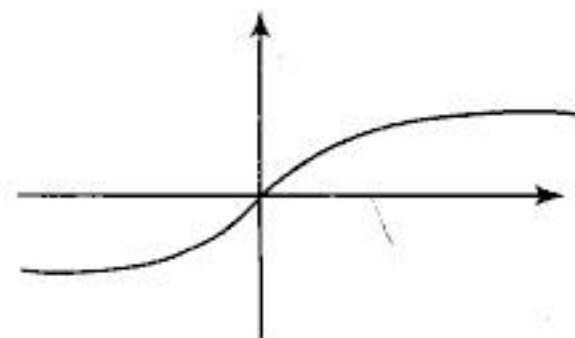


Fig. 1.27

Modulus Function

A function of the form, $y = f(x) = |x|$ is called Modulus function where

$$|x| = \begin{cases} +x, & \text{when } x > 0 \\ -x, & \text{when } x < 0 \\ 0, & \text{when } x = 0 \end{cases}$$

Graph of Modulus Function

$$y = f(x) = |x|$$

Domain, $x \forall R$

Range, $y \in [0, \infty)$

and graph is
symmetric about y-axis

Concept of Modulus

$$\text{Form I: } y = f(x) = \begin{cases} +x, & \text{when } x > 0 \\ -x, & \text{when } x < 0 \\ 0, & \text{when } x = 0 \end{cases}$$

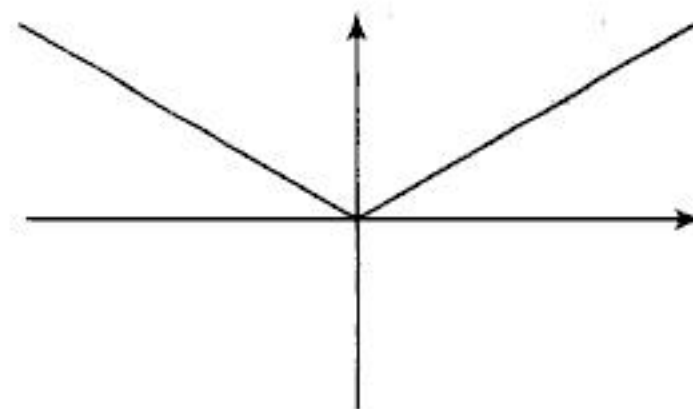


Fig. 1.28

Tip: Geometrically, $y = |x|$ represents the distance of real number x from the origin on real line.

Example 1.3 No. of solutions of equation $x^2 + 5|x| + 4 = 0$ is

- (a) 2 (b) 4 (c) 3 (d) None of these

Solution Given equation,

$$x^2 + 5|x| + 4 = 0$$

If $x > 0$ then $|x| = +x$

$$\therefore x^2 + 5|x| + 4 = 0$$

$$\Rightarrow x^2 + 5x + 4 = 0$$

$$\Rightarrow x = -1, -4 \quad (\text{Neglect})$$

Now, If $x < 0$ then $|x| = -x$

$$\therefore x^2 + 5|x| + 4 = 0$$

$$\Rightarrow x^2 - 5x + 4 = 0$$

$$\Rightarrow x = 1, 4 \quad (\text{Neglect})$$

Hence given equation has no solution.

$$\text{Form II: } y = f(x) = |x - a| = \begin{cases} +(x - a), & \text{when } x > a \\ -(x - a), & \text{when } x < a \\ 0, & \text{when } x = a \end{cases}$$

Tips: Geometrically $y = |x - a|$ represents distance between x and a on real line.

Example 1.4 Number of solutions of equation $x^2 - 3|x - 1| - 1 = 0$ are

- (a) 1 (b) 2 (c) 3 (d) 4

Solution Given equation $x^2 - 3|x - 1| - 1 = 0$

If $x > 1$ then $|x - 1| = +(x - 1)$

$$\therefore x^2 - 3|x - 1| - 1 = 0$$

$$\Rightarrow x^2 - 3(x - 1) - 1 = 0$$

$$\Rightarrow x^2 - 3x + 2 = 0$$

$$\Rightarrow x = 1, 2$$

If $x < 1$ then $|x - 1| = -(x - 1)$

$$\therefore x^2 - 3|x - 1| - 1 = 0$$

$$\Rightarrow x^2 + 3(x - 1) - 1 = 0$$

$$\Rightarrow x^2 + 3x - 3 - 1 = 0$$

$$\Rightarrow x^2 + 3x - 4 = 0$$

$$\Rightarrow x = \frac{-3 \pm \sqrt{9 + 16}}{2}$$

$$= \frac{-3 \pm 5}{2} = -4, 1$$

Hence, number of solutions of given the equation is 3.

Properties of Modulus

1. If $|f(x)| < a$ then $-a < f(x) < a$
2. If $|f(x)| \leq a$ then $-a \leq f(x) \leq a$
3. If $|f(x)| > a$ then $f(x) > a$ and $f(x) < -a$
4. If $|f(x)| \geq a$ then $f(x) \geq a$ and $f(x) \leq -a$
5. If $|f(x)| = +f(x)$ then $f(x) \geq 0$
6. If $|f(x)| = -f(x)$ then $f(x) \leq 0$
7. If $|f(x)| + |g(x)| = |f(x) + g(x)|$ then $f(x) \cdot g(x) \geq 0$
8. $|f(x) \pm g(x)| \leq |f(x)| + |g(x)|$
9. $|f(x) \pm g(x)| \geq ||f(x)| - |g(x)||$
10. $|f(x)|^{2n} = (f(x))^{2n}$, $n \in I$

Example 1.5 The sum of all the real roots of the equation $|x - 2|^2 + |x - 2| - 2 = 0$ is
 (a) 4 (b) 3 (c) 0 (d) 1

Solution Given equation

$$|x - 2|^2 + |x - 2| - 2 = 0$$

If $x > 2$ then $|x - 2| = +(x - 2)$

$$\therefore |x - 2|^2 + |x - 2| - 2 = 0$$

$$\Rightarrow (x - 2)^2 + (x - 2) - 2 = 0$$

$$\Rightarrow x^2 - 3x = 0$$

$$\Rightarrow x = 0 \text{ (neglect) and } x = 3$$

Now, If $x < 2$ then $|x - 2| = -(x - 2)$

$$\therefore |x - 2|^2 + |x - 2| - 2 = 0$$

$$\Rightarrow (x - 2)^2 - (x - 2) - 2 = 0$$

$$\Rightarrow x^2 - 5x + 4 = 0$$

$$\Rightarrow x = 1 \text{ and } x = 4 \text{ (neglect)}$$

Hence, the sum of all roots of given equation is $(1+3) = 4$

Signum Function

The form of function

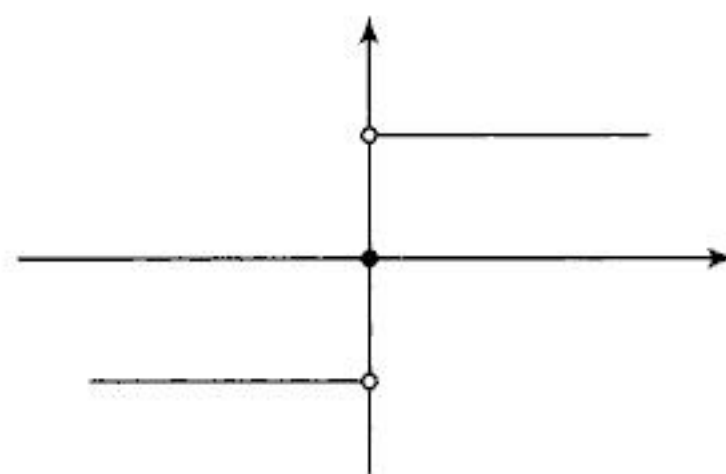
$$y = f(x) = \begin{cases} \frac{|x|}{x} \text{ or } \frac{x}{|x|} & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$$

is called signum function denoted as $\text{sgn}(x)$ where $\text{sgn}(x) = \begin{cases} +1, & \text{when } x > 0 \\ -1, & \text{when } x < 0 \\ 0, & \text{when } x = 0 \end{cases}$

Graph of Signum Function

$$\text{As } \text{sgn}(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$= \begin{cases} +1, & x > 0 \\ -1, & x < 0 \\ 0, & x = 0 \end{cases}$$

**Fig. 1.29**

Domain $x \in \mathbb{R}$

Range, $y \in \{-1, 0, 1\}$

And graph (Fig. 1.29) is symmetric about origin or in opposite quadrant.

Greatest Integral Function

If x be any real satisfy $n \leq x < n + 1$ $n \in I$ then form $y = [x]$ is called greatest integral function where $[x] = n$

e.g. $y = f(x) = [x]$

Put $x = 2.5$, $f(2.5) = [2.5] = 2$

Put $x = 2.9$, $f(2.9) = [2.9] = 2$

Put $x = 1.2$, $f(-1.2) = [-1.2] = -2$

Put $x = -3$, $f(-3) = [-3] = -3$

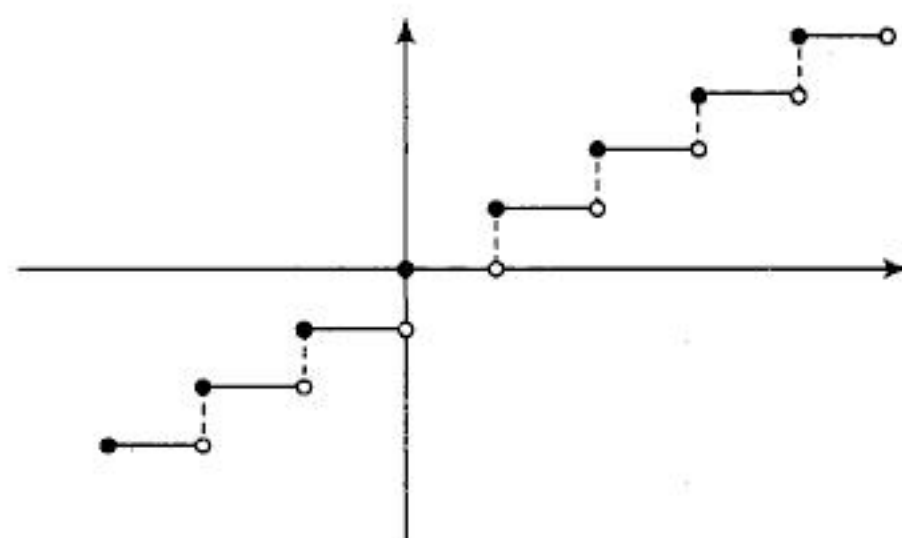
Tips: Greatest integral function is also called step wise function.

Graph of Greatest Integral Function

$$y = [x] = f(x)$$

Domain, $x \in \mathbb{R}$

Range $y \in I$

**Fig. 1.30****Properties of Greatest Integral Function**

1. $[[x]] = [x]$
2. $[x] = x$, when $x \in \mathbb{Z}$
3. $[x + a] = [x] + a$, when $a \in \mathbb{Z}$

$$4. [-x] = \begin{cases} -[x], & \text{when } x \in \mathbb{Z} \\ -[x] - 1, & \text{when } x \notin \mathbb{Z} \end{cases}$$

$$5. [-x] + [x] = \begin{cases} 0, & \text{when } x \in \mathbb{Z} \\ -1, & \text{when } x \notin \mathbb{Z} \end{cases}$$

$$6. [a + b] = \begin{cases} [a] + [b], & \text{when } 0 \leq a_f + b_f \leq 1 \\ [a] + [b] + 1, & \text{when } 0 \leq a_f + b_f \leq 2 \end{cases}$$

$$7. \text{ For every positive real } x, \left[\frac{x}{2} \right] + \left[\frac{x+1}{2} \right] = [x]$$

$$8. [x + y] \geq [x] + [y]$$

$$9. \text{ If } [f(x)] \geq L \text{ then } f(x) \geq L$$

$$10. \text{ If } [f(x)] \leq L \text{ then } f(x) \leq L + 1$$

$$11. x - 1 < [x] \leq x \text{ for all real } x.$$

$$12. \text{ For any positive real number } n, \left[\frac{n+1}{2} \right] + \left[\frac{n+2}{2} \right] + \left[\frac{n+4}{2} \right] + \left[\frac{n+8}{16} \right] + \dots + \dots = [n]$$

$$13. \text{ For positive real numbers } x \text{ and } n \in \mathbb{N}$$

$$[x] + \left[x + \frac{1}{n} \right] + \left[x + \frac{2}{n} \right] + \left[x + \frac{3}{n} \right] + \dots + \left[x + \frac{n-1}{n} \right] = [nx]$$

$$\text{e.g. } \left[\frac{6}{5} + \frac{1}{50} \right] + \left[\frac{6}{5} + \frac{2}{50} \right] + \dots + \left[\frac{6}{5} + \frac{49}{50} \right] = [60] = 60$$

Example 1.6 $[x] + \left[x + \frac{1}{50} \right] + \left[x + \frac{2}{50} \right] + \left[x + \frac{3}{50} \right] + \dots + \left[x + \frac{49}{50} \right] = 7$ then find x

Solution Given $[x] + \left[x + \frac{1}{50} \right] + \left[x + \frac{2}{50} \right] + \left[x + \frac{3}{50} \right] + \dots + \left[x + \frac{49}{50} \right] = 7$

$$\Rightarrow [50x] = 7 \Rightarrow 7 \leq 50x < 8 \Rightarrow \frac{7}{50} \leq x < \frac{8}{50} \Rightarrow x \in \left[\frac{7}{50}, \frac{8}{50} \right)$$

Least Integral Function

If x be any real satisfy $n < x < n + 1, n \in \mathbb{I}$

Then $y = (x) = n + 1$ is called least integral function.

e.g. $f(x) = (x)$

Put $x = 1.2, f(1.2) = (1.2) = 2$

$x = 1.9, f(1.9) = (1.9) = 2$

$x = 1.5, f(-1.5) = (-1.5) = -1$

$x = 2, f(2) = (2) = 2$

Graph of Least Integral Function

$$y = f(x) = (x)$$

Domain, $x \forall R$

Range, $y \forall I$ (Fig. 1.31)

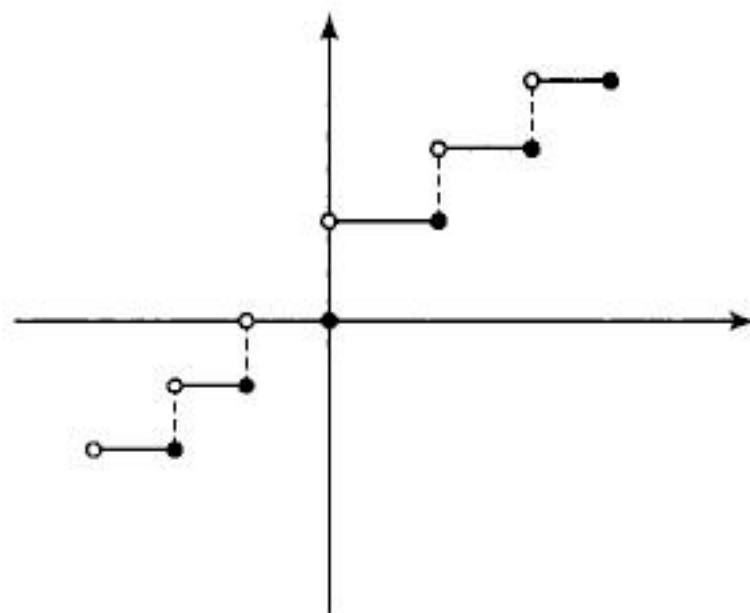


Fig. 1.31

Tips: Least integral function is also called ceiling function.

Properties of Least Integral Function

1. $(x) = [x]$, when x is integer
2. $(x) = [x] + 1$, when x is fractional.
3. $(x + n) = n + (x)$, when n is integer
4. $(-x) = -(x)$, when x is integer
5. $(-x) = -(x) + 1$, when x is fractional.

Fractional Part of Function

Form of function $y = f(x) = \{x\}$

where $\{x\} = x - [x]$ is called fractional part of function.

e.g. $y = f(x) = \{x\} = x - [x]$

Put $x = 1.2$, $f(1.2) = \{1.2\} = 1.2 - [1.2] = 1.2 - 1 = 0.2$

Put $x = -1.4$, $f(-1.4) = \{-1.4\} = -1.4 - [-1.4] = -1.4 + 2 = 0.6$

Put $x = -2$, $f(-2) = \{-2\} = -2 - [-2] = -2 - 0 = -2$

Graph of Fractional Part of Function

$$y = f(x) = \{x\} = x - [x]$$

Domain, $x \forall R$

Range, $y \in [0, 1)$ (Fig. 1.32)

Geometrical Function

The form of single-valued or multi-valued function which represents specified graph is called geometrical function.

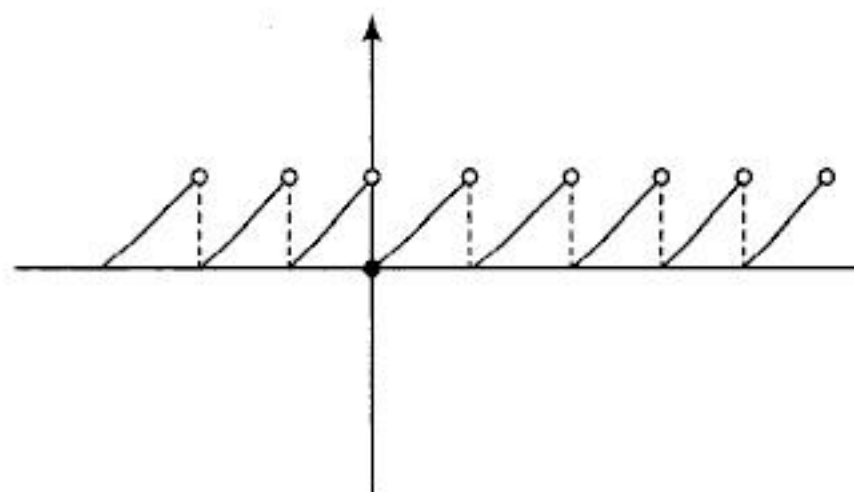


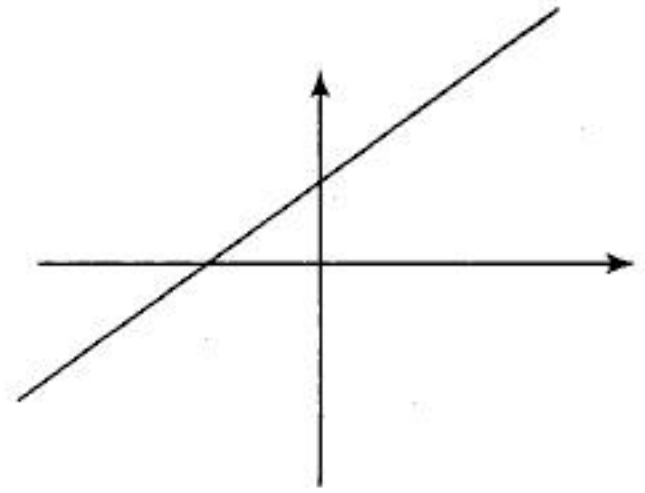
Fig. 1.32

Some Geometrical Functions**(a) Straight line**

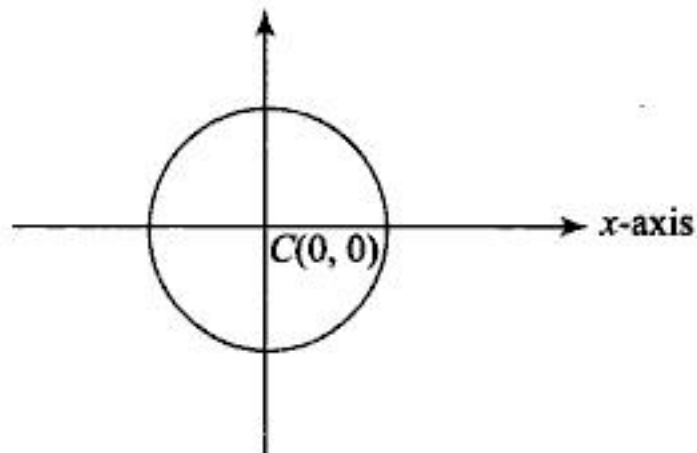
Form $ax + by + c = 0$ represents straight line

Domain, $x \forall R$

Range, $y \forall R$ (Fig. 1.33)

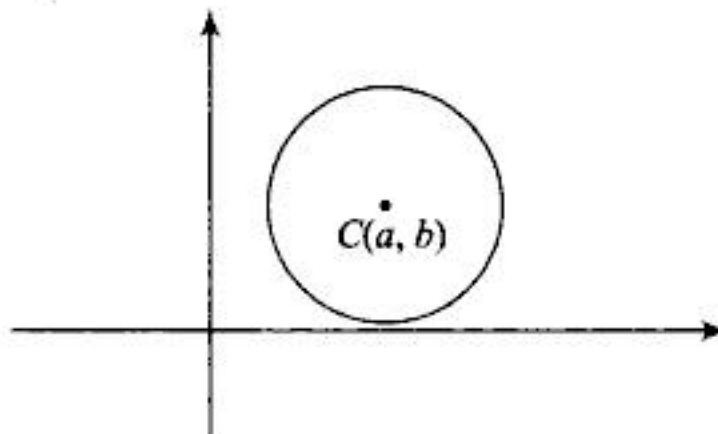
**Fig. 1.33****(b) Circle**

(i) $x^2 + y^2 = r^2$ represent a circle having centre $(0, 0)$ and radius r (Fig. 1.34)

**Fig. 1.34**

(ii) $(x - a)^2 + (y - b)^2 = r^2$

Represent circle having centre (a, b) and radius r .

**Fig. 1.35**

(iii) $x^2 + y^2 + 2gx + 2fy + c = 0$

Represent a circle having centre $C(-g, -f)$ and radius r

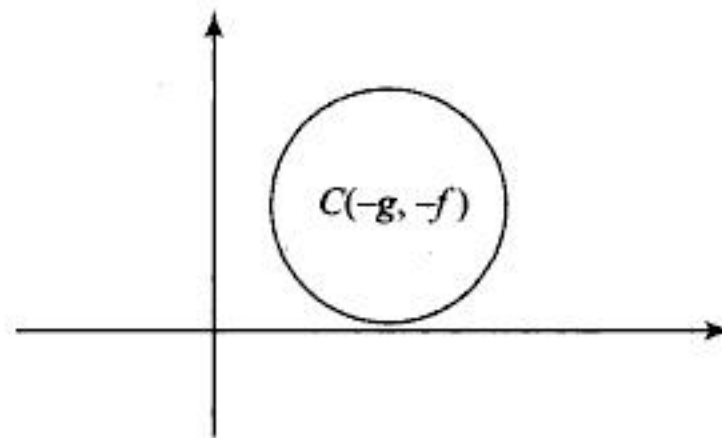


Fig. 1.36

(iv) Form $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$ represents circle having centre

$$C \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

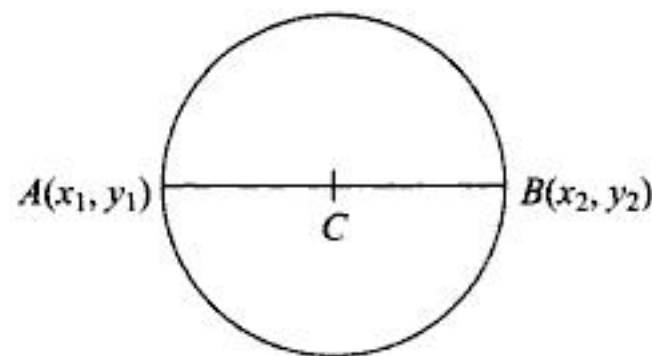


Fig. 1.37

(c) Parabola

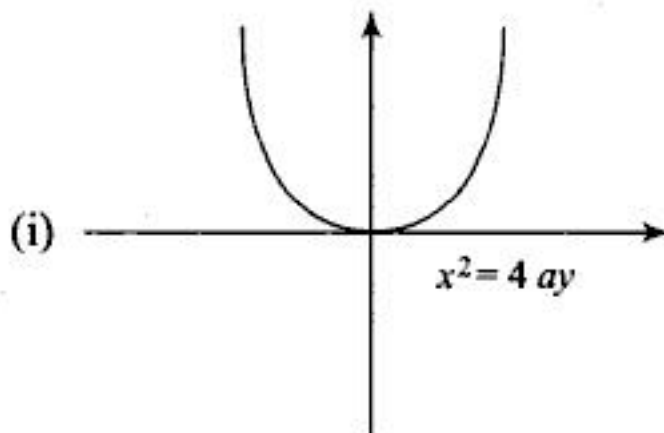


Fig. 1.38

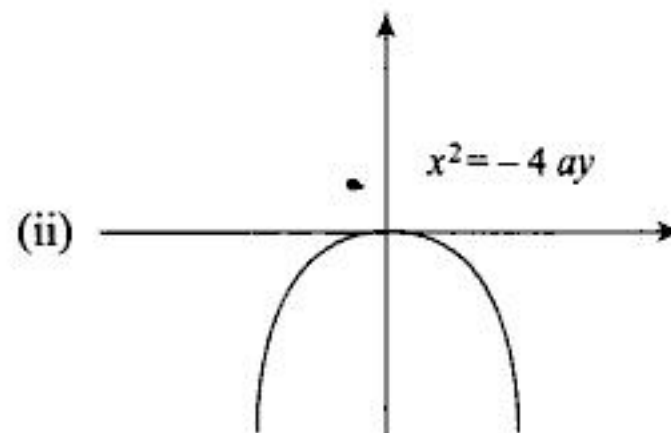


Fig. 1.39

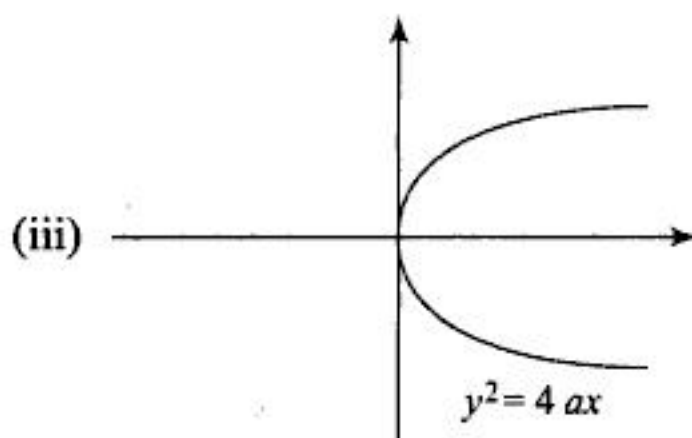


Fig. 1.40

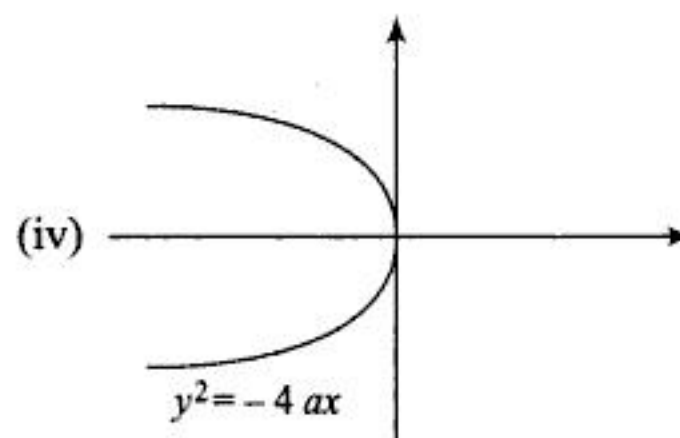


Fig. 1.41

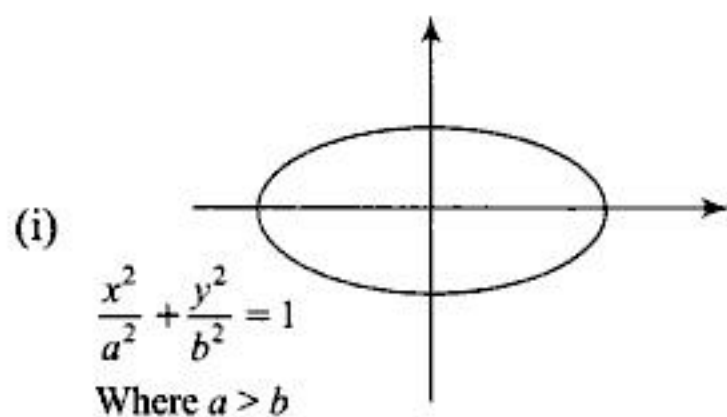
(d) Ellipse

Fig. 1.42

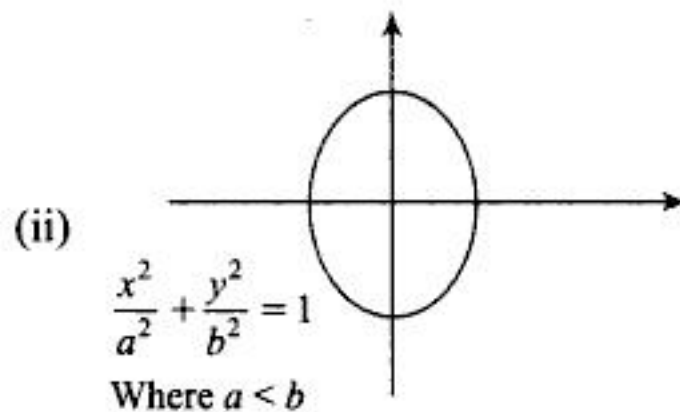


Fig. 1.43

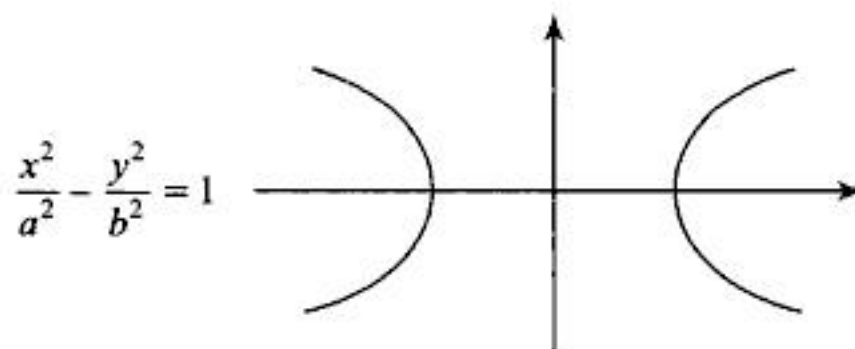
(e) Hyperbola

Fig. 1.44

Transcendental Function

The form of function in which independent variable is defined with mathematical term is called transcendental function.

e.g. $y = \log(x)$

Logarithmic Function

The form of transcendental function in which independent variable is defined with mathematical term logarithm, is called transcendental function.

Concept of Logarithm

Log of any positive real number x with base 'a' is defined as the power on base 'a' to be raised to get that number i.e.

$$\text{Log } a^x = y \Rightarrow a^y = x$$

e.g. $\log_2(x^2 - 3x + 10) = 3$

$$\Rightarrow x^2 - 3x + 10 = 2^3 = 8$$

$$\Rightarrow x^2 - 3x + 2 = 0$$

$$\Rightarrow x = 1, 2$$

There are three types of logarithms

(i) Napier's logarithm:- If base of logarithm is 'e' where $e > 2$ then logarithm is called Napier's logarithm. It is denoted as $\log e^x$ or $\log x$ or $\ln x$.

(ii) Common logarithm: If base of logarithm is '10' then logarithm is called common logarithm. It is denoted as $\log_{10} x$ or $\log x$ or $\ln x$.

(iii) Multi-logarithm: If form of logarithm is $y = \log_a x$

Where $0 < a < 1$ or $a > 1$ and $x > 0$ but $a \neq 10$, $a \neq e$ then logarithm is called multi-logarithm.

Tips: Value of logarithm $\log_a x$ may be positive, negative or zero depending on value of a and x .

As $y = \log_a x$

Here, $0 < a < 1$ or, $a > 1$ and $x > 0 \Rightarrow 0 < x < 1$ or $x = 1$ or $x > 1$

Case-I. If $a > 1$ then

$$\log_a x = \begin{cases} \text{positive,} & \text{when } x > 1 \\ \text{negative,} & \text{when } 0 < x < 1 \\ \text{zero,} & \text{when } x = 1 \end{cases}$$

Case-II. If $0 < a < 1$ then

$$\log_a x = \begin{cases} \text{positive,} & \text{when } 0 < x < 1 \\ \text{negative,} & \text{when } x > 1 \\ \text{zero,} & \text{when } x = 1 \end{cases}$$

Graph

(i) $y = \log_a x$ when $a > 1$

Domain, $x \in (0, \infty)$

Range, $y \forall R$

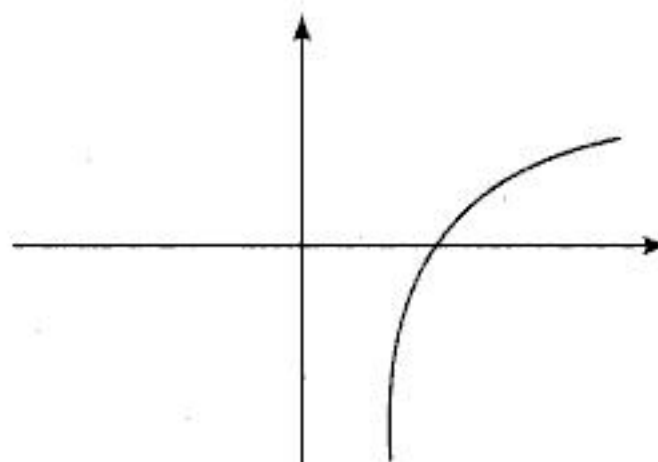


Fig. 1.45

(ii) $y = \log_a x$ when $0 < a < 1$

Domain, $x \in (0, \infty)$

Range, $y \forall R$

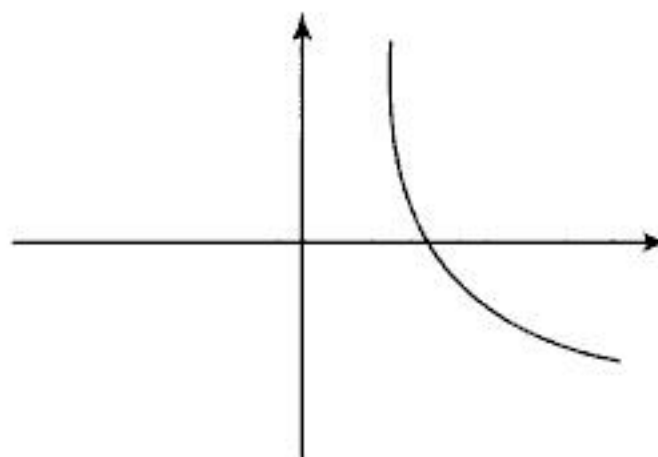


Fig. 1.46

Formulae Related to Logarithm

- (i) $\log_a (m \cdot n) = \log_a m + \log_a n$
- (ii) $\log_a (m + n) \neq \log_a m + \log_a n$
- (iii) $\log_a \left(\frac{m}{n} \right) = \log_a m - \log_a n$
- (iv) $\log_a (m - n) \neq \log_a m - \log_a n$
- (v) $\log_a (x^n) = n \log_a x$
- (vi) $\log_{a^n} x = \frac{1}{n} \log_a x$
- (vii) $\log_{(a^n)} x^n = \log_a x$
- (viii) $\log_{(a^\alpha)} (x^\beta) = \left(\frac{\beta}{\alpha} \right) \log_a x$
- (ix) $\log_a (1/x) = -\log_a x$
- (x) $\log_{\left(\frac{1}{a} \right)} x = -\log_a x$
- (xi) $\log_{\left(\frac{1}{a} \right)} \left(\frac{1}{x} \right) = \log_a x$
- (xii) $a^{(\log_a f(x))} = f(x)$
- (xiii) $a^{(\log b^c)} = c^{(\log b^a)}$
- (xiv) $\log a^a = 1$
- (xv) $\log a^{(1)} = 0$
- (xvi) $\log_a (\text{negative no.}) = \text{Undefined}$
- (xvii) $\log_a f(x) > 0$ Equivalent to $\begin{cases} f(x) > 1, & \text{when } a > 1 \\ \text{or, } 0 < f(x) < 1, & \text{when } 0 < a < 1 \end{cases}$
- (xviii) $\log_a f(x) \geq 0$ Equivalent to $\begin{cases} f(x) \geq 1, & \text{when } a > 1 \\ \text{or, } 0 < f(x) \leq 1, & \text{when } 0 < a < 1 \end{cases}$
- (xix) $\log_a f(x) < 0$ Equivalent to $\begin{cases} f(x) > 1, & \text{when } 0 < a < 1 \\ \text{or, } 0 < f(x) < 1, & \text{when } a > 1 \end{cases}$
- (xx) $\log_a f(x) \leq 0$ Equivalent to $\begin{cases} f(x) \geq 1, & \text{when } 0 < a < 1 \\ \text{or, } 0 < f(x) \leq 1, & \text{when } a > 1 \end{cases}$

$$\begin{aligned}
 \text{(xxi) } \log_a f(x) \geq \log_a g(x) \text{ Equivalent to } & \begin{cases} f(x) \geq g(x) & \text{when } a > 1 \\ \text{or,} \\ f(x) \leq g(x) & \text{when } 0 < a < 1 \end{cases} \\
 \text{(xxii) } \log_a f(x) \leq \log_a g(x) \text{ Equivalent to } & \begin{cases} f(x) \leq g(x) & \text{when } a > 1 \\ \text{or,} \\ f(x) \geq g(x), & \text{when } 0 < a < 1 \end{cases}
 \end{aligned}$$

Exponential Function

The form of function

$y = f(x) = (\text{constant})^{\text{variable}} = a^x$ is called exponential function

Graph

(i) $y = f(x) = a^x$, where $a > 1$

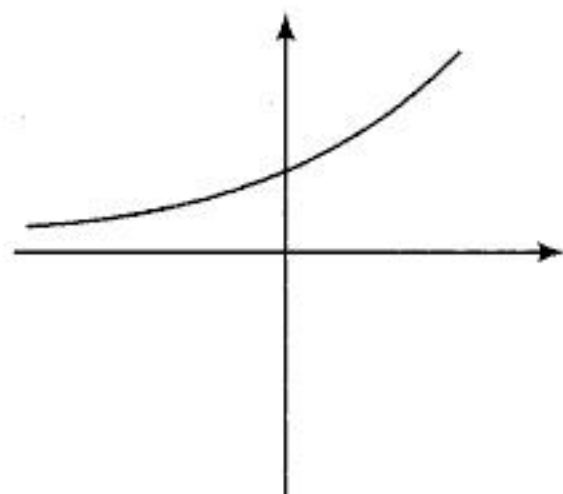


Fig. 1.47

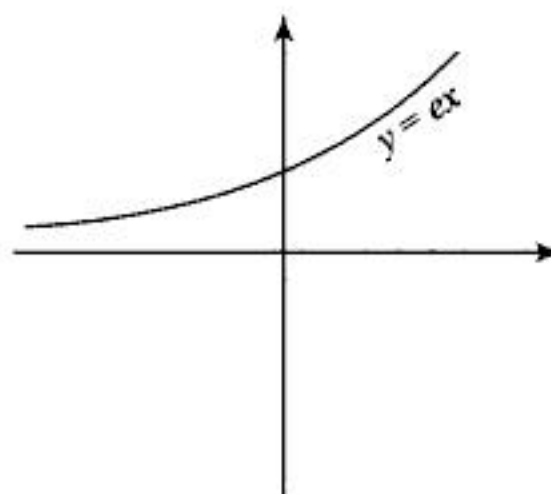


Fig. 1.48

(ii) $y = f(x) = a^x$, where $0 < a < 1$

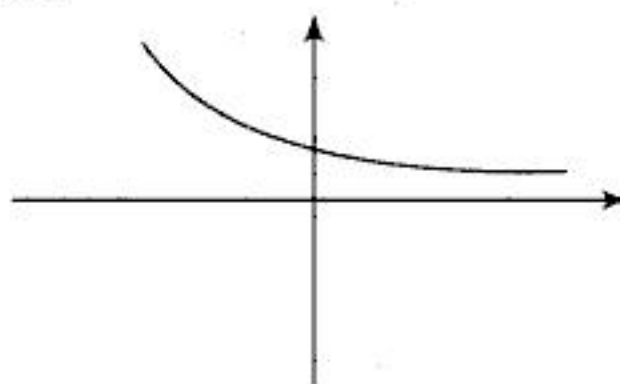


Fig. 1.49

1.3.2 Transcendental Function

Trigonometrical Function

The form of transcendental function in which independent variable is defined with trigonometrical ratio is called trigonometrical function.

(i) $y = f(x) = \sin x$

Domain, $x \in \mathbb{R}$ Range, $y \in [-1, 1]$

and graph is symmetric about origin.

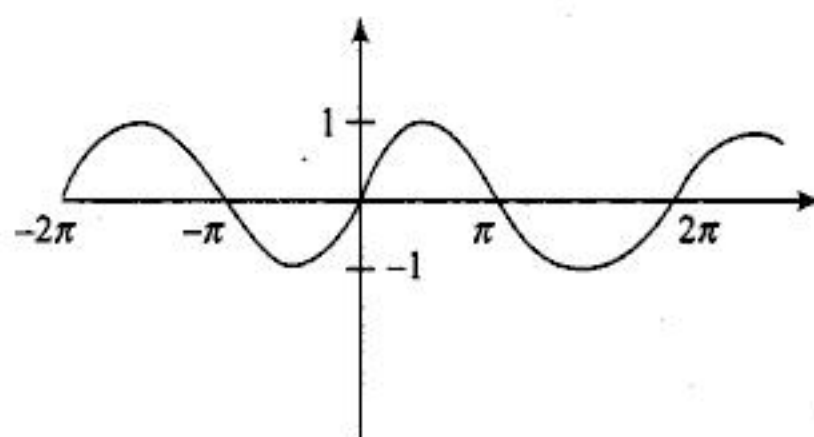


Fig. 1.50

(ii) $y = \cos x$

Domain, $x \in \mathbb{R}$ Range, $y \in [-1, 1]$

and graph is symmetric about y-axis.

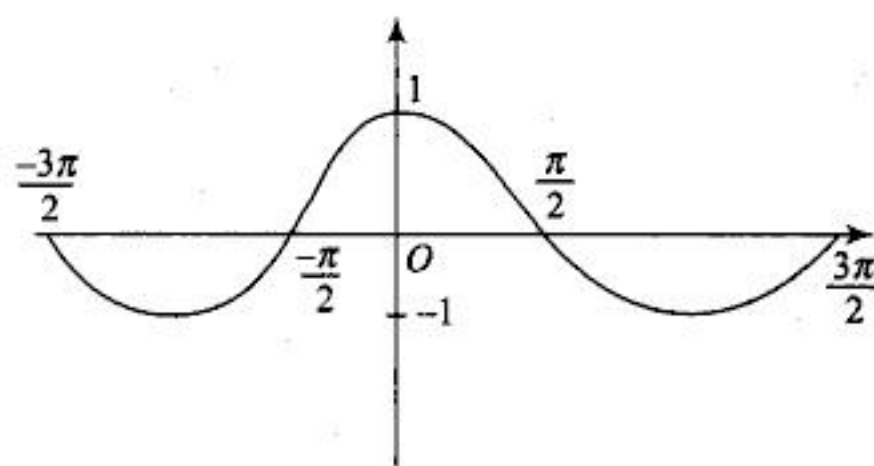


Fig. 1.51

(iii) $y = \tan x$

Domain, $x \in \mathbb{R} - (2n + 1)\frac{\pi}{2}, n \in \mathbb{I}$ Range, $y \in \mathbb{R}$

and graph is symmetric about origin.

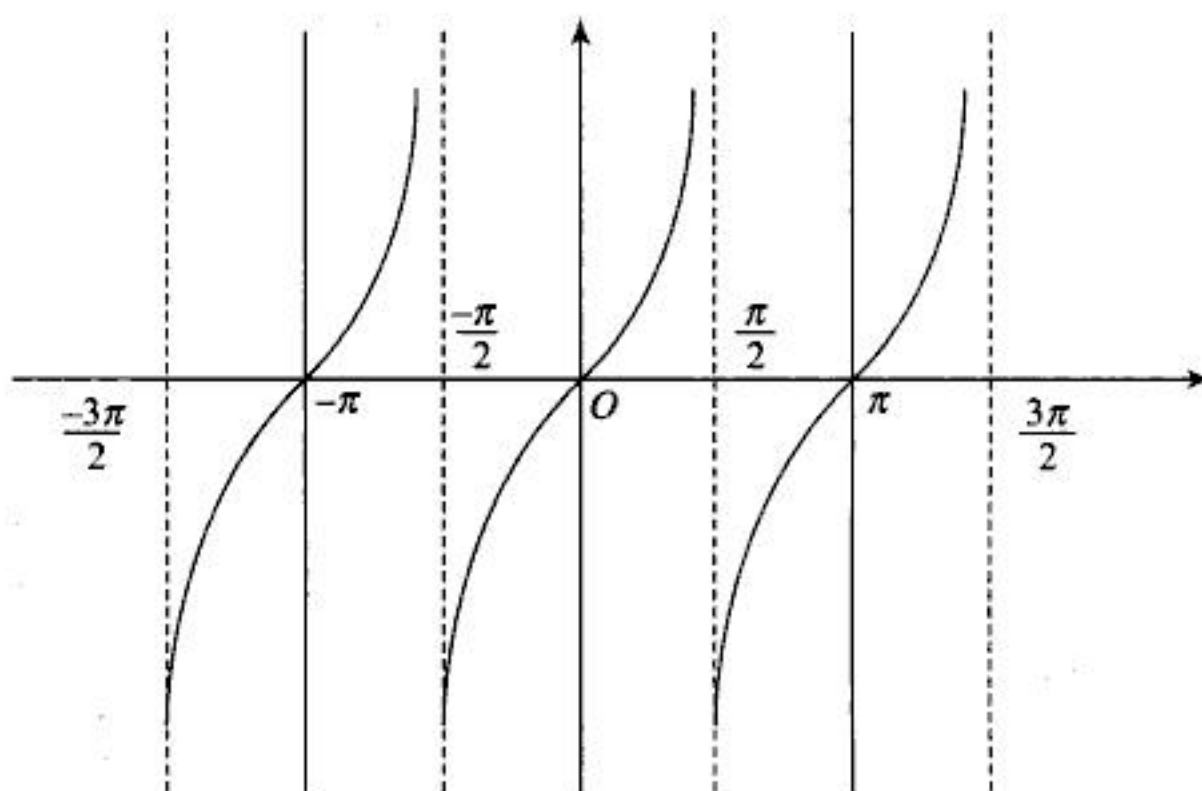


Fig. 1.52

(iv) $y = \cot x$

Domain, $x \in \mathbb{R} - (n\pi), n \in \mathbb{I}$ Range, $y \in \mathbb{R}$

and graph is symmetric about origin.

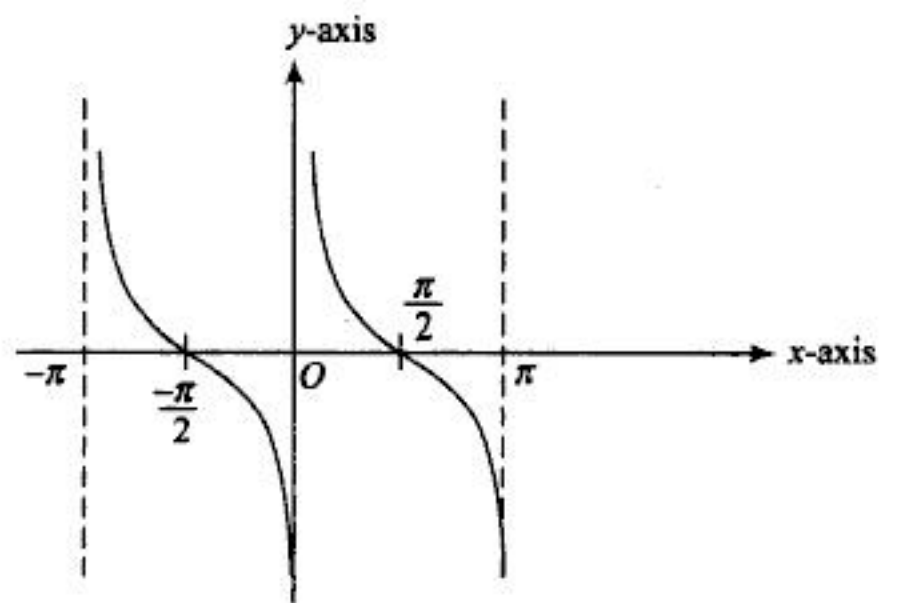


Fig. 1.53

(v) $y = \operatorname{cosec} x$

Domain, $x \in \mathbb{R} - (n\pi), n \in \mathbb{I}$ Range, $y \in \mathbb{R} - (-1, 1)$

and graph is symmetric about origin.

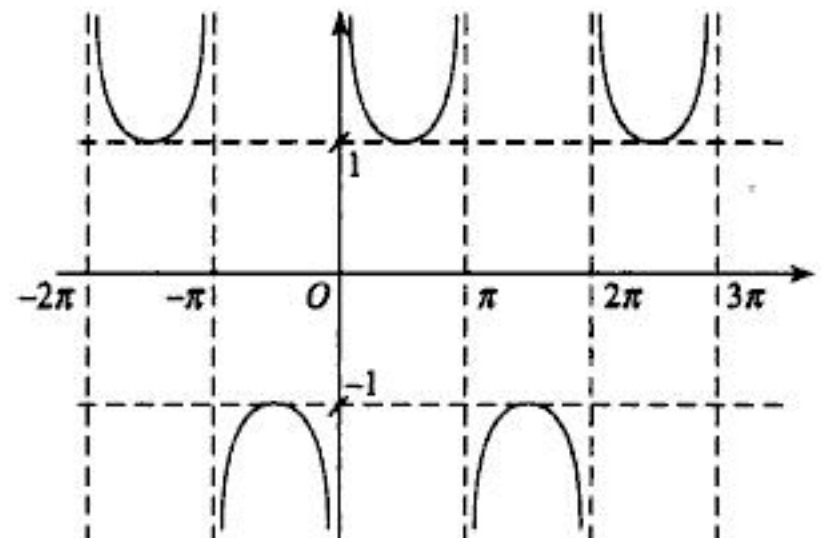


Fig. 1.54

(vi) $y = \sec x$

Domain, $x \in \mathbb{R} - (2n + 1)\frac{\pi}{2}$ Range, $y \in \mathbb{R} - (-1, 1)$

and graph is symmetric about y-axis.

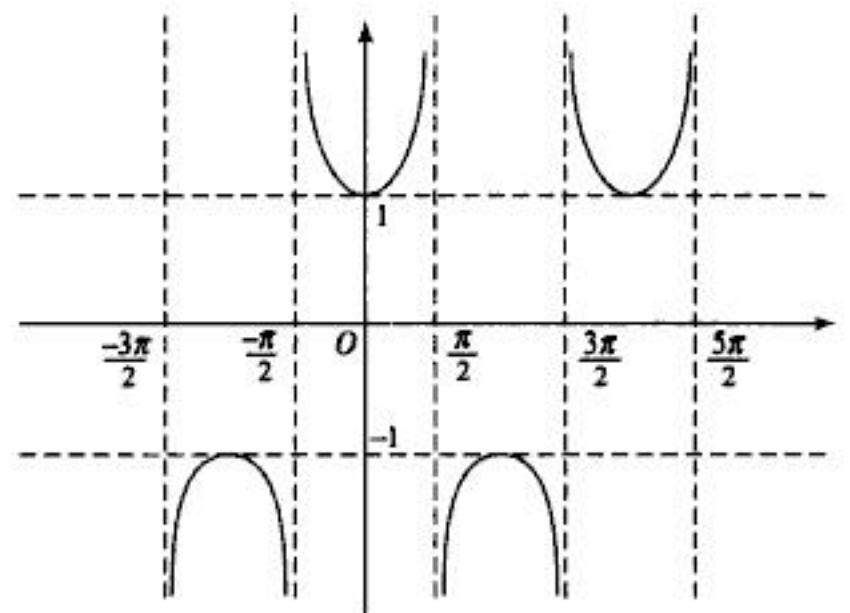


Fig. 1.55

Inverse Trigonometrical Function

The form of transcendental function in which independent variable is held with inverse form of trigonometrical ratio, is called inverse trigonometrical function.

(i) $y = f(x) = \sin^{-1} x$

Domain, $x \in [-1, 1]$

Range, $y \in R$

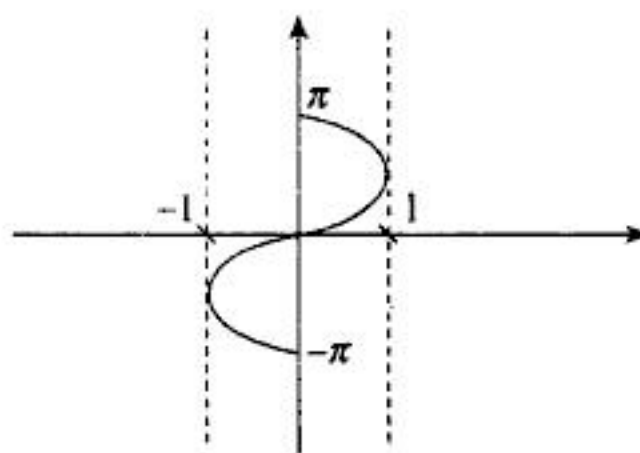


Fig. 1.56

(ii) $y = f(x) = \cos^{-1} x$

Domain, $x \in [-1, 1]$

Range, $y \in R$

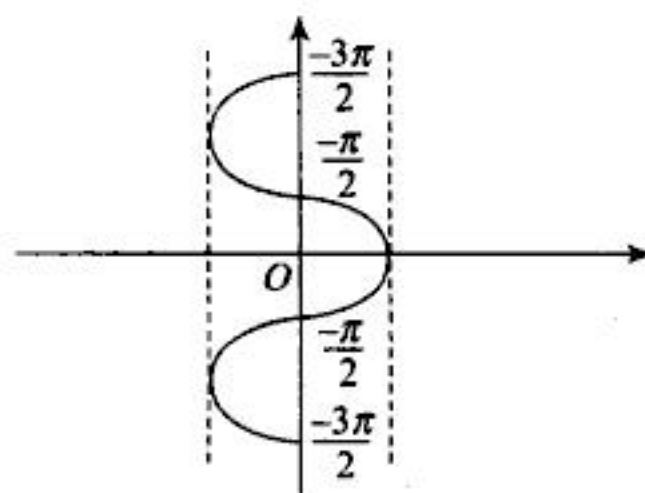


Fig. 1.57

(iii) $y = f(x) = \tan^{-1} x$

Domain, $x \in R$

Range, $y \in R - (2n + 1)\frac{\pi}{2}$

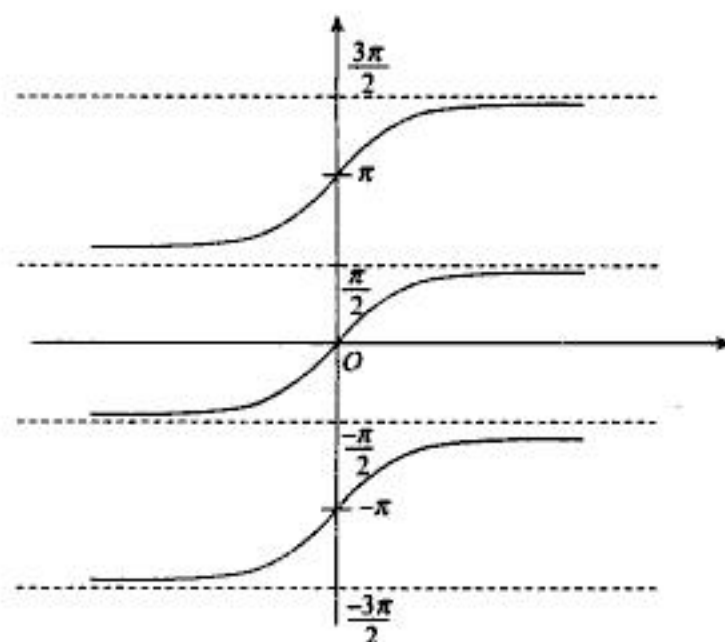


Fig. 1.58

(iv) $y = f(x) = \cot^{-1} x$

Domain, $x \in \mathbb{R}$

Range, $y \in \mathbb{R} - (n\pi), n \in \mathbb{I}$

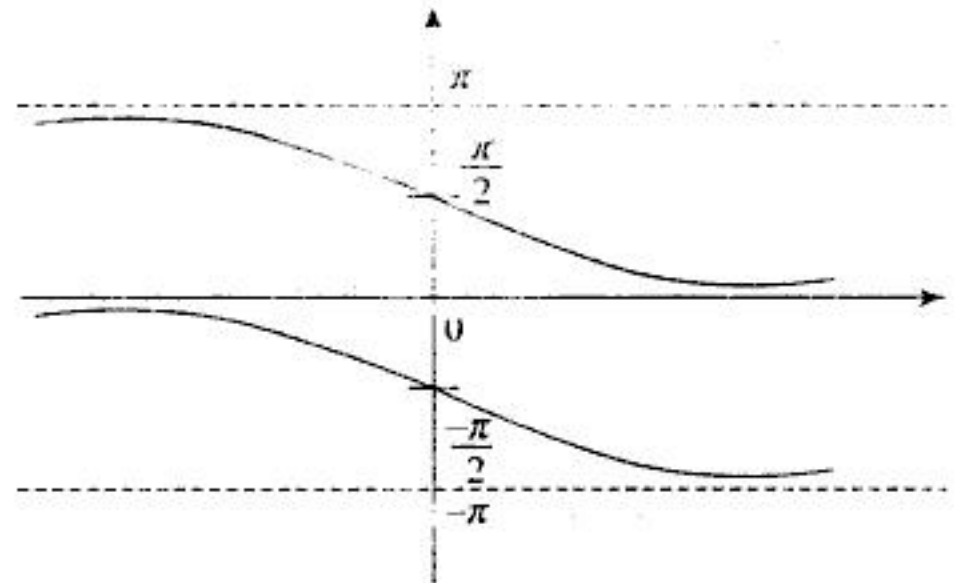


Fig. 1.59

(v) $y = f(x) = \sec^{-1} x = \cos^{-1} \left(\frac{1}{x} \right)$

Domain, $x \in \mathbb{R} - (-1, 1)$

Range, $y \in \mathbb{R} - (2n+1)\frac{\pi}{2}, n \in \mathbb{I}$

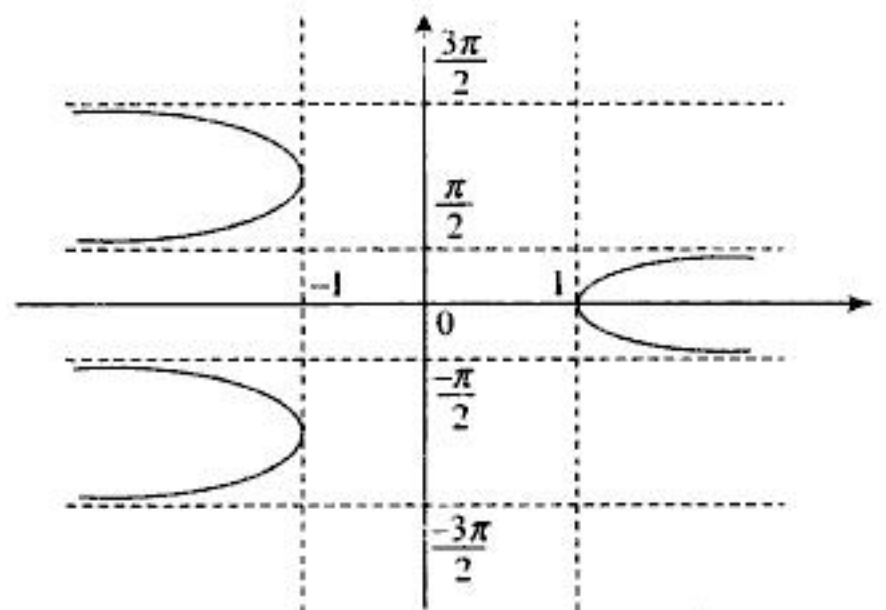


Fig. 1.60

(vi) $y = f(x) = \operatorname{cosec}^{-1} x = \sin^{-1} \left(\frac{1}{x} \right)$

Domain, $x \in \mathbb{R} - (-1, 1)$

Range, $y \in \mathbb{R} - (n\pi), n \in \mathbb{I}$

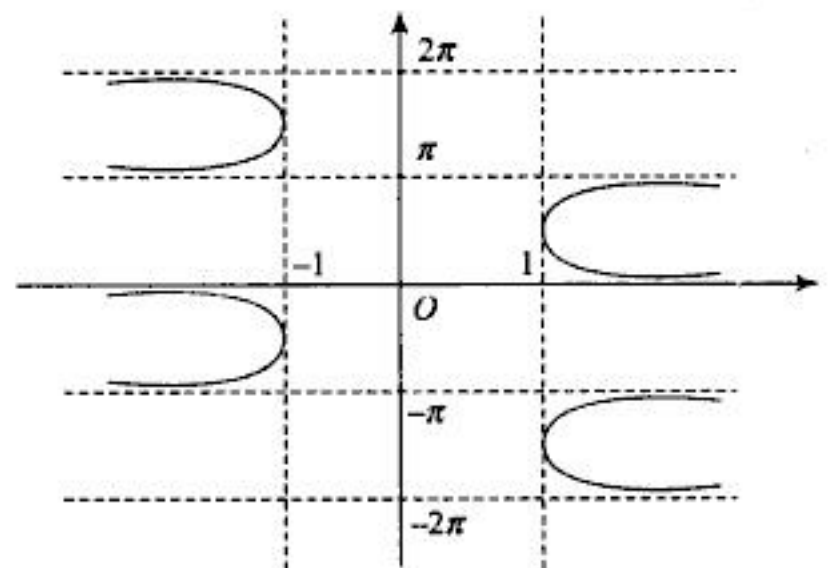


Fig. 1.61

1.3.3 Multi Function

Composite Function

The composition of two or more functions with respect to independent variable, is called composite function.

If $f(x)$ and $g(x)$ are any two functions then

$\text{fog}(x) = f\{g(x)\} \equiv$ Composition of function $g(x)$ in $f(x)$ w.r.t. x

similarly, $\text{gof} = g\{f(x)\} \equiv$ Composition of function $f(x)$ in $g(x)$ w.r.t. x

Tips: The domain of $\text{fog}(x)$ consist of the numbers x in the domain of $g(x)$ for which $g(x)$ lies in the domain of $f(x)$.

Example 1.7 If $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$ then for all x find $\text{fog}(x)$. IIT-JEE 2001

Solution Given,

$$g(x) = 1 + x - [x] = 1 + \{x\} > 1$$

$\Rightarrow g(x)$ is greater than 1

Hence, $\text{fog}(x) = f(g(x)) = 1$

Example 1.8 If $f(x) = \frac{1}{1-x}$ then find $\text{fofof}(x)$.

Solution Given $f(x) = \frac{1}{1-x}$

$$\begin{aligned} \text{As } f\{f(x)\} &= \frac{1}{1-f(x)} = \frac{1}{1-\frac{1}{1-x}} \\ &= \frac{1-x}{-x} \end{aligned}$$

$$\begin{aligned} \text{Hence, } \text{fofof}(x) &= f\{f\{f(x)\}\} \\ &= \frac{1}{1-f\{f(x)\}} = \frac{1}{1-\left(\frac{1-x}{-x}\right)} = x \end{aligned}$$

Example 1.9 let $f(x) = \frac{x-1}{x+1}$, $f^2(x) = f\{f(x)\}$, $f^3(x) = f\{f^2(x)\}$, then find $f^{2010}(x)$

Solution $f(x) = \frac{x-1}{x+1}$

$$f^2(x) = f\{f(x)\} = \frac{f(x)-1}{f(x)+1} = \frac{\left(\frac{x-1}{x+1}\right)-1}{\left(\frac{x-1}{x+1}\right)+1} = -\frac{1}{x}$$

$$f^3(x) = f\{f^2(x)\} = \frac{f^2(x)-1}{f^2(x)+1} = \frac{x+1}{1-x}$$

$$f^4(x) = f\{f^3(x)\} = \frac{f^3(x)-1}{f^3(x)+1} = x$$

$$f^5(x) = f\{f^4(x)\} = \frac{f^4(x)-1}{f^4(x)+1} = \frac{x-1}{x+1}$$

Here $f^k(x)$ repeats itself at an interval of $k = 4$

$$f^{2010}(x) = f^2(x) = -\frac{1}{x} \left\{ \begin{array}{l} 2010 = 502 \times 4 + 2 \end{array} \right.$$

Example 1.10 let $f: \{2, 3, 4, 5\} \rightarrow \{3, 4, 5, 9\}$ and $g: \{3, 4, 5, 9\} \rightarrow \{7, 11, 15\}$ be function defined as $f(2) = 3$, $f(3) = 4$, $f(4) = f(5) = 5$ and $g(3) = g(4) = 7$, $g(5) = g(9) = 11$ Find $\text{gof}(x)$

Solution Given

$$\begin{aligned} \text{As } \text{gof}(2) &= g\{f(2)\} = g(3) = 7 \\ \text{gof}(3) &= g\{f(3)\} = g(4) = 7 \\ \text{gof}(4) &= g\{f(4)\} = g(5) = 11 \\ \text{gof}(5) &= g\{f(5)\} = g(5) = 11 \\ \therefore \text{gof}(x) &= g\{(2, 7), (3, 7), (4, 11), (5, 11)\} \end{aligned}$$

Properties of Composite Function

1. If $f(x)$ and $g(x)$ are any two functions, then $\text{fog}(x)$ exists if the range of $g(x)$ is domain of $f(x)$.
2. If $\text{gof}(x)$ is one-one function, then $f(x)$ is one-one function.
3. If $\text{gof}(x)$ is onto function then $g(x)$ is also onto function.
4. If $f(x)$ and $g(x)$ are one-one and onto function then $\text{gof}(x)$ is one-one onto function.
5. If $f(x)$ and $g(x)$ are any two function then $\text{fog}(x) \neq \text{gof}(x)$ but if $\text{fog}(x) = \text{gof}(x)$ then $\text{fog}(x) = x$.

Periodic Function

Graphically, if graph of function repeats with certain interval then function is called periodic function and the magnitude of certain interval is called periodic function.

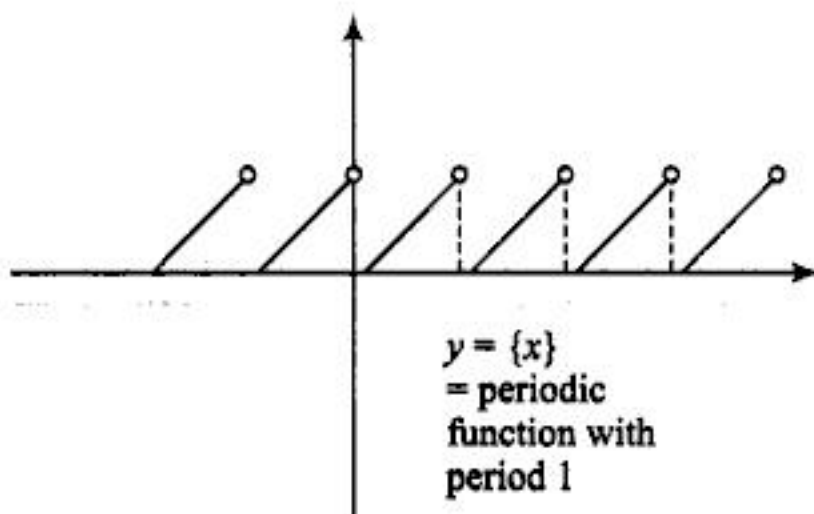


Fig. 1.62

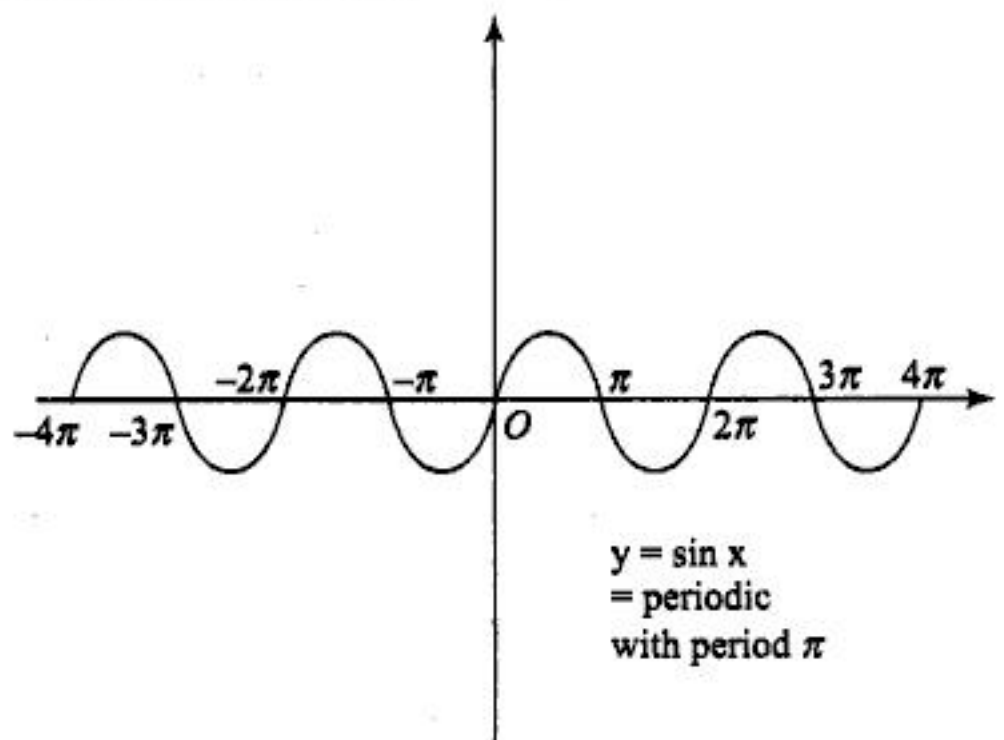


Fig. 1.63

Mathematically. If function $f(x)$ is called periodic function then $f(T + x) = f(x)$ where $|T|$ is called period of $f(x)$.

e.g. $f(x) = \sin x$

$$f(2\pi + x) = \sin(2\pi + x) = \sin x = f(x)$$

$\therefore f(x) = \sin x$ is periodic function with period 2π

For evaluate period of periodic function use:

- (i) write given function $f(x)$
- (ii) for periodic function $f(T + x) = f(x)$
then use following formula which ever to be required
 - (a) If $\sin \theta = \sin \alpha$ then $\theta = n\pi + (-1)^n \alpha$
 - (b) If $\cos \theta = \cos \alpha$ then $\theta = 2n\pi + \alpha$
 - (c) If $\tan \theta = \tan \alpha$ then $\theta = n\pi + \alpha$
- (iii) Put least positive integer n for which variable to be eliminate to get period T of given function $f(x)$

e.g. $f(x) = \sqrt{\sin x}$

for periodic function

$$f(T + x) = f(x)$$

$$\Rightarrow \sqrt{\sin(T + x)} = \sqrt{\sin x}$$

$$\Rightarrow \sin(T + x) = \sin x$$

$$\Rightarrow \sin(T + x) = n\pi + (-1)^n x$$

$$\text{Put } n = 2, T + x = 2\pi + x$$

$$\Rightarrow T = 2\pi$$

$\therefore f(x) = \sqrt{\sin x}$ is periodic function with period 2π

Properties of Periodic Function

1. If a constant quantity added, subtracted, multiplied or divided in periodic function, the period remains same i.e.

If $f(x)$ is periodic function with period T then

(a) $k \cdot f(x)$ is periodic with period T

(b) $f(x \pm k)$ is periodic with period T

(c) $f(x) \pm k$ is periodic with period T

e.g. $f(x) = \sin(x + 2009) - 2010 \equiv$ Periodic with period 2π

2. If $f(x)$ is periodic function with period T then $f(kx)$ is also periodic function with period $\frac{T}{|k|}$

where $k \neq 0$

e.g. $f(x) = \tan(2010x - 5) \equiv$ Periodic with period $\left(\frac{\pi}{2010}\right)$

3. Function $\sin^n x$, $\cos^n x$, $\sec^n x$ and $\operatorname{cosec}^n x$ are periodic then their period is defined as

$$\begin{cases} \pi, & \text{when } n \text{ is even integer} \\ \text{or,} \\ 2\pi, & \text{when } n \text{ is odd integer or fractional} \end{cases}$$

e.g. $f(x) = \sin^{2009} x$
 $\equiv$ Periodic with period 2π

But $f(x) = \sin^{2010} x$
 $\equiv$ Periodic with period π

4. Function $\tan^n x$ and $\cot^n x$ are periodic with period π for any real value of n

e.g. $f(x) = \tan^{2010} x$
 $\equiv$ Periodic with period π

5. Function $|\sin x|$, $|\cos x|$, $|\tan x|$, $|\cot x|$, $|\sec x|$ and $|\operatorname{cosec} x|$ are periodic with period π .

6. Algebraic function is periodic but period does not exist.

7. A constant function is periodic but does not have fundamental period.

8. If $f(x)$ is trigonometrical function ($\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, $\operatorname{cosec} x$) then $\{f(x)\}^0$ is also periodic with period π

9. If $f(x)$ and $g(x)$ are periodic function with period T_1 and T_2 respectively then period of the function $y = f(x) \pm g(x)$ is defined as

$$\begin{cases} \text{L.C.M. of } (T_1, T_2), & \text{when at least one of } f(x) \text{ \& } g(x) \text{ be odd function} \\ \text{or,} \\ \frac{1}{2}(\text{L.C.M. of } T_1 \text{ and } T_2) & \text{when } f(x) \text{ and } g(x) \text{ are even function} \end{cases}$$

Note:

(i) L.C.M of T_1 and T_2 means

$$\frac{\text{l.c.m. of numerator of } T_1 \text{ and } T_2}{\text{h.c.f. of denominator of } T_1 \text{ and } T_2}$$

(ii) L.C.M of rational with rational is possible.

(iii) L.C.M. of irrational with irrational is possible.

(iv) L.C.M rational and irrational is not possible.

e.g. $f(x) = 3 \sin(2009x - 5) - 4 \tan(2010x - 7)$

here, period of $3 \sin(2009x - 5) = \frac{2\pi}{2009}$

and period of $4 \tan(2010x - 7) = \frac{\pi}{2010}$

$$\therefore \text{period of } f(x) = \text{l.c.m. of } \frac{2\pi}{2009} \text{ and } \frac{2\pi}{2010}$$

$$= \frac{\text{l.c.m. of } 2\pi \text{ and } \pi}{\text{h.c.f. of } 2009 \text{ and } 2010}$$

$$= 2\pi$$

e.g. $f(x) = \cos 6\pi x + \{2x\}$

Here, period of $\cos 6\pi x = \frac{2\pi}{6\pi} = \frac{1}{3}$

and period of $\{2x\} = \frac{1}{2}$

$$\therefore \text{period of } f(x) = \text{l.c.m. of } \left(\frac{1}{3} \text{ and } \frac{1}{2}\right) = \frac{\text{l.c.m. of } 1 \text{ and } 1}{\text{h.c.f. of } 3 \text{ and } 2} = 1$$

e.g. $f(x) = \cos 3\pi x + \sec 7\pi x$

Here, period of $\cos 3\pi x = \frac{2\pi}{3\pi} = \frac{2}{3}$

and period of $\sec 7\pi x = \frac{2\pi}{7\pi} = \frac{2}{7}$

$$\therefore \text{period of } f(x) = \frac{1}{2} \left\{ \text{l.c.m. of } \frac{2}{3} \text{ \& } \frac{2}{7} \right\} = 1$$

10. If $f(x)$ and $g(x)$ are two periodic functions with period T_1 and T_2 respectively then $y = f(x) \cdot g(x)$ may be periodic with period defined as

$$\begin{cases} \text{l.c.m. of } T_1 \text{ and } T_2, \text{ when } f(x) \text{ and } g(x) \text{ are odd function} \\ \text{or} \\ \frac{1}{2}(\text{l.c.m. of } T_1 \text{ and } T_2) \text{ when at least one of } f(x), g(x) \text{ be even function.} \end{cases}$$

e.g. $f(x) = 2 \sin x \cdot \cos x$

Here, $g(x) = \sin x \equiv \text{period } 2\pi$

$$\equiv \text{odd function}$$

and $h(x) = \cos x \equiv \text{period } 2\pi$

$$\equiv \text{even function}$$

$\therefore$ Period of $f(x) = 2 \sin x \cos x$

$$= \frac{1}{2} (\text{l.c.m. of } 2\pi \text{ and } 2\pi) = \pi$$

Alternative

$$\begin{aligned} f(x) &= 2 \sin x \cos x \\ &= \sin 2x \\ &\equiv \text{Periodic with period} \end{aligned}$$

$$\frac{2\pi}{2} = \pi$$

11. If $f(x)$ and $g(x)$ are periodic function with period T_1 and T_2 then $h(x) = \frac{f(x)}{g(x)}$ may be periodic.

If periodic then period of $h(x) = \frac{f(x)}{g(x)}$ is defined as

$$\begin{cases} \text{L.C.M. of } T_1 \text{ and } T_2, \text{ when } f(x) \text{ and } g(x) \text{ are odd function} \\ \text{or} \\ \frac{1}{2}(\text{l.c.m of } T_1 \text{ and } T_2) \text{ when at least one of } f(x), g(x) \text{ be even function.} \end{cases}$$

Tips: Product and division of two periodic functions may or may not be a periodic function.

e.g. $f(x) = \frac{\sin x}{\cos x}$

Here, $g(x) = \sin x \equiv \text{periodic with period } 2\pi$
and $h(x) = \cos x \equiv \text{periodic with period } 2\pi$

$\therefore f(x) = \frac{\sin x}{\cos x} \equiv \text{periodic with period}$

$$= \frac{1}{2}(\text{l.c.m of } 2\pi \text{ and } 2\pi)$$

$$= \pi$$

Alternative

$$f(x) = \frac{\sin x}{\cos x} = \tan x \equiv \text{Periodic with period } \pi$$

12. There is an exception to the l.c.m. rule. In function where modulus is involved, the l.c.m. rules gives the period of the function but it may not be a fundamental period.

e.g. $f(x) = |\sin x| + |\cos x|$

$$\equiv \text{periodic with period defined as l.c.m. of } \pi \text{ and } \pi = \pi$$

Which is indeed period of $|\sin x| + |\cos x|$ but it is not fundamental period

Since $\left| \sin \left(x + \frac{\pi}{2} \right) \right| + \left| \cos \left(x + \frac{\pi}{2} \right) \right| = |\sin x| + |\cos x|$

Which show that fundamental period of $f(x)$ is $\frac{\pi}{2}$

Tips:

$$(i) \text{ Period of } a |\sin x| + b |\cos x| = \begin{cases} \pi, & \text{when } a \neq b \\ \frac{\pi}{2}, & \text{when } a = b \end{cases}$$

$$(ii) \text{ Period of } |\sin mx| + |\cos nx| = \begin{cases} \text{l.c.m of } \left(\frac{\pi}{m}, \frac{\pi}{n}\right) & \text{when } m \neq n \\ \text{or} \\ \frac{1}{2} \text{l.c.m of } \left(\frac{\pi}{m}, \frac{\pi}{n}\right) & \text{when } m = n \end{cases}$$

e.g. $f(x) = |\sin 2x| + |\cos 2x|$

Here, period of $|\sin 2x| = \frac{\pi}{2}$

and period of $|\cos 2x| = \frac{\pi}{2}$

$\therefore$ Period of $f(x) = |\sin 2x| + |\cos 2x|$

$$= \frac{1}{2} \left(\text{l.c.m of } \frac{\pi}{2} \text{ and } \frac{\pi}{2} \right) = \frac{\pi}{4}$$

e.g. $f(x) = 2 |\sin 2x| - 3 |\cos 2x|$

Here, period of $|\sin 2x| = \frac{\pi}{2}$

and Period of $|\cos 2x| = \frac{\pi}{2}$

$\therefore$ Period of $f(x)$ is

$$\text{L.c.m. of } \left(\frac{\pi}{2}, \frac{\pi}{2}\right) = \frac{\pi}{2}$$

e.g. $f(x) = |\sin 2x| + |\cos 3x|$

Here, period of $|\sin 2x| = \frac{\pi}{2}$

and period of $|\cos 3x| = \frac{\pi}{3}$

$\therefore$ Period of given function $f(x)$

$$= \text{L.c.m of } \frac{\pi}{2} \text{ and } \frac{\pi}{3}$$

$$= \frac{\text{L.c.m of } \pi \text{ and } \pi}{\text{H.c.f. of } 2 \text{ and } 3} = \pi$$

13. If $f(x)$ and $g(x)$ are periodic functions then $f \circ g(x)$ and $g \circ f(x)$ are periodic functions with period defined as

$$y = f\{g(x)\} = \begin{cases} \text{period of } g(x), & \text{when } f(x) \text{ is odd function} \\ \text{or} \\ \frac{\text{period of } g(x)}{2}, & \text{when } f(x) \text{ is even function} \end{cases}$$

e.g. $f(x) = \cos(\sin x)$
 $\equiv$ Periodic function

With period π

Because $\cos x$ is even function

But $f(x) = \sin(\cos x)$

$\equiv$ periodic function with period 2π because $\sin x$ is *odd function*

14. If $f(x)$ and $g(x)$ are non periodic functions then $f \circ g(x)$ and $g \circ f(x)$ are not periodic functions.
 15. If $f(x)$ is periodic function but $g(x)$ is not periodic function then $f \circ g(x)$ is not periodic but $g \circ f(x)$ is periodic function.

e.g. $f(x) = \sin x \equiv$ periodic with period 2π and $g(x) = \sqrt{x} \equiv$ non periodic

$\therefore f \circ g(x) = f\{g(x)\} = \sin \sqrt{x}$
 $\equiv$ non periodic

but $g \circ f(x) = g\{f(x)\} = \sqrt{\sin x}$
 $\equiv$ Periodic with period 2π

16. If $f(x)$ is any periodic function with period T then first order derivative $f'(x)$ may be periodic function with period T

e.g. $f(x) = \sin^2 x \equiv$ periodic with period π

and $f'(x) = 2 \sin x \cos x = \sin 2x$
 $=$ periodic with period π

17. If $f(x)$ is any periodic function with period T then $\int f(x) dx$ may or may not be periodic function.

e.g. $f(x) = \sin x \equiv$ Periodic with period 2π

$\Rightarrow \int f(x) dx = \int \sin x dx = \cos x + c$
 $\equiv$ Periodic with period 2π

but $f(x) = \sin^2 x$
 $\equiv$ periodic with period π

$\Rightarrow \int f(x) dx = \int \sin^2 x dx$

$$= \int \frac{1 - \cos 2x}{2} dx = \int \frac{1}{2} dx - \frac{1}{2} \int \cos 2x dx$$

$$= \frac{x}{2} - \frac{\sin 2x}{4} + c$$

$=$ Non periodic function.

18. Function $f(x) = x - [x] = \{x\}$ is periodic with period 1 and $f(x) = \{nx\}$ is periodic with period $\left(\frac{1}{n}\right)$ where $n \neq 0$
19. If $f(x)$ is any periodic function with period T then $g(x) = a^{f(x)}$ is also periodic function with period T
 e.g. $f(x) = (2010)^{\cos x}$
 $\equiv$ Periodic with period 2π
20. If $f(x)$ is any function and its graph is symmetric about the lines $x = a$ and $x = b$ where $a < b$ then $f(x)$ is periodic function.
21. Periodic function may not be one-one function.

Example 1.11 Find the period of function $f(x)$ if it satisfies the condition

(a) $f(x+1) - f(x-2) = 0$

(b) $f(x+1) + f(x) = 0$

(c) $f(x+p) = 1 + \{1 - 3f(x) + 3f^2(x) - f^3(x)\}^{\frac{1}{3}}$

Solution

(a) $f(x+1) - f(x-2) = 0$

Put $(x+2)$ in place of x

$$f(x+2+1) = f(x+2-2)$$

$$\Rightarrow f(x+3) = f(x)$$

$\therefore f(x)$ is periodic function or with period 3.

(b) $f(x+1) + f(x) = 0$ (i)

Put $(x+1)$ in place of x

$$f(x+2) + f(x+1) = 0$$
 (ii)

Subtract (i) and (ii)

$$f(x+2) = f(x)$$

$\therefore f(x)$ is periodic with period 2.

(c) $f(x+p) = 1 + \{1 - 3f(x) + 3f^2(x) - f^3(x)\}^{\frac{1}{3}}$

$$\Rightarrow f(x+p) = 1 + \{(1 - f(x))^3\}^{\frac{1}{3}}$$

$$\Rightarrow f(x+p) = 1 + 1 - f(x)$$

$$\Rightarrow f(x+p) + f(x) = 2$$
 (1)

Put $(x+p)$ in place of x

$$f(x+2p) + f(x+p) = 2$$
 (2)

Subtract (1) and (2)

$$f(x) - f(2p+x) = 0$$

$$\Rightarrow f(x+2p) = f(x)$$

$\therefore f(x)$ is periodic function with period $2p$

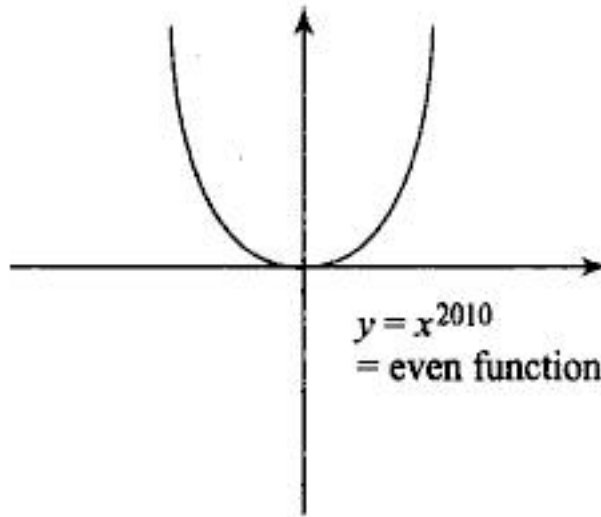


Fig. 1.64

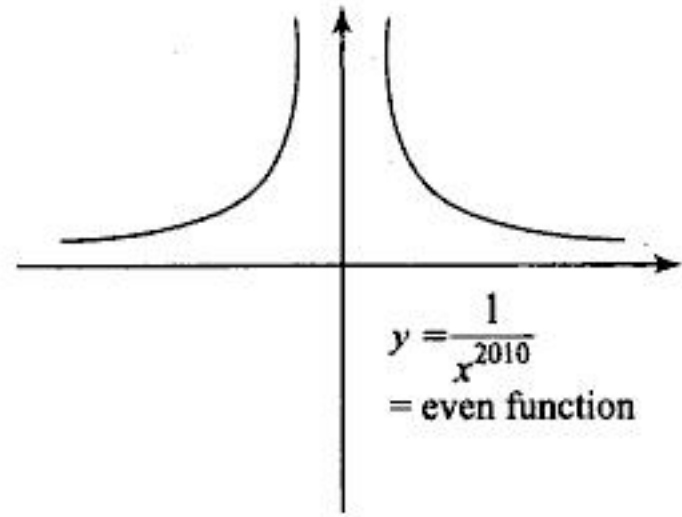


Fig. 1.65

Even Function

Graphically, if graph of any function is symmetrical about y-axis then function is called even function.

Mathematically, function $f(x)$ is called even function if and only if $f(-x) = f(x)$

$$\Rightarrow f(-x) - f(x) = 0$$

e.g. Show that function $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$ is even function.

Solution: Given $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1 \Rightarrow f(-x) = \frac{-x}{e^{-x} - 1} - \frac{x}{2} + 1$

As, $f(x) - f(-x) = \left(\frac{x}{e^x - 1} + \frac{x}{2} + 1 \right) - \left(\frac{-x}{e^{-x} - 1} - \frac{x}{2} + 1 \right) = 0$

Hence, given function is even function.

Odd Function

Graphically, if graph of function is symmetrical about the origin or opposite quadrant then function is called odd function.

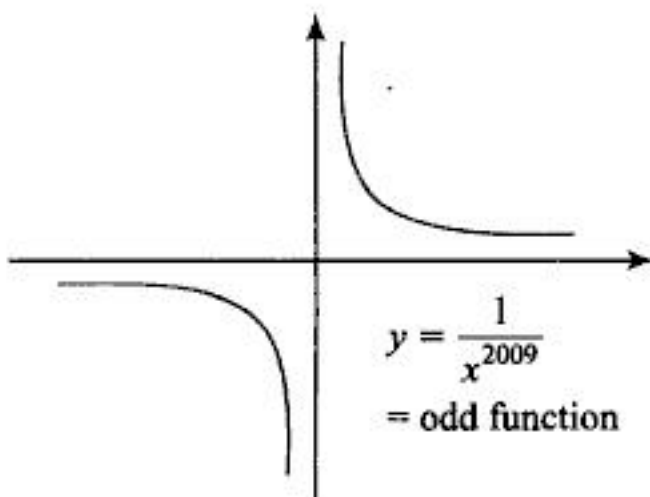


Fig. 1.66

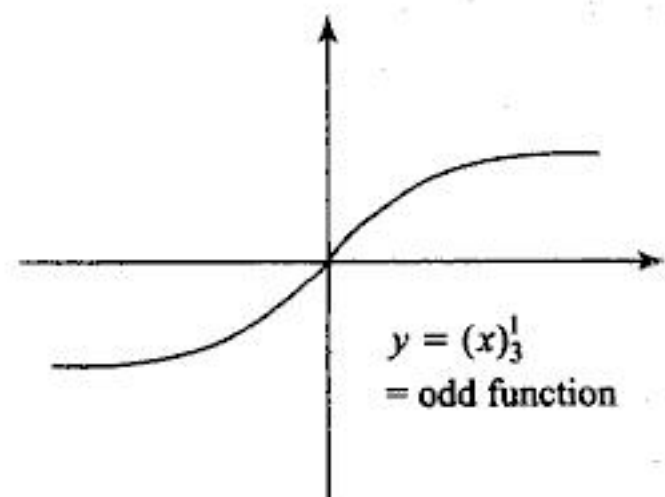


Fig. 1.67

Mathematically, Function $f(x)$ is called odd function if and only if $f(-x) = -f(x)$

$$\Rightarrow f(x) + f(-x) = 0$$

Example 1.12 Show that function $f(x) = \log(x + \sqrt{x^2 + 1})$ is odd function.

Solution Given $f(x) = \log(x + \sqrt{x^2 + 1})$ and $f(-x) = \log(-x + \sqrt{x^2 + 1})$

$$\begin{aligned}\therefore f(-x) + f(x) &= \log(\sqrt{x^2 + 1} - x) + \log(\sqrt{x^2 + 1} + x) \\ &= \log\{(\sqrt{x^2 + 1} + x)(\sqrt{x^2 + 1} - x)\} \\ &= \log\{x^2 + 1 - x^2\} = \log 1 = 0\end{aligned}$$

Hence, $f(x)$ is odd function.

Properties of Odd and Even Function

1. Sum of two odd functions is odd function.
2. Sum of two even functions is an even function.
3. Sum of an odd function and an even function is neither even nor odd function.
4. Difference between two odd functions is odd function.
5. Difference between two even functions is even function.
6. Difference between an odd function and an even function is neither odd nor even function.
7. The products of multiplication of two odd functions is an even function.
8. The products of multiplication of two even functions is an even function.
9. The products of multiplication of an odd and an even function is an odd function.
10. Division of two odd functions gives an even function.
11. Division of two even functions gives an even function.
12. Division of an odd and an even function gives an odd function.
13. If $f(x)$ is any ordinary function then $f(-x) + f(x)$ represent an even function and $f(-x) - f(x)$ represents an odd function.

Hence, $f(x)$ can be expressed as a sum of an even function and an odd function as

$$f(x) = \frac{1}{2}\{f(x) + f(-x)\} + \frac{1}{2}\{f(x) - f(-x)\}$$

e.g. Express function $f(x) = e^x$ as a sum of an odd and an even function.

Solution: Given $f(x) = e^x$, $f(-x) = e^{-x}$

$$\begin{aligned}\therefore f(x) &= \frac{1}{2}\{f(x) + f(-x)\} + \frac{1}{2}\{f(x) - f(-x)\} \\ &= \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} \\ &= \text{even function} + \text{odd function}\end{aligned}$$

14. If $f(x)$ is any odd function then its first order derivative $f'(x)$ is even function

e.g. $f(x) = \sin x + x^3$
 $= \text{odd} + \text{odd} = \text{odd function}$

and $f'(x) = \cos x + 3x^2$
 $= \text{even} + \text{even} = \text{even function}$

15. If $f(x)$ is any even function then its first order derivative $f'(x)$ represents odd function.
16. When the function is either of an odd or an even then their square becomes an even function.

17. If $f(x)$ is even function then $y = a^{f(x)}$ is also even function but if $f(x)$ is any odd function then $y = a^{f(x)}$ is *orbitaly function*.

e.g. $f(x) = (2010)^{\cos x}$

= even function

but $f(x) = (2010)^{\sin x}$

= neither even nor odd function

18. If $f(x)$ is any odd function then $\int_a^x f(x) dx$ is *even function*.

19. If $f(x)$ is any even function then $\int_a^x f(x) dx$ is an *odd function*.

20. If $f(x)$ and $g(x)$ are even functions then $f \circ g(x)$ and $g \circ f(x)$ are *even functions*.

21. If $f(x)$ and $g(x)$ are odd functions then $f \circ g(x)$ and $g \circ f(x)$ are *odd functions*.

22. If $f(x)$ is odd and $g(x)$ is even function then $f \circ g(x)$ and $g \circ f(x)$ are *even function*.

Tips: As even function is symmetric about y-axis due to this even function is not one-one function.

Even Extensive Function

If $f(x)$ is even function in $[0, a]$ then form of function

$$g(x) = \begin{cases} f(x), & \text{when } 0 \leq x \leq a \\ f(-x), & \text{when } -a \leq x < 0 \end{cases}$$

called even extensive function.

Example 1.13 If $f(x) = \begin{cases} |x|, & -2 \leq x \leq 2 \\ \sin^2 x + \tan x, & 2 < x \leq \pi \\ \log(x + \sqrt{x^2 + 1}), & \pi < x \leq 10 \end{cases}$

Find the even extension of the function.

Solution Given $f(x) = \begin{cases} |x|, & -2 \leq x \leq 2 \\ \sin^2 x + \tan x, & 2 < x \leq 5 \\ \log(x + \sqrt{x^2 + 1}), & \pi < x \leq 10 \end{cases}$

Then even extension of $f(x) = \begin{cases} |x|, & -2 \leq x \leq 2 \\ \sin^2 x - \tan x, & -5 \leq x < -2 \\ \log(-x + \sqrt{x^2 + 1}), & 10 \leq x < -\pi \end{cases}$

Odd Extensive Function

If $f(x)$ is odd function in $[0, a]$ then form of function

$$g(x) = \begin{cases} f(x), & 0 \leq x \leq a \\ -f(-x), & -a \leq x < 0 \end{cases}$$

called odd extensive function.

Example 1.14 If $f(x) = \sin x + x^{2007} + e^x$ in $[0, 4]$ then find its odd extensive form.

Solution $\sin x + x^{2007} + e^x, x \in [0, 4]$

$$\therefore \text{odd extensive form} = \begin{cases} f(x), & 0 \leq x \leq a \\ -f(-x), & -a \leq x < 0 \end{cases} = \begin{cases} \sin x + x^{2007} + e^x, & 0 \leq x \leq 4 \\ \sin x + x^{2007} - e^{-x}, & -4 \leq x < 0 \end{cases}$$

Implicit Function

The form of function in which independent variable and dependent variable can not to be defined, is called implicit function.

e.g. $x^2 y^3 = \log(x + y)$

e.g. $xy = \sin(x^2 + y^3)$

Explicit Function

The form of function in which dependent and independent variable to be defined i.e. dependent variable can be expressed in the form of independent variable then function is called explicit function

e.g. $10^x + 10^y = 10$

$$\Rightarrow 10^y = 10 - 10^x$$

$$\Rightarrow \log_{10} 10^y = \log_{10}(10 - 10^x)$$

$$\Rightarrow y = \log_{10}(10 - 10^x) \equiv \text{Explicit function}$$

Monotonic Function

If function is increasing or decreasing in a certain interval then the function is called monotonic function.

Increasing Function

Graphically, if graph of function increases with increasing value of the independent variable and decreases with decreasing value of independent variable then function is called increasing function.

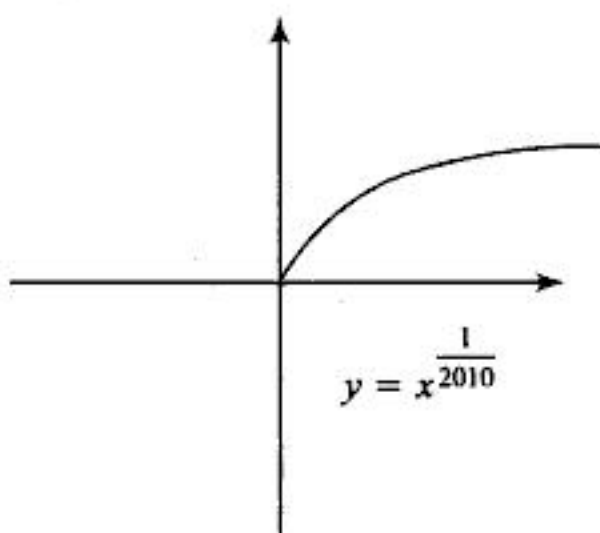


Fig. 1.68

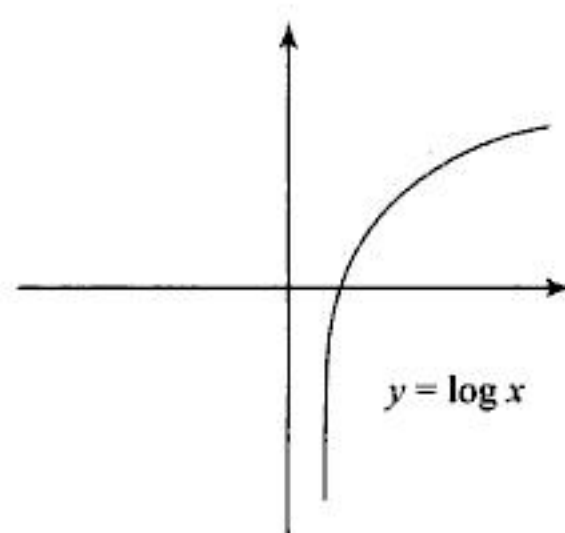


Fig. 1.69

Mathematically, function $f(x)$ is called increasing function.

If $\alpha > \beta$ then $f(\alpha) > f(\beta)$ and If $\alpha < \beta$ then $f(\alpha) < f(\beta)$

Example 1.15 Function $f(x) = x + \sin x$ is increasing in $x \in [0, \pi/2]$. True or false.

Solution Given function $f(x) = x + \sin x$

$$\text{Here, at } x = \frac{\pi}{6}, f\left(\frac{\pi}{6}\right) = \frac{\pi}{6} + \frac{1}{2} = \frac{\pi+3}{6}$$

$$\text{and at } x = \frac{\pi}{2}, f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + 1 = \frac{\pi+2}{2}$$

$$\text{As, } \frac{\pi}{6} < \frac{\pi}{2} \Rightarrow f\left(\frac{\pi}{6}\right) < f\left(\frac{\pi}{2}\right)$$

$$\therefore f(x) = x + \sin x, x \in \left[0, \frac{\pi}{2}\right] \text{ is increasing function}$$

Ans. True.

Calculus Method Function $f(x)$ is called increasing function in $[a, b]$ if and only if $f'(x) \geq 0$ $x \in [a, b]$

Example 1.16 If $f(x) = x \cdot e^{x(1-x)}$ then $f(x)$ is

IIT-JEE 2001

(a) Increasing on $\left[-\frac{1}{2}, 1\right]$

(b) Decreases on $\mathbb{R}$

(c) Increases on $\mathbb{R}$

(d) Decreases on $\left[-\frac{1}{2}, 1\right]$

Solution Given

$$f(x) = x e^{x(1-x)}$$

Differentiate w.r.t. x

$$f'(x) = e^{x(1-x)}[-2x^2 + x + 1]$$

for increasing function $f'(x) \geq 0$

$$\Rightarrow e^{x(1-x)}[-2x^2 + x + 1] \geq 0$$

$$\because e^{x(1-x)} > 0 \therefore -2x^2 + x + 1 \geq 0$$

$$\Rightarrow 2x^2 - x - 1 \leq 0 \Rightarrow (x-1)(2x+1) \leq 0 \Rightarrow f(x) \text{ is increasing on } \left[-\frac{1}{2}, 1\right]$$

Ans. (a)

Example 1.17 Consider the following S and R

S : both $\sin x$, $\cos x$ are decreasing function in $(\pi/2, \pi)$

R : If a differentiable function decreases in the interval (a, b) then its derivative also decreases in (a, b)

IIT-JEE 2000

Which of the following is true?

(a) Both S and R are wrong.

(b) Both S and R are correct but R is not the correct explanation of S .

(c) S is correct and R is the correct explanation for S .

(d) S is correct and R is wrong.

Solution GivenSin x and $\cos x$ are decreases in $(\pi/2, \pi)$ Now, If $f(x) = \frac{1}{x}, x \in (1, 2)$ Then $f'(x) = \frac{-1}{x^2} < 0$ $\Rightarrow$ If $f(x)$ decreases then $f'(x)$ is increasing function in $(1, 2)$ $\therefore R$ is not true but S is true.

Ans. (d)

Example 1.18 The function $f(x) = \sin^4 x + \cos^4 x$ increases if

IIT-JEE 1999

- (a) $0 < x < \frac{\pi}{8}$ (b) $\frac{\pi}{4} < x \leq \frac{3\pi}{8}$ (c) $\frac{3\pi}{8} < x < \frac{5\pi}{8}$ (d) $\frac{5\pi}{8} < x < \frac{3\pi}{4}$

Solution Given

$$f(x) = \sin^4 x + \cos^4 x$$

$$\Rightarrow f'(x) = 4\sin^3 x \cos x - 4\cos^3 x \sin x$$

$$= -\sin 4x$$

 $\therefore$ For increasing function

$$f'(x) \geq 0 \Rightarrow -\sin 4x \geq 0$$

$$\Rightarrow \sin 4x \leq 0$$

$$\text{As, } \sin z \leq 0$$

$$\text{Then } \pi \leq z \leq 2\pi$$

$$\therefore \text{For } \sin 4x \leq 0$$

$$\Rightarrow \pi \leq 4x \leq 2\pi$$

$$\Rightarrow \frac{\pi}{4} \leq x \leq \frac{\pi}{2}$$

$$\text{Ans (b), } \frac{\pi}{4} < x < \frac{3\pi}{8}$$

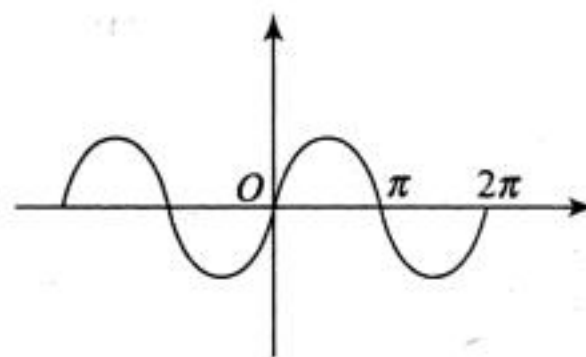


Fig. 1.69b

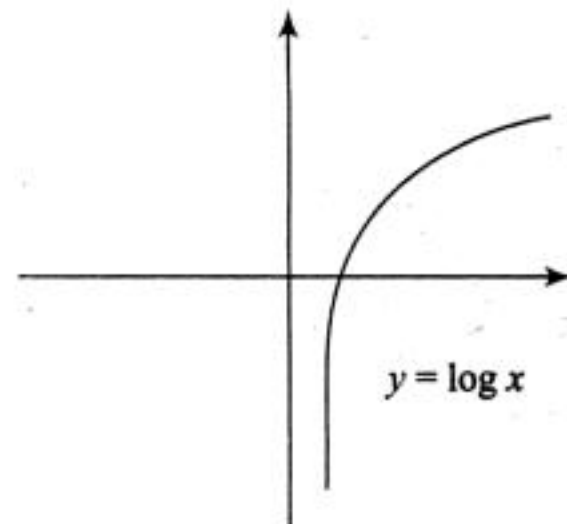


Fig. 1.70

Example 1.19 If $f(x) = \int_{-1}^x (t^2 + 2t)(t^2 - 1)dt$ Then find interval for which $f(x)$ increases

$$\text{Solution } f(x) = \int_{-1}^x (t^2 + 2t)(t^2 - 1)dt$$

Use Newton Leibniz theorem

$$f'(x) = (x^2 + 2x)(x^2 - 1) \cdot \frac{dx}{dx} - (1 - 2)(1 - 1) \cdot 0$$

$$\Rightarrow f'(x) = x(x + 2)(x + 1)(x - 1)$$

or increasing function

$$f'(x) \geq 0 \Rightarrow x(x+2)(x+1)(x-1) \geq 0$$

$$x \in (-\infty, -2) \cup (-1, 0) \cup (1, \infty)$$

Tips: Newton Leibniz theorem

If $f(x) = \int_{g(x)}^{h(x)} w(x) dx$ then

$$\frac{d(f(x))}{dx} = w(h(x)) \cdot \frac{d(h(x))}{dx} - w(g(x)) \cdot \frac{d(g(x))}{dx}$$

Strictly Increasing Function Function $f(x)$ is called strictly increasing function if and only if $f'(x) > 0$ i.e. tangent to the curve of function cannot be parallel to x -axis

Example 1.20 Let the function $g : (-\infty, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

be given by $g(x) = 2 \tan^{-1}(e^x) - \frac{\pi}{2}$ then $g(x)$ is

- (a) Even and is strictly increasing in $(0, \infty)$
- (b) Odd and is strictly decreasing in (∞, ∞)
- (c) Odd and is strictly increasing in $(-\infty, \infty)$
- (d) Neither even nor odd, but is strictly increasing in $(-\infty, \infty)$

IIT-JEE 2008

Solution $g(u) = 2 \tan^{-1}(e^u) - \frac{\pi}{2}$

$$= 2 \tan^{-1}(e^u) - \tan^{-1}(e^u) - \cot^{-1}(e^u)$$

$$= \tan^{-1}(e^u) - \cot^{-1}(e^u)$$

$$\Rightarrow g(-u) = \tan^{-1}(e^{-u}) - \cot^{-1}(e^{-u})$$

$$= -\{\tan^{-1}(e^u) - \cot^{-1}(e^u)\} = -g(u)$$

$\therefore g(u)$ is odd function

$$\text{Now, } g(u) = \tan^{-1}(e^u) - \cot^{-1}(e^u)$$

$$\Rightarrow g(u) = \tan^{-1}(e^u) - \tan^{-1}\left(\frac{1}{e^u}\right)$$

$$\tan^{-1}\left(\frac{e^u - \frac{1}{e^u}}{1 + e^u \cdot \frac{1}{e^u}}\right)$$

$$\Rightarrow g(u) = \frac{1}{1 + \left(\frac{e^{2u} - 1}{e^u}\right)} > 0$$

$\therefore g(u)$ is strictly increasing

Ans: (c)

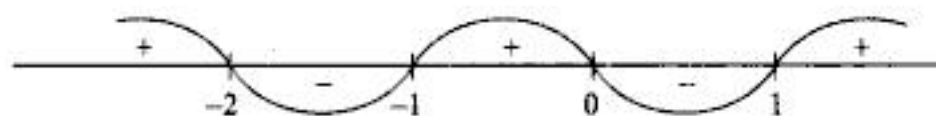


Fig. 1.71

Decreasing Function

Graphically, if graph of function decreases with increasing value of independent variable of function and increases with decreasing value of independent variable then function is called decreasing function.

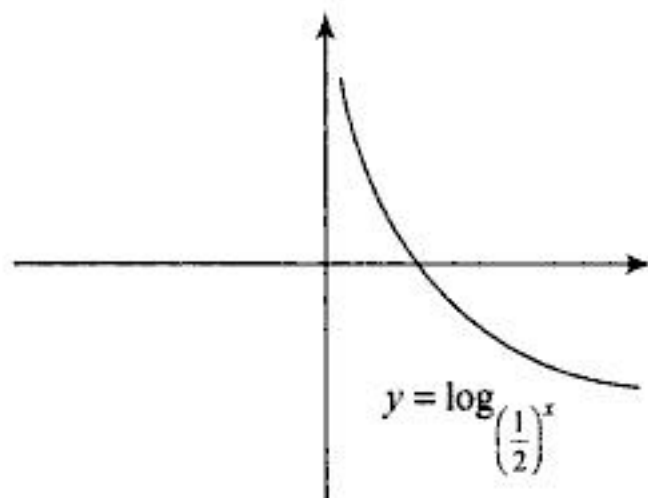


Fig. 1.72

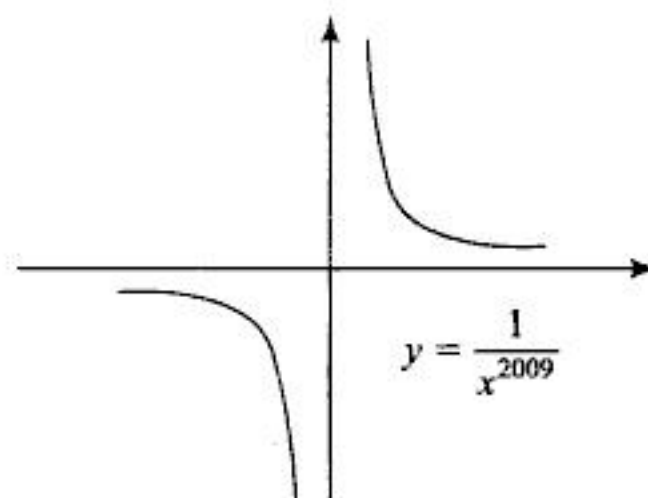


Fig. 1.73

Mathematically, function $f(x)$ is called increasing function
If $\alpha < \beta$ then $f(\alpha) > f(\beta)$ and if $\alpha > \beta$ then $f(\alpha) < f(\beta)$

e.g. $f(x) = \frac{1}{x^2 + x + 1}$

at $x = 1, f(1) = \frac{1}{3}$

and at $x = 2, f(2) = \frac{1}{4 + 2 + 1} = \frac{1}{7}$

Here $1 < 2$ but $f(1) > f(2)$

$\therefore f(x)$ is decreasing function.

Calculus Method Function $f(x)$ is called decreasing function if and only if $f'(x) \leq 0$.

Example 1.21 The function $f(x) = \frac{\log(x + \pi)}{\log(e + x)}$ is

IIT-JEE 1995

(a) Increasing on $(0, \infty)$

(b) decreasing on $(0, \infty)$

(c) Neither increasing nor decreasing

(d) none of these

Solution $f(x) = \frac{\log(x + \pi)}{\log(e + x)}$

$$\Rightarrow f'(x) = \frac{\left(\frac{1}{\pi + x}\right)\log(e + x) - \frac{1}{(e + x)}\log(\pi + x)}{\{\log(e + x)\}^2}$$

$$= \frac{(e+x)\log(e+x) - (\pi+x)\log(\pi+x)}{(e+x)(\pi+x)\{\log(e+x)\}^2}$$

For decreasing function $f'(x) \leq 0$

$$\Rightarrow \frac{(e+x)\log(e+x) - (\pi+x)\log(\pi+x)}{(e+x)(\pi+x)\{\log(e+x)\}^2} \leq 0$$

$$\Rightarrow x \in (0, \infty)$$

Strictly Decreasing Function Function $f(x)$ is called strictly decreasing function if and only if $f'(x) < 0$ i.e. tangent to the curve of function cannot parallel to x - axis.

Example 1.22 At what values of 'a' does the function $f(x) = 2 + ax - x^3$ strictly decreased for all real x .

Solution $f(x) = 2 + ax - x^3$

$$\Rightarrow a - 3x^2 < 0 \Rightarrow a < 3x^2$$

Here, $\min^m$ value of $3x^2 = 0$

Hence, $a < 0$

Properties of Monotonic Functions

1. Sum of two decreasing functions is a decreasing function.
2. Sum of two increasing functions is increasing function.
3. Sum of increasing function and decreasing functions is neither increasing nor decreasing function.
4. Difference between two increasing functions is neither increasing nor decreasing function.
5. Difference between two decreasing functions is neither increasing nor decreasing function.
6. If $f(x)$ is increasing function and $g(x)$ is decreasing function then $f(x) - g(x)$ is increasing function but $g(x) - f(x)$ is a decreasing function.
7. Reciprocal of increasing function is decreasing function.
8. Reciprocal of decreasing function is an increasing function.
9. Negative of an increasing function is negative decreasing function.
10. Negative of decreasing function is negative increasing function.
11. Product of two positive decreasing functions is a decreasing function.
12. Product of two positive increasing functions is an increasing function.
13. Product of an increasing function and a decreasing function is neither increasing function nor decreasing function.
14. Division of two positive increasing functions is neither an increasing function nor a decreasing function.
15. Division of two positive decreasing functions is neither increasing nor decreasing function.
16. If $f(x)$ is increasing function and $g(x)$ is decreasing function then $\frac{f(x)}{g(x)}$ is increasing function

and $\frac{g(x)}{f(x)}$ is a decreasing function.

17. If $f(x)$ and $g(x)$ are two increasing functions then $f \circ g(x)$ or $g \circ f(x)$ is also an increasing function.

18. If $f(x)$ and $g(x)$ are two decreasing functions then $f \circ g(x)$ or $g \circ f(x)$ is an increasing function.
19. If $f(x)$ is an increasing function and $g(x)$ is a decreasing function then $f \circ g(x)$ or $g \circ f(x)$ is decreasing function.
20. Polynomial function of an odd function may be an increasing function.

One-One Function

Graphically, if a straight line parallel to x -axis, intersects the graph of function at only one point then function is called one-one or injective function.

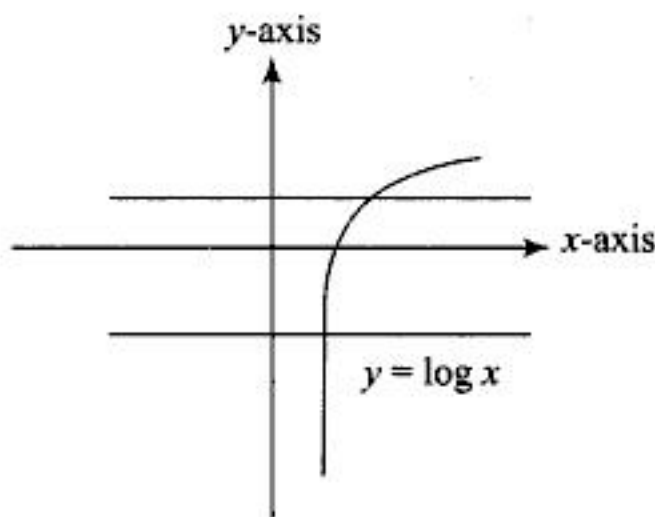


Fig. 1.74

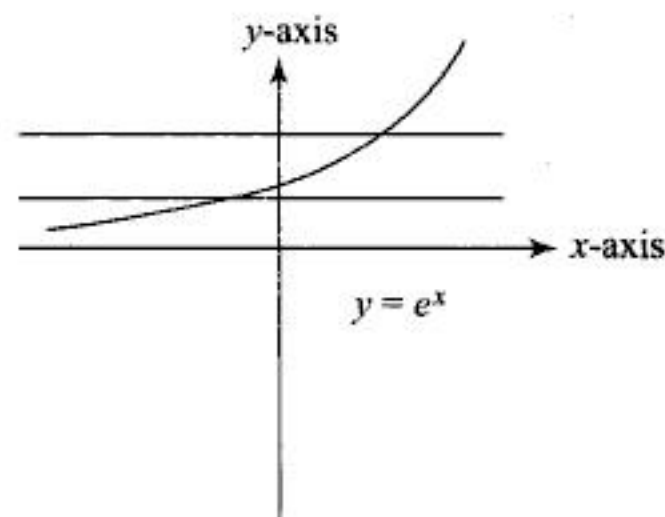


Fig. 1.75

Mathematically, function $f(x)$ is called one-one function, if $f(\alpha) = f(\beta)$ then $\alpha = \beta$ and other relation between α and β represents value of α or β be imaginary value.

Example 1.23 $f(x) = x^3 + 3x + 1$

Solution let $f(\alpha) = f(\beta)$

$$\Rightarrow \alpha^3 + 3\alpha + 1 = \beta^3 + 3\beta + 1$$

$$\Rightarrow (\alpha^3 - \beta^3) + 3(\alpha - \beta) = 0$$

$$\Rightarrow (\alpha - \beta) \{\alpha^2 + \alpha\beta + \beta^2 + 3\} = 0$$

$$\Rightarrow \alpha = \beta \text{ or } \alpha^2 + \beta^2 + \alpha\beta + 3 = 0$$

$$\Rightarrow \alpha = \beta \text{ or } \alpha^2 + \beta^2 + \alpha\beta + 3 = 0$$

Here, at $\alpha = 0, \beta^2 + 3 = 0$

$$\Rightarrow \beta = \pm \sqrt{-3} = \pm \sqrt{3}i = \text{imaginary}$$

$$f(x) = x^3 + 3x + 1 \text{ is one - one function}$$

Calculus Method Function $f(x)$ is called one - one function if and only if $f'(x) > 0$ or $f'(x) < 0$ in certain interval.

Example 1.24 Let function $f: R \rightarrow R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$ then $f(x)$ is one - one function. True or false.

IIT-JEE 2002

Solution Given

$$f(x) = 2x + \sin x$$

$$\Rightarrow f'(x) = 2 + \cos x > 0$$

$$\Rightarrow f'(x) > 0$$

$\therefore f(x) = 2x + \sin x$ is one – one function.

Example 1.25 Function $f(x) = \frac{x}{x+1}$ is one- one function. True or false.

Solution Given function

$$f(x) = \frac{x}{x+1}, x \neq -1$$

Here,
$$f'(x) = \frac{(x+1) \frac{dx}{dx} - x \frac{d(x+1)}{dx}}{(x+1)^2}$$

$$= \frac{1}{(x+1)^2} > 0$$

$\therefore f(x) = \frac{x}{x+1}$ is one – one function.

Ans. True.

Note: if A and B are any two non empty sets have m and n numbers of elements respectively then of one – one function

$$f: A \rightarrow B = \begin{cases} {}^n P_m, & \text{when } n \geq m \\ 0, & \text{when } n < m \end{cases}$$

Example 1.26 If $A = \{1, 2, 3, 4\}$ and $B = \{a, b\}$ then find number of one – one function $f: B \rightarrow A$

Solution Given

$$A = \{1, 2, 3, 4\}, n(A) = 4$$

$$B = \{a, b\}, n(B) = 2$$

$\therefore$ No. of one – one function $F: A \rightarrow B$ is ${}^2P_4 = 0$

Many-One Function

Graphically, if a straight line parallel to x – axis intersecting the curve of function at more than one points then function is called many-one function.

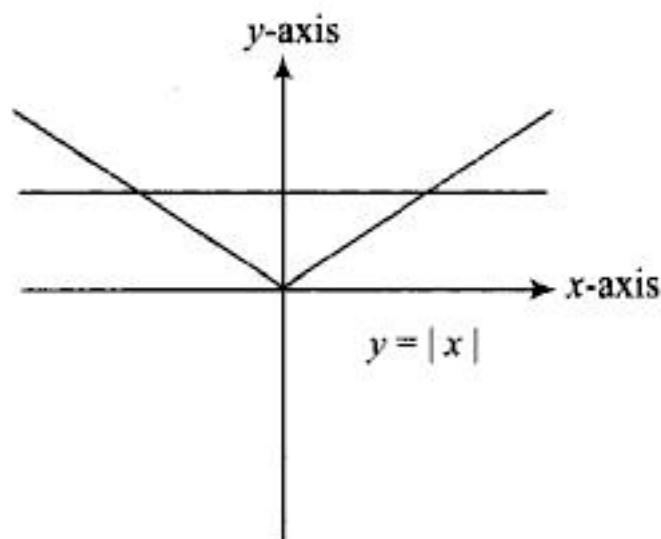


Fig. 1.76

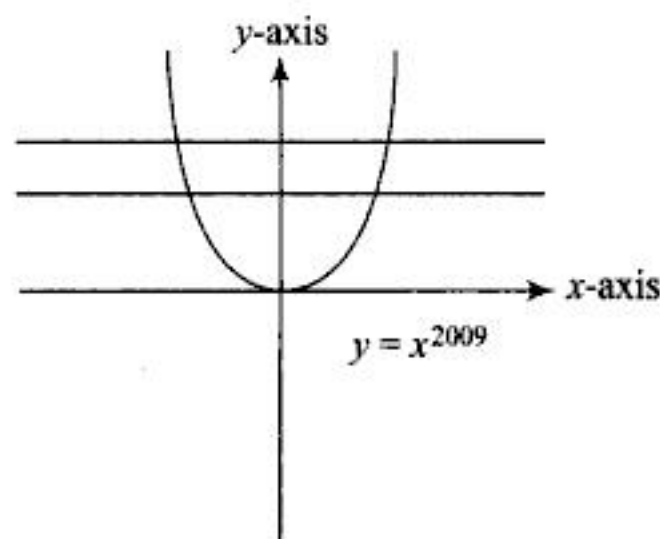


Fig. 1.77

Mathematically function $f(x)$ is called many-one function if $f(\alpha) = f(\beta)$ then $\alpha \neq \beta$ or if $\alpha = \beta$ but other relation between α and β represent real values of α and β .

Example 1.27 $f(x) = x^2 + 5x + 7$

Solution Let $f(\alpha) = f(\beta)$

$$\Rightarrow \alpha^2 + 5\alpha + 7 = \beta^2 + 5\beta + 7$$

$$\Rightarrow (\alpha^2 - \beta^2) + 5(\alpha - \beta) = 0$$

$$\Rightarrow (\alpha - \beta)(\alpha + \beta + 5) = 0$$

$$\Rightarrow \alpha = \beta \text{ or } \alpha + \beta + 5 = 0$$

Here, α and β always represent real values $\therefore f(x) = x^2 + 5x + 7$ is many-one function.

Calculus Method Function $f(x)$ is called many-one function if and only if $f'(x) > 0$ or $f'(x) < 0$ cannot be defined in certain interval.

e.g. $f(x) = x^2 + 5x + 7$

$$\Rightarrow f'(x) = 2x + 5 = \begin{cases} \geq 0, & \text{when } x > \frac{-5}{2} \\ < 0, & \text{when } x < \frac{-5}{2} \end{cases}$$

$\therefore f(x) = x^2 + 5x + 7$ is many-one function.

Onto Function or Surjective Function

If the function $f: A \rightarrow B$ such that each element of B is the f image of at least one element in A then function $f: A \rightarrow B$ is called onto function.

Tips: If $f: A \rightarrow B$ then Co-domain of function is B but values of y for which their independent variable is defined called range of function.

Tips: If $y = f(x)$ be given function then Co-domain of function is $y \forall R$

Example 1.28 If $f: [0, \infty) \rightarrow [0, \infty)$ and $f(x) = \frac{x}{1+x}$ then $f(x)$ is

- (a) one-one and onto
(c) onto but not one-one

- (b) one-one but not onto
(d) none

IIT-JEE 2003

Solution Given

$$f(x) = \frac{x}{1+x}$$

$$\Rightarrow f'(x) = \frac{1}{(1+x)^2} > 0$$

$\Rightarrow f(x)$ is one-one function.

Now, $f: [0, \infty) \rightarrow [0, \infty)$

Here, Co-domain, $y \in [0, \infty)$

$$\text{As } \frac{x}{x+1} = y \Rightarrow y + yx = x$$

$$\Rightarrow x = \frac{y}{1-y}$$

Here, x to be defined when $1 - y \neq 0 \Rightarrow y \neq 1$

$\Rightarrow$ range $y \in [0, \infty) - (1)$

$\Rightarrow$ range $\neq$ co-domain

Hence, $f(x)$ is one - one but not onto function.

Example 1.29 If $f: R \rightarrow \left[\frac{\pi}{6}, \frac{\pi}{2}\right]$, $f(x) = \sin^{-1}\left(\frac{x^2 - a}{x^2 + 1}\right)$ is a onto function then set of values of a is

- (a) $\left\{-\frac{1}{2}\right\}$ (b) $\left[-\frac{1}{2}, -1\right]$ (c) $(-1, \infty)$ (d) none

Solution Given $f(x) = \sin^{-1}\left(\frac{x^2 - a}{x^2 + 1}\right)$ and $f: R \rightarrow \left[\frac{\pi}{6}, \frac{\pi}{2}\right]$

$\therefore f(x)$ is onto function

$$\therefore \frac{\pi}{6} \leq f(x) \leq \frac{\pi}{2}$$

$$= \frac{\pi}{6} \leq \sin^{-1}\left(\frac{x^2 - a}{x^2 + 1}\right) \leq \frac{\pi}{2}$$

$$\frac{1}{2} \leq 1 - \frac{a+1}{x^2+1} < 1$$

$$\Rightarrow a + 1 > 0$$

$$\Rightarrow a \in (-1, \infty)$$

Example 1.30 $f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$ and $g(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$ then $f - g$ is

IIT-JEE 2005

(a) neither one - one nor onto

(b) one - one and onto

(c) one - one and into

(d) many - one and onto

Solution Given

$$f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$$

Here,
$$f(x) - g(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ -x, & \text{if } x \text{ is irrational} \end{cases}$$

$\Rightarrow f(x) - g(x)$ has range, $y \forall R$ also $f(x) - g(x)$ co-domain $y \forall R$ and $\frac{dy}{dx} = f'(x) - g'(x) > 0$ when x is rational

and $\frac{dy}{dx} = f'(x) - g'(x) < 0$ when x is irrational.

$\therefore f(x) - g(x)$ is one-one and onto function.

Tips: If A and B are two sets having m and n elements respectively such that $1 \leq n \leq m$ then number

$$\text{of onto function} = \sum_{r=1}^n (-1)^{n-r} \cdot {}^n C_r (r^n)$$

Note: Total number of mapping from A to B is n^m then total no. of onto function(s)
 $=$ total No. of mapping – total No. of elements in B
 $= n^m - n$

Example 1.31 Find number of surjection from A to B where

$$A = \{1, 2, 3, 4\}, B = \{a, b\}$$

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Solution $A = \{1, 2, 3, 4\}, B = \{a, b\} \Rightarrow n(A) = m = 4$ and $n(B) = n = 2$
 Number of surjection from A to B

$$\begin{aligned} &= \sum_{r=1}^n (-1)^{n-r} \cdot {}^n C_r (r^n) \\ &= (-1)^{2-1} {}^2 C_1 (1)^4 + (-1)^{2-2} \times {}^2 C_2 (2)^4 \\ &= -2 + 16 = 14 \end{aligned}$$

Into Function

A function $f: A \rightarrow B$ is an into function, if there exist an element in B having no pre-image in A .
 Or, Function $f: A \rightarrow B$ is not onto then function is into.

Tips: If range of function is the subset of their co-domain then the function is called into function.

One-One onto Function

The function represent one – one and onto behavior is called one-one onto function.

Example 1.32 Let the function defined in column –I have domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and range $(-\infty, \infty)$

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Column- I

- (i) $1 + 2x$
- (ii) $\tan x$

Column-II

- (a) onto but not one-one
- (b) one – one but not onto
- (c) one-one and onto

Solution Given $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow (-\infty, \infty)$

Here, co-domain, $y \in (-\infty, \infty)$

- (i) $f(x) = 1 + 2x$
 $\Rightarrow f'(x) = 2 > 0$
 $\Rightarrow f(x)$ is one- one function

$$\text{and } y = 1 + 2x \Rightarrow \frac{y-1}{2} = x$$

$$\Rightarrow \text{range } y \forall R$$

$\therefore f(x) = 1 + 2x$ is one-one and onto function.

- (ii) $f(x) = \tan x$
 $\Rightarrow f'(x) = \sec^2 x \geq 0$
 $\Rightarrow f(x)$ is one – one function
 Now, $f(x) = \tan x$
 Range of $f(x)$, $y \forall R$
 $\Rightarrow \text{range} = \text{Co-domain}$
 $\therefore f(x)$ is one – one onto function.

Tips: If A and B are two non empty sets each having n number of elements then numbers of one – one onto function $= n!$

e.g. If $A = \{l, m, n\}$ and $B = \{a, b, c\}$ then no. of one – one onto functions $f: A \rightarrow B$ is $3! = 3 \times 2 \times 1 = 6$

Partial Function

If function $f(x)$ can be obtained from a relation then function $f(x)$ will be called partial function.

Example 1.33 Find $f(x)$ which satisfies relation $f(x) + 2f(1 - x) = x + 2$

Solution Given

$$f(x) + 2f(1 - x) = x^2 + 2 \quad (i)$$

put $(1 - x)$ in place of x

$$\therefore f(1 - x) + 2f\{1 - (1 - x)\} = (1 - x)^2 + 2$$

$$\Rightarrow f(1 - x) + 2f(x) = 1 + x^2 - 2x + 2 \quad (ii)$$

$$(i) - 2 \times (ii)$$

$$f(x) + 2f(1 - x) = x^2 + 2$$

$$2f(1 - x) + 4f(x) = 2x^2 - 4x + 6$$

$$f(x) - 4f(x) = (x^2 + 2) - (2x^2 - 4x + 6)$$

$$\Rightarrow -3f(x) = -x^2 + 4x - 4$$

$$\Rightarrow f(x) = \frac{x^2 - 4x + 4}{3}$$

= required partial function

Identical Function

Two functions $f(x)$ and $g(x)$ are called identical when

- $f(x)$ can be expressed into $g(x)$
- Range of $f(x)$ is equal to range of $g(x)$ and domain of $f(x)$ is equal to domain of $g(x)$.

Example 1.34 $f(x) = 2^{\log_{\sqrt{2}}(x^2 + 1)^{1/2}}$ and $g(x) = x^2 + 1$ then $f(x)$ and $g(x)$ are identical. True or False?

Solution Given

$$f(x) = 2^{\log_{\sqrt{2}}(x^2 + 1)^{1/2}}$$

$$= 2^{\frac{(1/2)}{(1/2)} \log_2(x^2 + 1)}$$

$$= (x^2 + 1)$$

$$= g(x)$$

$\therefore f(x)$ and $g(x)$ two identical functions.

Bounded Function

If $f(x)$ is any function satisfying $|f(x)| \leq m$ where m is positive real or $a \leq f(x) \leq b$ then function $f(x)$ is called bounded function.

Example 1.35 $f(x) = \sin x + \sqrt{3} \cos x + 5$

Solution $f(x) = 2 \left\{ \frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x \right\} + 5$

$$= 2 \sin \left(x + \frac{\pi}{3} \right) + 5$$

Here, $f(x)_{\max} = 7$ and

$$f(x)_{\min} = 3$$

$$\Rightarrow 7 \geq f(x) \geq 3$$

Hence $f(x)$ is bounded function.

Inverse Function

Graphically, if graph of function $f(x)$ and $g(x)$ are mirror image to each other with respect to line mirror $y = x$ then both functions $f(x)$ and $g(x)$ are called inverse to each other.

Mathematically, for inverse of function use,

- Let $f: A \rightarrow B$ where $y = f(x)$
- First establish that $f: A \rightarrow B$ is bijective and hence invertible.
- Simplify $y = f(x)$ to express it as $x = g(y)$ now, $g(x)$ is the inverse of $f(x)$.

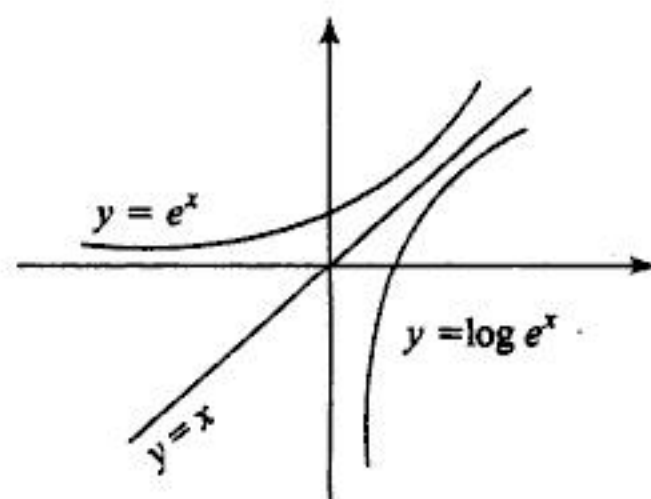


Fig. 1.78

Tips: If $g(x)$ is the inverse of $f(x)$ then $f\{g(x)\} = x$

Tips: If only $y = f(x)$ is given without mentioning the set A and B , then A and B are taken as the natural domain and in range of $f(x)$ respectively.

Tips: If $f(x)$ is increasing function then $f^{-1}(x)$ is also increasing function.

Tips: If $f(x)$ is decreasing function then $f^{-1}(x)$ is also decreasing function

Tips: If $f(x)$ is even function then $f^{-1}(x)$ is also even function

Tips: If $f(x)$ is odd function then $f^{-1}(x)$ is also odd function

Example 1.36 If $f: [1, \infty) \rightarrow [2, \infty]$ is given by $f(x) = x + \frac{1}{x}$ then $f^{-1}(x)$ is equal to

- (a) $\frac{x + \sqrt{x^2 - 4}}{2}$ (b) $\frac{x}{1 + x^2}$ (c) $\frac{x - \sqrt{x^2 - 4}}{4}$ (d) $1 + \sqrt{x^2 - 4}$

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Solution Given $f: [1, \infty) \rightarrow [2, \infty]$

Here, co-domain $y \in [2, \infty]$

and domain, $x \in [1, \infty)$

As $y = f(x) = x + \frac{1}{x}$

$$\Rightarrow y = \frac{x^2 + 1}{x} \Rightarrow yx = x^2 + 1$$

$$\Rightarrow x^2 - yx + 1 = 0$$

$$\Rightarrow x = \frac{y \pm \sqrt{y^2 - 4}}{2}$$

Here, at $y = 3, x = \frac{3 + \sqrt{9 - 4}}{2} = \frac{3 + \sqrt{5}}{2} > 1$ or $\frac{3 - \sqrt{5}}{2} < 1$

$$\therefore x = \frac{y + \sqrt{y^2 - 4}}{2}$$

$$\Rightarrow f^{-1}(x) = \frac{x + \sqrt{x^2 - 4}}{2}$$

Example 1.37 Let $f(x) = \frac{\alpha x}{x+1}, x \neq -1$, then for what value of α is $f(f(x)) = x$ is

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(a) $\sqrt{2}$

(b) $-\sqrt{2}$

(c) 1

(d) -1

Solution Given $f(x) = \frac{\alpha x}{x+1}$

$$\Rightarrow f\{f(x)\} = \frac{\alpha f(x)}{f(x)+1} = \frac{\alpha^2 x}{(\alpha+1)x+1}$$

But $f\{f(x)\} = x$

$$\Rightarrow \frac{\alpha^2 x}{(\alpha+1)x+1} = x$$

Put $x = 1$ to set $\alpha^2 = (\alpha+1) + 1$

$$\Rightarrow \alpha = -1 \text{ or } 2$$

But if $\alpha = -1$ then

$$f(x) = \frac{-x}{1+x}$$

and $f(f(x)) = f\left(\frac{-x}{1+x}\right) = x$

Ans: (d)

Example 1.38 Suppose $f(x) = (x+1)^2$, for $x \geq -1$. If $g(x)$ is the function whose graph is the reflection of the graph of $f(x)$ with respect to the line $y = x$ then $g(x)$ is equal to

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(a) $-\sqrt{x} - 1, x \geq 0$

(b) $\frac{1}{(x+1)^2}, x > -1$

(c) $\sqrt{x+1}, x \geq -1$

(d) $\sqrt{x} - 1, x \geq 0$

Solution Given

$$f(x) = (x + 1)^2, x \geq -1$$

As we know if $g(x)$ is the reflection of $f(x)$ in the line $y = x$ then $g(x)$ is the inverse of $f(x)$.

$$\therefore y = (x + 1)^2$$

$$\Rightarrow \sqrt{y} = x + 1 \Rightarrow x = \sqrt{y} - 1$$

$$\Rightarrow f^{-1}(x) = \sqrt{x} - 1, x \geq 0$$

Ans: (d).

Example 1.39 If the function $f : [1, \infty) \rightarrow [1, \infty]$ is defined by $f(x) = 2^{x(x-1)}$ then $f^{-1}(x)$ is,

- (a) $\left(-\frac{1}{2}\right)^{x(x-1)}$ (b) $\frac{1}{2}(1 + \sqrt{1 + 4 \log_2 x})$ (c) $\frac{1}{2}(1 - \sqrt{1 + 4 \log_2 x})$ (d) none of above

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Solution Given

$$y = 2^{x(x-1)}$$

$$\Rightarrow \log_2 y = x(x-1) = x^2 - x$$

$$\Rightarrow x^2 - x \log_2 y = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{1 + 4 \log_2 y}}{2}$$

For $y \geq 1, (1 - \sqrt{1 + 4 \log_2 y}) \leq 0$

$$\therefore x = \frac{1}{2} \left(\frac{1 + \sqrt{1 + 4 \log_2 y}}{2} \right)$$

$$\Rightarrow f^{-1}(x) = \frac{1}{2} (1 + \sqrt{1 + 4 \log_2 x})$$

1.4 SOLUTION OF EQUATION

Use:-

- Write given equation $f(x) = g(x)$.
- Draw graphs of $y = f(x)$ and $y = g(x)$ in a plane.
- Number of solutions of given equation is the point of intersection of the curves.

Example 1.40 Number of solutions of equation $|x| = e^x$ is

- (a) 2 (b) 1
(c) 3 (d) 4

Solution Given equation

$$|x| = e^x$$

Here, No. of points of intersection of curve = 2 = No. of solutions (fig. 1.78).

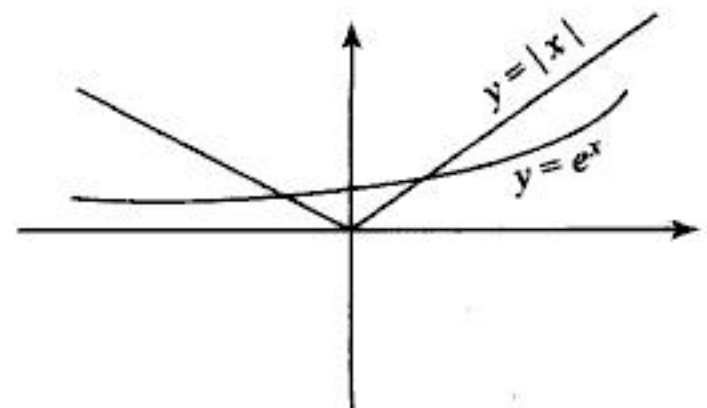


Fig. 1.79



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$$\therefore \sin z < \frac{1}{\sqrt{2}} \text{ (fig. 1.84)}$$

$$\text{Here, } \frac{3\pi}{4} < z < \frac{9\pi}{4}$$

$\therefore$ the solution of the given inequation is

$$\sin z < \frac{1}{\sqrt{2}} \text{ is}$$

$$2n\pi + \frac{3\pi}{4} < z < 2n\pi + \frac{9\pi}{4}$$

Hence, the solution of the given inequation is

$$\begin{aligned} 2n\pi + \frac{\pi}{4} &< \frac{3\pi}{2} + \frac{\pi}{12} < 2n\pi + \frac{9\pi}{4} \\ &= \frac{4}{9}\pi + \frac{4}{3}n\pi < x < \frac{13}{9}\pi + \frac{4}{3}n\pi, n \in \mathbb{Z} \end{aligned}$$

Example 1.47 Solve the inequalities $\cos 2x - \sin 2x \geq 0$

Solution Given $\cos 2x - \sin 2x \geq 0$

$$\Rightarrow \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos 2x - \frac{1}{\sqrt{2}} \sin 2x \right) \geq 0$$

$$\Rightarrow \cos \left(2x + \frac{\pi}{4} \right) \geq 0$$

$$\text{Let } z = 2x + \frac{\pi}{4}$$

$$\therefore \cos z \geq 0$$

$$\text{Here, } -\frac{\pi}{2} \leq z \leq \frac{\pi}{2}$$

$$\begin{aligned} \Rightarrow 2n\pi - \frac{\pi}{2} &\leq 2x + \frac{\pi}{4} \leq 2n\pi + \frac{\pi}{2} \\ &= n\pi - \frac{3\pi}{8} \leq x \leq n\pi + \frac{\pi}{8} \end{aligned}$$

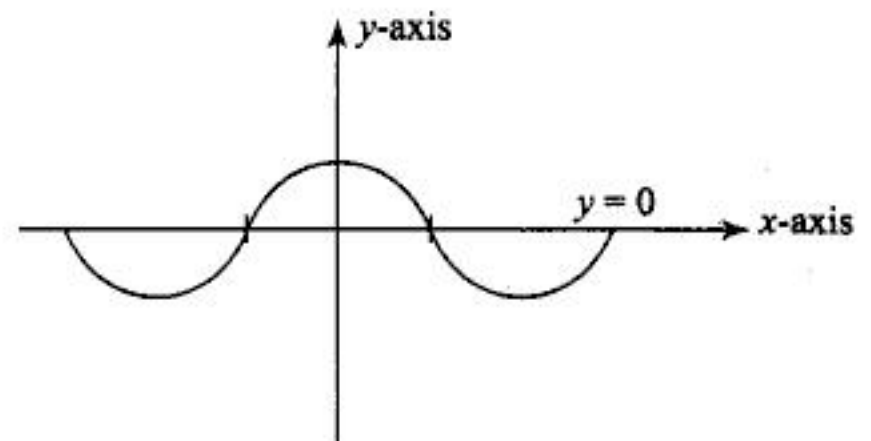


Fig. 1.86

1.6 DOMAIN

The values of independent variable of any function for which their dependent variable is defined, is called a domain of the function.

For evaluating domain of function use the following concepts:-

1.6.1 Concept I

Form of function $y = \frac{f(x)}{g(x)}$ then for domain use, y to be defined when denominator $g(x) \neq 0$



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Here, $\log_2 \left(\frac{x^2}{2} \right) \geq -1$

$$\Rightarrow \log_2 \frac{x^2}{2} + 1 \geq 0$$

$$\Rightarrow \log_2 \frac{x^2}{2} + \log_2 2 \geq 0 \Rightarrow \log_2 x^2 \geq 0$$

$$\Rightarrow x^2 \geq 1 \Rightarrow x^2 - 1 \geq 0 \Rightarrow (x-1)(x+1) \geq 0$$

$$x \in (-\infty, -1] \cup [1, \infty)$$

(i)



Fig. 1.88

and (ii) $\log_2 \left(\frac{x^2}{2} \right) \leq 1$

$$\Rightarrow \log_2 \left(\frac{x^2}{2} \right) - \log_2 2 \leq 0$$

$$\Rightarrow 0 < \frac{x^2}{4} \leq$$

$$\Rightarrow 0 < x^2 \leq 4$$

$$\Rightarrow x^2 - 4 \leq 0$$

$$(x-2)(x+2) \leq 0$$

$$x \in [-2, 2]$$

(ii)



Fig. 1.89

From (i) and (ii)

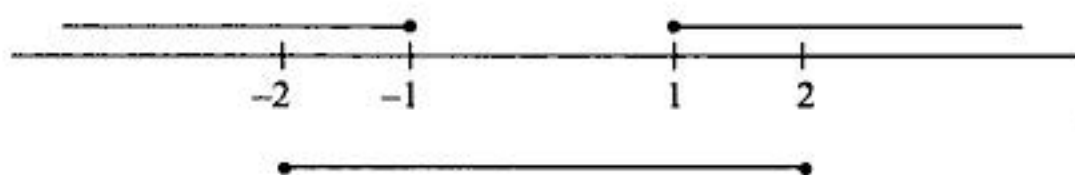


Fig. 1.90

$\therefore$ Domain of $f(x)$

$$x \in [-2, -1] \cup [1, 2]$$

1.6.6 Concept VI

If function is an algebraic function, then use the following points

(a) If $a > 0$ and $D = b^2 - 4ac < 0$ then $ax^2 + bx + c$ is always positive.

(b) If $a < 0$ and $D = b^2 - 4ac < 0$ then $ax^2 + bx + c$ is always negative.

Example 1.55 Find the domain of $f(x) = (x^2 + x + 1)^{\frac{-2009}{2}}$

Solution Given



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$$\Rightarrow \cos x \leq \frac{1}{2}$$

$$-\frac{11\pi}{6} \leq x \leq -\frac{\pi}{3} \text{ or } \frac{11\pi}{6} \leq x \leq \frac{\pi}{3} \quad (\text{ii})$$

From (i) & (ii)

$$x \in \left(-\frac{\pi}{2}, -\frac{\pi}{3} \right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2} \right)$$

$\therefore$ The domain of $f(x)$

$$x \in \left(2n\pi - \frac{\pi}{2}, 2n\pi - \frac{\pi}{3} \right) \cup \left(2n\pi + \frac{\pi}{3}, 2n\pi + \frac{\pi}{2} \right)$$

1.6.9 Concept IX

If the function is related to the greatest integer, least integer modulus or fractional part then use their properties for domain.

Example 1.60 Find the domain of $f(x) = \log_{[x^2]}(4 - |x|)$

Solution Given $f(x) = \log_{[x^2]}(4 - |x|)$

Here, $f(x)$ is defined for

$$(i) \quad 4 - |x| \geq 0 \Rightarrow |x| \leq 4$$

$$\Rightarrow -4 \leq x \leq 4$$

$$(ii) \quad [x^2] > 0, \neq 1$$

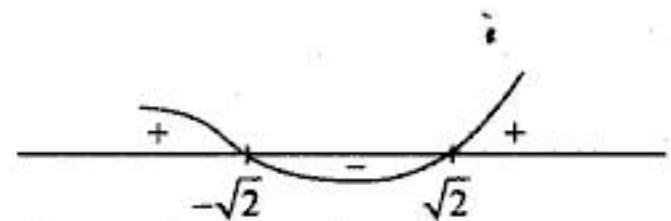
$$\Rightarrow [x^2] \geq 2 \quad \{ \because [x] \text{ is integer} \}$$

$$\Rightarrow x^2 \geq 2 \Rightarrow (x - \sqrt{2})(x + \sqrt{2}) \geq 0$$

$$x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$$

from (i) & (ii)

(i)



(ii)

Fig. 1.97

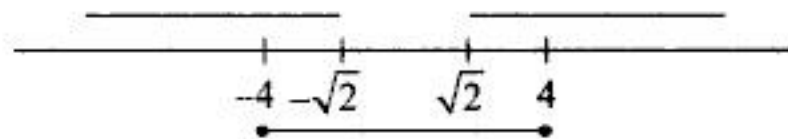


Fig. 1.98

Domain of $f(x)$

$$x \in [-4, -\sqrt{2}] \cup (\sqrt{2}, 4)$$

Tips

(a) $|x| \geq x$ when $x \leq 0$ and $|x| \leq x$ cannot define for any real x

(b) $[x] \leq x, x \forall R$ but $[x] > x$ not defined.



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Using AM $\geq$ GM, we get

$$\frac{\frac{1}{|\sin x|} + \frac{1}{|\cos x|}}{2} \geq \left(\frac{1}{|\sin x| |\cos x|} \right)^{1/2}$$

$$\Rightarrow \frac{1}{|\sin x|} + \frac{1}{|\cos x|} \geq 2(2 |\operatorname{cosec} 2x|)^{1/2}$$

{Where $(\operatorname{cosec} 2x) \geq 1$ }

$$\therefore \frac{1}{|\sin x|} + \frac{1}{|\cos x|} \geq 2\sqrt{2}$$

$\therefore$ The range of $f(x) \in [2\sqrt{2}, \infty) \forall x \in R$

Example 1.68 Find the range of the function $f(x) = 3 \sin x + 4 \cos x - 1$

Solution Given function

$$f(x) = 3 \sin x + 4 \cos x - 1$$

Divide and multiply right side by 5

$$f(x) = 5 \left(\frac{3}{5} \sin x + \frac{4}{5} \cos x - \frac{1}{5} \right) \Rightarrow f(x) = 5 \left(\cos \alpha \sin x + \sin \alpha \cos x - \frac{1}{5} \right)$$

$$\Rightarrow f(x) = 5 \sin(x + \alpha) - 1 \quad \{\text{let } \cos \alpha = 3/5 \text{ then } \sin \alpha = 4/5\}$$

Here, maximum value of $f(x) = 4$ and minimum value is -6

Hence, the range of the given function is $[-6, 4]$

1.7.3 Method-III Substitution Method

If the domain of the function does not contain interval, then find values of y putting the values of x from the domain. The set of all values of y will be the range.

Example 1.69 Find the range of the function $f(x) = {}^4C_x$

Solution Given function $f(x) = {}^4C_0 = \frac{4!}{(4-x)!x!}$

Here, (i) $x!$ is defined for $x = \{0, 1, 2, 3, \dots\}$

(ii) $(4-x)!$ is defined for $x = \{\dots, -1, 0, 1, 2, 3, 4\}$

From (i) and (ii) the domain of given function is $x = \{0, 1, 2, 3, 4\}$

Now, at $x = 0$, $f(0) = {}^4C_0 = \frac{4!}{(4-0)!0!} = 1$

at $x = 1$, $f(1) = {}^4C_1 = \frac{4!}{(4-1)!1!} = 4$

at $x = 2$, $f(2) = {}^4C_2 = \frac{4!}{(4-2)!2!} = 6$



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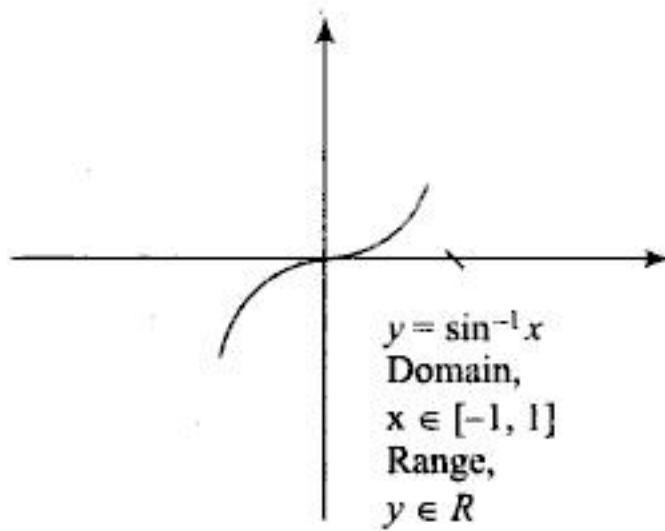


Fig. 1.115

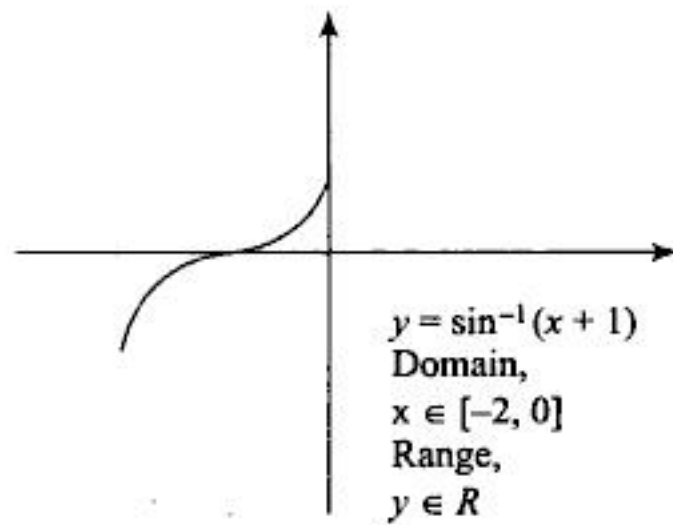


Fig. 1.116

1.8.4 Concept 4

If $f(x)$ is any function then for graph of function $f(x - k)$ where $k > 0$ use:-

- Draw the graph of function $y = f(x)$.
- Then shift the graph of function through ' k ' units towards the right side the y -axis.

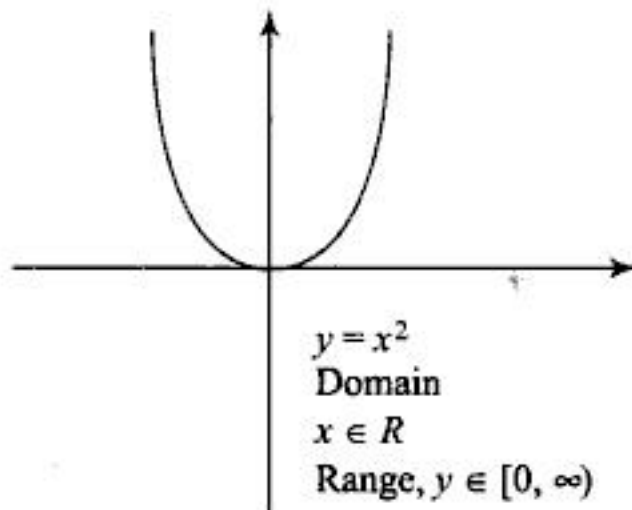


Fig. 1.117

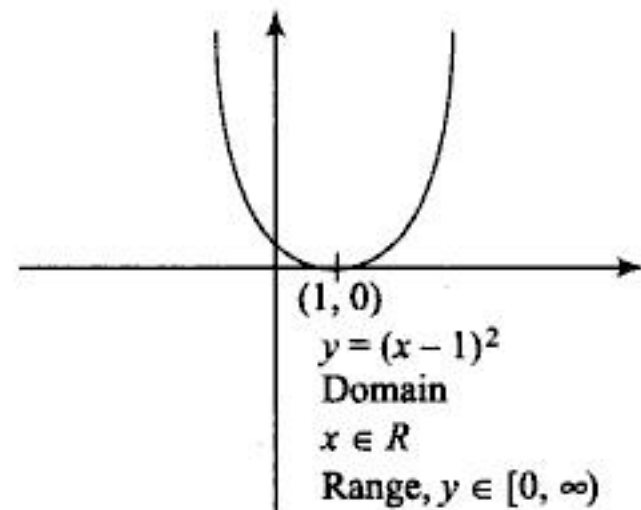


Fig. 1.118

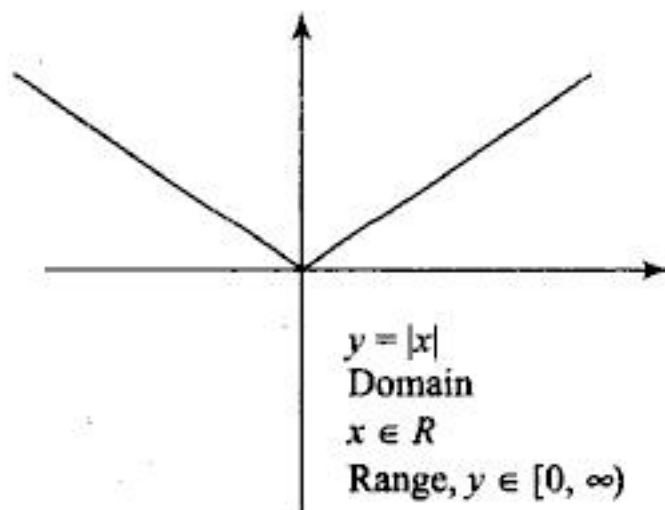


Fig. 1.119

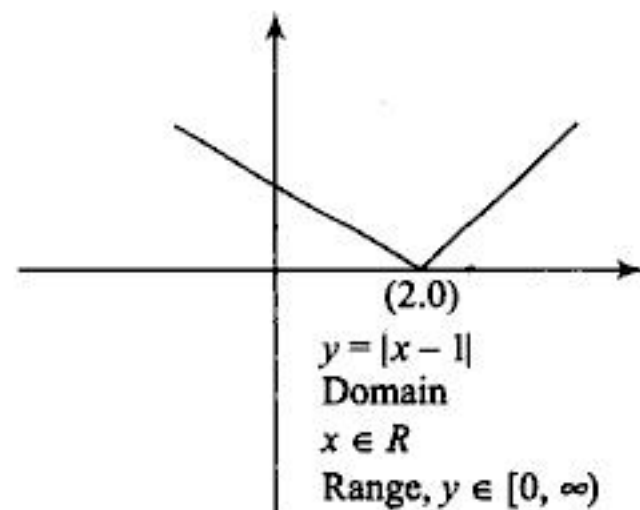


Fig. 1.120



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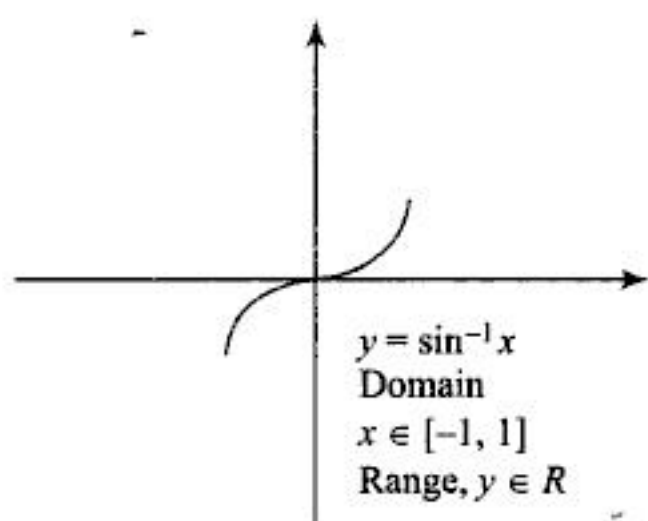


Fig. 1.129

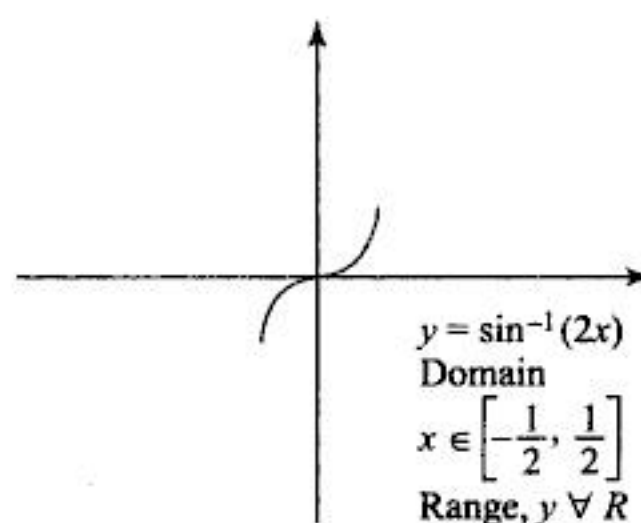


Fig. 1.130

1.8.8 Concept 8

If $f(x)$ is any function then for graph of function $f\left(\frac{1}{k} \cdot x\right)$ where $k > 1$ use:-

- Draw the graph of function $y = f(x)$
- Stretch the graph of function through ' k ' times along x -axis.

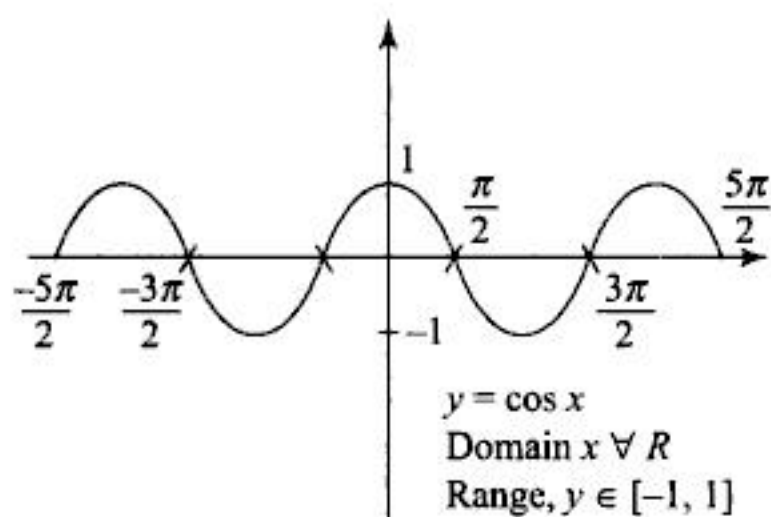


Fig. 1.131

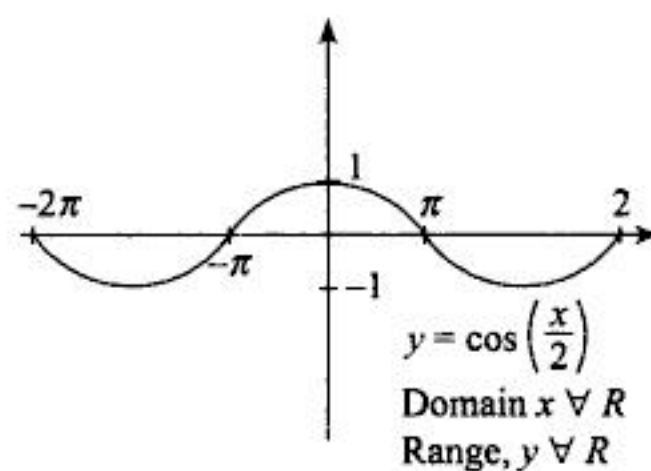


Fig. 1.132

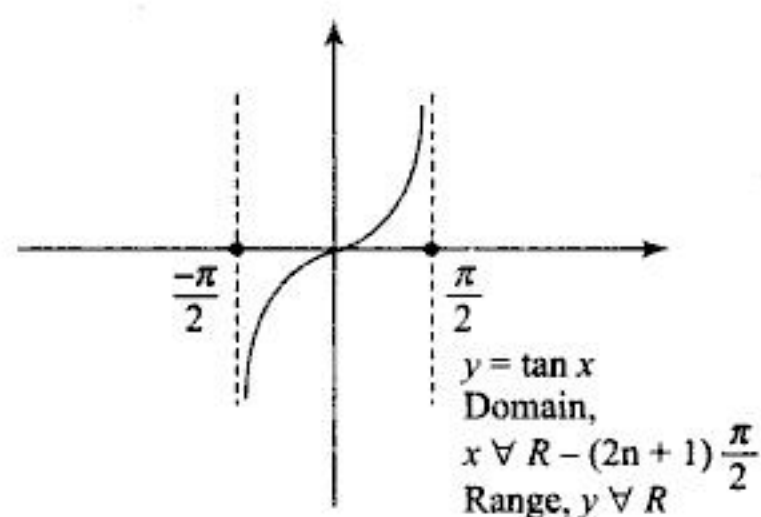


Fig. 1.133

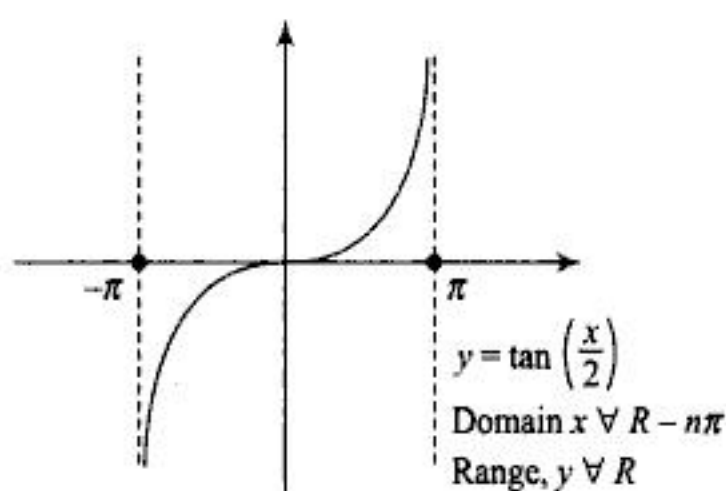


Fig. 1.134



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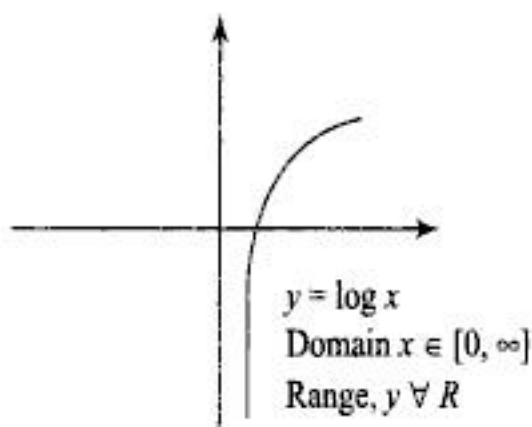


Fig. 1.154

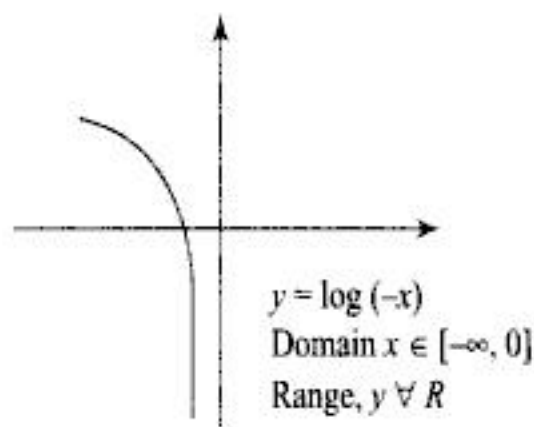


Fig. 1.155

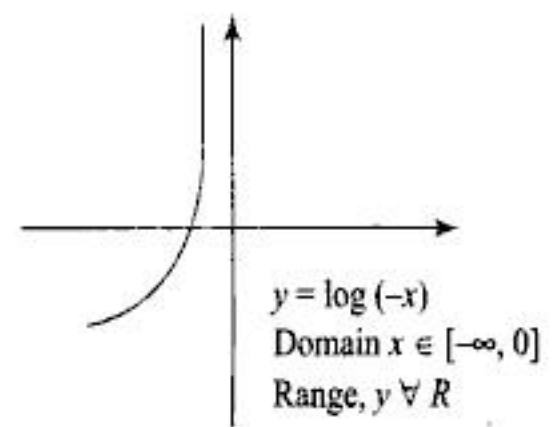


Fig. 1.156

1.8.12 Concept 12

If $f(x)$ is any function then for graph of function $|f(x)|$ use:-

- Draw the graph of function $y = f(x)$
- Take the part of $f(x)$ above the x -axis and take mirror image of the portion of graph which lies below the x -axis with respect to x -axis as plane mirror to get graph of function $y = |f(x)|$

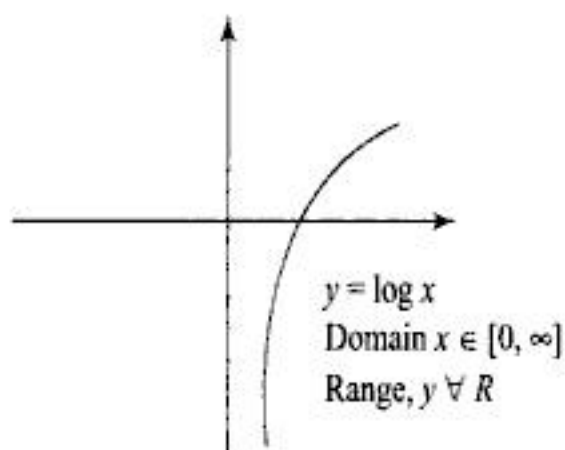


Fig. 1.157

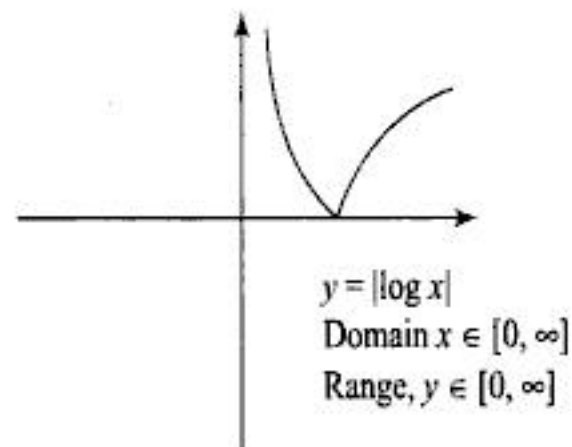


Fig. 1.158

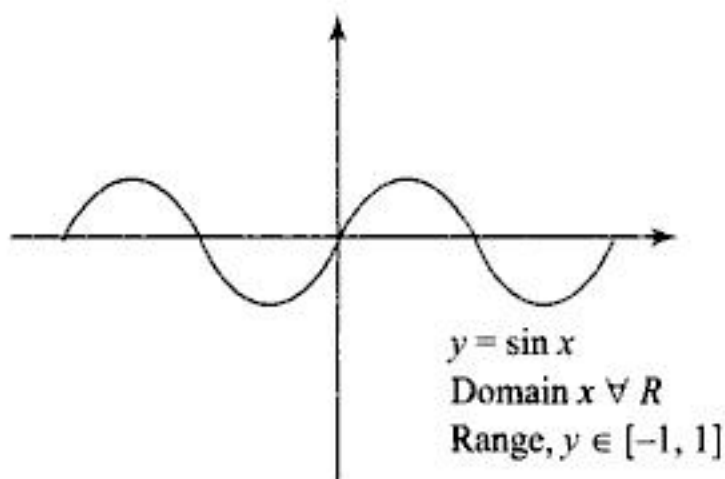


Fig. 1.159

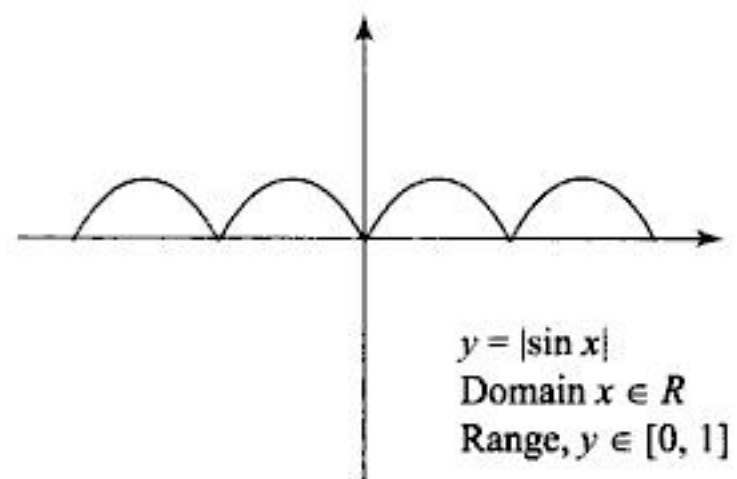


Fig. 1.160



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1.8.16 Concept 16

If $y = f(x)$ is any function then for graph of function $|y| = |f(x)|$ use:-

- (i) Draw the graph of function $y = f(x)$.
- (ii) Take the part of $f(x)$ which lies above the x -axis and its mirror image with respect to x -axis as plane mirror to get graph of function $|y| = f(x)$.
- (iii) Then take the part of $|y| = f(x)$ above the x -axis and take the mirror image of the portion of graph which lies below the x -axis with respect to x -axis as plane mirror to get graph of function $|y| = |f(x)|$.

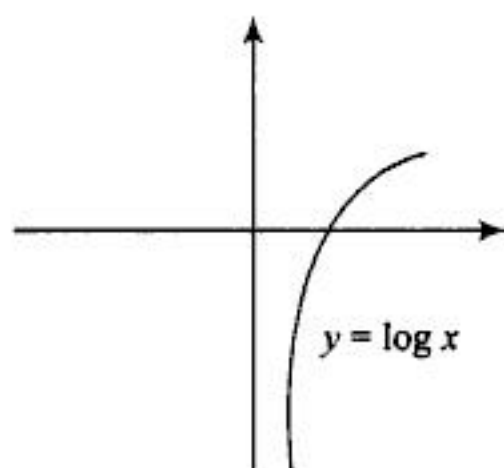


Fig. 1.184

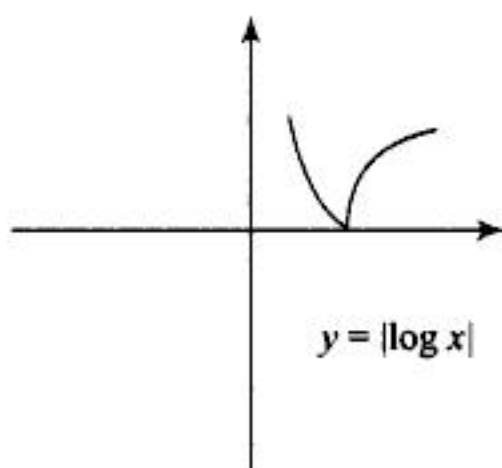


Fig. 1.185

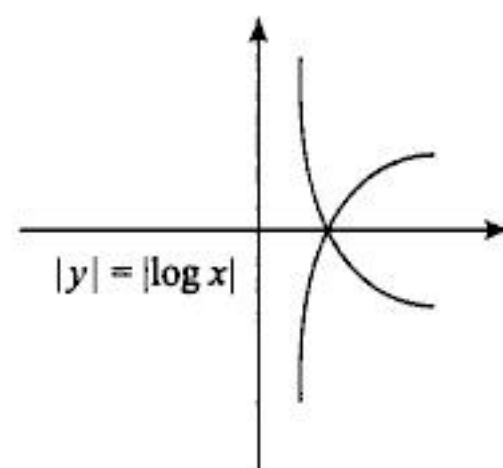


Fig. 1.186

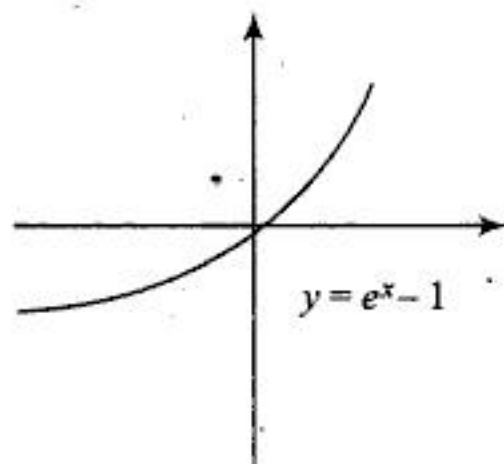


Fig. 1.187

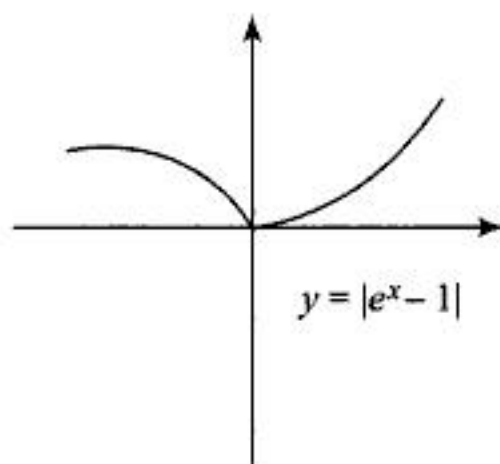


Fig. 1.188

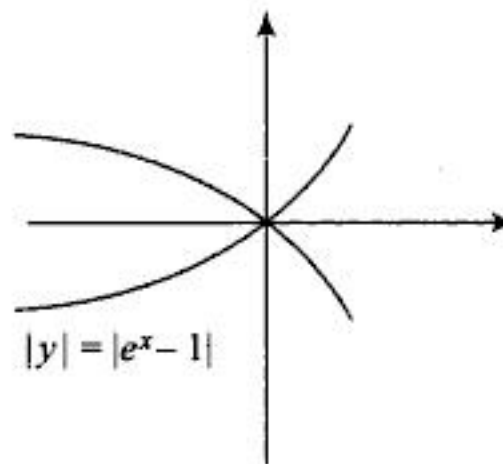


Fig. 1.189

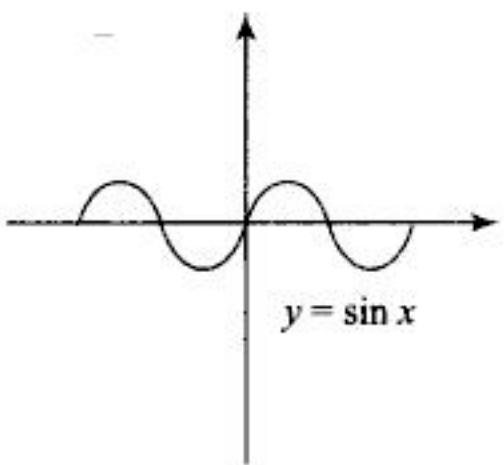


Fig. 1.190

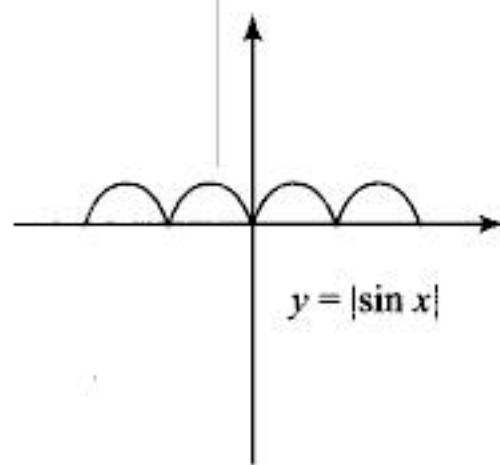


Fig. 1.191

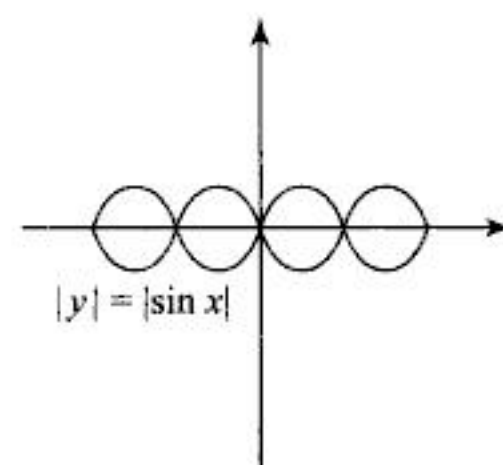


Fig. 1.192



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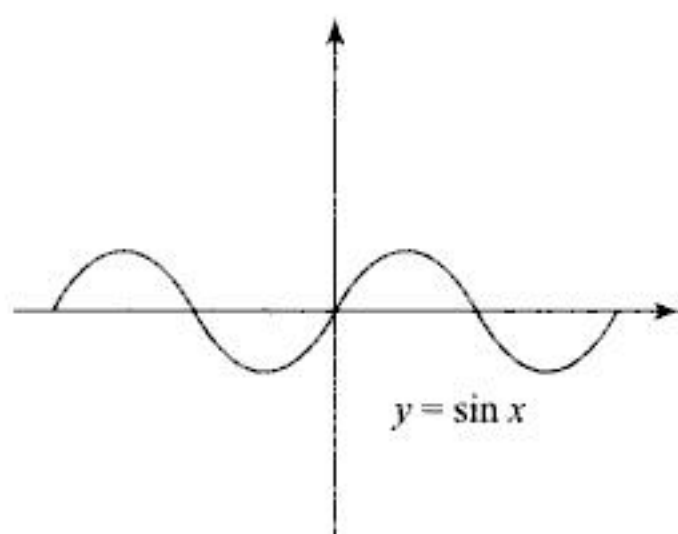


Fig. 1.209

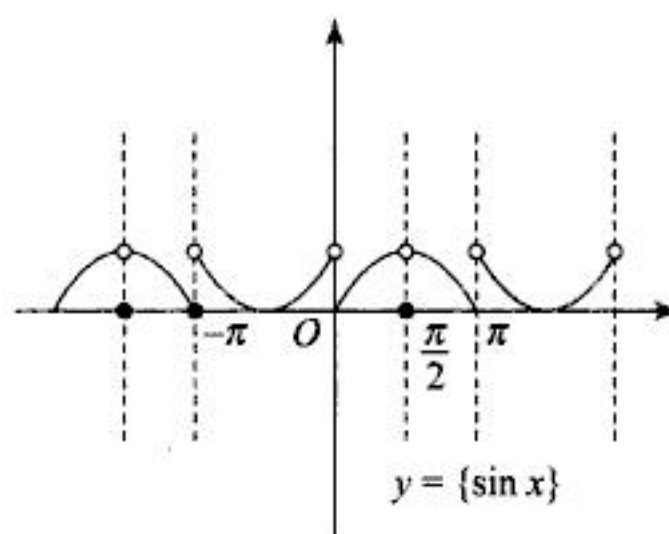


Fig. 1.210

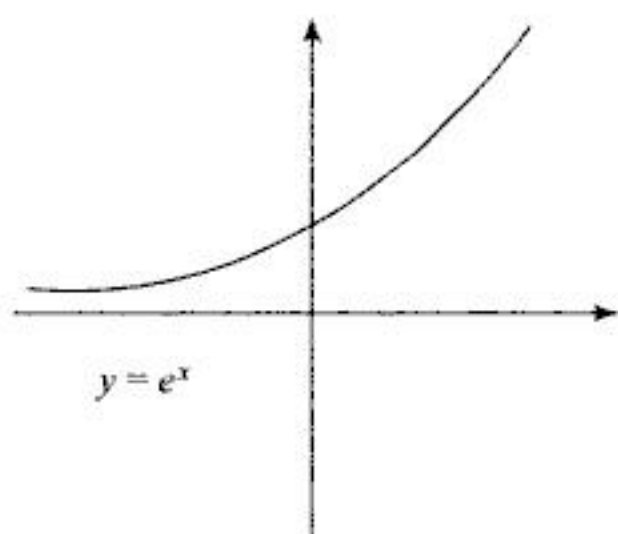


Fig. 1.211

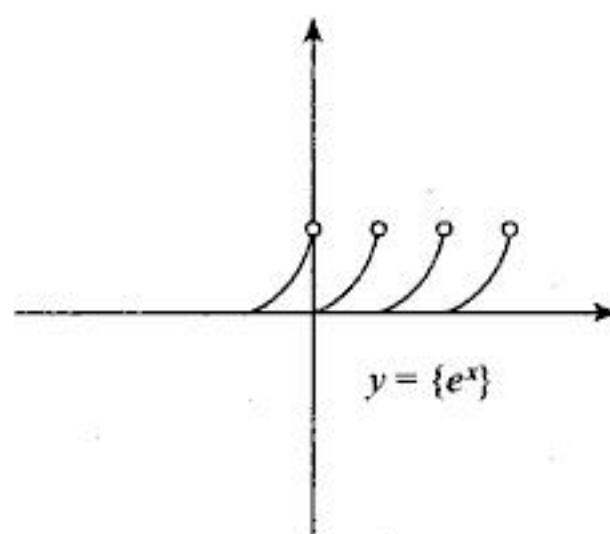


Fig. 1.212

1.8.21 Concept 21

If $f(x)$ and $g(x)$ are two functions then for graph of the function $h(x) = \text{Maximum } \{f(x), g(x)\} = \text{Max. } \{f(x), g(x)\}$ use:-

- (i) Draw the graph of function $y = f(x)$ and $y = g(x)$ and defined their points of intersection.
- (ii) Between the two points of intersection use $h(x) = \text{Maximum } \{f(x), g(x)\} = \text{Max. } \{f(x), g(x)\} =$

$$\begin{cases} f(x), & \text{when } f(x) \geq g(x) \\ g(x), & \text{when } g(x) \geq f(x) \end{cases}$$

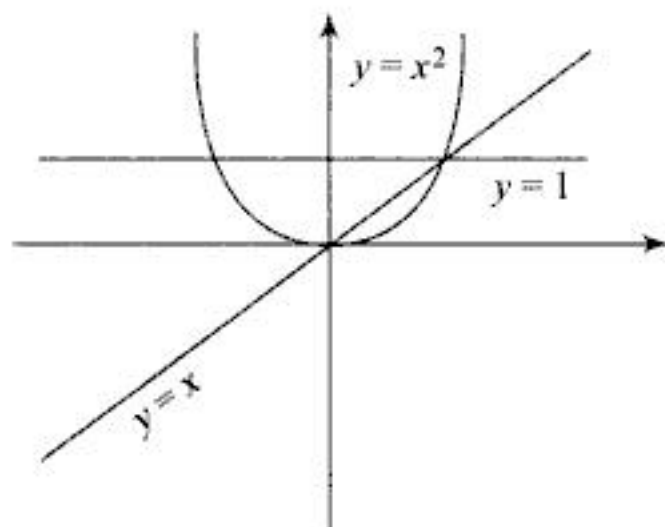


Fig. 1.213

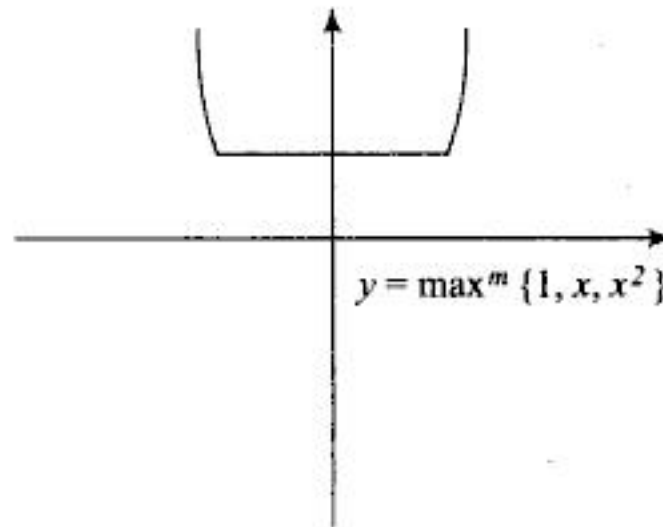


Fig. 1.214



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1.8.26 Concept 26

If $f(x)$ and $g(x)$ are two functions then for graph of function $h(x) = \frac{f(x)}{g(x)}$ use:-

- (i) Find maximum and minimum value of $f(x)$.
- (ii) If $a \leq f(x) \leq b$ then divide all sides by $g(x)$ i.e. $\frac{a}{g(x)} \leq \frac{f(x)}{g(x)} \leq \frac{b}{g(x)} \Rightarrow \frac{a}{g(x)} \leq y \leq \frac{b}{g(x)}$

- (iii) Draw the graph of $y = \frac{a}{g(x)}$ and $y =$

$\frac{b}{g(x)}$ separately on the plane then graph

of function $h(x) = \frac{f(x)}{g(x)}$ lies between them.

e.g. $y = \frac{\sin x}{e^x}$

$\therefore -1 \leq \sin x \leq 1$

Divide all three sides by e^x

$$= \frac{-1}{e^x} \leq \frac{\sin x}{e^x} \leq \frac{1}{e^x}$$

$$= -e^{-x} \leq y \leq e^{-x}$$

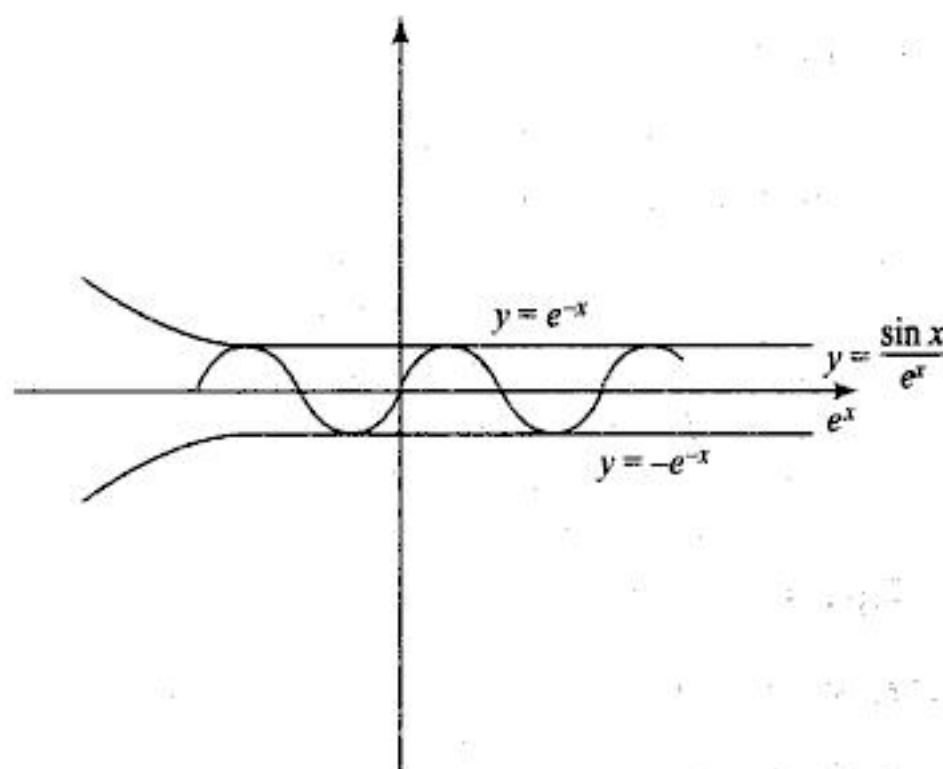


Fig. 1.223

1.8.27 Concept 27

If $f(x)$ be any non zero function then for graph of function $\frac{1}{f(x)}$ use:-

- (i) Find critical points where $f(x) = 0$ and represent critical points on x-axis.
- (ii) If $x = a$ is critical point then draw graph of $x = a$ and use

- (a) When $f(x)$ increases then $\frac{1}{f(x)}$ decreases.

- (b) When $f(x)$ decreases then $\frac{1}{f(x)}$ increases

- (c) When $f(x) \rightarrow 0$ then $\frac{1}{f(x)} \rightarrow \infty$ or, $-\infty$

according as $f(x) > 0$ or $f(x) < 0$ in the neighbourhood.

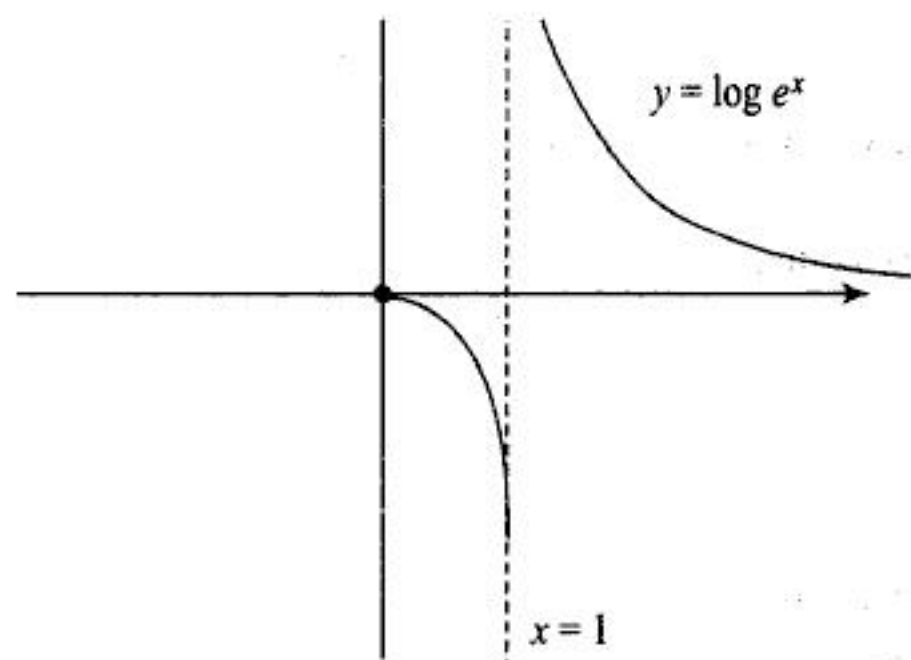


Fig. 1.224



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TYPEWISE SOLVED EXAMPLES

Type-I: Problems Based on Value of Function at a Point

Example 1.70 If $f(x) = \frac{1+x}{1-x}$ then the value of $\frac{f(x) \cdot f(x^2)}{1+\{f(x)\}^2}$ is equal to

- (a) 1 (b) $1/2$ (c) $x-1$ (d) None of these

Solution Given $f(x) = \frac{1+x}{1-x} \therefore f(x^2) = \frac{1+x^2}{1-x^2}$

$$\text{As, } \frac{f(x) \cdot f(x^2)}{1+\{f(x)\}^2} = \frac{\left(\frac{1+x}{1-x}\right) \cdot \left(\frac{1+x^2}{1-x^2}\right)}{1+\left(\frac{1+x}{1-x}\right)^2} = \frac{1+x^2}{2(1+x^2)} = \frac{1}{2}$$

Answer: (b)

Type-II: Problems Based on Composite Function

Example 1.71 If $f(x) = \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cdot \cos\left(x + \frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right) = 1$ then the value of $g \circ f(x)$ is equal to

- (a) $1/3$ (b) 0 (c) 5 (d) 1

Solution Given $f(x) = \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cdot \cos\left(x + \frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right) = 1$

$$\begin{aligned} \text{As } f(x) &= \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cdot \cos\left(x + \frac{\pi}{3}\right) \\ &= \sin^2 x + \frac{1}{4}(\sin x + \sqrt{3} \cos x)^2 + \frac{1}{2} \cos x (\cos x + \sqrt{3} \sin x) \\ &= \frac{5}{4}(\sin^2 x + \cos^2 x) = \frac{5}{4} \therefore g \circ f(x) = g\{f(x)\} = g\left(\frac{5}{4}\right) = 1 \end{aligned}$$

Answer: (d)

Example 1.72 If $f(x) = (a - x^n)^{1/n}$ where $a > 0$ and n is positive integer then $f\{f(x)\} = x$. True or False.

Solution Given

$$y = f(x) = (a - x^n)^{1/n}$$

$$\therefore f\{f(x)\} = f\{(a - x^n)^{1/n}\} = \{a - \{(a - x^n)^{1/n}\}^n\}^{1/n} = x$$

Answer: True.



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(a) Even extension of the function $f(x)$ is

$$\begin{cases} f(x), & \text{when } 0 \leq x < 1 \text{ or } x \geq 1 \\ f(-x), & \text{when } -1 \leq x < 0 \text{ or } x \leq -1 \end{cases}$$

$$= \begin{cases} \begin{cases} x^2, & \text{when } 0 \leq x < 1 \\ 2x, & \text{when } x \geq 1 \end{cases} \\ \begin{cases} x^2, & \text{when } -1 \leq x < 0 \\ -2x, & \text{when } x \leq -1 \end{cases} \end{cases}$$

$$= \begin{cases} x^2, & \text{when } |x| < 1 \\ 2|x|, & \text{when } |x| \geq 1 \end{cases}$$

(b) Odd extension of the function $f(x)$ is

$$\begin{cases} f(x), & \text{when } 0 \leq x < 1 \text{ or } x \geq 1 \\ -f(-x), & \text{when } -1 \leq x < 0 \text{ or } x \leq -1 \end{cases}$$

$$= \begin{cases} \begin{cases} x^2, & \text{when } 0 \leq x < 1 \\ 2x, & \text{when } x \geq 1 \end{cases} \\ \begin{cases} -x^2, & \text{when } -1 \leq x < 0 \\ 2x, & \text{when } x \leq -1 \end{cases} \end{cases}$$

$$= \begin{cases} x|x|, & \text{when } |x| < 1 \\ 2x, & \text{when } |x| \geq 1 \end{cases}$$

Example 1.83 Let the function $f(x) = 4 \sin x + 3 \cos x + \log(|x| + \sqrt{1+x^2})$ be defined on the interval $[0, 1]$.

The odd extension of $f(x)$ to the interval $[-1, 1]$ is

(a) $4 \sin x + 3 \cos x + \log(|x| + \sqrt{1+x^2}), -1 \leq x < 0$

(b) $4 \sin x - 3 \cos x - \log(|x| + \sqrt{1+x^2}), -1 \leq x < 0$

(c) $4 \sin x + 3 \cos x + \log(|x| + \sqrt{1+x^2}), -1 \leq x < 0$

(d) None of these

Solution The odd extension of $f(x)$ on $[-1, 1]$ is given by

$$g(x) = \begin{cases} f(x), & 0 \leq x \leq 1 \\ -f(-x), & -1 \leq x < 0 \end{cases}$$



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$$\therefore f^{-1}(x) = \log(x + \sqrt{x^2 + 1})$$

Answer. (a)

Example 1.94 The inverse of the function $y = [1 - (x^3)^4]^{1/7}$ is

- (a) $3 + (1 - x^7)^{1/4}$ (b) $3 - (1 - x^7)^{1/4}$ (c) $3 - (1 + x^7)^{1/4}$ (d) None of these.

Solution Clearly y is one-one and onto function.

$$\text{We have } y = [1 - (x^3)^4]^{1/7}$$

$$\Rightarrow (x^3)^4 = 1 - y^7$$

$$\Rightarrow x = 3 + (1 - y^7)^{1/4}$$

$$\Rightarrow f^{-1}(y) = 3 + (1 - y^7)^{1/4} \Rightarrow f^{-1}(x) = 3 + (1 - x^7)^{1/4}.$$

Answer. (a)

Example 1.95 The inverse of the function $f(x) = \frac{a^x - a^{-x}}{a^x + a^{-x}}$ is

- (a) $\frac{1}{2} \log_a \left(\frac{1-x}{1+x} \right)$ (b) $\frac{1}{2} \log_a \left(\frac{1+x}{1-x} \right)$ (c) $\log_a \left(\frac{1+x}{1-x} \right)$ (d) None of these

$$\text{Solution Let } y = \frac{a^x - a^{-x}}{a^x + a^{-x}} = \frac{a^{2x} - 1}{a^{2x} + 1}$$

$$\Rightarrow \frac{y-1}{y+1} = \frac{(a^{2x}-1) - (a^{2x}+1)}{(a^{2x}-1) + (a^{2x}+1)}$$

$$\Rightarrow \frac{y-1}{y+1} = \frac{-2}{a^{2x}} \Rightarrow a^{2x} = \frac{y+1}{y-1}$$

$$\Rightarrow 2x \log_a a = \log_a \left(\frac{y+1}{y-1} \right) \Rightarrow x = \frac{1}{2} \log_a \left(\frac{y+1}{y-1} \right) \therefore f^{-1}(x) = \frac{1}{2} \log_a \left(\frac{x+1}{x-1} \right)$$

Answer: (b)

Type-XII: Problems Based on Domain of Function

Example 1.96 If $f(x) = \sin \left\{ \log \left(\frac{\sqrt{4-x^2}}{1-x} \right) \right\}$ then the domain of $f(x)$ is

- (a) $[-1, 1]$ (b) $[2, 3]$ (c) $[-1, 2]$ (d) $(-2, 1)$

$$\text{Solution Given function } f(x) = \sin \left\{ \log \left(\frac{\sqrt{4-x^2}}{1-x} \right) \right\}$$

Here, $f(x)$ is defined for $\frac{\sqrt{4-x^2}}{1-x} > 0 \Rightarrow 1-x > 0$ and $\sqrt{4-x^2} > 0$



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And $x > 0, x^2 - 1 > 0$ and $x^2 - 1 \neq 1$

$$\Rightarrow x > 0, x \in (-\infty, -1) \cup (1, \infty) \text{ and } x \neq \pm \sqrt{2}$$

$$\Rightarrow x \in (1, \sqrt{2}) \cup (\sqrt{2}, \infty)$$

(2)

From equation (1) and (2), we get

$$x \in (\sqrt{2}, \infty)$$

Hence, domain of $f(x)$ is $(\sqrt{2}, \infty)$

Answer. (a)

Example 1.104 The domain of the function $f(x) = \tan^{-1} \sqrt{x(x+1)} + \sqrt{x^2+x+1}$ is

- (a) $[-1, 0]$ (b) $\{-1, 0\}$ (c) $(-\infty, -1] \cup [0, \infty)$ (d) $(-\infty, \infty)$

Solution $\tan^{-1} \sqrt{x(x+1)}$ is defined, if

$$x(x+1) \geq 0 \text{ or } x \leq -1 \text{ or } x \geq 0 \quad (1)$$

$\sin^{-1} \sqrt{x^2+x+1}$ is defined, if $0 \leq x^2+x+1 \leq 1$

$$\text{Now } x^2+x+1 = \left(x+\frac{1}{2}\right)^2 + \frac{3}{4} \geq 0, \text{ for all real } x \quad (2)$$

$$\text{Also, } x^2+x+1 \leq 1 \Rightarrow x(x+1) \leq 0 \Rightarrow -1 \leq x \leq 0 \quad (3)$$

From, (1), (2) and (3), we get domain of $f = \{-1, 0\}$.

Answer. (b)

Example 1.105 The domain of the function $f(x) = \sin^{-1} \left(\frac{4}{3+2\cos x} \right)$ is

$$(a) \ 2n\pi - \frac{\pi}{6} \leq x \leq 2n\pi + \frac{\pi}{6}, n \in I \quad (b) \ 0 \leq x \leq 2n\pi + \frac{\pi}{6}, n \in I$$

$$(c) \ 2n\pi - \frac{\pi}{6} < x < 2n\pi + \frac{\pi}{6}, n \in I \quad (d) \ 2n\pi - \frac{\pi}{6} \leq x \leq 0, n \in I$$

Solution $f(x)$ is defined, if $-1 \leq \frac{4}{3+2\cos x} \leq 1$

Since $3+2\cos x > 0$ for all real x ,

$$\therefore \frac{4}{3+2\cos x} \leq 1 \Rightarrow \cos x \geq \frac{1}{2}$$

$$\Rightarrow 2n\pi - \frac{\pi}{6} \leq x \leq 2n\pi + \frac{\pi}{6}, n \in I.$$

Answer. (a)



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Since $\sin x$ is an increasing function on $\left[0, \frac{\pi}{4}\right]$

$$\text{Therefore, } \sin 0 \leq \sin \sqrt{\frac{\pi^2}{16} - x^2} \leq \sin \frac{\pi}{4}$$

$$\Rightarrow 0 \leq 3 \sin \sqrt{\frac{\pi^2}{16} - x^2} \leq \frac{3}{\sqrt{2}} \Rightarrow 0 \leq y \leq \frac{3}{\sqrt{2}}$$

$$\therefore \text{Range of } y = \left[0, \frac{3}{\sqrt{2}}\right].$$

Answer. (a)

Type-XIV: Problems Based on Algebra of Function

Example 1.113 A cubic expression $f(x)$ satisfies the condition

$$f(x) + f\left(\frac{1}{x}\right) = f(x) f\left(\frac{1}{x}\right), \text{ then prove that } f(x) = 1 + x^3 \text{ or } 1 - x^3.$$

If $f(3) = 28$ then prove that $f(2) = 9$.

$$\text{Solution } f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\therefore f(x) \left\{ f\left(\frac{1}{x}\right) - 1 \right\} = f\left(\frac{1}{x}\right) - 1 + 1$$

$$\text{or } \left\{ f\left(\frac{1}{x}\right) - 1 \right\} \{f(x) - 1\} = 1$$

Above shows that $f(x) - 1$ and $f\left(\frac{1}{x}\right) - 1$ are reciprocal of each other and so are x and $1/x$.

Above is possible when

$$f(x) - 1 = x^n \text{ or } -x^n \text{ so that } f\left(\frac{1}{x}\right) - 1 = \frac{1}{x^n} \text{ or } \frac{1}{x^n}.$$

Hence $f(x) = 1 + x^n$ or $1 - x^n$, i.e., $1 \pm x^n$

Here $f(x)$ is a cubic $\therefore f(x) = 1 + x^3$

Given $f(3) = 28 \therefore f(3) = 1 + 3^3 = 28$

$\therefore f(x) = 1 + x^3$ Hence $f(2) = 1 + 2^3 = 9$.

Example 1.114 Let $f(x)$ be a polynomial function satisfying $f(x)f(y) = f(x) + f(y) + f(xy) - 2 \forall x, y \in R$. If $f(2) = 5$ then prove that $f(5) = 26$.

Solution We are given $f(2) = 5$, so we put $x = 2, y = 1$

$$f(2)f(1) = f(2) + f(1) + f(2) - 2$$



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- (a) $f(x) = x^2 - x + \sin x$
 (b) $f(x) = x^2 - x - \sin x$
 (c) $f(x) = -x^2 + x - \sin x$
 (d) None of these
2. Let $f(x) = x^3 - \sin x + \sec x$, $2 \leq x \leq 4$ then odd extension of $f(x)$ in $[-4, -2]$ is given by
 (a) $f(x) = -x^3 - \sin x + \sec x$
 (b) $f(x) = x^3 - \sin x + \sec x$
 (c) $f(x) = -x^3 \sin x - \sec x$
 (d) None of these
3. If $f: [-4, 0] \rightarrow R$ is defined by $f(x) = e^x + \sin x$, its even extension to $[-4, 4]$ is given by
 (a) $f: [-4, 0] \rightarrow R$, $f(x) = e^x + \sin x$
 (b) $f: [-4, 0] \rightarrow R$, $f(x) = e^x - \sin x$
 (c) $f: [-4, 0] \rightarrow R$, $f(x) = e^{-x} - \sin x$
 (d) None of these

Type-VI: Problems Based on Implicit and Explicit Functions

1. Function $2^y - 2^{-y} = 2x$ is
 (a) Implicit function
 (b) Explicit function
 (c) Periodic function
 (d) None
2. Function $x^2 - \cos^{-1} x = \pi$ is
 (a) Implicit function
 (b) Explicit function
 (c) Periodic function
 (d) None
3. Function $10^y + 10^x = 5$ is
 (a) Implicit function
 (b) Explicit function
 (c) Periodic function
 (d) None

Type-VII: Problems Based on Monotonic Function

1. Let $f(x) = \int e^x(x-1)(x-2)dx$ then $f(x)$ decreases in the interval
 (a) $(-\infty, -2)$ (b) $(-2, -1)$
 (c) $(1, 2)$ (d) $(2, \infty)$

2. The function $f(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$ is
 (a) Increasing
 (b) Decreasing and odd
 (c) Even
 (d) Increasing and even
3. If $f(x) = \frac{x}{\sin x}$ and $g(x) = \frac{x}{\tan x}$, $0 < x \leq 1$ then in this interval
 (a) Both $f(x)$ and $g(x)$ are increasing functions.
 (b) Both $f(x)$ and $g(x)$ are decreasing functions.
 (c) $f(x)$ is an increasing function.
 (d) $g(x)$ is an increasing function.
4. If $f(x) = kx^3 - 9x^2 + 9x + 3$ is monotonically increasing in each interval then
 (a) $k < 3$ (b) $k \leq 3$
 (c) $k > 3$ (d) None of these
5. Value of a does the curve $f(x) = x(a^2 - 2a - 2) + \cos x$ always remain monotonic for all real x is
 (a) $a \in R$
 (b) $a > 0$
 (c) $1 - \sqrt{2} < a < 1 + \sqrt{2}$
 (d) None of these
6. Function $f(x) = \int_0^x (e^t - 1)(t - 1)(\sin t - \cos t) \sin t dt$, $\forall x \in (-\pi/2, 2\pi)$ then $f(x)$ is decreasing in
 (a) $(-\pi/2, 0) \cup (\pi/4, 1) \cup (\pi, \pi/4)$
 (b) $(-\pi/2, \pi/4) \cup (1, \pi) \cup (5\pi/4, 2\pi)$
 (c) $(\pi/4, 1) \cup (\pi, 5\pi/4)$
 (d) None of these
7. If $f(x) = (ab - b^2 - 2)x + \int_0^x (\sin^4 t + \cos^4 t) dt$ is decreasing for all real x , $b \in R$ and b is independent of x , then
 (a) $a \in (0, \sqrt{6})$
 (b) $a \in (-\sqrt{6}, \sqrt{6})$



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LEVEL-II SUBJECTIVE TOPIC- WISE PROBLEMS

Type-I: Problems Based on Value of Function at a Point

$$1. \text{ Let } f(x) = \begin{cases} 3^x, & \text{when } -1 < x < 0 \\ 4, & \text{when } 0 \leq x < 1 \\ 3x-1 & \text{when } 1 \leq x \leq 3 \end{cases}$$

then find the value of $f(2)$, $f(0)$, $f(0.5)$, $f(-0.5)$ and $f(3)$

$$2. \text{ Find a function of the form } f(x) = a + bc^x \text{ (} c > 0 \text{) when } f(0) = 15, f(2) = 30, f(4) = 90$$

$$3. \text{ Let } f(x) = \begin{cases} 2x^2 + 1, & \text{when } x \leq 2 \\ \frac{1}{x-2}, & \text{when } 2 \leq x \leq 3 \\ 2x-5 & \text{when } x > 3 \end{cases}$$

then find the value of $f(2\sqrt{2})$, $f(\sqrt{8})$, $f(\sqrt{\log_2 1024})$

$$4. \text{ Let } f(x) = \begin{cases} 3^{-x} - 1, & \text{when } -1 \leq x < 0 \\ \tan(x/2), & \text{when } 0 \leq x < \pi \\ \frac{x}{x^2 - 2} & \text{when } \pi \leq x \leq 6 \end{cases}$$

then find the value of $f(-1)$, $f(\pi/2)$, $f(2\pi/3)$, $f(4)$ and $f(6)$

$$5. \text{ Given, the function } f(x) = \frac{x+1}{x^3-1}. \text{ Find } f(-1), f(a+1), f(a)+1$$

Type-II: Problems Based on Composite Function

$$1. \text{ Let } f: [0, 1] \rightarrow [0, 1] \text{ where } f(x) = \frac{1-x}{1+x}$$

and $g: [0, 1] \rightarrow [0, 1]$ where $g(x) = 4x(1-x)$ then find (i) $f \circ g(x)$ (ii) $g \circ f(x)$

$$2. \text{ Let } f(x) = \frac{x-1}{x+1}, f^2(x) = f\{f(x)\}, f^3\{f(x)\} =$$

$$f\{f^2(x)\}, \dots, f^{k+1}(x) = f\{f^k(x)\} \text{ for } k = 1, 2, 3, \dots, \text{ find } f^{2010}(x).$$

$$3. \text{ Let } f(x) = \sin^2 x + \sin^2 \left(x + \frac{\pi}{3} \right) + \cos x \cdot$$

$$\cos \left(x + \frac{\pi}{3} \right) \text{ and } g(x) = \begin{cases} 2x, & 0 \leq x < 1 \\ x + \frac{3}{4}, & 1 \leq x < 2 \end{cases}$$

then find $g\{f(x)\}$.

$$4. \text{ If } f(x) = \frac{1-x}{1+x}, x \neq 1 \text{ then show that } f\{f(x)\}$$

$$+ f\left\{f\left(\frac{1}{x}\right)\right\} > 2, x > 0$$

$$5. \text{ If } f(x) = \cos(\log x), \text{ show that } 2 \cdot f(x) \cdot f(y) - \left\{ f(xy) + f\left(\frac{x}{y}\right) \right\} = 0$$

Type-III : Problems Based on Periodic Function

$$1. \text{ Let } f(x) \text{ be a real valued function defined for all real } x \text{ such that for some fixed } a >$$

$$0 \quad f(x+a) = \frac{1}{2} + \sqrt{f(x) - \{f(x)\}^2} \text{ for real } x.$$

Show that $f(x)$ is periodic function with period $2a$.

$$2. \text{ Let } f(x, y) \text{ be a periodic function satisfy } f(x, y) = f(2x+2y, 2y-2x) \text{ for all real } x \text{ and } y. \text{ Let } g(x) = f(2^x, 0) \text{ show that } g(x) \text{ is a periodic function with period } 12.$$

$$3. \text{ Find the period of } f(x) \text{ which satisfy}$$

$$(a) \quad f(x-1) + f(x+1) = \sqrt{3} f(x).$$

$$(b) \quad f(x+p) = 1 + \{1 - 3f(x) + 3f^2(x) - f^3(x)\}^{1/3}$$

$$(c) \quad f(x-1) + f(x+3) = f(x+1) + f(x+5)$$

$$4. \text{ If } a > 1 \text{ find the period of following functions}$$

$$(i) \quad f(x) = a^{\cos x} \quad (ii) \quad f(x) = a^{|\cos x|}$$



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$$(v) f(x) = \frac{\log(\sin^{-1} \sqrt{x^2 + x + 1})}{\log(x^2 - x + 1)}$$

$$(vi) f(x) = \sin^{-1} \frac{1}{|x^2 - 1|} +$$

$$\frac{1}{\sqrt{\sin^2 x + \sin x + 1}}$$

$$(vii) f(x) = \sqrt{\sin^{-1}(3x - 4x^3)} + \sqrt{\cos^{-1} x}$$

$$(viii) f(x) = \frac{\log_{2x} 3}{\cos^{-1}(2x - 1)}$$

3. Find the domain of the following algebraic function:

$$(i) f(x) = \sqrt{\frac{2x + 1}{x^3 - 3x^2 + 2x}}$$

$$(ii) f(x) = \sqrt{2 - |x|} + \sqrt{1 + |x|}$$

$$(iii) f(x) = \sqrt{1 - \sqrt{1 - \sqrt{1 - x^2}}}$$

$$(iv) f(x) = \sqrt{x^{12} - x^9 + x^4 - x + 1}$$

$$(v) f(x) = (x^2 + x + 1)^{1/2008}$$

4. Find the domain of the following function which related to factorial:

$$(i) f(x) = {}^3C_x$$

$$(ii) f(x) = {}^{16-x}C_{2x-1} + {}^{20-3x}P_{4x-5}$$

$$(iii) f(x) = \sqrt{x^2 + 4x} C_{2x^2+3}$$

5. Find the domain of the following greatest integral, least integral, fractional part function:

$$(i) f(x) = \sqrt{\frac{x-1}{x-2\{x\}}}$$

$$(ii) f(x) = \log\{x\}$$

$$(iii) f(x) = \sin^{-1}\left(\frac{2-3[x]}{4}\right)$$

$$(iv) f(x) = \sin^{-1} \lfloor (2x^2 - 3) \rfloor$$

$$(v) f(x) = \sqrt{2\{x\}^2 - 3\{x\} + 1}, \quad x \in [-1, 1]$$

6. If $f(x) = \frac{1}{\sqrt{[x] - x}}$, where $[.]$ denotes the

greatest integral function less than equal to x , then find the domain of the function.

7. The domain $f(x)$ is defined in $[0, 1]$, then find the domain of the function $f(\tan x)$.

8. Find the domain of the function

$$f(x) = \sin^{-1}\left(\frac{2-|x|}{4}\right) + \cos^{-1}\left(\frac{2-|x|}{4}\right) + \tan^{-1}\left(\frac{2-|x|}{4}\right) \text{ is:}$$

9. The domain of the function $f(x)$ is $(0, 1)$, then find the domain of the function $f(e^x) + f(\ln|x|)$

10. Find the domain of the function $f(x) =$

$$\sqrt{\left\{\sin^{-1}(2x) + \frac{\pi}{6}\right\}}$$

11. If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$, then find $g \circ f(x)$ is invertible in the domain.

12. If $0 < \lambda < 1$ and $f(x) = \sqrt{\frac{\log_{\lambda}|x-2|}{|x|}}$,

then find domain of $f(x)$.

13. If $f\{g(x)\}$ is defined in interval $[0, 1]$, then find range of function $g(x)$.

Type-XIII: Problems Based on Range of Function

1. Find the range of the function

$$(a) f(x) = 3 \sin x + 4 \cos x$$

$$(b) f(x) = {}^3C_x$$



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- R:** Function $f(x) = \log x$ is the inverse of function $g(x) = e^x$.
4. **A:** Let $f: \mathbb{R} - \{1, 2, 3\} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{1}{x-1} + \frac{2}{x-2} + \frac{3}{x-3}$ then $f(x)$ is many-one function.
R: If either of $f'(x) > 0$ or $f'(x) < 0$, $\forall x \in$ domain of $f(x)$ then $y = f(x)$ is one-one function.
5. **A:** $y = -x + \sin x$ and its inverse intersect the curve $y = -x$.
R: Let $f(x)$ be an invertible function and $f^{-1}(x)$ be the inverse of $f(x)$ then $y = f(x)$ and $y = f^{-1}(x)$ intersect either of the line $y = -x + c$ or $y = x$.
6. **A:** $f(x) = |x|$ is not one-one function.
R: The negative real number are not the images of any real numbers.
7. **A:** $y = f(x)$ is defined by $x^2 - \cos^{-1} y = \pi$ then domain of $y = f(x)$ is $\mathbb{R}$
R: $\cos^{-1} y \in [0, \pi]$.
8. **A:** Every even function $y = f(x)$ are not one-one function for all $x \in \text{Domain}$.
R: Every even function is symmetrical about y-axis.
9. **A:** $f(x) = \sin x + \cos ax$ is a periodic function.
R: a is rational number.
10. **A:** $f(x) = x^5 - 16x + 2$ then $f(x) = 0$ has only one root in $[-1, 1]$
R: $f(-1)$ and $f(1)$ are opposite signs.
11. **A:** The domain of the function $f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x$ is $[-1, 1]$
R: $\sin^{-1} x$ and $\cos^{-1} x$ is defined in $|x| \leq 1$ and $\tan^{-1} x$ defined for all x .
12. **A:** Every equation $f(x) = x^3 + 2x^2 + 3x + 2$ has at least one negative root.
B: Every equation of odd degree has at least one real root whose sign is opposite to that of its constant term.
13. **A:** A continuous one-one onto function is always monotonic function.
B: A continuous function is one-one only when it is monotonic.
14. **A:** If $f'(x) > 0 \forall x \in \mathbb{R}$, $f'(3) = 0$ and $g(x) = f(\tan^2 x - 2 \tan x + 4)$ then $g(x)$ is increasing in $(\pi/4, \pi/2)$.
R: $g(x) = f((\tan x - 1)^2 + 3) \Rightarrow g'(x) > 0$ always true.
15. **A:** For $a_1 < a_2 < a_3 < \dots < a_{2n}$ the function. $f(x) = |x - a_1| + |x - a_2| + |x - a_3| + \dots + |x - a_{2n}|$ will have no extreme points.
R: $f(x)$ is monotonically decreasing from $x \in (-\infty, a_n)$.
 $f(x)$ is constant for $x \in [a_n, a_{n+1}]$.
 $f(x)$ is monotonically increasing from $x \in [a_{n+1}, \infty)$.

ANALYTICAL TEST

1. Which of the following is the correct option?
 The domain of function $f(x) = {}^{7-x}P_{x-3}$ is
 (a) $\{3, 4, 5\}$ (b) $\{1, 2, 3\}$
 (c) $[1, 2, 3]$ (d) $[3, 4, 5]$
2. The function $f(x) = \cos\left(\frac{\pi}{x}\right)$ is increasing or decreasing in the interval
 (a) $(2n+1, 2n), n \in \mathbb{N}$
- (b) $\left(\frac{1}{2n+1}, \frac{1}{2n}\right), n \in \mathbb{N}$
 (c) $\left(\frac{1}{2n+2}, \frac{1}{2n+1}\right), n \in \mathbb{N}$
 (d) None
3. Which of the following is correct
 (a) $f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$ is one-one function.



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Type-II: Problem Based on Composite Function

$$1. (a) \text{ fog}(x) = \frac{x^2 - x + 1}{-x^2 + x + 1}, x \in [0, 1]$$

$$(b) \text{ gof}(x) = \frac{2x(1-x)}{(1+x)^2}, x \in [0, 1]$$

$$2. -\frac{1}{x}$$

$$3. 2$$

Type-III: Problems Based on Periodic Function

$$3. (a) 12 \quad (b) 2p \quad (c) 8$$

$$4. (i) 2\pi \quad (ii) \pi$$

$$5. (i) 2\pi|k| \quad (ii) 2\pi \quad (iii) 1 \quad (iv) 2\pi \quad (v) 2\pi \quad (vi) \frac{1}{2}$$

Type-IV: Problems Based on Odd and Even Functions

$$3. (i) \frac{(x+1)^{2010} + (1-x)^{2010}}{2} + \frac{(x+1)^{2010} - (1-x)^{2010}}{2}$$

$$(ii) \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} \quad 5. [-x]$$

$$6. (a) \text{ Odd function} \quad (b) \text{ even function} \quad (c) \text{ even function} \quad (d) \text{ odd function} \\ (e) \text{ even when } n \text{ is even and odd when } n \text{ is odd} \quad (f) \text{ even function}$$

Type-V: Problems Based on Odd Extensive and Even Extensive Functions

$$1. \text{ even extensive} = \begin{cases} \log(-x + \sqrt{x^2 + 1}) & , \text{ when } -10 \leq x < -\pi \\ \sin^2 x - \tan x & , \text{ when } -\pi \leq x < -2 \text{ and odd extensive not define} \\ |x| & , \text{ when } -2 \leq x \leq 2 \end{cases}$$

$$2. \begin{cases} -f(-x) = -x^2 + x + \sin x + \cos x - \log(1 + |x|), -1 \leq x < 0 \\ f(x) = x^2 + x + \sin x - \cos x + \log(1 + |x|), 0 \leq x \leq 1 \end{cases}$$

Type-VII: Problems Based on Monotonic Function

$$1. (a) (-\infty, -1) \cup (3, \infty) \quad (b) (-1, 3)$$

$$2. (a) f(x) \text{ is increases } x \in (-\infty, -2] \cup [-1, 0] \cup [1, \infty) \\ (b) f(x) \text{ is decreases } x \in [-2, -1] \cup [0, 1]$$



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2.1.2 Right Hand Limit (R.H.L.)

Right tendency of $f(x)$ at $x = a$ is called its right hand limit i.e. if x approaches a from the right that is from larger values of x than a , the limit of $f(x)$ is called right hand limit. It is denoted as $\lim_{x \rightarrow a^+} f(x)$

or $\lim_{x \rightarrow a^+} f(x)$ and calculated as $\lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h)$

If L.H.L = R.H.L i.e. $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$ then $\lim_{x \rightarrow a} f(x)$ exist.

Example 2.1 The value of $\lim_{x \rightarrow 1} \frac{\sqrt{1 - \cos 2(x-1)}}{(x-1)}$ is

(IIT (JEE)-1998)

- (a) 1 (b) 2 (c) 4 (d) none

Solution Given $\lim_{x \rightarrow 1} \frac{\sqrt{1 - \cos 2(x-1)}}{(x-1)} = \lim_{x \rightarrow 1} \frac{\sqrt{2} |\sin(x-1)|}{(x-1)}$

$$= \lim_{x \rightarrow 1} \frac{\sqrt{2} \left| \sin \frac{(x-1)}{(x-1)} \right|}{(x-1)} = \lim_{x \rightarrow 1} \sqrt{2} \frac{|x-1|}{(x-1)}$$

$$\text{Here, L.H.L.} = \lim_{x \rightarrow 1-h} \sqrt{2} \frac{|x-1|}{(x-1)} = \lim_{h \rightarrow 0} \frac{\sqrt{2} |1-h-1|}{1-h-1} = \lim_{h \rightarrow 0} \frac{\sqrt{2}(h)}{(-h)} = -\sqrt{2}$$

$$\text{And R.H.L} = \lim_{x \rightarrow 1+h} \sqrt{2} \frac{|x-1|}{(x-1)} = \lim_{h \rightarrow 0} \frac{\sqrt{2} |1+h-1|}{1+h-1} = \lim_{h \rightarrow 0} \frac{\sqrt{2}|h|}{h} = \sqrt{2}$$

Here, L.H.L $\neq$ R.H.L

Hence, $\lim_{x \rightarrow 1} \frac{\sqrt{1 - \cos 2(x-1)}}{(x-1)}$ does not exist.

The correct option is (d)

Example 2.2 If $\lim_{x \rightarrow a} \{f(x) \cdot g(x)\}$ exist then $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist. State whether true or false.

(IIT (JEE)-1981)

Solution Let $f(x) = \frac{|x-a|}{x-a}$ and $g(x) = \frac{x-a}{|x-a|}$

$$\text{Here, } \lim_{x \rightarrow a} f(x) \cdot g(x) = \lim_{x \rightarrow a} \left\{ \frac{|x-a|}{(x-a)} \times \frac{(x-a)}{|x-a|} \right\} = \lim_{x \rightarrow a} 1 = 1$$

But $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ does not exist.

Hence, the above statement is false.



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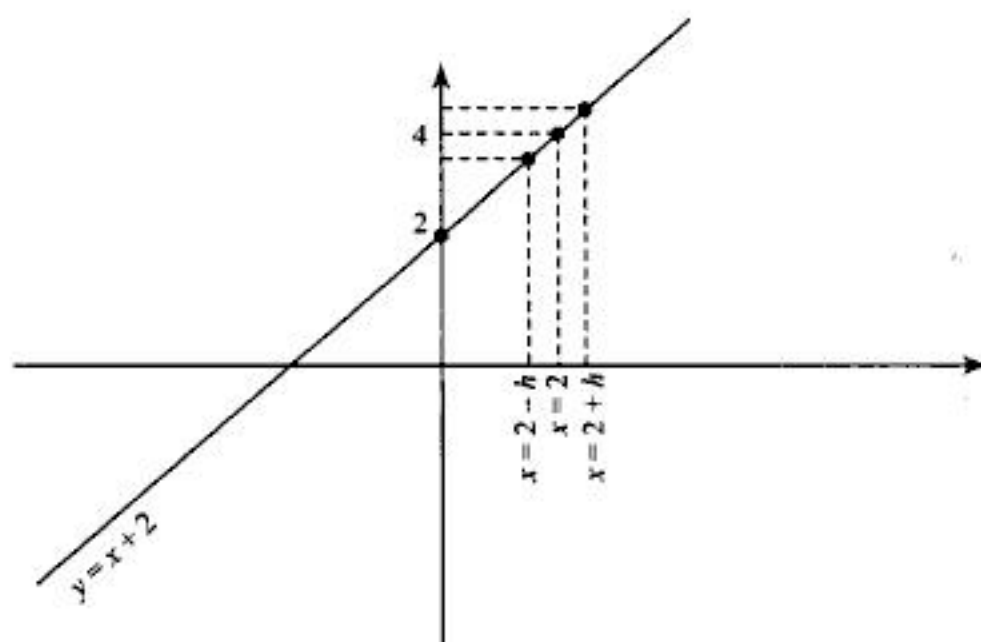


Fig. 2.3

Here, $x \rightarrow 2 - h$, $f(x) \rightarrow 4$ and $x \rightarrow 2 + h$, $f(x) \rightarrow 4$
 $f(2 - h) = f(2 + h) = f(2) = 4$

Possibility IV $\lim_{x \rightarrow a} f(x)$ and $f(a)$ both exist but are unequal:-

Mathematical examples:- for function $f(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x < 2 \\ 3, & x = 2 \end{cases}$

Here left hand limit of $f(x)$ at $x = 2$ is

$$\lim_{x \rightarrow 2-h} f(x) = \lim_{x \rightarrow 2-h} \left(\frac{x^2 - 4}{x - 2} \right) = \lim_{h \rightarrow 0} \frac{(2-h)^2 - 4}{(2-h) - 2} = 4$$

And right hand limit of $f(x)$ at $x = 2$ is

$$\lim_{x \rightarrow 2+h} f(x) = \lim_{x \rightarrow 2+h} \left(\frac{x^2 - 4}{x - 2} \right) = \lim_{h \rightarrow 0} \frac{(2+h)^2 - 4}{(2+h) - 2} = 4$$

And the value of function $f(x)$ at $x = 2$ is $f(2) = 3$

Hence, left hand limit of $f(x)$ at $x = 2$ is equal to right hand limit of $f(x)$ at $x = 2$ but not equal to middle hand limit of $f(x)$ at $x = 2$ i.e. $\lim_{x \rightarrow 2+h} f(x) = \lim_{x \rightarrow 2} f(x) \neq \lim_{x \rightarrow 2} f(x)$ i.e.

$\lim_{x \rightarrow a} f(x)$ and $f(a)$ both exist but are unequal

Graphically, from figure



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Example 2.13 Evaluate the limit $\lim_{x \rightarrow 1} \frac{\sqrt{x+8} - \sqrt{7x+2}}{\sqrt{x+3} - \sqrt{5-x}}$

Solution Given $\lim_{x \rightarrow 1} \frac{\sqrt{x+8} - \sqrt{7x+2}}{\sqrt{x+3} - \sqrt{5-x}}$

$$= \lim_{x \rightarrow 1} \frac{(x+8) - (7x+2)}{\sqrt{(x+8)+\sqrt{7x+2}} \times (\sqrt{x+3} + \sqrt{5-x})} \times \frac{\sqrt{x+3} + \sqrt{5-x}}{(x+3) - (5-x)}$$

$$= \lim_{x \rightarrow 1} \frac{-6(x-1)}{2(x-1)} \times \frac{\sqrt{x+3} + \sqrt{5-x}}{\sqrt{x+8} + \sqrt{7x+2}} = -2$$

Example 2.14 Evaluate the limit $\lim_{x \rightarrow 2} \frac{\sqrt[3]{3x+2} - \sqrt[3]{x+6}}{\sqrt{x+7} - 3\sqrt{x-1}}$

Solution Given $\lim_{x \rightarrow 2} \frac{\sqrt[3]{3x+2} - \sqrt[3]{x+6}}{\sqrt{x+7} - 3\sqrt{x-1}}$

Let $a = \sqrt[3]{3x+2} \Rightarrow a^3 = 3x+2$ and $b = \sqrt[3]{x+6} \Rightarrow b^3 = x+6$

$$\therefore \lim_{x \rightarrow 2} \frac{(3x+2) - (x+6)}{(3x+2)^{2/3} + (3x+2)^{1/2}(x+6)^{1/3} + (x+6)^{2/3}} \times \frac{\sqrt{x+7} + 3\sqrt{x-1}}{(x+7) - 9(x-1)} \left\{ \begin{array}{l} a-b = \frac{a^3-b^3}{a^2+ab+b^2} \end{array} \right.$$

$$= \lim_{x \rightarrow 2} \frac{2(x-2)}{-8(x-2)} \cdot \frac{\sqrt{x+7} + 3\sqrt{x-1}}{(3x+2)^{2/3} + (3x+2)^{1/2}(x+6)^{1/3} + (x+6)^{2/3}} = -\frac{1}{8}$$

Direct Use Formula

$$\lim_{f(x) \rightarrow a} \frac{\{f(x)\}^n - a^n}{f(x) - a} = na^{n-1}$$

Example 2.15 Evaluate the limit $\lim_{x \rightarrow -1} \left(\frac{1 + \sqrt[3]{x}}{1 + \sqrt[5]{x}} \right)$

Solution Given $\lim_{x \rightarrow -1} \frac{1 + \sqrt[3]{x}}{1 + \sqrt[5]{x}} = \lim_{x \rightarrow -1} \frac{x^{1/3} - (-1)}{x^{1/5} - (-1)}$

$$= \frac{\lim_{x \rightarrow -1} \left(\frac{x^{1/3} - (-1)}{x - (-1)} \right)}{\lim_{x \rightarrow -1} \left(\frac{x^{1/5} - (-1)}{x - (-1)} \right)} = \frac{\lim_{x \rightarrow -1} \left(\frac{x^{1/3} - (-1)^{1/3}}{x - (-1)} \right)}{\lim_{x \rightarrow -1} \left(\frac{x^{1/5} - (-1)^{1/5}}{x - (-1)} \right)} = \frac{\frac{1}{3}(-1)^{1/3-1}}{\frac{1}{5}(-1)^{1/5-1}} = \frac{5}{3}$$



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Example 2.23 Evaluate the limit $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$

(IIT(JEE) -2001)

Solution Given $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$

$$= \lim_{x \rightarrow 0} \frac{\sin\{\pi(1 - \sin^2 x)\}}{x^2} = \lim_{x \rightarrow 0} \frac{\sin\{\pi - \pi \sin^2 x\}}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin(\pi \sin^2 x)}{\pi \sin^2 x} \cdot \pi \sin^2 x}{x^2} = \lim_{x \rightarrow 0} \frac{\pi \left(\sin \frac{x}{x} \cdot x\right)^2}{x^2} = \pi$$

Substitution Method

Form (i): If $f(x)$ is any function and a is non-zero real then $\lim_{x \rightarrow a} f(x)$

can be solved by putting $x = a + h$ where $h \rightarrow 0$ i.e. $\lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a + h)$

Example 2.24 Evaluate the limit $\lim_{x \rightarrow -1} \frac{\cos\left(\frac{5\pi}{2}\right)}{x+1}$

Solution Given $\lim_{x \rightarrow -1} \frac{\cos\left(\frac{5\pi}{2}\right)}{x+1}$

Put $x = -1 + h$ where $h \rightarrow 0$

$$\begin{aligned} \therefore \lim_{x \rightarrow -1} \frac{\cos\left(\frac{\pi x}{2}\right)}{x+1} &= \lim_{h \rightarrow 0} \frac{\cos\left\{\frac{\pi(-1+h)}{2}\right\}}{-1+h+1} \\ &= \lim_{h \rightarrow 0} \frac{\cos\left\{-\frac{\pi}{2} + \frac{h}{2}\right\}}{h} = \lim_{h \rightarrow 0} \frac{\cos\left\{\frac{\pi}{2} - \frac{h}{2}\right\}}{h} \quad \{\because \cos(-\theta) = \cos \theta\} \\ &= \lim_{h \rightarrow 0} \frac{\sin\left(\frac{h/2}{h/2}\right) \cdot \frac{h}{2}}{h} = \frac{1}{2} \end{aligned}$$

Example 2.25 Evaluate the limit $\lim_{x \rightarrow \pi/4} \frac{\sin\left(x - \frac{\pi}{4}\right)}{1 - \sqrt{2} \cos x}$

Solution Given $\lim_{x \rightarrow \pi/4} \frac{\sin\left(x - \frac{\pi}{4}\right)}{1 - \sqrt{2} \cos x}$



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$$\begin{aligned}
 &= - \lim_{h \rightarrow 0} \frac{\log \left\{ \frac{1+2h}{1+2h} + \frac{h^2}{1+2h} \right\}}{h^2} = - \lim_{h \rightarrow 0} \frac{\log \left\{ 1 + \left(\frac{h^2}{1+2h} \right) \right\}}{\left(\frac{h^2}{1+2h} \right) (1+2h)} \left(\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \right) \\
 &= - \lim_{h \rightarrow 0} \left(\frac{1}{1+2h} \right) = -1
 \end{aligned}$$

2.5.4 Exponential Limits

If $f(x)$ is any exponential function i.e. $f(x) = (\text{constant})^{\text{variable}}$ then $\lim_{x \rightarrow a} f(x)$ is called exponential limit.

To solve exponential limit use

Form (i)

$$\lim_{x \rightarrow \infty} (a^x) = \begin{cases} 0, & \text{when } 0 < a < 1 \\ 1, & \text{when } a = 1 \\ \infty, & \text{when } a > 1 \end{cases}$$

Example 2.32 Evaluate the limit $\lim_{n \rightarrow \infty} \left(\frac{3^n + 2^n}{3^n - 2^n} \right)$

Solution Given $\lim_{n \rightarrow \infty} \left(\frac{3^n + 2^n}{3^n - 2^n} \right)$

Divide Numerator and Denominator by 3^n

$$\therefore \lim_{n \rightarrow \infty} \left(\frac{3^n + 2^n}{3^n - 2^n} \right) = \lim_{n \rightarrow \infty} \left(\frac{\frac{3^n}{3^n} + \frac{2^n}{3^n}}{\frac{3^n}{3^n} - \frac{2^n}{3^n}} \right) = \lim_{n \rightarrow \infty} \left(\frac{1 + \left(\frac{2}{3} \right)^n}{1 - \left(\frac{2}{3} \right)^n} \right) = 1$$

Form (ii)

$$\lim_{x \rightarrow \infty} \{f(x)\}^{g(x)}$$

Where $\lim_{x \rightarrow \infty} f(x) = \text{finite value} = l$ and $\lim_{x \rightarrow \infty} g(x) = \text{finite value} = m$

Then $\lim_{x \rightarrow \infty} \{f(x)\}^{g(x)} = (l)^m$



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Example 2.42 Evaluate the limit $\lim_{x \rightarrow 0} \left\{ (\sin x)^{1/x} + \left(\frac{1}{x} \right)^{\sin x} \right\}$

(IIT (JEE)-2006)

Solution Given $\lim_{x \rightarrow 0} \left\{ (\sin x)^{1/x} + \left(\frac{1}{x} \right)^{\sin x} \right\}$

$$\because |\sin x| < 1 \Rightarrow \lim_{x \rightarrow 0} (\sin x)^{1/x} = 0$$

$$\therefore \lim_{x \rightarrow 0} \left\{ (\sin x)^{1/x} + \left(\frac{1}{x} \right)^{\sin x} \right\} = \lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{\sin x} = e^{\lim_{x \rightarrow 0} \sin x \log \left(\frac{1}{x} \right)} = e^{\lim_{x \rightarrow 0} \frac{\log \left(\frac{1}{x} \right)}{\csc x}}$$

Use L' Hospital's rules

$$e^{\lim_{x \rightarrow 0} \frac{\log \left(\frac{1}{x} \right)}{\csc x}} = e^{\lim_{x \rightarrow 0} \frac{x \cdot \left(-\frac{1}{x^2} \right)}{-\csc x \cdot \cot x}} = e^0 = 1$$

2.6 L' HOSPITAL'S RULE

Use following steps:-

(i) First change given limit into form $\left(\frac{0}{0} \right)$ or $\left(\frac{\infty}{\infty} \right)$

(ii) L' Hospital's rule is

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)} = \lim_{x \rightarrow a} \frac{f'''(x)}{g'''(x)} = \dots$$

Tips

(I) L' Hospital's rule is applicable only when the form of limit is

$$\left(\frac{0}{0} \right) \text{ or } \left(\frac{\infty}{\infty} \right)$$

(II) If the form of limit is $(0 \cdot \infty)$ then bring suitable factor in the denominator

$$\text{For Example } \lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2} (0 \cdot \infty) = \lim_{x \rightarrow 1} \frac{(1-x)}{\cot \frac{5\pi}{2}} \left(\frac{0}{0} \right)$$

(III) When the form of limit is $(\infty - \infty)$ then simplified until it reduces to the form $\left(\frac{0}{0} \right)$ or $\left(\frac{\infty}{\infty} \right)$

(IV) When the form of limit is 1^∞ , ∞^0 , 0^0 then let required limit is P , then take logarithm and proceed.



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Divide all side by n^2 then take limit $\lim_{n \rightarrow \infty}$ all side,

$$\Rightarrow \lim_{n \rightarrow \infty} \left\{ \frac{\frac{nx(n+1)}{2} - n}{n^2} \right\} < \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n [rx]}{n^2} \leq \lim_{n \rightarrow \infty} \frac{\frac{nx(n+1)}{2}}{n^2} \Rightarrow \frac{x}{2} < \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{[rx]}{n^2} \leq \frac{x}{2}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{[rx]}{n^2} = \lim_{n \rightarrow \infty} \frac{[x] + [2x] + [3x] + \dots + [nx]}{n^2} = \frac{x}{2}$$

Example 2.51 Evaluate $\lim_{n \rightarrow \infty} \frac{[1^2 \cdot x] + [2^2 \cdot x] + [3^2 \cdot x] + \dots + [n^2 \cdot x]}{n^3}$

Solution Given $\lim_{n \rightarrow \infty} \frac{[1^2 \cdot x] + [2^2 \cdot x] + [3^2 \cdot x] + \dots + [n^2 \cdot x]}{n^3}$

As, $1^2 \cdot x - 1 < [1^2 \cdot x] \leq 1^2 \cdot x$

$\Rightarrow 2^2 \cdot x - 1 < [2^2 \cdot x] \leq 2^2 \cdot x$

$\Rightarrow 3^2 \cdot x - 1 < [3^2 \cdot x] \leq 3^2 \cdot x$

.....

.....

$\Rightarrow n^2 \cdot x - 1 < [n^2 \cdot x] \leq n^2 \cdot x$

Add, $(1^2x + 2^2x + \dots + n^2x) - n < [1^2x] + [2^2x] + \dots + [n^2x] \leq (1^2x + 2^2x + \dots + n^2x)$

$\Rightarrow \frac{nx(n+1)(2n+1)}{6} - n < \sum_{r=1}^n [r^2x] \leq \frac{nx(n+1)(2n+1)}{6}$

Divide all side by n^3 then take limit $\lim_{n \rightarrow \infty}$ all side,

$\Rightarrow \lim_{n \rightarrow \infty} \left\{ \frac{\frac{nx(n+1)(2n+1)}{6} - n}{n^3} \right\} < \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n [r^2x]}{n^3} \leq \lim_{n \rightarrow \infty} \frac{\frac{nx(n+1)(2n+1)}{6}}{n^3} \Rightarrow \frac{x}{3} < \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{[r^2x]}{n^3} \leq \frac{x}{3}$

$\Rightarrow \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{[r^2x]}{n^3} = \lim_{n \rightarrow \infty} \frac{[1^2 \cdot x] + [2^2 \cdot x] + [3^2 \cdot x] + \dots + [n^2 \cdot x]}{n^3} = \frac{x}{3}$

2.8 NEWTON -LEIBNIZ'S FORMULA

If $f(x) = \int_{h(x)}^{g(x)} f(t) dt$ then $\frac{d\{f(x)\}}{dx} = f\{g(x)\} \cdot g'(x) - f\{h(x)\} \cdot h'(x)$



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Solution $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$

$$= \lim_{x \rightarrow 0} \frac{(1+x) - (1-x)}{x[\sqrt{1+x} + \sqrt{1-x}]}$$

$$= \lim_{x \rightarrow 0} \frac{2}{\sqrt{1+x} + \sqrt{1-x}} = \frac{2}{1+1} = 1.$$

Example 2.56 Evaluate $\lim_{x \rightarrow \infty} [x - \sqrt{x^2 + x}]$

Solution Given $\lim_{x \rightarrow \infty} [x - \sqrt{x^2 + x}]$

Put $x = 1/h$, then $h \rightarrow 0$ as $x \rightarrow \infty$.

Then $\lim_{x \rightarrow \infty} [x - \sqrt{x^2 + x}]$

$$= \lim_{h \rightarrow 0} \left\{ \frac{1}{h} - \sqrt{\left(\frac{1}{h^2} + \frac{1}{h}\right)} \right\} = \lim_{h \rightarrow 0} \frac{1 - \sqrt{1+h}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1 - (1+h)}{h[1 + \sqrt{1+h}]} = \lim_{h \rightarrow 0} \frac{-h}{h[1 + \sqrt{1+h}]}$$

$$= \lim_{h \rightarrow 0} \left(-\frac{1}{1 + \sqrt{1+h}} \right) = -\frac{1}{1+1} = -\frac{1}{2}$$

Example 2.57 Evaluate $\lim_{x \rightarrow \infty} \frac{2\sqrt{x} + 3\sqrt[3]{x} + 5 \cdot \sqrt[5]{x}}{\sqrt{5x-2} + \sqrt[3]{3x-2}}$

Solution Given $\lim_{x \rightarrow \infty} \frac{2\sqrt{x} + 3\sqrt[3]{x} + 5 \cdot \sqrt[5]{x}}{\sqrt{5x-2} + \sqrt[3]{3x-2}}$

Put $x = 1/y$, when $x \rightarrow \infty$, $y \rightarrow 0$.

$$\therefore \lim_{x \rightarrow \infty} \frac{2\sqrt{x} + 3\sqrt[3]{x} + 5 \cdot \sqrt[5]{x}}{\sqrt{5x-2} + \sqrt[3]{3x-2}} = \lim_{y \rightarrow 0} \frac{\frac{2}{y^{1/2}} + \frac{3}{y^{1/3}} + \frac{5}{y^{1/5}}}{\frac{1}{y^{1/2}} \sqrt{(5-2y)} + \frac{1}{y^{1/3}} (3-2y)^{1/3}}$$

Multiply above and below by $y^{1/2}$

$$= \lim_{y \rightarrow 0} \frac{2 + 3y^{1/6} + 5y^{3/10}}{\sqrt{(5-2y)} + y^{1/6} (3-2y)^{1/3}} = \frac{2}{\sqrt{5}}$$



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Solution Given $\lim_{x \rightarrow 4} \frac{x^3 - 2x^2 - 9x + 4}{x^2 - 2x - 8}$

$$= \lim_{x \rightarrow 4} \frac{(x-4)(x^2 + 2x - 1)}{(x-4)(x+2)} = \lim_{x \rightarrow 4} \frac{(x^2 + 2x - 1)}{(x+2)} = \frac{23}{6}$$

TYPE-II: PROBLEMS BASED ON TRIGONOMETRIC LIMITS

Example 2.67 Find the limits using equivalent infinitesimals:

(a) $\lim_{x \rightarrow 0} \frac{\sin \sqrt{x} \sqrt{x}}{\ln(1+5x)}$

(b) $\lim_{x \rightarrow 0} \frac{\sin 3x + (\sin^{-1} x)^2 - (\tan^{-1} x)^2}{e^{5x} - 1}$

(c) $\lim_{x \rightarrow 0} \frac{(1 - \cos x)(e^x - \cos x)}{x^n}$

(d) $\lim_{x \rightarrow 0} \frac{(\sin x - \tan x)^2 - x(1 - \cos 2x)^2}{8 \tan^8 x + 5 \sin^5 x}$

(e) $\lim_{x \rightarrow 0} \frac{\sin \sqrt[3]{x} \ln(1-5x)}{(\tan^{-1} \sqrt{x})^2 (e^{\sqrt{x}} - 1)}$

(f) $\lim_{x \rightarrow 0} \frac{1 - \cos x + 2 \sin^{-1} x - \sin^3 x}{3 \tan^{-1} x - x^2 + x^5}$

Solution

(a) $\lim_{x \rightarrow 0} \frac{\sin \sqrt{x} \sqrt{x}}{\ln(1+5x)} = \lim_{x \rightarrow 0} \frac{x^{3/4}}{5x} = \infty.$

Hence, limit does not exist.

(b) $\lim_{x \rightarrow 0} \frac{\sin 3x + (\sin^{-1} x)^2 - (\tan^{-1} x)^2}{e^{5x} - 1} = \lim_{x \rightarrow 0} \frac{3x + x^2 - x^2}{5x} = \frac{3}{5}.$

(c) $\lim_{x \rightarrow 0} \frac{(1 - \cos x)(e^x - \cos x)}{x^n} = \lim_{x \rightarrow 0} \frac{(x^2/2)[(e^x - 1) - (1 - \cos x)]}{x^n}$

$$= \lim_{x \rightarrow 0} \frac{x - \frac{x^2}{2}}{2x^{n-2}} = \begin{cases} 0 & \text{if } n < 3 \\ 1/2 & \text{if } n = 3. \\ \infty & \text{if } n > 3 \end{cases}$$

(d) $\lim_{x \rightarrow 0} \frac{(\sin x - \tan x)^2 - x(1 - \cos 2x)^2}{8 \tan^8 x + 5 \sin^5 x} = \lim_{x \rightarrow 0} \frac{(x - x)^2 - x(4x^2/2)^2}{8x^8 + 5x^5}$

$$= \lim_{x \rightarrow 0} \frac{-4x^5}{8x^8 + 5x^5} = \frac{-4}{5}.$$

(e) $\lim_{x \rightarrow 0} \frac{\sin \sqrt[3]{x} \ln(1-5x)}{(\tan^{-1} \sqrt{x})^2 (e^{\sqrt{x}} - 1)} = \frac{(\sqrt[3]{x})(-5x)}{(\sqrt{x})^2 (\sqrt[3]{x})} = -5.$



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$$\text{i.e.} \quad \lim_{x \rightarrow 0} e^{\frac{\ln[1+x+g(x)]}{x}} = e^3 \quad \left[\text{putting } \frac{f(x)}{x} = g(x) \right]$$

$$\text{i.e.} \quad \lim_{x \rightarrow 0} \frac{\ln[1+x+g(x)]}{x} = 3.$$

Since the denominator approaches zero, the numerator should also approach zero for a finite limit to exist.

Thus, we have $\lim_{x \rightarrow 0} g(x) = 0$.

Now, using L'Hospital's Rule the above equation reduces to $\lim_{x \rightarrow 0} \frac{1+g'(x)}{1+x+g(x)} = 3$

$$\text{i.e.} \quad 1 + g'(0) = 3, \text{ gives } g'(0) = 2$$

$$\begin{aligned} \text{Hence, we have} \quad \lim_{x \rightarrow 0} \left[1 + \frac{f(x)}{x} \right]^{1/x} &= \lim_{x \rightarrow 0} e^{\frac{\ln[1+g(x)]}{x}} = \lim_{x \rightarrow 0} e^{\frac{g'(x)}{1+g(x)}} \\ &= \lim_{x \rightarrow 0} e^{g'(0)} = e^2. \end{aligned}$$

Example 2.77 Evaluate $\lim_{x \rightarrow 1} \left(\frac{1+2x^2+x+x^3}{x^2+2x+3} \right)^{\frac{1-\cos(x-1)}{(x-1)^2}}$

Solution Given $\lim_{x \rightarrow 1} \left(\frac{1+2x^2+x+x^3}{x^2+2x+3} \right)^{\frac{1-\cos(x-1)}{(x-1)^2}}$

Put $x = t + 1$ where $t \rightarrow 0$ as $x \rightarrow 1$

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{1-\cos t}{t^2} &= \lim_{t \rightarrow 0} \frac{2 \sin^2(t/2)}{t^2} \\ &= \lim_{t \rightarrow 0} \frac{2}{t^2} \left(\frac{t}{2} \right)^2 = \frac{1}{2} \end{aligned}$$

$$\therefore \quad \lim_{x \rightarrow 1} \left(\frac{1+2x^2+x+x^3}{x^2+2x+3} \right)^{1/2} = \sqrt{\frac{5}{6}}$$

Example 2.78 Evaluate $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4}$

Solution Given $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4}$



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2. $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\sin x}{x - \sin x}}$ is equal to
 (a) e (b) e^2
 (c) $1/e$ (d) none of these

3. $\lim_{x \rightarrow 0} \left(\frac{1 + \tan x}{1 + \sin x} \right)^{\operatorname{cosec} x}$ is equal to
 (a) 1 (b) e^2
 (c) $1/e$ (d) none of these

4. $\lim_{x \rightarrow 0} \left(\frac{1 + \sin x}{1 - \tan x} \right)^{\operatorname{cosec} x}$ is equal to
 (a) 1 (b) e^2
 (c) $1/e$ (d) none of these

5. $\lim_{x \rightarrow \infty} \left(\frac{x-2}{ax-1} \right)^x = e^2$ then a is equal to
 (a) 1 (b) 2
 (c) 0 (d) cannot be found

6. $\lim_{n \rightarrow \infty} (0.2)^{\log_{\sqrt{2}} (1/4 + 1/8 + 1/16 + \dots \text{to } n \text{ terms})}$ is equal to
 (a) 2 (b) 4
 (c) 8 (d) 0

7. $\lim_{x \rightarrow 0} (ax+1)^{b/x} = e^2$ then (a, b) is
 (a) $(1/2, 4)$ (b) $(8, 4)$
 (c) $(16, 8)$ (d) None of these

8. If x_1, x_2 are the distinct roots of the equation $ax^2 + bx + c = 0$ then $\lim_{x \rightarrow x_1} \{1 + \sin \frac{1}{(ax^2 + bx + c)^{x-x_1}}\}$ is
 (a) e^{x-x_1} (b) $e^{x_2-x_1}$
 (c) $e^{a(x_1-x_2)}$ (d) None of these

9. The value of $\lim_{x \rightarrow \infty} \left[\frac{a_1^{1/x} + a_2^{1/x} + \dots + a_n^{1/x}}{n} \right]$ where $a_i > 0, i = 1, 2, 3, \dots$ is

(a) $a_1 a_2 a_3 \dots a_n$ (b) $e^{a_1 a_2 a_3 \dots a_n}$
 (c) 1 (d) 0

10. If $f(x)$ is a polynomial of least degree such

that $\lim_{x \rightarrow 0} \left(1 + \frac{x^2 + f(x)}{x^2} \right)^{1/x} = e^2$ then $f(x)$

is

(a) $x^2 + 2x^3$
 (b) $-2x^2 + x^3$
 (c) $-x^2 + 2x^3$
 (d) Cannot be defined

Type-IV: Problems Based on Logarithmic Limit

1. Value of limit $\lim_{x \rightarrow 0} \left\{ \frac{\ln(1+2h)}{-2\ln(1+h)} \right\}$ is equal to

(a) 0 (b) 1
 (c) -1 (d) None of these

2. If $f(x) = \frac{\sin(e^{x-2} - 1)}{\log_e(x-1)}$, then $\lim_{x \rightarrow 2} f(x)$ is equal to

(a) 1 (b) 2
 (c) -1 (d) 0

3. $\lim_{x \rightarrow 0} \log_{\tan^2 x} (\tan^2 2x)$ is equal to

(a) 1 (b) e^2
 (c) $1/e$ (d) none of these

4. If $f(x) = \begin{cases} x, & x \leq 0 \\ -x, & x > 0 \end{cases}$ and $g(x) = |x| + f(x)$

then $\lim_{x \rightarrow 0^+} (\log_{\sin x} x)^{g(x)}$ is

(a) 1 (b) 0
 (c) dne (d) none of these

5. The value of $\lim_{x \rightarrow 0} \cos \left\{ \log \left(\frac{x-1}{x} \right) \right\}$ is equal to



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3. Find value of following limits

$$(i) \lim_{x \rightarrow \infty} 3x \sin\left(\frac{1}{x}\right)$$

$$(ii) \lim_{x \rightarrow \infty} 2n\pi \sin\left(\frac{\pi}{x}\right)$$

$$(iii) \lim_{x \rightarrow \infty} \frac{\cos x}{x}$$

$$(iv) \lim_{x \rightarrow \infty} \frac{x + \sin x + 1}{x + \cos x}$$

Type-III: Problems Based on Logarithmic Limit

Find value of following limits

$$(i) \lim_{x \rightarrow 1} \frac{\log x}{x-1} \quad (ii) \lim_{x \rightarrow a} \frac{\log\{1+(x-a)\}}{(x-a)}$$

$$(iii) \lim_{x \rightarrow 0} \frac{\log(5+x) - \log(5-x)}{x}$$

Type-IV: Problems Based on Exponential Limit

1. Find value of following limits

$$(i) \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$$

$$(ii) \lim_{x \rightarrow 0} \frac{a^x - 1}{x}$$

$$(iii) \lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$$

$$(iv) \lim_{x \rightarrow 0} \frac{(ab)^x - a^x - b^x + 1}{x^2}$$

2. Find value of following limits

$$(i) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x \quad (ii) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{2n}\right)^n$$

$$(iii) \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^{2x}$$

3. Find value of following limits

$$(i) \lim_{x \rightarrow 0} \left(\tan\left(\frac{\pi}{4} + x\right)\right)^{1/x}$$

$$(ii) \lim_{x \rightarrow 0} \left(\frac{x-3}{x+2}\right)^x$$

$$(iii) \lim_{x \rightarrow 0} \left(\frac{1+5x^2}{1+3x^2}\right)^{\frac{1}{x^2}}$$

$$(iv) \lim_{x \rightarrow \infty} \left(\frac{n+1}{n+2}\right)^{2n+1}$$

4. For $x > 0$ find value of

$$\lim_{x \rightarrow 0} \{(\sin)^{1/x} + (1/x)^{\sin x}\}$$

5. If a, b, c, d are positive then find the value

$$\text{of } \lim_{x \rightarrow \infty} \left(1 + \frac{1}{ax+b}\right)^{cx+d}$$

Type-V: Problems Based on Miscellaneous Limit

1. If $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sin x^2 dx}{x^n}$ is a non zero definite number, then find value of n .

2. Find value of $\lim_{x \rightarrow 0} \frac{\int_0^x (\tan^{-1} t) dt}{\sqrt{x^2+1}-1}$.

3. Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x^5} \int_0^x e^{-t^2} dt - \frac{1}{x^4} + \frac{1}{3x^2} \right)$



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$$4. \lim_{x \rightarrow 0} f(x) = -e/2$$

$$5. \lim_{x \rightarrow 0} f(x) < -1$$

3. Match Column I with Column II

Column I

(a) If $\lim_{x \rightarrow 0} (\cos x + a \sin bx)^{1/x} = e^2$, then

(b) If $(1 + ax + bx^2)^{2/x} = e^3$, then

(c) If $\lim_{x \rightarrow 0} \frac{ae^x - b}{x} = 2$ then

Column II

1. $ab = 2$

2. $ab = 3$

3. $ab = 4$

4. $a + b = 3$

5. $a + b = 4$

4. Match Column I with Column II

Column I

(a) If $f(n) = \frac{(n+1)^2}{2n^2}$, then

(b) If $f(n) = \frac{n^3 - 100n^2 + 1}{100n^2 + 15n}$, then

(c) $f(n) = \frac{\sqrt[3]{n^3 + 2n - 1}}{n + 2}$

(d) $f(n) = \frac{n!}{(n+1)! - n!}$

Column II

1. $\lim_{n \rightarrow \infty} f(n) \leq 1$

2. $\lim_{n \rightarrow \infty} f(n) \leq 0$

3. $\lim_{n \rightarrow \infty} f(n) \geq 1/2$

4. $\lim_{n \rightarrow \infty} f(n) < \infty$

Type III: Assertion- Reason

Directions: The following questions contains of two statements, one labelled as **Assertion (A)** and the other as **Reason (R)**. Examine these two statements carefully and decide the correct answer using the options given below:

- Both A and R are true and R is the correct explanation of A
- Both A and R are true but R is not the correct explanation of A
- A is true but R is false
- A is false but R is true.

- Let $f(x)$ and $g(x)$ is real valued function defined on $(-1, 1)$ such that $g''(x)$ is

continuous $g(0) \neq 0$, $g'(0) = 0$, $g''(0) \neq 0$ and $f(x) = g(x) \sin x$

A: $\lim_{n \rightarrow 0} \{g(x) \cos x - g(0) \operatorname{cosec} x\} = f''(0)$

R: $f'(0) = g(0)$

$$2. \text{ A: } \lim_{x \rightarrow \infty} \left(\frac{1}{x^2} + \frac{2}{x^2} + \frac{3}{x^2} + \dots + \frac{x}{x^2} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{1}{x^2} \right) + \lim_{x \rightarrow \infty} \left(\frac{2}{x^2} \right) + \lim_{x \rightarrow \infty} \left(\frac{3}{x^2} \right)$$

$$+ \dots + \lim_{x \rightarrow \infty} \left(\frac{x}{x^2} \right) = 0$$



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LEVEL II: TOPIC-WISE SUBJECTIVE QUESTIONS**Type-I: Problems Based on Algebraic Limit**

- | | | | |
|---------------|------------|---------------|-------------|
| 1. (i) $7/2$ | (ii) 0 | (iii) 0 | (iv) $-2/3$ |
| (v) $2/3$ | (vi) $-a$ | | |
| 2. (i) $3/2$ | (ii) $1/2$ | (iii) $1/4$ | (iv) $1/3$ |
| 3. (i) $5a^2$ | (ii) $1/4$ | (iii) $1/n$ | (iv) $2/3$ |
| 4. (i) $1/3$ | (ii) 1 | (iii) $-1/10$ | |

Type-II: Problems Based on Trigonometrical Limit

- | | | | |
|------------------------|------------------------|-----------------------|-----------------------------|
| 1. (i) 4 | (ii) 2 | (iii) $\frac{k^2}{2}$ | (iv) $1/2$ |
| (v) 2 | (vi) $\frac{\pi}{180}$ | | |
| 2. (i) $\frac{2}{\pi}$ | (ii) does not exist | (iii) $\frac{1}{4}$ | (iv) $\frac{1}{16\sqrt{2}}$ |
| 3. (i) 3 | (ii) $2\pi^2$ | (iii) 0 | (iv) 1 |

Type-III: Problems Based on Exponential Limit

- | | | |
|-------|--------|-------------|
| (i) 1 | (ii) 1 | (iii) $2/5$ |
|-------|--------|-------------|

Type-IV: Problems Based on Exponential Limit

- | | | | |
|--------------|-----------------|----------------------|---------------------------------|
| 1. (i) $1/2$ | (ii) $\log_e a$ | (iii) $\log_e (a/b)$ | (iv) $\log_e a \times \log_e b$ |
| 2. (i) e | (ii) $\sqrt{e}$ | (iii) e^{-2} | |
| 3. (i) e^2 | (ii) e^{-5} | (iii) e^2 | (iv) e^{-2} |
| 4. 1 | | 5. $e^{c/a}$ | |

Type-V: Problems Based on Miscellaneous Limit

- | | | | | | |
|------|------|-----------|------|--------------------|------|
| 1. 6 | 2. 1 | 3. $1/10$ | 4. 0 | 5. $\frac{\pi}{4}$ | 6. 0 |
| 7. 2 | | | | | |

LEVEL III: MISCELLANEOUS QUESTIONS**Type-I: Passage****Passage-I**

- | | | | | |
|--------|--------|--------|--------|--------|
| 1. (c) | 2. (b) | 3. (b) | 4. (a) | 5. (a) |
|--------|--------|--------|--------|--------|



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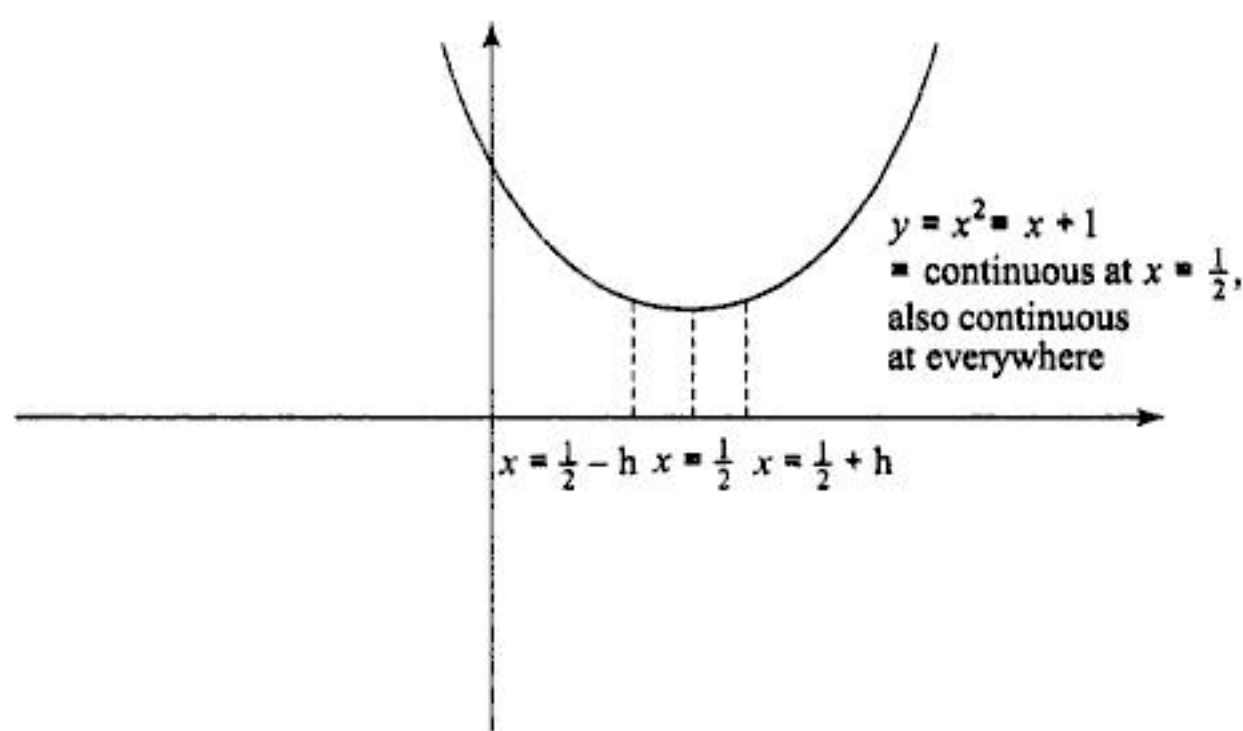


Fig. 3.5

Example 3.2 If a function $f(x)$ is defined as $f(x) = \begin{cases} 2\operatorname{sgn}(x), & \text{when } x > 0 \\ 2, & \text{when } x = 0 \\ 2 - 2x & \text{when } x < 0 \end{cases}$ then

- (a) $f(x)$ is continuous at $x = 0$ (b) $f(x)$ is discontinuous at $x = 0$
 (c) $f(x)$ is bounded function (d) None of these

Solution Given function $f(x) = \begin{cases} 2\operatorname{sgn}(x), & \text{when } x > 0 \\ 2, & \text{when } x = 0 \\ 2 - 2x & \text{when } x < 0 \end{cases}$

$$\text{L.H.L of } f(x) \text{ at } (x = 0) = \lim_{x \rightarrow 0-h} f(x) = \lim_{x \rightarrow 0-h} (2 - 2x) = \lim_{h \rightarrow 0} \{2 - 2(0 - h)\} = 2$$

$$\text{R.H.L of } f(x) \text{ at } (x = 0) = \lim_{x \rightarrow 0+h} f(x) = \lim_{x \rightarrow 0+h} 2\operatorname{sgn}(x) = \lim_{h \rightarrow 0} 2\operatorname{sgn}(0 + h) = 2 \times 1 = 2$$

$$\text{M.H.L of } f(x) \text{ at } (x = 0) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} 2 = 2 \quad \left\{ \because \operatorname{sgn} x = \begin{cases} +1, & x > 0 \\ -1, & x < 0 \\ 0, & x = 0 \end{cases} \right.$$

Here, L.H.L = M.H.L. = R.H.L.

Hence, $f(x)$ is continuous at $x = 0$

Example 3.3 Let $f(x + y) = f(x) + f(y)$ for all x and y if $f(x)$ is continuous at $x = a$ then show that $f(x)$ is continuous for all x

Solution Given $f(x + y) = f(x) + f(y)$ for all x and y .

And $f(x)$ is continuous at $x = 0$ i.e. $\lim_{x \rightarrow 0} f(x) = f(0) \Rightarrow f(0 + h) = f(0 - h) = f(0) = 0$ then prove that $f(x)$ is

continuous at $x = c$

$$\text{i.e. } \lim_{x \rightarrow c} f(x) = f(c) = \lim_{x \rightarrow c-h} f(x) = \lim_{x \rightarrow c+h} f(x)$$



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And R.H.L of $f(x)$ at $(x = 1)$

$$= \lim_{x \rightarrow 1+h} f(x) = \lim_{x \rightarrow 1+h} (x^3 + 8) = \lim_{h \rightarrow 0} \{(1+h)^3 + 8\} = 9$$

$\therefore f(x)$ is discontinuous at $x = 1$

Hence, function $f(x)$ is continuous in $[0, 2]$ except at $x = 1$

3.3 DISCONTINUOUS FUNCTION

(A) Graphically—If Graph of any function is break at a point or in interval then function is called discontinuous function at that point or in interval.

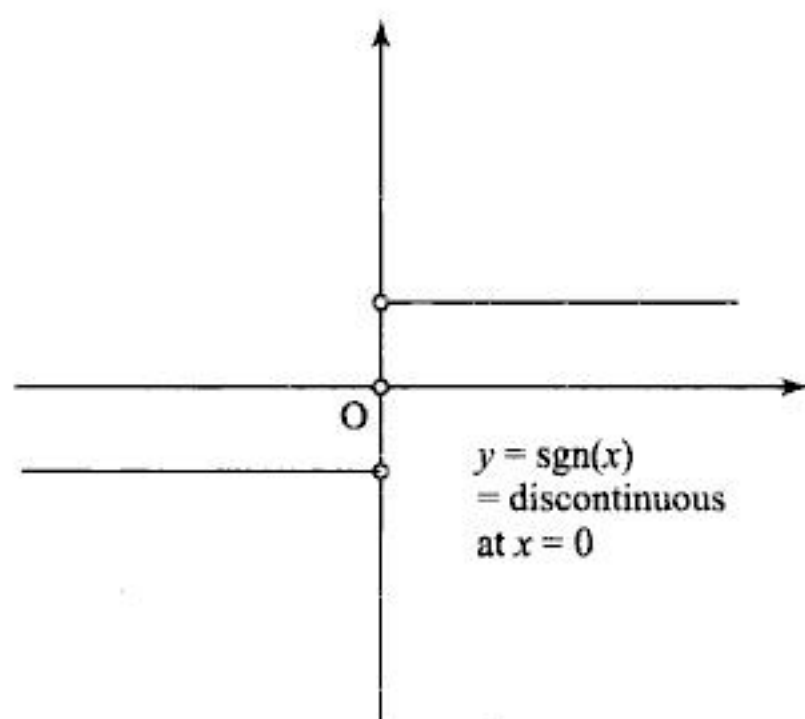


Fig. 3.10

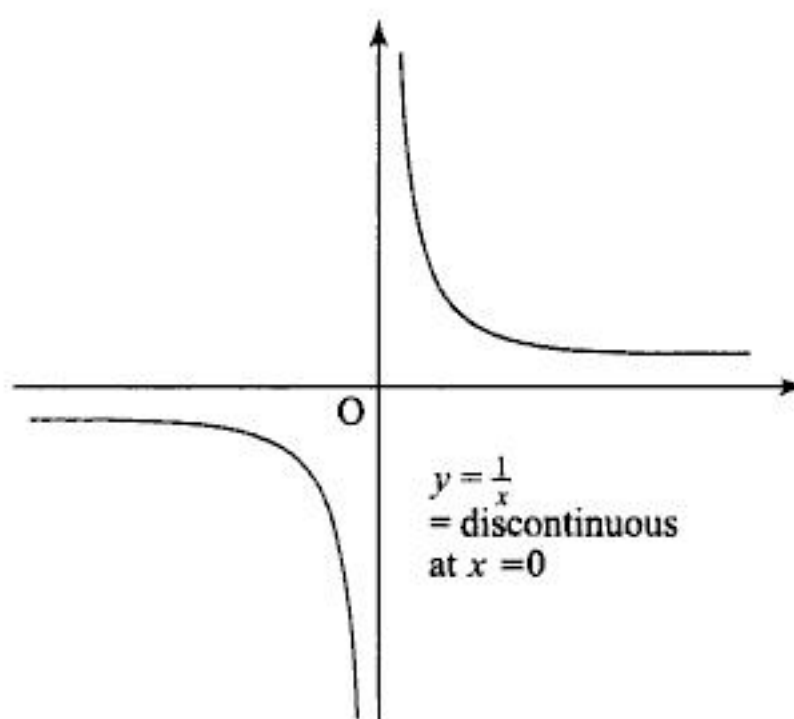


Fig. 3.11

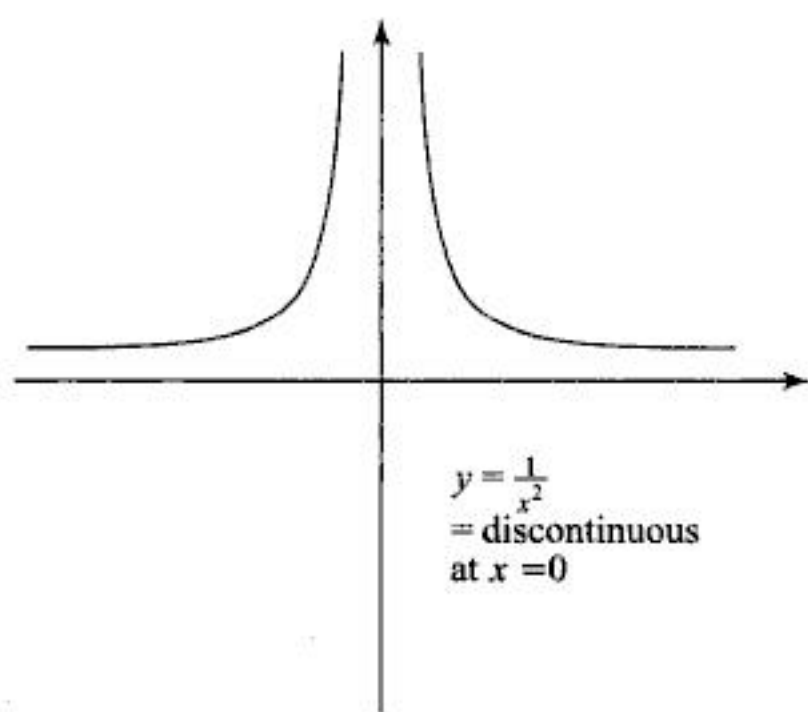


Fig. 3.12

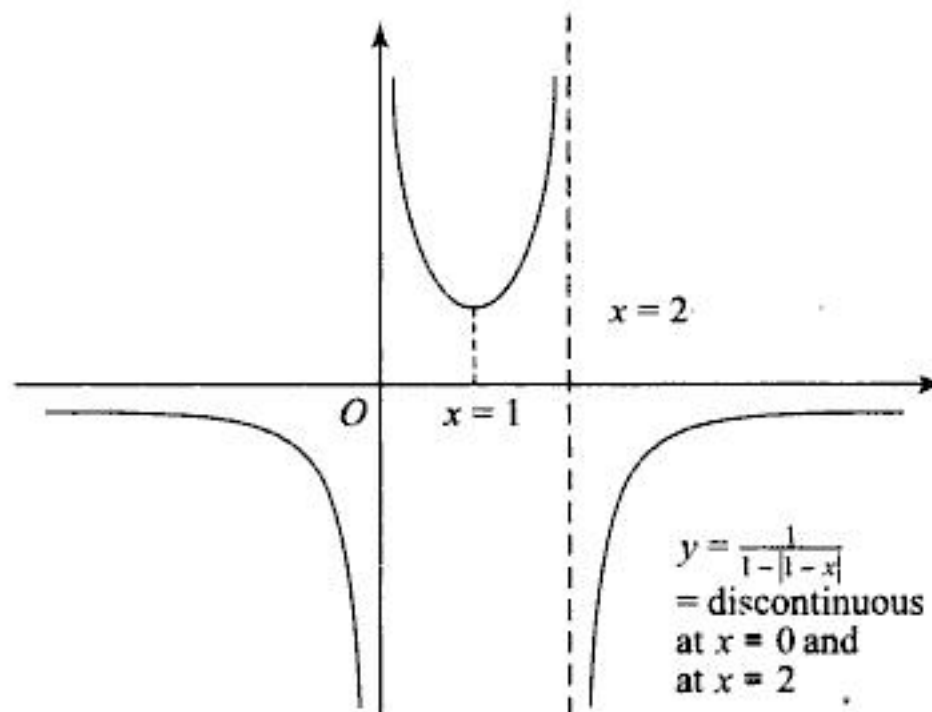


Fig. 3.13



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$$= \lim_{x \rightarrow 0-h} f(x) = \lim_{x \rightarrow 0-h} \left(\frac{1}{x} \right) = \lim_{h \rightarrow 0} \frac{1}{(0-h)} = \lim_{h \rightarrow 0} \left(\frac{1}{-h} \right) = -\infty$$

M.H.L. of function $f(x)$ at $(x = 0)$

$$= \lim_{x \rightarrow 0} f(x) = 0$$

And R.H.L. of function $f(x)$ at $(x = 0)$

$$= \lim_{x \rightarrow 0+h} f(x) = \lim_{x \rightarrow 0+h} \frac{1}{x} = \lim_{h \rightarrow 0} \left(\frac{1}{0+h} \right) = \lim_{h \rightarrow 0} \left(\frac{1}{h} \right) = \infty$$

Here, L.H.L. $\neq$ M.H.L. $\neq$ R.H.L.

Hence, function $f(x)$ is infinite discontinuous at $x = 0$

Removal Discontinuous Function

If value of L.H.L. of function $f(x)$ at $x = a$, M.H.L. of function $f(x)$ at $x = a$ and R.H.L. of function $f(x)$ at $x = a$, represent

finite value but $\lim_{x \rightarrow a-h} f(x) = \lim_{x \rightarrow a+h} f(x) \neq \lim_{x \rightarrow c} f(x)$, then function $f(x)$ is called removal discontinuous function at $x = a$.

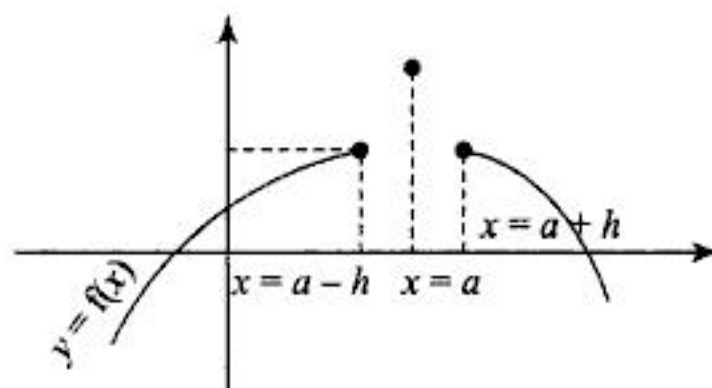


Fig. 3.15e

Example 3.11 Show that function $f(x)$ is removal discontinuous at $x = \frac{1}{2}$ where

$$f(x) = \begin{cases} x, & \text{when } 0 \leq x \leq \frac{1}{2} \\ 1-x, & \text{when } \frac{1}{2} < x < 1 \\ 1, & \text{when } x = \frac{1}{2} \end{cases}$$

Solution Given $f(x) = \begin{cases} x, & \text{when } 0 \leq x \leq \frac{1}{2} \\ 1-x, & \text{when } \frac{1}{2} < x < 1 \\ 1, & \text{when } x = \frac{1}{2} \end{cases}$

L.H.L. of function $f(x)$ at $(x = \frac{1}{2})$

$$= \lim_{x \rightarrow \frac{1}{2}-h} f(x) = \lim_{x \rightarrow \frac{1}{2}-h} (x) = \lim_{h \rightarrow 0} \left(\frac{1}{2} - h \right) = \frac{1}{2}$$

M.H.L. of function $f(x)$ at $(x = \frac{1}{2})$



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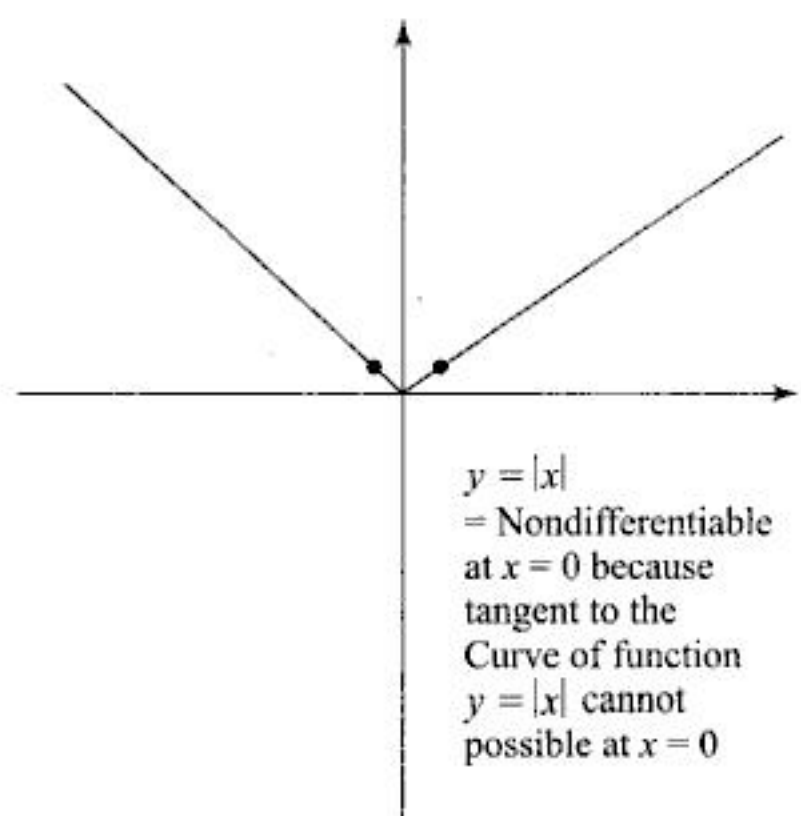


Fig. 3.18

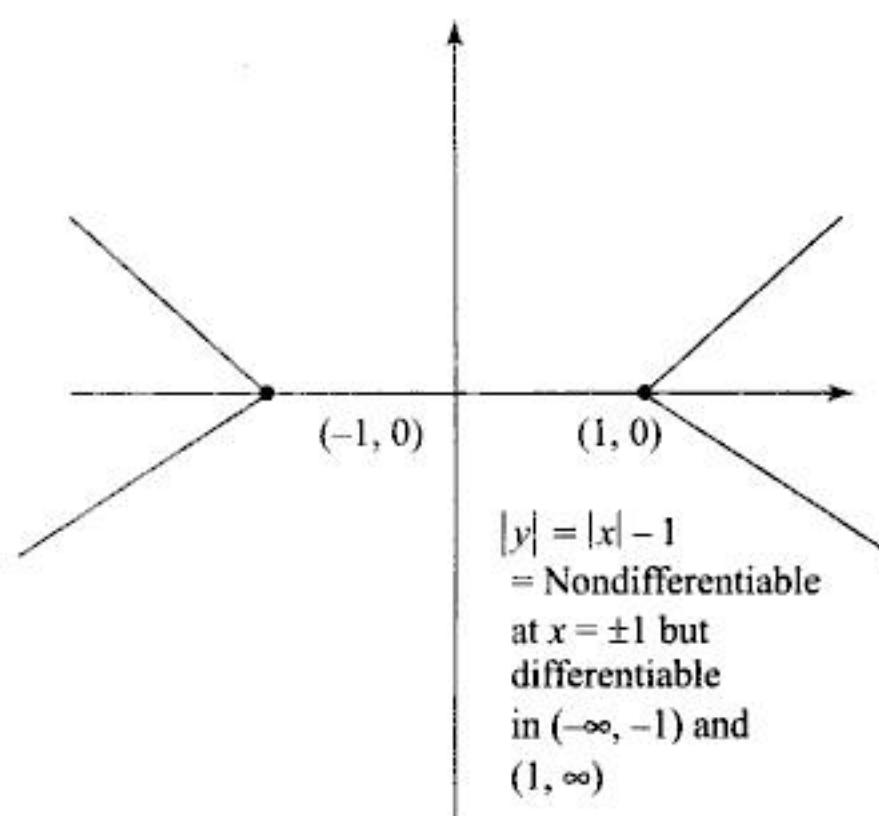


Fig. 3.19

(B) Mathematically—Function $f(x)$ is called differentiable at a point $x = a$ if and only if, L.H.D. of

$f(x)$ at $x = a$ is equal to R.H.D of $f(x)$ at $x = a$ i.e. $\lim_{h \rightarrow 0} \frac{f(a) - f(a-h)}{h} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$

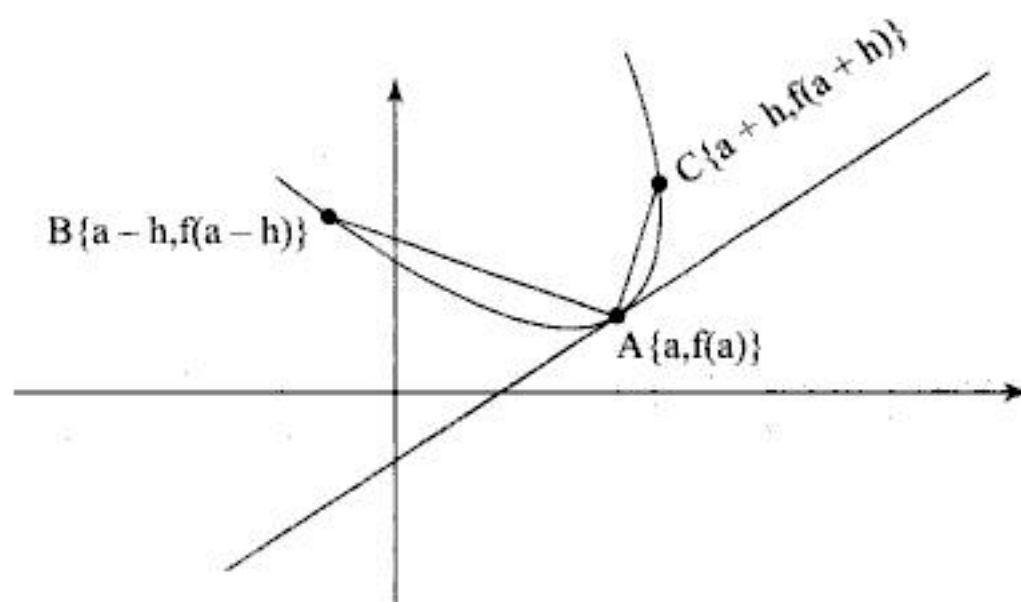


Fig. 3.20

Here, L.H.D. of $f(x)$ at $x = a$ is equal to $\lim_{h \rightarrow 0} \frac{f(a) - f(a-h)}{h} = f'(a-h)$ or $f'(a^-)$

And R.H.D of $f(x)$ at $x = a$ is equal to $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = f'(a+h)$ or $f'(a^+)$



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16. If $f(x)$ is differentiable at $x = a$ and $g(x)$ is not differentiable at $x = a$ then $f(x) + g(x)$ is not differentiable at $x = a$.
17. If $f(x)$ and $g(x)$ are not differentiable at $x = a$ then $f(x) + g(x)$ may be differentiable function at $x = a$.
18. Composition of two differentiable functions is also differentiable function.

TYPEWISE SOLVED EXAMPLES

TYPE I: PROBLEMS BASED ON EXISTENCE OF LIMIT

Example 3.18 Does $\lim_{x \rightarrow 2} \frac{x^2 - 4}{|x - 2|}$ exist?

Solution Given $\lim_{x \rightarrow 2} \frac{x^2 - 4}{|x - 2|}$

$$\text{L.H. Limit: } \lim_{x \rightarrow 2-h} \frac{x^2 - 4}{|x - 2|} = \lim_{h \rightarrow 0} \frac{(2-h)^2 - 4}{|(2-h) - 2|} = -4$$

$$\text{and R.H. Limit: } \lim_{x \rightarrow 2+h} \frac{x^2 - 4}{|x - 2|} = \lim_{h \rightarrow 0} \frac{(2+h)^2 - 4}{|(2+h) - 2|} = 4$$

Here, $\text{L.H.L} \neq \text{R.H.L.}$

Hence, $\lim_{x \rightarrow 2} \frac{x^2 - 4}{|x - 2|}$ does not exist

Example 3.19 If possible, find the value of $\lim_{x \rightarrow 0} \frac{|x^3|}{x}$

Solution Given $\lim_{x \rightarrow 0} \frac{|x^3|}{x}$

$$\text{L.H.L: } \lim_{x \rightarrow 0-h} \frac{|x^3|}{x} = \lim_{h \rightarrow 0} \frac{|(0-h)^3|}{(0-h)} = \lim_{h \rightarrow 0} \frac{|h^3|}{-h} = \lim_{h \rightarrow 0} \{-h^2\} = 0$$

$$\text{R.H.L: } \lim_{x \rightarrow 0+h} \frac{|x^3|}{x} = \lim_{h \rightarrow 0} \frac{|(0+h)^3|}{0+h} = \lim_{h \rightarrow 0} h^2 = 0$$

Here, $\text{L.H.L} = \text{R.H.L.}$

Hence, Limit $\lim_{x \rightarrow 0} \frac{|x^3|}{x}$ exists and is equal to 0.



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Solution Given $f(x) = \begin{cases} 1+x, & 0 \leq x \leq 2 \\ 3-x, & 2 < x \leq 3 \end{cases}$

$$\therefore g(x) = f\{f(x)\} = \begin{cases} 2+x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \\ 4-x, & 2 < x \leq 3 \end{cases}$$

Here, $g(x)$ is defined in interval $[0, 3]$

The only doubtful points for continuity of $g(x)$ are 1 and 2 and so we shall test the continuity of $g(x)$ at $x = 1$ and $x = 2$

For, $x = 1$

$$\text{L.H.L: } \lim_{x \rightarrow 1-h} g(x) = \lim_{x \rightarrow 1-h} (2+x) = \lim_{h \rightarrow 0} \{2+1-h\} = 3$$

$$\text{R.H.L: } \lim_{x \rightarrow 1+h} g(x) = \lim_{x \rightarrow 1+h} (2-x) = \lim_{h \rightarrow 0} \{2-1-h\} = 1$$

Here, $\text{L.H.L} \neq \text{R.H.L.}$

Hence, $g(x)$ is discontinuous at $x = 1$

Now, For $x = 2$

$$\text{L.H.L: } \lim_{x \rightarrow 2-h} g(x) = \lim_{x \rightarrow 2-h} (2-x) = \lim_{h \rightarrow 0} \{2-2+h\} = 0$$

$$\text{R.H.L: } \lim_{x \rightarrow 2+h} g(x) = \lim_{x \rightarrow 2+h} (4-x) = \lim_{h \rightarrow 0} \{4-2-h\} = 2$$

Here, $\text{L.H.L} \neq \text{R.H.L.}$

Hence, $g(x)$ is discontinuous at $x = 2$

Hence, $g(x)$ is continuous at all points in $[0, 3]$ except at $x = 1$ and $x = 2$

TYPE V: PROBLEMS BASED ON EVALUATION OF ARBITRARY CONSTANT, WHEN FUNCTION IS CONTINUOUS

Example 3.30 The values of a , b and c which make the

$$\text{Function } f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x}, & x < 0 \\ c, & x = 0 \\ \frac{\sqrt{x+bx^2} - \sqrt{x}}{bx^{3/2}}, & x > 0 \end{cases} \quad \text{continuous at } x = 0, \text{ are}$$

(a) $a = -\frac{3}{2}, c = \frac{1}{2}, b = 0$

(b) $a = \frac{3}{2}, c = \frac{1}{2}, b \neq 0$

(c) $a = -\frac{3}{2}, c = \frac{1}{2}, b \neq 0$

(c) none of these

Solution In the definition of the function, $b \neq 0$ then $f(x)$ will be undefined in $x > 0$,

$\therefore f(x)$ is continuous at $x = 0$,

$\therefore \text{LHS} = \text{RHS} = f(0)$



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$$= f'(1) [f(1) + 4] = f'(1) (4 + 4) \\ = 8f'(1).$$

$$[\because f(1) = 4]$$

Hence (a) is the correct answer.

Example 3.37 If $f(x) = \frac{x}{1+x} + \frac{x}{(1+x)(2x+1)} + \frac{x}{(2x+1)(3x+1)} + \dots \infty$, then at $x = 0$, $f(x)$

- (a) has no limit (b) is discontinuous
(c) is continuous but not differentiable (d) is differentiable

Solution We have, for $x \neq 0$,

$$f(x) = \frac{x}{1+x} + \frac{x}{(1+x)(2x+1)} + \frac{x}{(2x+1)(3x+1)} + \dots \infty \\ = \sum_{n=1}^{\infty} \frac{x}{[(n-1)x+1][nx+1]} \\ = \sum_{n=1}^{\infty} \left[\frac{1}{(n-1)x+1} - \frac{1}{nx+1} \right] = \lim_{n \rightarrow \infty} \left[1 - \frac{1}{nx+1} \right] = 1$$

For $x = 0$, $f(x) = 0$. thus, $f(x) = \begin{cases} 1, & x \neq 0 \\ 0, & x = 0 \end{cases}$

Clearly, $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0^+} f(x) = 1 \neq f(0)$.

So, $f(x)$ is discontinuous at $x = 0$

Hence (b) the correct answer.

Example 3.38 If $f(x) = \begin{cases} 3, & x < 0 \\ 2x+1, & x \geq 0 \end{cases}$, then

- (a) Both $f(x)$ and $f(|x|)$ are differentiable at $x = 0$
(b) $f(x)$ is differentiable but $f(|x|)$ is not differentiable at $x = 0$
(c) $f(|x|)$ is differentiable but $f(x)$ is not differentiable at $x = 0$
(d) both $f(x)$ and $f(|x|)$ are not differentiable at $x = 0$

Solution We have, $Lf'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{3-1}{-h} = -\infty$

$\therefore f(x)$ is not differentiable at $x = 0$.

Also, if $x < 0$ or $x \geq 0$, then $|x| \geq 0$

$\therefore f(|x|) = 2|x| + 1$ for all x .

$\therefore Rf'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{2h+1-1}{h} = 2$

and $Lf'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{2(-h)+1-1}{-h} = 2$

$\therefore f(|x|)$ is differentiable at $x = 0$.

Hence (c) is the correct answer.



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10. The value of p for which the function $f(x) = \frac{(4^x - 1)^3}{\sin(x/p) \ln\left(1 + \frac{x^2}{3}\right)}$, $x \neq 0$, $12(\ln 4)^3$, $x = 0$ may be continuous at $x = 0$ is
- (a) 1 (b) 2
(c) 3 (d) 4

Type VI: Problems Based on Verification of Differentiability of Function

- The left hand derivative of $f(x) = [x] \sin(\pi x)$ at $x = k$, an integer is:
(a) $(-1)^k (k-1)\pi$ (b) $(-1)^{k-1} (k-1)\pi$
(c) $(-1)^k k\pi$ (d) $(-1)^{k-1} k\pi$
- The set of all points, where function $f(x) = x/(1 + |x|)$ is differentiable, is:
(a) $(-\infty, \infty)$ (b) $(0, \infty)$
(c) $(-\infty, 0) \cup (0, \infty)$ (d) none of these
- The value of the derivative of $|x - 1| + |x - 3|$ at $x = 2$ is:
(a) -2 (b) 0
(c) 2 (d) not defined
- The number of points at which the function $f(x) = |x - 0.5| + ||x - 1|\tan x$ does not have derivative in the interval $[0, 2]$ is:
(a) 1 (b) 2
(c) 3 (d) 4
- Let $[.]$ denote the greatest integer function and $f(x) = [\tan^2 x]$. then:
(a) $\lim_{x \rightarrow 0} f(x)$ does not exist
(b) $f(x)$ is continuous at $x = 0$
(c) $f(x)$ is not differentiable at $x = 0$
(d) $f'(0) = 1$
- Let $f(2) = 4$ and $f'(2) = 1$ then

$$\lim_{x \rightarrow 2} \left(\frac{xf(2) - 2f(x)}{x - 2} \right) \text{ is given by:}$$

- (a) 2 (b) -2
(c) -4 (d) 3

7. At the point $x = 1$, the function $f(x) = \begin{cases} x^3 - 1 : 1 < x < \infty \\ x - 1 : -\infty < x \leq 1 \end{cases}$

- (a) Continuous and differentiable
(b) Continuous and not differentiable
(c) Discontinuous and differentiable
(d) Discontinuous and not differentiable

8. Let $f: R \rightarrow R$ be a function defined by $f(x) = \max\{x, x^3\}$. The set of all points where $f(x)$ is not differentiable is:

- (a) $\{-1, 1\}$ (b) $\{-1, 0\}$
(c) $\{0, 1\}$ (d) $\{-1, 0, 1\}$

9. Let $f: R \rightarrow R$ be such that $f(1) = 3$ and $f'(1) = 6$. Then

$$\lim_{x \rightarrow 0} \left(\frac{f(x+1)}{f(x)} \right)^{1/x} \text{ is equal to:}$$

- (a) 1 (b) $e^{1/2}$
(c) e^2 (d) e^3

10. Let $f: R \rightarrow R$ is a differentiable function and $f(1) = 4$. Then the value of

$$\lim_{x \rightarrow 1} \int_4^{f(x)} \frac{2t}{x-1} dt \text{ is:}$$

- (a) $8f'(1)$ (b) $4f'(1)$
(c) $2f'(1)$ (d) $f'(1)$

11. $g(x) = xf(x)$, where $f(x) = \begin{cases} x \sin(1/x) : x \neq 0 \\ 0 : x = 0, \end{cases}$ At $x = 0$, consider the

following statements:

- (i) g is differentiable but g' is not continuous
(ii) g is differentiable while f is not differentiable
(iii) both f and g are differentiable
(iv) g is differentiable but g' is continuous.

Then correct statements are:

- (a) (i), (ii) (b) (i), (iii)
(c) (ii), (iii) (d) all four



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- (a) $s = \phi$
 (b) $s = \{ax + (1-a)x^2 \mid \forall a \in (0, \infty)\}$
 (c) $s = s = \{ax + (1-a)x^2 \mid \forall a \in \mathbb{R}\}$
 (d) $s = \{ax + (1-a)x^2 \mid \forall a \in (0, 2)\}$
8. If f is real-valued differentiable function satisfying $|f(x) - f(y)| \leq (x - y)^2$, $x, y \in \mathbb{R}$ and $f(0) = 0$, then $f(1) =$
 (a) 1 (b) 2
 (c) 0 (d) -1
9. Let $f''(x) = -f(x)$ and $g(x) = f'(x)$ and $f(x) = (f(x/2))^2 + (g(x/2))^2$. If $f(5) = 5$, then $f(10) =$
 (a) 5 (b) 10
 (c) 15 (d) 0
10. Let $f(x) = x^3 + x$ be function and $g(x) =$

$$\begin{cases} f(|x|), & x \geq 0 \\ f(-|x|), & x < 0 \end{cases}$$
, then
 (a) $g(x)$ is continuous $\forall x \in \mathbb{R}$
 (b) $g(x)$ is continuous $\forall x \in \mathbb{R}^-$ only
 (c) $g(x)$ is continuous $\forall x \in \mathbb{R}^+$ only
 (d) $g(x)$ is discontinuous $\forall x \in \mathbb{R}^-$ only

LEVEL II: TOPIC-WISE SUBJECTIVE TYPE QUESTIONS

Type I: Problems Based on Existence of Limit

1. If $f(x) = \begin{cases} \frac{\sin x}{x}, & \text{when } x > 0 \\ \frac{\tan x}{x}, & \text{when } x < 0 \end{cases}$ then find $\lim_{x \rightarrow 0} f(x)$

2. If $\lim_{x \rightarrow 0} f(x)$ exist then find $\lim_{x \rightarrow 0} f(x)$ where

$$f(x) = \begin{cases} x, & \text{when } x < 0 \\ x^2, & \text{when } x < 0 \\ 1, & \text{when } x = 0 \end{cases}$$

3. If possible find the value of $\lim_{x \rightarrow 0} \frac{|x^3|}{x}$.

4. Does $\lim_{x \rightarrow 2} \frac{x^2 - 4}{|x - 2|}$ exist?

Type II: Problems based on verification of continuity of function at a point.

1. Let $f(x) = \begin{cases} 5x - 4 & \text{when } 0 \leq x \leq 1 \\ 4x^3 - 3x & \text{when } 1 < x \leq 2 \end{cases}$
 then show that $f(x)$ is continuous at $x = 1$ and find $\lim_{x \rightarrow 1} f(x)$
2. If $f(x) = x^2 + 1$, $0 \leq x < 1$
 $= 3x^2 - 1$, $1 \leq x \leq 2$
 Then test the continuity of $f(x)$ at $x = 0$, 1 and 2
3. Is $f(x) = (1 + x)^{\frac{1}{x}}$, when $x \neq 0$
 $= e$, when $x = 0$
 continuous at $x = 0$?
4. Let $f(x) = 5x - 4$, when $0 \leq x \leq 1$
 $= 4x^3 - 3x$, when $1 < x \leq 2$
 If $f(x)$ continuous at $x = 1$?
5. By defining $f\left(\frac{\pi}{4}\right)$ suitably obtain a continuous extension of the function f given by $f(x) = \frac{\sqrt{2} \cos x - 1}{\cot x - 1}$, $x \neq \frac{\pi}{4}$

Type III: Problems Based on Verification of Continuity of Function in Interval

1. Show that the function $f(x) =$

$$\begin{cases} 2x + 3, & \text{when } -3 \leq x < -2 \\ x + 1 & \text{when } -2 \leq x < 0 \\ x + 2 & \text{when } 0 \leq x \leq 1 \end{cases}$$
 is discontinuous at $x = 0$ and continuous at every point in the interval $[-3, 1]$
2. Test the continuity of function $f(x) = x - |x - x^2|$ in the interval $[-1, 1]$.



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Type IV: Problems Based on Evaluation of Arbitrary Constant, when Function is Continuous

1. $\pi/6$ and $-\pi/12$
2. $A = -1$ and $B = 1$
3. $a - b = 2$
4. $A = -4, B = 5, 20$

Type V: Problems Based on Discontinuous Function

1. Discontinuous
2. Discontinuous

Type VI: Problems Based on Differentiable Function

1. $f(x)$ is continuous at $x = 0$
- (b) $f(x)$ is not differentiable at $x = 0$
2. Not differentiable at $x = 0, 1$
3. $g(x)$ is continuous and differentiable at all x in $(0, 2)$ except $g(x)$ is not differentiable at $x = 1$
4. $\{0, 1\}$
5. $\{-1, 0, 1\}$
6. $m = -1, b = \pi$

LEVEL III: MISCELLANEOUS**Type I: Passage****Passage I**

1. (b)
2. (a)
3. (a)

Passage II

1. (d)
2. (c)
3. (d)

Passage III

1. (c)
2. (a)
3. (b)

Passage IV

1. (c)
2. (a)
3. (a)

Passage V

1. (b)
2. (a)
3. (c)

Type II: Column-Matching

1. (a) $-\{i, ii, iii\}$, (b) $-\{ii, iv\}$, (c) $-\{iii, iv\}$, (d) $-\{i, ii\}$
2. (a) $-\{iii\}$, (b) $-\{iv\}$, (c) $-\{ii\}$, (d) $-\{ii\}$
3. (a) $-\{i, ii\}$, (b) $-\{iv, v\}$, (c) $-\{iii\}$

Type III: Assertion-Reason

1. (a)
2. (a)
3. (c)
4. (c)
5. (c)
6. (b)
7. (b)
8. (a)
9. (a)
10. (b)

Differentiation

4.1 INTRODUCTION

In this chapter, we will investigate the derivative as a function derived from $f(x)$ by considering the limit at each point of domain of $f(x)$. The process for calculating derivative of a function is called differentiation of function.

There are two principles to evaluate differentiation of any function:

- (I) Differentiation of function by first principle
- (II) Differentiation of function by second principle

4.1.1 Differentiation of Function by First Principle

Use following steps for differentiation:

- (i) Write expression for $f(x)$ and $f(x + h)$
- (ii) Expand $\frac{f(x + h) - f(x)}{h}$ and simplify.
- (iii) Using simplified form and find $f'(x) = \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$, to get differentiation of function $f(x)$ w.r.t. x by first principle.

Example 4.1 Find the differentiation of x w.r.t. $\sin x$.

Solution Given function $f(x) = \sin x$ and $f(x + h) = \sin(x + h)$

$$\begin{aligned} \therefore f'(x) &= \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x + h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 \sin\left(\frac{x + h - x}{2}\right) \cdot \cos\left(\frac{x + h + x}{2}\right)}{h} = \lim_{h \rightarrow 0} \frac{2 \sin\left(\frac{h}{2}\right) \cdot \cos\left(\frac{2x + h}{2}\right)}{h} = \cos x \end{aligned}$$

Hence, differentiation of $f(x) = \sin x$ w.r.t. x is $\cos x$

Example 4.2 Find the differentiation of e^x w.r.t. x .

Solution Given function $f(x) = e^x$

$$\text{and } f(x + h) = e^{x + h}$$

$$\therefore f'(x) = \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = \lim_{h \rightarrow 0} \frac{(e^h - 1)e^x}{h} = e^x \left(\lim_{x \rightarrow 0} \frac{(e^x - 1)}{x} \right)$$

Hence, differentiation of $f(x) = e^x$ w.r.t. x is e^x

Example 4.3 Find the differentiation of $\log_e x$ w.r.t. x .

Solution Given function $f(x) = \log_e x$ and $f(x+h) = \log_e (x+h)$

$$\begin{aligned} \therefore f'(x) &= \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\log_e (x+h) - \log_e x}{h} = \lim_{h \rightarrow 0} \frac{\log_e \left(1 + \frac{h}{x}\right)}{\frac{h}{x}} = \frac{1}{x} \quad \left(\lim_{x \rightarrow 0} \frac{\log_e (1+x)}{x} = 1 \right) \end{aligned}$$

Hence, differentiation of $f(x) = \log_e x$ w.r.t. is $\frac{1}{x}$

Example 4.4 Find the differentiation of $\sin^{-1} x$ w.r.t. x .

Solution Given $y = \sin^{-1} x \Rightarrow f(y) = \sin y = x$ and $f(y+h) = \sin (y+h)$

$$\begin{aligned} \therefore f'(y) &= \frac{dx}{dy} = \lim_{h \rightarrow 0} \frac{f(y+h) - f(y)}{h} = \lim_{h \rightarrow 0} \frac{\sin(y+h) - \sin y}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin\left(\frac{y+h-y}{2}\right) \cdot \cos\left(\frac{y+h+y}{2}\right)}{h} = \lim_{h \rightarrow 0} \frac{\sin\left(\frac{h}{2}\right) \cdot \cos\left(\frac{y+h+y}{2}\right)}{h} \\ \Rightarrow \frac{dy}{dx} &= f'(x) = \frac{1}{\left(\frac{dx}{dy}\right)} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}} \end{aligned}$$

Hence, differentiation of $f(x) = \sin^{-1} x$ w.r.t. x is $\frac{1}{\sqrt{1-x^2}}$.

4.1.2 Differentiation of Function by Second Principle

In this principle we can directly use differential co-efficient of standard functions as formula.

- (i) If $y = \sin x$ then $\frac{d(\sin x)}{dx} = \cos x$
- (ii) If $y = \cos x$ then $\frac{d(\cos x)}{dx} = -\sin x$
- (iii) If $y = \tan x$ then $\frac{d(\tan x)}{dx} = \sec^2 x$

$$(iv) \text{ If } y = \sec x \text{ then } \frac{d(\sec x)}{dx} = \sec x \cdot \tan x$$

$$(v) \text{ If } y = \cot x \text{ then } \frac{d(\cot x)}{dx} = -\operatorname{cosec}^2 x$$

$$(vi) \text{ If } y = \operatorname{cosec} x \text{ then } \frac{d(\operatorname{cosec} x)}{dx} = -\operatorname{cosec} x \cdot \cot x$$

$$(vii) \text{ If } y = \sin^{-1} x \text{ then } \frac{d(\sin^{-1} x)}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$(viii) \text{ If } y = \cos^{-1} x \text{ then } \frac{d(\cos^{-1} x)}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

$$(ix) \text{ If } y = \tan^{-1} x \text{ then } \frac{d(\tan^{-1} x)}{dx} = \frac{1}{1+x^2}$$

$$(x) \text{ If } y = \cot^{-1} x \text{ then } \frac{d(\cot^{-1} x)}{dx} = -\frac{1}{1+x^2}$$

$$(xi) \text{ If } y = \sec^{-1} x \text{ then } \frac{d(\sec^{-1} x)}{dx} = \frac{1}{|x|\sqrt{x^2-1}}$$

$$(xii) \text{ If } y = \operatorname{cosec}^{-1} x \text{ then } \frac{d(\operatorname{cosec}^{-1} x)}{dx} = -\frac{1}{|x|\sqrt{x^2-1}}$$

$$(xiii) \text{ If } y = \sin^{-1}(\sin x) \text{ then } \frac{d(\sin^{-1}(\sin x))}{dx} = \begin{cases} 1, & \text{when } \left(2n - \frac{1}{2}\right)\pi < x < \left(2n + \frac{1}{2}\right)\pi \\ -1 & \text{when } \left(2n + \frac{1}{2}\right)\pi < x < \left(2n + \frac{3}{2}\right)\pi \\ \text{does not define,} & \text{when } x = \left\{(2n + 1)\frac{\pi}{2}, n \in I\right\} \end{cases}$$

$$(xiv) \text{ If } y = \cos^{-1}(\cos x) \text{ then } \frac{d(\cos^{-1}(\cos x))}{dx} = \begin{cases} 1, & \text{when } 2n\pi < x < (2n + 1)\pi \\ -1 & \text{when } (2n + 1)\pi < x < (2n + 2)\pi \\ \text{does not define,} & \text{when } x = \{n\pi, n \in I\} \end{cases}$$

$$(xv) \text{ If } y = \tan^{-1}(\tan x) \text{ then } \frac{d(\tan^{-1}(\tan x))}{dx} = \begin{cases} 1, & \text{when } \left(2n - \frac{1}{2}\right)\pi < x < \left(2n + \frac{1}{2}\right)\pi \\ \text{does not define,} & \text{when } x = \left\{(2n + 1)\frac{\pi}{2}, n \in I\right\} \end{cases}$$

$$(xvi) \text{ If } y = \sin(\sin^{-1} x) \text{ then } \frac{d(\sin(\sin^{-1} x))}{dx} = 1, \text{ where } -1 < x < 1$$

$$(xvii) \text{ If } y = \cos(\cos^{-1} x) \text{ then } \frac{d(\cos(\cos^{-1} x))}{dx} = 1, \text{ where } -1 < x < 1$$

$$(xviii) \text{ If } y = \tan(\tan^{-1} x) \text{ then } \frac{d(\tan(\tan^{-1} x))}{dx} = 1$$

$$(xix) \text{ If } y = \operatorname{cosec}(\operatorname{cosec}^{-1} x) \text{ then } \frac{d(\operatorname{cosec}(\operatorname{cosec}^{-1} x))}{dx} = 1, \text{ where } |x| > 1$$

$$(xx) \text{ If } y = \sec(\sec^{-1} x) \text{ then } \frac{d(\sec(\sec^{-1} x))}{dx} = 1, \text{ where } |x| > 1$$

$$(xxi) \text{ If } y = \cot(\cot^{-1} x) \text{ then } \frac{d(\cot(\cot^{-1} x))}{dx} = 1$$

$$(xxii) \text{ If } y = e^x \text{ then } \frac{dy}{dx} = e^x$$

$$(xxiii) \text{ If } y = a^x \text{ then } \frac{dy}{dx} = a^x \log a.$$

$$(xxiv) \text{ If } y = \log_e x \text{ then } \frac{dy}{dx} = \frac{1}{x}$$

$$(xxv) \text{ If } y = \log_a x \Rightarrow y = \log_a e \cdot \log_e x \text{ then } \frac{dy}{dx} = \frac{1}{x} \cdot \log_a e$$

$$(xxvi) \text{ If } y = x^n \text{ then } \frac{dy}{dx} = nx^{n-1}.$$

$$(xxvii) \text{ If } y = \sqrt{x} \text{ then } \frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

$$(xxviii) \text{ If } y = \frac{1}{x} \text{ then } \frac{dy}{dx} = -\frac{1}{x^2}$$

(xxix) If $y = |x|$ then $\frac{dy}{dx} = \begin{cases} 1, & \text{when } x > 0 \\ -1, & \text{when } x < 0 \end{cases}$

(xxx) If $y = \text{vers}^{-1}\left(\frac{x}{a}\right)$ then $\frac{dy}{dx} = \frac{1}{\sqrt{2ax - x^2}}$ (As, $1 - \cos x = \text{vers } x$, $1 - \sin x = \text{covers } x$).

4.2 FUNDAMENTAL RULES FOR DIFFERENTIATION

Rule-I

If $y = k.f(x)$ then $\frac{dy}{dx} = k \frac{d\{f(x)\}}{dx}$ where k is real constant.

Rule-II

If $y = k$ then $\frac{dy}{dx} = 0$ where k is real constant.

Rule-III

If $y = f(x) + g(x)$ then $\frac{dy}{dx} = \frac{d\{f(x) + g(x)\}}{dx} = \frac{df(x)}{dx} + \frac{dg(x)}{dx}$

Rule-IV

If $y = f(x) - g(x)$ then $\frac{dy}{dx} = \frac{d\{f(x) - g(x)\}}{dx} = \frac{df(x)}{dx} - \frac{dg(x)}{dx}$

Rule-V

If $y = f(x) \cdot g(x)$ then $\frac{dy}{dx} = \frac{d\{f(x) \cdot g(x)\}}{dx} = g(x) \frac{df(x)}{dx} + f(x) \frac{dg(x)}{dx}$

Rule-VI

If $y = \frac{f(x)}{g(x)}$ then $\frac{dy}{dx} = \frac{d\left\{\frac{f(x)}{g(x)}\right\}}{dx} = \frac{g(x) \frac{df(x)}{dx} - f(x) \frac{dg(x)}{dx}}{\{g(x)\}^2}$

Example 4.5 Find the differentiation of function $y = \frac{5x^4 + 6x^2 - x + 1}{x}$.

Solution Given function $y = \frac{5x^4 + 6x^2 - x + 1}{x}$

$$\Rightarrow y = 5x^3 + 6x - 1 + \frac{1}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{d\left(5x^3 + 6x - 1 + \frac{1}{x}\right)}{dx} = 5\frac{dx^3}{dx} + 6\frac{dx}{dx} - \frac{d(1)}{dx} + \frac{d(1/x)}{dx} = 15x^2 + 6 - \frac{1}{x^2}$$

Example 4.6 Find the differentiation of function $y = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2$

Solution Given function $y = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2$

$$\Rightarrow y = x + \frac{1}{x} + 2 \Rightarrow \frac{dy}{dx} = 1 - \frac{1}{x^2} = \frac{x^2 - 1}{x^2}$$

Example 4.7 Find the differentiation of function $y = x^4 \log x$

Solution Given function $y = x^4 \log x$

$$\Rightarrow \frac{dy}{dx} = \frac{d(x^4 \log x)}{dx} = x^4 \frac{d(\log x)}{dx} + \log x \frac{d(x^4)}{dx} = 4x^3 \log x + x^3$$

Example 4.8 Find the differentiation of function $f(x) = \frac{1 - \cos x}{1 + \cos x}$ then find $f'\left(\frac{\pi}{2}\right)$.

Solution Given $f(x) = \frac{1 - \cos x}{1 + \cos x}$

$$\Rightarrow f'(x) = \frac{(1 + \cos x) \frac{d(1 - \cos x)}{dx} - (1 - \cos x) \frac{d(1 + \cos x)}{dx}}{(1 + \cos x)^2} = \frac{2 \sin x}{(1 + \cos x)^2}$$

$$\therefore f'\left(\frac{\pi}{2}\right) = 2$$

Example 4.9 If $y = (x - a)(x - b)$ then find x for which $\frac{dy}{dx} = 0$.

Solution Given function $y = (x - a)(x - b)$

$$\Rightarrow y = x^2 - x(a + b) + ab \Rightarrow \frac{dy}{dx} = 2x - (a + b)$$

$$\text{A/Q, } \frac{dy}{dx} = 0 \Rightarrow 2x - (a + b) = 0 \Rightarrow x = \frac{a + b}{2}$$

Example 4.10 If $f(x) = \frac{x - 4}{2\sqrt{x}}$ then find $f'(4)$

Solution Given function $f(x) = \frac{x - 4}{2\sqrt{x}}$

$$\Rightarrow f'(x) = \frac{1}{2} \left[\frac{\sqrt{x} \frac{d(x - 4)}{dx} - (x - 4) \frac{d(\sqrt{x})}{dx}}{(\sqrt{x})^2} \right] = \frac{1}{2} \left[\frac{\sqrt{x} - \frac{(x - 4)}{2\sqrt{x}}}{x} \right]$$

$$\therefore f'(4) = \frac{1}{2} \left[\frac{\sqrt{4} - \frac{(4 - 4)}{2\sqrt{4}}}{4} \right] = \frac{1}{4}$$

4.3 COMPOSITE FUNCTION

Composite function If $f(x)$ and $g(x)$ are any two function then

$\text{fog}(x) = f\{g(x)\}$ = composition of $g(x)$ in $f(x)$ w.r.t. x and

$\text{gof}(x) = g\{f(x)\}$ = composition of $f(x)$ in $g(x)$ w.r.t. x

4.3.1 Differentiation of Composite Function

To differentiate composite function use:

(a) Write given function $y = \text{fog}(x) = f\{g(x)\}$

(b) Differentiate both side w.r.t. x

$$\text{i.e. } \frac{dy}{dx} = \frac{df\{g(x)\}}{dg(x)} = \frac{df\{g(x)\}}{dg(x)} \cdot \frac{dg(x)}{dx} = [(\text{d.c. of } f\{g(x)\} \text{ w.r.t. } g(x))(\text{d.c. of } g(x) \text{ w.r.t. } x)]$$

Example 4.11 Find the differentiation of function $y = e^{\sqrt{\ln(ax^2 + bx + c)}}$

Solution Given function $y = e^{\sqrt{\ln(ax^2 + bx + c)}}$

$$\Rightarrow \frac{dy}{dx} = \frac{d\{e^{\sqrt{\ln(ax^2+bx+c)}}\}}{dx} = \frac{d\{e^{\sqrt{\ln(ax^2+bx+c)}}\}}{d\{\ln(ax^2+bx+c)\}} \cdot \frac{d\{\ln(ax^2+bx+c)\}}{(ax^2+bx+c)} \cdot \frac{d(ax^2+bx+c)}{dx}$$

$$\Rightarrow \frac{dy}{dx} = e^{\sqrt{\ln(ax^2+bx+c)}} \cdot \frac{1}{ax^2+bx+c} \cdot (2ax+b) = \frac{(2ax+b)e^{\sqrt{\ln(ax^2+bx+c)}}}{ax^2+bx+c}$$

Example 4.12 Find the differentiation of function $y = \sin^3 \sqrt{ax^2+bx+c}$

Solution Given function $y = \sin^3 \sqrt{ax^2+bx+c}$

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= \frac{d\{\sin^3 \sqrt{ax^2+bx+c}\}}{dx} \\ &= \frac{d\{\sin \sqrt{ax^2+bx+c}\}^3}{d\{\sin \sqrt{ax^2+bx+c}\}} \cdot \frac{d\{\sin \sqrt{ax^2+bx+c}\}}{d\{\sqrt{ax^2+bx+c}\}} \cdot \frac{d\{\sqrt{ax^2+bx+c}\}}{d\{ax^2+bx+c\}} \cdot \frac{d\{ax^2+bx+c\}}{dx} \\ &= \frac{3\sin^2 \sqrt{ax^2+bx+c} \cdot \cos \sqrt{ax^2+bx+c} \cdot (2ax+b)}{2\sqrt{ax^2+bx+c}} \end{aligned}$$

Example 4.13 Find the differentiation of function $y = \cos(\sin^{-1} x)$

Solution Given function $y = \cos(\sin^{-1} x)$

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= \frac{d\{\cos(\sin^{-1} x)\}}{d\{\sin^{-1} x\}} \cdot \frac{d\{\sin^{-1} x\}}{dx} \\ &= -\sin(\sin^{-1} x) \cdot \frac{1}{\sqrt{1-x^2}} = \frac{-x}{\sqrt{1-x^2}} \end{aligned}$$

Example 4.14 Find the differentiation of function $y = \tan^{-1}(\log x)$

Solution Given function $y = \tan^{-1}(\log x)$

$$\Rightarrow \frac{dy}{dx} = \frac{d\{\tan^{-1}(\log x)\}}{d(\log x)} \cdot \frac{d\{\log x\}}{dx} = \frac{1}{1+(\log x)^2} \cdot \frac{1}{x}$$

Example 4.15 Find the differentiation of function $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$

Solution Given function $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$

Put $x = \tan \alpha \Rightarrow \alpha = \tan^{-1} x$

$$\begin{aligned} \therefore y &= \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right) = \tan^{-1} \left(\frac{\sqrt{1+\tan^2 \alpha}-1}{\tan \alpha} \right) = \tan^{-1} \left(\frac{1-\cos \alpha}{\sin \alpha} \right) \\ &= \tan^{-1} \left(\tan \frac{\alpha}{2} \right) = \frac{\alpha}{2} = \frac{\tan^{-1} x}{2} \Rightarrow \frac{dy}{dx} = \frac{1}{2(1+x^2)} \end{aligned}$$

Example 4.16 Find the differentiation of function $y = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$

Solution Given function $y = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$

Put, $x = \tan \alpha \Rightarrow \alpha = \tan^{-1} x$

$$\therefore y = \tan^{-1} \left(\frac{2 \tan \alpha}{1 - \tan^2 \alpha} \right) = \tan^{-1} (\tan 2\alpha) = 2\alpha = 2 \tan^{-1} x$$

$$\Rightarrow \frac{dy}{dx} = 2 \frac{d\{\tan^{-1} x\}}{dx} = \frac{2}{1+x^2}$$

Example 4.17 Find the differentiation of function $y = \sin^{-1} \sqrt{1-x^2}$

Solution Given function $y = \sin^{-1} \sqrt{1-x^2}$

Put, $x = \cos \alpha \Rightarrow \alpha = \cos^{-1} x$

$$\therefore y = \sin^{-1} \sqrt{1-x^2} = \sin^{-1} \sqrt{1-\cos^2 \alpha} = \sin^{-1} \{\sin \alpha\} = \alpha$$

$$\Rightarrow y = \cos^{-1} x \Rightarrow \frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}}$$

4.3.2 Differentiation of Implicit Function

Implicit Function

Form of function in which independent and dependent variables are not defined i.e. dependent variable cannot express in the form of independent variable then the function is called implicit function.

For differentiate implicit function use:

(a) Differentiate the given relation between x and y w.r.t. x

(b) Bring all the terms containing $\frac{dy}{dx}$ on left hand side and remaining term on right hand side then

$$\text{find } \frac{dy}{dx}$$

(c) Use the given relation between x and y to get the result in simplified form.

Example 4.18 If $x = y \log_e(xy)$ then find $\frac{dy}{dx}$.

Solution Given function $x = y \log_e(xy)$

$$\Rightarrow \frac{dx}{dx} = y \frac{d\{\log_e xy\}}{dx} + \log_e xy \frac{dy}{dx} \Rightarrow 1 = y \frac{d\{\log_e x + \log_e y\}}{dx} + \log_e xy \frac{dy}{dx}$$

$$\Rightarrow 1 = y \left\{ \frac{1}{x} + \frac{1}{y} \cdot \frac{dy}{dx} \right\} + \log_e xy \frac{dy}{dx} \Rightarrow 1 = \frac{y}{x} + \frac{dy}{dx} + \log xy \frac{dy}{dx}$$

$$\Rightarrow \frac{dx}{dx} = \frac{x-y}{x\{1+\log xy\}} = \frac{x-y}{x\{1+x/y\}} = \frac{y}{x} \left(\frac{x-y}{x+y} \right)$$

Example 4.19 If $y \cdot \log_e(x+y)$ then find $\frac{dx}{dy}$.

Solution Given function $y \log_e(x+y) = \sin(xy)$

$$\Rightarrow \log_e(x+y) \frac{dy}{dx} + y \frac{d\{\log_e(x+y)\}}{d\{x+y\}} \cdot \frac{d(x+y)}{dx} = \frac{d\{\sin xy\}}{d(xy)} \cdot \frac{d(xy)}{dx}$$

$$\Rightarrow \log_e(x+y) \frac{dy}{dx} + y \frac{1}{\{x+y\}} \left\{ \frac{d(x)}{dx} + \frac{d(y)}{dx} \right\} = \cos(xy) \left\{ \frac{yd(x)}{dx} + \frac{xd(y)}{dx} \right\}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y \cos(xy) - \frac{y}{x+y}}{\frac{y}{x+y} + \log(x+y) - x \cos(xy)}$$

Example 4.20 If $xy = x^3 + y^3$ then find $\frac{dx}{dy}$.

Solution Given function $xy = x^3 + y^3$

$$\Rightarrow \frac{d\{xy\}}{dx} = \frac{d(x^3 + y^3)}{dx} \Rightarrow x \frac{dy}{dx} + y \frac{dx}{dx} = \frac{d\{x^3\}}{dx} + \frac{dy^3}{dy} \cdot \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{3x^2 - y}{x - 3y^2}$$

Example 4.21 If $xe^{xy} = y + \sin^2 x$, then at $x = 0$, $\frac{dy}{dx}$ is equal to

(a) -1

(b) 1

(c) 0

(d) none of these

Solution When $x = 0$, we get $y = 0$.

On differentiating both sides of the given equation w.r.t. x , we get

$$e^{xy} + xe^{xy} \left(x \frac{dy}{dx} + y \right) = \frac{dy}{dx} + 2 \sin x \cos x$$

On putting $x = 0, y = 0$, we get

$$e^0 + 0 \cdot e^0 \left(0 \cdot \left[\frac{dy}{dx} \right]_{(0,0)} + 0 \right) = \left[\frac{dy}{dx} \right]_{(0,0)} + 2 \sin 0 \cos 0$$

$$\Rightarrow \left[\frac{dy}{dx} \right]_{(0,0)} = 1$$

Example 4.22 If $y = f(x)$ and it follows the relation $x \cos y + y \cos x = \pi$, then $y''(0)$ is equal to:

(IIT-JEE-2005)

- (a) -1 (b) π (c) $-\pi$ (d) 1

Solution Given $x \cos y + y \cos x = \pi$

Put, $x = 0, y(0) = \pi$

Differentiate both side w.r.t. x

$$-x \sin y \frac{dy}{dx} + \cos y - y \sin x + \cos x \cdot \frac{dy}{dx} = 0 \quad (I)$$

Put, $x = 0, y'(0) = 1$

Again, differentiate (I) both side w.r.t. x , then put $x = 0$

To get, $y''(0) = \pi$

Example 4.23 If y is a function of x and $\log(x + y) = 2xy$ then value of $y'(0)$ is equal to:

(IIT-JEE-2004)

- (a) 1 (b) -1 (c) 2 (d) 0

Solution Given $\log(x + y) = 2xy$

Differentiate both side w.r.t. x

$$\frac{1}{x+1} \left(1 + \frac{dy}{dx} \right) = 2y + 2x \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{\frac{1}{x+1} - 2y}{2x - \frac{1}{x+1}} \Rightarrow y'(0) = 1$$

4.3.3 Differentiation of Parametric Function

Parametric Function: Form of function in which x and y are defined with any parametric, then function is called parametric function.

To differentiate parametric function use:

- (a) Write given parametric function $y = f(t)$ and $x = g(t)$ where t is parameter.
 (b) Differentiate $y = f(t)$ and $x = g(t)$ w.r.t. parameter t i.e.

$$\text{Evaluate } \frac{dy}{dt} = \frac{df(t)}{dt} \text{ and } \frac{dx}{dt} = \frac{d\{g(t)\}}{dt}.$$

Hence, differentiation of parametric function w.r.t. x i.e. $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

Note: $\frac{d^2 y}{dx^2} \neq \frac{d^2 y}{d^2 x} = \frac{(d^2 y/dt^2)}{(d^2 x/dt^2)}$

Example 4.24 The derivative of $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ w.r.t. $\cot^{-1} \sqrt{\frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}}$ is

- (a) 1 (b) -1 (c) $\frac{1}{2}$ (d) none of these

Solution Let $y = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ and $t = \cot^{-1} \sqrt{\frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}}$

Put $x = \tan \theta$

$$\begin{aligned} \therefore y &= \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) = \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) \\ &= \tan^{-1} \tan \frac{\theta}{2} = \frac{\theta}{2} \end{aligned}$$

and
$$\begin{aligned} t &= \cos^{-1} \sqrt{\frac{1 + \sec \theta}{2 \sec \theta}} = \cos^{-1} \sqrt{\frac{1 + \cos \theta}{2}} \\ &= \cos^{-1} \cos \frac{\theta}{2} = \frac{\theta}{2} \end{aligned}$$

$$\therefore y = \frac{\theta}{2} = t; \quad \frac{dy}{dt} = 1.$$

Example 4.25 The derivative of $f(\tan x)$ w.r.t. $g(\sec x)$ at is

- (a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$ (c) 1 (d) none of these

Solution Let $y = f(\tan x)$ and $u = g(\sec x)$

$$\Rightarrow \frac{dy}{dx} = f'(\tan x) \sec^2 x$$

and
$$\frac{du}{dx} = g'(\sec x) \cdot \sec x \tan x$$

$$\therefore \frac{dy}{du} = \frac{dy/dx}{du/dx} = \frac{f'(\tan x) \sec^2 x}{g'(\sec x) \sec x \tan x}$$

$$\therefore \left[\frac{dy}{du} \right]_{x=\frac{\pi}{4}} = \frac{f'(\tan \frac{\pi}{4})}{g'(\sec \frac{\pi}{4}) \sin \frac{\pi}{4}} = \frac{f'(1) \cdot \sqrt{2}}{g'(\sqrt{2})}$$

$$= \frac{2 \cdot \sqrt{2}}{4} = \frac{1}{\sqrt{2}}.$$

Example 4.26 If $x = t^t$ and $y = t^t$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{t^t \cdot \left[(1 + \log t) \log t + \frac{1}{t} \right]}{(1 + \log t)^2}$

(b) $\frac{t^t \left[1 + \log t + \frac{1}{t} \right]}{(1 + \log t)}$

(c) $\frac{t^t \left[(1 + \log t) \log t + \frac{1}{t} \right]}{(1 + \log t)^2}$

(d) none of these

Solution We have $x = t^t = e^{t \log t}$

$$\Rightarrow \frac{dx}{dt} = e^{t \log t} (1 + \log t) = t^t (1 + \log t)$$

Also, $y = t^t \Rightarrow \log y = t \log t = e^{t \log t} \cdot \log t$

$$\Rightarrow \frac{1}{y} \frac{dy}{dt} = e^{t \log t} (1 + \log t) \log t + e^{t \log t} \cdot \frac{1}{t}$$

$$\frac{dy}{dt} = t^t \left\{ t^t (1 + \log t) \log t + \frac{t^t}{t} \right\}, \therefore \frac{dy}{dx} = \frac{t^t \left[(1 + \log t) \log t + \frac{1}{t} \right]}{(1 + \log t)}$$

Example 4.27 $\frac{d^2x}{dy^2}$ equals to

(IIT (JEE)-2007)

(a) $\left(\frac{d^2y}{dx^2} \right)^{-1}$

(b) $-\left(\frac{d^2y}{dx^2} \right)^{-1} \left(\frac{dy}{dx} \right)^{-3}$

(c) $\frac{d^2y}{dx^2} \left(\frac{dy}{dx} \right)^{-2}$

(d) $-\left(\frac{d^2y}{dx^2} \right) \left(\frac{dy}{dx} \right)^{-3}$

Solution Given $\frac{dy}{dx} = \left(\frac{dx}{dy} \right)^{-1} \Rightarrow \frac{d \left(\frac{dy}{dx} \right)}{dx} = \frac{d \left(\frac{dx}{dy} \right)}{dx}$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{d \left(\frac{dx}{dy} \right)^{-1}}{dx} \cdot \frac{dy}{dx} = - \left(\frac{dx}{dy} \right)^{-2} \frac{d^2x}{dy^2} \cdot \frac{dy}{dx} = - \frac{d^2y}{dx^2} \left(\frac{dy}{dx} \right)^{-3}$$

Example 4.28 Let $f(x)$ be differentiable function on the interval $(0, \infty)$ such that $f(1) = 1$, and

$$\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1 \text{ for each } x > 0 \text{ then } f(x) \text{ is}$$

(IIT (JEE)-2007)

(a) $\frac{1}{3x} + \frac{2x^2}{3}$

(b) $-\frac{1}{3x} + \frac{4x^2}{3}$

(c) $-\frac{1}{x} + \frac{2}{x^2}$

(d) $\frac{1}{x}$

Solution Given

$$\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$$

Use, L'Hospital Rule

$$\lim_{t \rightarrow x} \frac{2tf(x) - x^2 f'(t)}{1} = 1 = 1 \Rightarrow 2xf(x) - x^2 f'(x) = 1 \Rightarrow f'(x) = \frac{dy}{dx} = \frac{2xy - 1}{x^2}$$

Solve differential equation, to get

$$f(x) = -\frac{1}{3x} + \frac{4x^2}{3}$$

4.3.4 Differentiation of Logarithmic and Exponential Function

Logarithmic Function

Form of function $y = (\text{variable})^{\text{variable}}$ is called logarithmic function and $y = (\text{constant})^{\text{variable}}$ is called exponential function.

Differentiation

To differentiate function use:

- Write given function.
- Take log both side and simplify.
- Differentiate simplify form w.r.t. x

Example 4.29 If $f(x) = \log_5 \log_3 x$, then $f'(e)$ is equal to

- (a) $e \log 5$ (b) $-e \log 5$ (c) $\frac{1}{e \log 5}$ (d) none of these

Solution Given

$$\begin{aligned} f(x) &= \log_5 \log_3 x = \log_5 \left(\frac{\log x}{\log 3} \right) \\ &= \log_5 (\log x) - \log_5 (\log 3) \\ &= \log_5 (\log x) - \log_5 (\log 3) \\ &= \frac{\log (\log x)}{\log 5} - \log_5 (\log 3) \end{aligned}$$

$$\phi'(x) = \frac{1}{\log 5} \cdot \frac{1}{\log x} \cdot \frac{1}{x} = 0$$

$$\therefore \phi'(e) = \frac{1}{\log 5} \cdot \frac{1}{\log e} \cdot \frac{1}{e} = \frac{1}{e \log 5}$$

Example 4.30 If $y = x^{(\log x)^{\log \log x}}$, then $\frac{dy}{dx}$ is equal to

- (a) $\frac{y \log y}{x \log x} (2 \log \log x + 1)$ (b) $\frac{x \log x}{y \log y} (2 \log \log x + 1)$
 (c) $\frac{2y \log y}{x \log x} (\log \log x + 1)$ (d) none of these

Solution We have, $y = x^{(\log x)^{\log \log x}}$

$$\Rightarrow \log y = (\log x)^{\log \log x} \cdot \log x$$

$$\Rightarrow \log \log y = \log \log x \cdot \log \log x + \log \log x$$

Differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{y \log y}{x \log x} (2 \log \log x + 1).$$

Example 4.31 Let $g(x) = \log \{f(x)\}$ where $f(x)$ is a twice differentiable positive function. On $(0, \infty)$ such that $f(x+1) = xf(x)$ then for $N = 1, 2, 3, 4, \dots$ (IIT (JEE)-2008)

$$g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$$

$$(a) -4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$$

$$(b) 4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$$

$$(c) -4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$$

$$(d) -4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$$

Solution Given $f(x+1) = xf(x)$

$$\text{and } g(x) = \log \{f(x)\} \Rightarrow g(x+1) = \log \{f(x+1)\} \\ = \log \{xf(x)\} = \log x + \log f(x) = \log x + g(x)$$

$$\Rightarrow g(x+1) - g(x) = \log x$$

Differentiate two times both side w.r.t. x

$$g''(x+1) - g''(x) = -\frac{1}{x^2} \Rightarrow g''\left(x + \frac{1}{2}\right) - g''\left(x - \frac{1}{2}\right) = \frac{4}{(2x-1)^2}$$

Put, $x = 1, 2, 3, 4, 5, \dots, N$

$$g''\left(1 + \frac{1}{2}\right) - g''\left(1 - \frac{1}{2}\right) = \frac{4}{(2 \cdot 1 - 1)^2} = -\frac{4}{1^2}$$

$$g''\left(2 + \frac{1}{2}\right) - g''\left(2 - \frac{1}{2}\right) = \frac{4}{(2 \cdot 2 - 1)^2} = -\frac{4}{3^2}$$

$$g''\left(3 + \frac{1}{2}\right) - g''\left(3 - \frac{1}{2}\right) = \frac{4}{(2 \cdot 3 - 1)^2} = -\frac{4}{5^2}$$

.....
.....

$$g''\left(N + \frac{1}{2}\right) - g''\left(N - \frac{1}{2}\right) = \frac{4}{(2 \cdot N - 1)^2} = -\frac{4}{(2N-1)^2}$$

Adding all terms

$$g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) = -4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$$

TIPS AND TRICKS

1. If $f(x)$ is any polynomial and $f(a) = f(b)$ then at least one root of equation $f'(x) = 0$ lying in $[a, b]$.
2. If α is a repeated root of equation $f(x) = 0$, then α is also root of equation $f'(x) = 0$.
3. If α is repeated common root of the equation $f(x) = 0$ and $g(x) = 0$ then α is also a common root of the equation $f'(x) = 0$ and $g'(x) = 0$.
4. If $f(x) = 0$ is a polynomial equation has n number of real roots then the equation $f'(x) = 0$ will have $(n - 1)$ real roots, $f''(x) = 0$ will have $(n - 2)$ real roots But its converse may or may not be true.
5. If $f(x)$ is any polynomial function and sign of $f(a)$ and $f(b)$ are opposite then equation $f(x) = 0$ represents odd numbers of real root.

Graphically

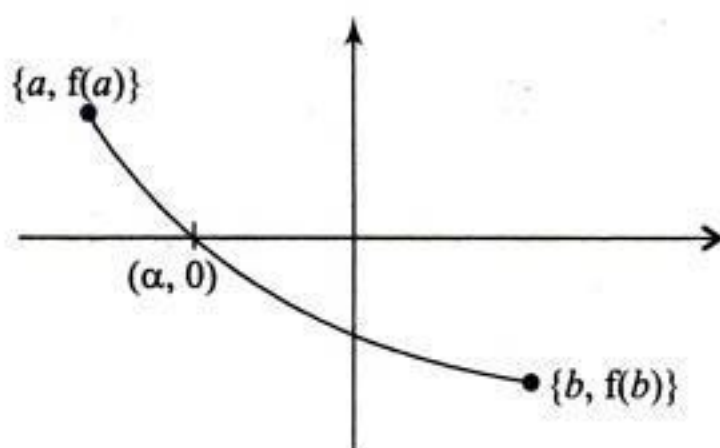


Fig. 14.1 Here, equation $f(x) = 0$ defined only one root

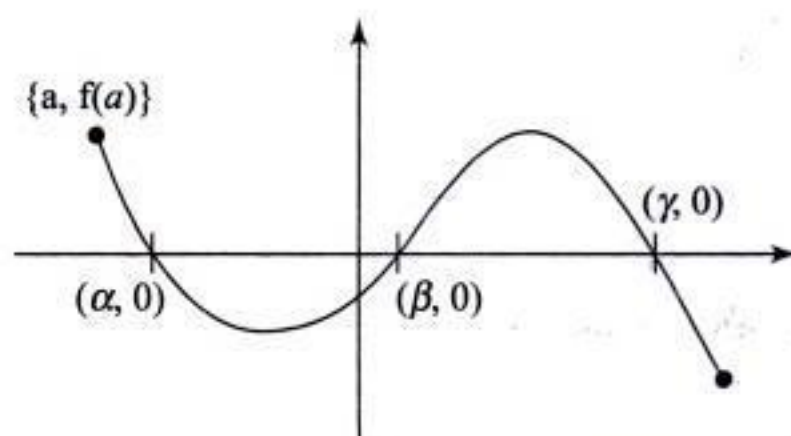


Fig. 14.2 Here, $f(x) = 0$ defined 3 real roots

6. If $f(x)$ is any polynomial function and sign of $f(a)$ and $f(b)$ are same then equation $f(x) = 0$ represents real or imaginary roots. In this case, if roots of equation $f(x) = 0$ are real then even number of real roots can be defined.

Graphically

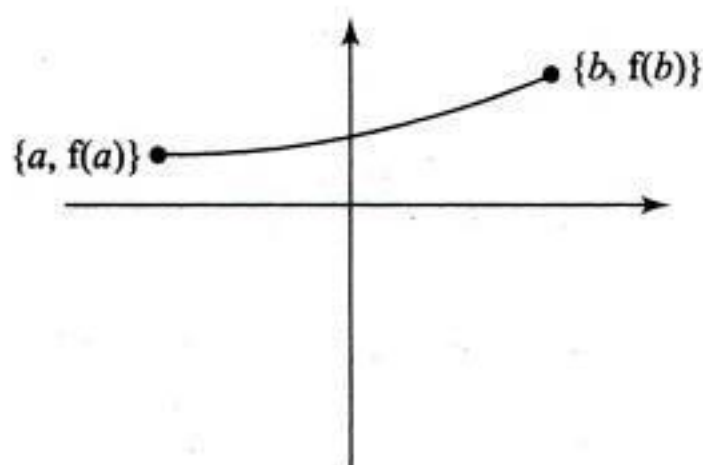


Fig. 14.3 Here, $f(x) = 0$ cannot defined real roots

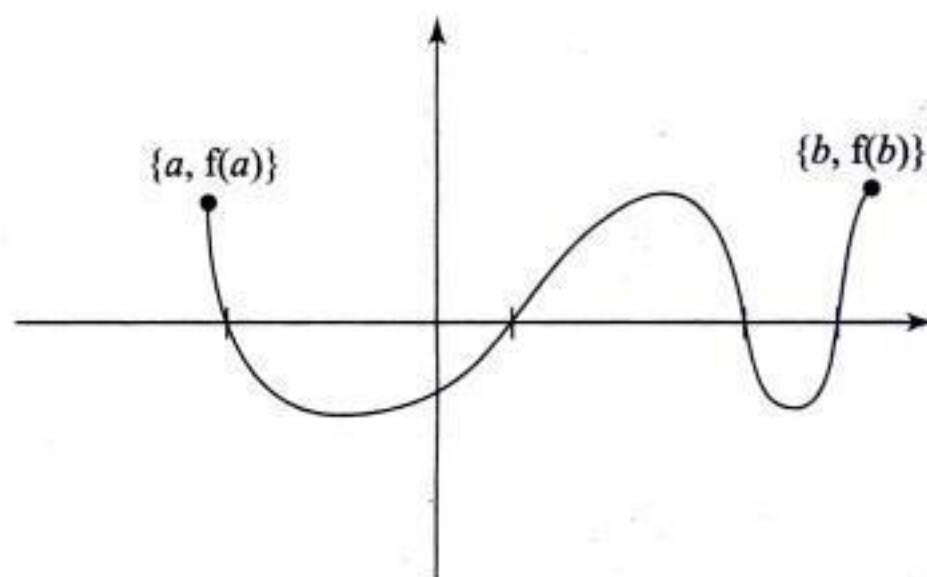


Fig. 14.4 Here, $f(x) = 0$ defined 4 real roots

7. Differentiation of a Determinant

$$\text{If } f(x) = \begin{vmatrix} f_1 & f_2 & f_3 \\ g_1 & g_2 & g_3 \\ h_1 & h_2 & h_3 \end{vmatrix}$$

Where $f_1, f_2, f_3; g_1, g_2, g_3; h_1, h_2, h_3;$ are functions of x , then

$$F'(x) = \begin{vmatrix} f_1' & f_2' & f_3' \\ g_1 & g_2 & g_3 \\ h_1 & h_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2 & f_3 \\ g_1' & g_2' & g_3' \\ h_1 & h_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2 & f_3 \\ g_1 & g_2 & g_3 \\ h_1' & h_2' & h_3' \end{vmatrix}$$

$$\text{Also } F'(x) = \begin{vmatrix} f_1' & f_2 & f_3 \\ g_1' & g_2 & g_3 \\ h_1' & h_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2' & f_3 \\ g_1 & g_2' & g_3 \\ h_1 & h_2' & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2 & f_3' \\ g_1 & g_2 & g_3' \\ h_1 & h_2 & h_3' \end{vmatrix}$$

8. Form of binomial co-efficient series in which natural numbers occurs as the product of binomial co-efficient then for sum of series use following steps

Step-(I): If natural co-efficient of last term of series is M , then divide M by n to get quotient Q and R be the remainder i.e. $M = nQ + R$

Step-(II): As, $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \dots + {}^nC_nx^n$

Put x^Q in place of x and multiply both sides by x^R , then differentiate both sides with respect to x , one times, two times, three times, according to numbers of natural numbers multiply in each term of given series.

Step-(III): Put $x = 1$ or -1 or i or $-i$ or etc. according to given series.

Example 4.32: If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$,

Show that $C_0 - 2 \cdot C_1 + 3 \cdot C_2 - \dots + (-1)^n (n+1)C_n = 0$

Solution: Given series is $C_0 - 2 \cdot C_1 + 3 \cdot C_2 - \dots + (-1)^n (n+1)C_n$

[Here numerical value of the last term is $(n+1)$. C_n and terms are alternately positive and negative.

Co-efficient of C_n is $(n+1)$]

$$\frac{n)n+1(1}{1}$$

Given, $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$

Multiplying both sides by x , we get

$$x \cdot (1+x)^n = C_0x + C_1x^2 + C_2x^3 + \dots + C_nx^{n+1}$$

Diff, both sides w. r. to x , we get

$$1 \cdot (1+x)^n + x \cdot n(1+x)^{n-1} = C_0 + C_1 \cdot 2x + C_2 \cdot 3x^2 + \dots + C_n \cdot (n+1)x^n$$

Putting $x = -1$, we get

$$\begin{aligned} C_0 - 2 \cdot C_1 + 3 \cdot C_2 - \dots + (-1)^n (n+1)C_n \\ = (1-1)^n + (-1)n(1-1)^{n-1} = 0 + 0 = 0 \end{aligned}$$

Example 4.33 If $y = \begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix}$, then $\frac{dy}{dx}$ equal to

- (a) 1 (b) -1 (c) 0 (d) none of these

Solution $\frac{dy}{dx} = \begin{vmatrix} \cos x & -\sin x & \cos x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix} + \begin{vmatrix} \sin x & \cos x & \sin x \\ -\sin x & -\cos x & -\sin x \\ x & 1 & 1 \end{vmatrix} + \begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ 1 & 0 & 0 \end{vmatrix}$

$$= 0 - \begin{vmatrix} \sin x & \cos x & \sin x \\ \sin x & \cos x & \sin x \\ x & 1 & 1 \end{vmatrix} + \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$= 0 + (\cos^2 x + \sin^2 x) = 1.$$

Example 4.34 If $f(x) = (\cos x + i \sin x)(\cos 2x + i \sin 2x)(\cos 3x + i \sin 3x) \dots (\cos nx + i \sin nx)$ and $f(1) = 1$, then $f''(1)$ is equal to

- (a) $\frac{n(n+1)}{2}$ (b) $\left(\frac{n(n+1)}{2}\right)^2$ (c) $-\left(\frac{n(n+1)}{2}\right)^2$ (d) none of these

Solution $f(x) = (\cos x + i \sin x)(\cos 2x + i \sin 2x)(\cos 3x + i \sin 3x) \dots (\cos nx + i \sin nx)$
 $= \cos(x + 2x + 3x + \dots + nx) + i \sin(x + 2x + 3x + \dots + nx)$
 $= \cos \frac{n(n+1)}{2}x + i \sin \frac{n(n+1)}{2}x$

$$\Rightarrow f'(x) = \frac{n(n+1)}{2} \left[-\sin \frac{n(n+1)}{2}x + i \cos \frac{n(n+1)}{2}x \right]$$

$$\Rightarrow f''(x) = -\left(\frac{n(n+1)}{2}\right)^2 \left(\cos \frac{n(n+1)}{2}x + i \sin \frac{n(n+1)}{2}x \right)$$

$$= -\left(\frac{n(n+1)}{2}\right)^2 f(x)$$

$$\therefore f''(1) = -\left(\frac{n(n+1)}{2}\right)^2 f(1) = -\left(\frac{n(n+1)}{2}\right)^2.$$

10. Differentiation of integral function

Let $f(x) = \int_{u(x)}^{v(x)} g(t)dt$ then $f'(x) = g\{v(x)\} \cdot \frac{d\{v(x)\}}{dx} - g\{u(x)\} \cdot \frac{d\{u(x)\}}{dx}$

Example 4.35 Let $f(x) = \int_{2x}^{x^2-x} \sin t dt$ then find $f'(x)$

Solution Given $f(x) = \int_{2x}^{x^2-x} \sin t dt$

$$f'(x) = \sin(x^2 - x) \times (2x - 1) - \sin 2x \times 2 = (2x - 1) \sin(x^2 - x) - 2 \sin 2x$$

11. From $\frac{d\{f(x)\}}{dx}$ at any point $x = a$ exist only when $x = a$ is in the domain of $f(x)$ as well as of $f'(x)$.

12. Let $y = f(x)$ is any differentiable function then $y_1 = \frac{dy}{dx} = f'(x)$ is also a differentiable function

of x . The derivative of $\frac{dy}{dx}$ is denoted as $\frac{d^2 y}{dx^2}$ or y_2 or $f''(x)$ which is also differentiable function in x .

Hence, If n is positive integer then n -th derivative of $y = f(x)$ is denoted as y_n or $f^n(x)$ or $\frac{d^n y}{dx^n}$

or $D^n(x)$.

13. **n -th derivative of some special function**

(a) $D^n(a)^x = (\log a)^n a^x$

(b) $D^n(e)^{ax+b} = (e)^{ax+b} a^n$

(c) (i) If $m > n$ then

$$D^n(ax + b)^m = m(m-1) \dots (m-n+1)a^n (ax + b)^{m-n} = \frac{m!}{(m-n)!} a^n (ax + b)^{m-n}$$

(ii) If $m < n$ then $D^n(ax + b)^m = 0$ and if $m = n$ then $D^n(ax + b)^m = a^n \cdot n!$

(d) $D^n(ax + b)^{-1} = (-1)^n \cdot n! \cdot a^n \cdot (ax + b)^{-1-n}$

(e) $D^n \log(ax + b) = \frac{(-1)^n (n-1)! a^n}{(ax + b)^n}$

(f) $D^n \sin(ax + b) = a^n \sin\left\{ax + b + \frac{n\pi}{2}\right\}$

(g) $D^n \cos(ax + b) = a^n \cos\left\{ax + b + \frac{n\pi}{2}\right\}$

(i) $D^n \{e^{ax} \sin(bx + c)\} = (a^2 + b^2)^{n/2} e^{ax} \sin\left\{bx + c + n \tan^{-1}\left(\frac{b}{a}\right)\right\}$

$$(j) D^n \{e^{ax} \cos(bx + c)\} = (a^2 + b^2)^{n/2} e^{ax} \cos \left\{ bx + c + n \tan^{-1} \left(\frac{b}{a} \right) \right\}$$

14. Leibniz's Theorem

If $f(x)$ and $g(x)$ are any two function in x then

$$D^n \{f(x) \cdot g(x)\} = {}^nC_0 D^{n-0} \{f(x)\} \cdot g(x) + {}^nC_1 D^{n-1} \{f(x)\} \cdot D\{g(x)\} + {}^nC_2 D^{n-2} \{f(x)\} \cdot D^2\{g(x)\} + \dots + {}^nC_n D^n \{g(x)\}$$

15. Centre of conics

As we know, second degree equation

$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0$ represents degenerated and non-degenerated curves (conics), then for centre of conic, use differentiation.

$$\text{Let } U = ax^2 + by^2 + 2gx + 2fy + 2hxy + c \quad (I)$$

Differentiate both side (I) w.r.to x

$$\frac{dU}{dx} = 2ax + 2g + 2hy$$

Again, differentiate both side (I) w.r.to y

$$\frac{dU}{dy} = 2by + 2f + 2hx$$

For centre, use

$$\frac{dU}{dx} = 2ax + 2g + 2hy = 0 \Rightarrow ax + hy + g = 0 \quad (II)$$

$$\frac{dU}{dy} = 2by + 2f + 2hx = 0 \Rightarrow hx + by + f = 0 \quad (III)$$

Solve, (II) and (III) to get centre of conic $C = (x, y)$

Example 4.36 Find the centre of conic $14x^2 + 11y^2 - 4yx - 44x - 58y + 71 = 0$

Solution Given equation of conic

$$14x^2 + 11y^2 - 4yx - 44x - 58y + 71 = 0$$

$$\text{Let } U = 14x^2 + 11y^2 - 4yx - 44x - 58y + 71$$

Differentiate both side (I) w.r.to x

$$\frac{dU}{dx} = 28x - 4y - 44$$

Again, differentiate both side (I) w.r.to y

$$\frac{dU}{dy} = -4x + 22y - 58$$

For centre, use

$$\frac{dU}{dx} = 28x - 4y - 44 = 0 \Rightarrow 7x - y - 11 = 0 \quad (II)$$

And $\frac{dU}{dy} = 4x + 22y - 58 = 0 \Rightarrow -2x + 11y = 29$ (III)

Solve, (II) and (III) to get centre of conic $C = (2, 3)$

TYPEWISE SOLVED EXAMPLES

TYPE I: PROBLEMS BASED ON FUNDAMENTAL RULES OF DIFFERENTIATION

Example 4.37 If $f(x) = \left(\frac{\sin^m x}{\sin^n x}\right)^{m+n} \cdot \left(\frac{\sin^n x}{\sin^p x}\right)^{n+p} \cdot \left(\frac{\sin^p x}{\sin^m x}\right)^{p+m}$ then $f'(x)$ is equal to

- (a) 0 (b) 1 (c) $\cos^{m+n+p}$ (d) none of these

Solution We have,

$$\begin{aligned} f(x) &= (\sin^{m-n} x)^{m+n} \cdot (\sin^{n-p} x)^{n+p} \cdot (\sin^{p-m} x)^{p+m} \\ &= \sin^{m^2-n^2} x \cdot \sin^{n^2-p^2} x \cdot \sin^{p^2-m^2} x \\ &= (\sin x)^{m^2-n^2+n^2-p^2+p^2-m^2} = (\sin x)^0 = 1 \end{aligned}$$

$\therefore f'(x) = 0.$

Example 4.38 If $y = \frac{1}{1+x^{\beta-\alpha}+1+x^{\gamma-\alpha}} + \frac{1}{1+x^{\alpha-\beta}+1+x^{\gamma-\beta}} + \frac{1}{1+x^{\alpha-\gamma}+1+x^{\beta-\gamma}}$ then $\frac{dy}{dx}$ is equal to

- (a) 0 (b) 1 (c) $(\alpha + \beta + \gamma)^{\alpha+\beta+\gamma-1}$ (d) none of these

Solution We have,

$$\begin{aligned} y &= \frac{1}{1+\frac{x^\beta}{x^\alpha}+\frac{x^\gamma}{x^\alpha}} + \frac{1}{1+\frac{x^\alpha}{x^\beta}+\frac{x^\gamma}{x^\beta}} + \frac{1}{1+\frac{x^\alpha}{x^\gamma}+\frac{x^\beta}{x^\gamma}} \\ &= \frac{x^\alpha}{x^\alpha+x^\beta+x^\gamma} + \frac{x^\beta}{x^\alpha+x^\beta+x^\gamma} + \frac{x^\gamma}{x^\alpha+x^\beta+x^\gamma} \\ &= \frac{x^\alpha+x^\beta+x^\gamma}{x^\alpha+x^\beta+x^\gamma} = 1 \end{aligned}$$

$\therefore \frac{dy}{dx} = 0.$

Example 4.39 If $f(x) = \frac{1-\cos x}{1-\sin x}$, then $f'\left(\frac{\pi}{2}\right)$ is equal to

- (a) 1 (b) 0 (c) ∞ (d) does not exist

Solution $f'(x) = \frac{(1 - \sin x)(\sin x) - (1 - \cos x)(-\cos x)}{(1 - \sin x)^2}$

$$= \frac{\sin x - \sin^2 x + \cos x - \cos^2 x}{(1 - \sin x)^2} = \frac{\sin x + \cos x - 1}{(1 - \sin x)^2}$$

$$\therefore f'\left(\frac{\pi}{2}\right) = \frac{1 + 0 - 1}{(1 - 1)^2} = \frac{0}{0}$$

Hence $f'\left(\frac{\pi}{2}\right)$ does not exist.

TYPE II: PROBLEMS BASED ON DIFFERENTIATION OF COMPOSITE FUNCTION

Example 4.40 If $y = f\left(\frac{2x-1}{x^2+1}\right)$ and $f'(x) = \sin x^2$, then $\frac{dy}{dx}$ is equal to

(a) $\sin\left(\frac{2x-1}{x^2+1}\right)^2 \cdot \left(\frac{2+2x+x^2}{(x^2+1)^2}\right)$

(b) $\sin\left(\frac{2x-1}{x^2+1}\right)^2 \cdot \left(\frac{2+2x-x^2}{(x^2+1)^2}\right)$

(c) $\sin\left(\frac{2x-1}{x^2+1}\right)^2 \cdot \left(\frac{2+2x-x^2}{(x^2+1)^2}\right)$

(d) None of the above

Solution We have, $y = f\left(\frac{2x-1}{x^2+1}\right)$

$$\Rightarrow \frac{dy}{dx} = f'\left(\frac{2x-1}{x^2+1}\right) \cdot \left[\frac{(x^2+1)2 - (2x-1) \cdot 2x}{(x^2+1)^2}\right]$$

$$= \sin\left(\frac{2x-1}{x^2+1}\right)^2 \cdot \left[\frac{2+2x-2x^2}{(x^2+1)^2}\right]$$

$$\left[\because f'(x) = \sin x^2, \therefore f'\left(\frac{2x-1}{x^2+1}\right) = \sin\left(\frac{2x-1}{x^2+1}\right)^2 \right]$$

Example 4.41 If $f(x) = \frac{x^2-x}{x^2+2x}$, then $\frac{df^{-1}}{dx}$ equal to

(a) $-\frac{3}{(1-x)^2}$

(b) $\frac{3}{(1-x)^2}$

(c) $\frac{1}{(1-x)^2}$

(d) none of these

Solution We have, $f(x) = \frac{x^2 - x}{x^2 + 2x}$

Here, $f(0)$ and $f(-2)$ are not defined.

So domain of $f = \mathbb{R} - \{0, -2\}$.

Then in this domain, we have

$$y = f(x) = \frac{x-1}{x+2} \Rightarrow yx + 2y = x - 1$$

$$\Rightarrow x = \frac{2y+1}{1-y} \Rightarrow f^{-1}(x) = \frac{2x+1}{1-x}$$

$$\therefore \frac{df^{-1}(x)}{dx} = \frac{2(1-x) + 2x+1}{(1-x)^2} = \frac{3}{(1-x)^2}$$

TYPE III: PROBLEMS BASED ON DIFFERENTIATION OF IMPLICIT FUNCTION

Example 4.42 If $2x^2 - 3x^2y^2 + 4x - y + 7 = 0$ and $y(1) = 1$, then the value of $y''(1)$ is equal to

(a) $-\frac{343}{474}$

(b) $-\frac{474}{343}$

(c) $\frac{474}{343}$

(d) $\frac{343}{474}$

Solution Differentiation the given equation w.r.t. x , we get

$$6x^2 - 6xy^2 - 6xy^2y' + 4 - y' = 0$$

On putting $x = 1$ and $y = 1$, we get

$$6 - 6 - 6y'(1) + 4 - y'(1) = 0$$

$$\Rightarrow y'(1) = \frac{4}{7}$$

Differentiation equation (1) w.r.t x , we have

$$12x - 6y^2 - 24xyy' - 6x^2y'^2 - 6x^2yy'' - y'' = 0$$

$$\Rightarrow y''(1) = \frac{474}{343}$$

Example 4.43 If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3(x^3 - y^3)$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{x^2}{y^2} \sqrt{\frac{1-y^6}{1-x^6}}$

(b) $\frac{y^2}{x^2} \sqrt{\frac{1-y^6}{1-x^6}}$

(c) $\frac{x^2}{y^2} \sqrt{\frac{1-x^6}{1-y^6}}$

(d) none of these

Solution On putting $x^3 = \sin \alpha$ and $y^3 = \sin \beta$, we have

$$\sqrt{1-\sin^2 \alpha} + \sqrt{1-\sin^2 \beta} = a^3(\sin \alpha - \sin \beta)$$

$$\Rightarrow (\cos \alpha + \cos \beta) = a^3(\sin \alpha - \sin \beta)$$



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TYPE VI: PROBLEMS BASED ON MISCELLANEOUS

Example 4.48 If $f(x) = |\cos x|$, then $f'\left(\frac{3\pi}{4}\right)$ is equal to

- (a) $-\frac{1}{\sqrt{2}}$ (b) $\frac{1}{\sqrt{2}}$ (c) 1 (d) none of these

Solution When $\frac{\pi}{2} < x < \pi$, $\cos x < 0$

$$\begin{aligned} \therefore |\cos x| &= -\cos x \\ \therefore f(x) &= -\cos x \Rightarrow f'(x) = \sin x \\ \therefore f'\left(\frac{3\pi}{4}\right) &= \sin \frac{3\pi}{4} = \frac{1}{\sqrt{2}} \end{aligned}$$

Example 4.49 If $f(x) = \log |2x|$, $x \neq 0$, then $f'(x)$ is equal to

- (a) $\frac{1}{x}$ (b) $-\frac{1}{x}$ (c) $\frac{1}{|x|}$ (d) none of these

$$\begin{aligned} \text{Solution } f(x) &= \log |2x| \\ &= \begin{cases} \log 2x, & 2x > 0 \text{ i.e., } x > 0 \\ \log (-2x), & 2x < 0 \text{ i.e., } x < 0 \end{cases} \\ \Rightarrow f'(x) &= \begin{cases} \frac{1}{2x} \times 2, & x > 0 \\ \frac{1}{-2x} \times -2, & x < 0 \end{cases} = \begin{cases} \frac{1}{x}, & x > 0 \\ \frac{1}{x}, & x < 0 \end{cases} \\ \therefore f'(x) &= \frac{1}{x}, x \neq 0. \end{aligned}$$

Example 4.50 If $8f(x) + 6f\left(\frac{1}{x}\right) = x + 5$ and $y = x^2 f(x)$, then $\frac{dy}{dx}$ at $x = -1$ is equal to

- (a) 0 (b) $\frac{1}{14}$ (c) $-\frac{1}{14}$ (d) none of these

Solution We have,

$$8f(x) + 6f\left(\frac{1}{x}\right) = x + 5 \text{ for all } x \quad (1)$$

Therefore,

$$8f\left(\frac{1}{x}\right) + 6f(x) = \frac{1}{x} + 5 \quad (2)$$

$$\left[\text{putting } x = \frac{1}{x} \right]$$

From equation (1) and (2), we have

$$f(x) = \frac{1}{28} \left(8x - \frac{6}{x} + 10 \right)$$



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- (a) $1/2$ (b) $1 \forall x$
 (c) -1 for $x \in (-\infty, 0) - \{-1\}$
 (d) none of these
7. If $y = \log_u v$, where u and v are functions of x , then dy/dx equal to:
 (a) $-\frac{\log v}{u(\log u)^2} \frac{du}{dx} + \frac{1}{v(\log u)} \cdot \frac{dv}{dx}$
 (b) $\frac{1}{v} \frac{dv}{dx} + \log v \frac{du}{dx}$
 (c) $\frac{1}{v} \frac{dv}{dx}$ (d) $\frac{1}{u} \frac{du}{dx} + \frac{1}{v} \frac{dv}{dx}$
8. The derivative of the function $f(x) = \log_x 3$ w.r.t x is:
 (a) $\frac{\ln 3}{2x(\ln x)^2}$ (b) $\frac{2(\ln 3)}{x(\ln x)^2}$
 (c) $-\frac{\ln 3}{2x(\ln x)^2}$ (d) $\frac{-2\ln 3}{x(\ln x)^2}$
9. If $x = 3\cos\theta - 2\cos^3\theta$ and $y = 3\sin\theta - 2\sin^3\theta$ then $\frac{dy}{dx} =$
 (a) $\sin\theta$ (b) $\cos\theta$
 (c) $\tan\theta$ (d) $\cot\theta$
10. If $x = \frac{t - e^{-t^2}}{2t^2}$, then $t \frac{dx}{dt} - e^{-t^2} =$
 (a) $\frac{1}{2t} - 2x$ (b) $\frac{1}{2t} - x$
 (c) $\frac{1}{t} - 2x$ (d) None
- (c) $x \log(\log x)$
 (d) $1/x (\log x) \cdot x^{\log x - 1}$
2. The value of $\frac{d}{dx}(x^x)$ is
 (a) xx^{x-1} (b) $x^x \log_e x$
 (c) $x^x \log$ (d) None
3. If $y = (1 + 1/x)^x$ then $\frac{dy}{dx} =$
 (a) $\left(1 + \frac{1}{x}\right)^x \left[\log\left(1 + \frac{1}{x}\right) - \frac{1}{1+x}\right]$
 (b) $\left(1 + \frac{1}{x}\right)^x \left[\log\left(1 + \frac{1}{x}\right)\right]$
 (c) $\left(1 + \frac{1}{x}\right)^x \left[\log(x+1) - \frac{x}{x+1}\right]$
 (d) $\left(1 + \frac{1}{x}\right)^x \left[\log\left(1 + \frac{1}{x}\right) + \frac{x}{x+1}\right]$
4. If $x^m y^n = (x+y)^{m+n}$ then $\frac{dy}{dx}$ is:
 (a) x/y (b) $1/y$
 (c) $1/x$ (d) y/x
5. If $y = (\sin x)^{\tan x}$ then $\frac{dy}{dx} =$
 (a) $(\sin x)^{\tan x} [1 + \sec^2 x \cdot \log(\sin x)]$
 (b) $\tan x (\sin x)^{\tan x - 1} \cos x$
 (c) $(\sin x)^{\tan x} \cdot \sec^2 x \log \sin x$
 (d) $\tan x \cdot (\sin x)^{\tan x - 1}$
6. If $y = \log_y x$, then $\frac{dy}{dx} =$
 (a) $\frac{1}{x + \log y}$ (b) $\frac{1}{\log x (1 + y)}$
 (c) $\frac{1}{x(1 + \log y)}$ (d) $\frac{1}{y + \log x}$

Type V: Problems Based on Logarithmic and Exponential Limit

1. If $y = x^{\log x}$ then $\frac{dy}{dx}$ equal to:
 (a) $\log x \cdot x^{\log x - 1}$
 (b) $2\log x \cdot x \log^{x-1}$



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- (III) $x = y \log(xy)$
 (IV) $x\sqrt{y} + y\sqrt{x} = 1$
 (V) $x^2y^2 = \sin(xy)$
 (VI) $x \cos y + y \sin x = \tan(x + y)$
 (VII) $y = x + y^2 \sin^3\left(\frac{x}{2}\right)$

2. Differentiate the following function w.r.t.x.

- (I) $x \cos y + y \cos x = \tan^{-1}(x^2)$
 (II) $x + y = \tan^{-1}(xy)$
 (III) $e^{xy} = \cos(x^2 + y^2)$
 (IV) $e^{xy} + \log(xy) + xy = 0$

- (V) $3y - \sqrt{9y^2 + 1} = 5x^2$
 (VI) $x^3y^2 = \log_e(x + y) + \sin(e^x)$

3. (a) If $\sin y = x \sin(y + a)$ then prove that

$$\frac{dy}{dx} = \frac{\sin^2(y + a)}{\sin a}.$$

(b) If $\cos y = x \cos(y + b)$ then prove that

$$\frac{dy}{dx} = \frac{\cos^2(y + b)}{\sin b}.$$

4. (a) If $y =$

$$\sqrt{\cos x + \sqrt{\cos x + \sqrt{\cos x + \dots \text{to } \infty}}}$$

$$\text{then prove that } (2y - 1) \frac{dy}{dx} + \sin x =$$

$$0.$$

(b) If $y =$

$$\sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \text{to } \infty}}}$$

$$\text{then prove that } \frac{dy}{dx} = \frac{\cos x}{2y - 1}.$$

(c) If $y = \sqrt{x + \sqrt{x + \sqrt{x + \dots \text{to } \infty}}}$

$$\text{then prove that } \frac{dy}{dx} = \frac{1}{2y - 1}$$

5. (a) If $y = x + \frac{1}{x}$ then prove that $x \frac{dy}{dx} + y = 2x$

(b) If $y = \frac{1}{x}$ then prove that

$$\frac{dy}{\sqrt{1 + y^4}} + \frac{dx}{\sqrt{1 + x^4}} = 0.$$

Type IV: Problems Based on Differentiation of Parametric Function

1. Differentiate the following function w.r.t.x.

(I) $x = a(\cos \theta + \theta \sin \theta), y = a(\sin \theta - \theta \cos \theta)$

(II) $x = a(\theta + \sin \theta), y = a(1 - \cos \theta)$

(III) $x = 3 \cos \theta - \cos^3 \theta, y = 3 \sin \theta - \sin^3 \theta$

(IV) $y = a(\theta + \sin \theta), x = a \sin \theta$

2. Differentiate the following function w.r.t.x.

(I) $x = \log t + \sin t, y = e^t + \cos t$

(II) $x = t + \frac{1}{t}, y = t - \frac{1}{t}$

(III) $x = a \left(\cos t + \log \tan \left(\frac{t}{2} \right) \right), y = a \sin t$

(IV) $x = \frac{2t}{1 + t^2}, y = \frac{1 - t^2}{1 + t^2}$

(V) $x = \frac{3at}{1 + t^3}, y = \frac{3at^2}{1 + t^3}$ at $t = \frac{1}{2}$

(VI) $\sin x = \frac{2t}{1 + t^2}, \tan y = \frac{2t}{1 - t^2}$

3. If $x = a(\cos \theta + \theta \sin \theta), y = a(\sin \theta - \theta \cos \theta)$ then find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{4}$.



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(d) $xy = (x + y)^p$ then

$$\frac{dy}{dx} = \frac{y}{x} \text{ then } p \quad (\text{iv}) \quad \frac{1}{2}$$

(e) If $y = \sin^{-1}\left(x - \frac{4}{27}x^3\right)$

$$\text{then } \frac{dy}{dx} = \quad (\text{v}) \quad 2$$

Type III: Assertion- Reasoning

Directions: Each of the following questions contains two statements – **Assertion (A)** and **Reason (R)**. Each question has four options—(a), (b), (c), and (d). Select the correct one.

- (a) **A** is true, **R** is true and **R** is the correct explanation for **A**
 (b) **A** is true, **R** is true and **R** is not the correct explanation for **A**
 (c) **A** is true, **R** is false
 (d) **A** is false and **R** is true

1. **A:** If $u = e^x$ and $v = \log x$ then $\frac{dy}{dx} = xe^x$

R: If $u = f(x)$, $v = g(x)$ then the derivative of $f(x)$ w.r.t. $g(x)$ is $\frac{du}{dv} = \frac{du/dx}{dv/dx}$.

2. **A:** If $x < 0$ then $\frac{d\{\log|x|\}}{dx} = -\frac{1}{x}$

R: If $x < 0$ then $|x| = -x$.

3. **A:** Differentiation of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r.t.

$\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is 1 when $0 < x < 1$.

R: $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ when $-1 \leq x \leq 1$

4. **A:** Let $f: [0, \infty) \rightarrow [0, \infty)$ be a function de-

fined by $y = x^2$ then $\left(\frac{d^2y}{dx^2}\right)\left(\frac{d^2x}{dy^2}\right) = 1$

R: $\left(\frac{dy}{dx}\right)\left(\frac{dx}{dy}\right) = 1$

5. **A:** If $f(x) = \sin^2x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x$

$\cos\left(x + \frac{\pi}{3}\right)$ then $f'(x) = 0$

R: Differentiation of constant function is zero.

6. Let $f(x) = 2 + \cos x$ for all real x .

A: For each value of t , there exist a point c in $[t, t + \pi]$ such that $f'(c) = 0$ because

R: $f(t) = f(t + 2\pi)$ for each value of t .

ANSWER KEY

LEVEL I: TOPIC-WISE OBJECTIVE QUESTIONS

Type I: Problems Based on Fundamental Rules of Differentiation

1. (b) 2. (d) 3. (b) 4. (d)

Type II: Problem Based on Differentiation of Composite Function

1. (d) 2. (b) 3. (a) 4. (d) 5. (d) 6. (a)
 7. (b) 8. (d) 9. (b) 10. (b) 11. (b)



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$$\Rightarrow \text{Slope of normal} = -\frac{1}{\text{slope of tangent}} = -\frac{1}{\frac{dy}{dx}(a, b)} = -\frac{1}{f'(a)} = m$$

Here $\frac{Dy}{Dx} = 3x^2$

$$\begin{aligned} \text{Slope of Normal} &= \frac{-1}{\text{Slope of tangent}} \\ &= \frac{-1}{3a^2} \end{aligned}$$

Example 5.4 If the normal to the curve $y = f(x)$ at the point $(3, 4)$ makes an angle $\frac{3\pi}{4}$ with positive direction of x-axis, then $f'(3)$ is equal to (IIT (JEE)-2000)

(a) -1 (b) -3/4 (c) 4/3 (d) 1

Solution Given function $y = f(x)$

Here, Slope of normal to the curve of given function at $(3, 4)$ is $-\frac{1}{f'(3)}$

$$\Rightarrow -\frac{1}{f'(3)} = \tan\left(\frac{3\pi}{2}\right) = \tan\left(\frac{\pi}{2} + \frac{\pi}{4}\right) = -\cot\left(\frac{\pi}{4}\right) = -1$$

Hence, $f'(3) = 1$.

5.2.2 Nature of Slope of Normal

(a) If normal to the curve of function $y = f(x)$ is parallel to x-axis or horizontal then slope of

tangent to the curve $m = \frac{dy}{dx} = \frac{1}{\infty}$

(b) If normal to the curve of function $y = f(x)$ is parallel to y-axis or vertical then slope of tangent

to the curve $m = \frac{dy}{dx} = 0$.

(c) If normal to the curve of function $y = f(x)$ is parallel to the line $ax + by + c = 0$ then $\frac{dy}{dx} = \frac{b}{a}$

(d) If normal to the curve of function $y = f(x)$ is perpendicular to the line $ax + by + c = 0$ then

$$\frac{dy}{dx} = -\frac{a}{b}$$

(e) If normal to the curve of function $y = f(x)$ is equally inclined with co-ordinate axes then $\frac{dy}{dx} = 1$ or -1

Example 5.5 The normal to the curve $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$ at the point $\theta = (2n + 1)\pi$, $n \in I$ are all

(a) parallel to x-axis

(b) parallel to y-axis

(c) parallel to the line $y = x$

(d) none of these



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Hence, angle between the two curves is

$$\theta = \pm \tan^{-1} \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right) = \pm \tan^{-1} \sqrt{3} = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}.$$

Hence (b) is the correct answer.

5.3 TANGENT

If two points A and B lie on the curve of function $y = f(x)$ then line joining the points represent a chord AB but if point B tends to A then chord AB becomes tangent to the curve of function $y = f(x)$ at point A .

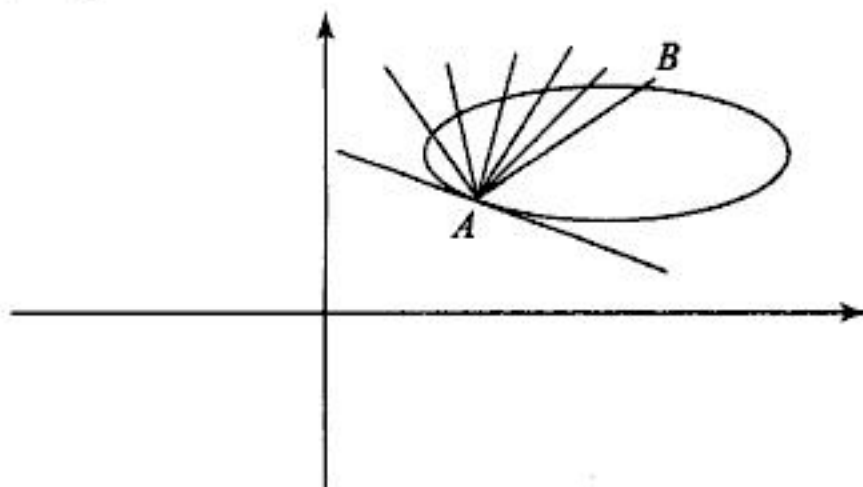


Fig. 5.7

5.3.1 Equation of Tangent

Equation of tangent to the curve of function $y = f(x)$ at the point (α, β) is $y - \beta = \frac{dy}{dx}(\alpha, \beta)(x - \alpha)$

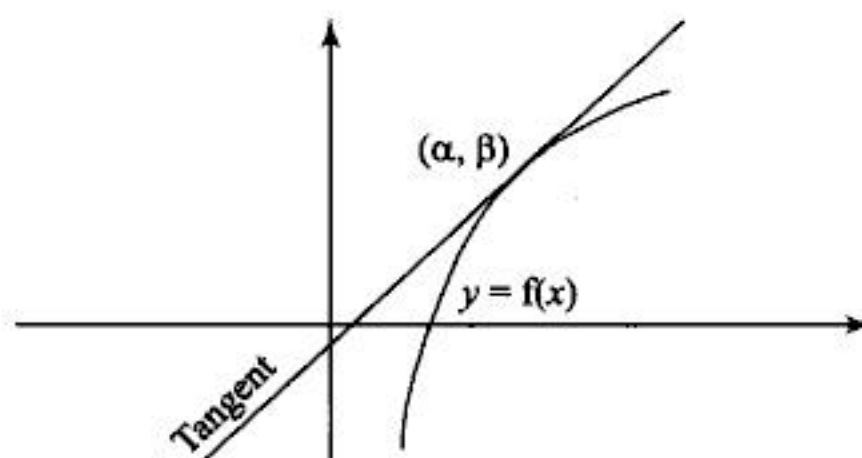


Fig. 5.8

Example 5.10 Tangent is drawn to the ellipse $\frac{x^2}{3^2} + y^2 = 1$ at $(3\sqrt{3} \cos \theta, \sin \theta)$ where $\theta \in (0, \pi/2)$ then the value of θ such that sum of intercepts on the axes made by this tangent is minimum, is (IIT-JEE (2003))

(a) $\pi/3$

(b) $\pi/6$

(c) $\pi/8$

(d) $\pi/4$

Solution Equation of tangent to the ellipse $\frac{x^2}{3^2} + y^2 = 1$ at $(3\sqrt{3} \cos \theta, \sin \theta)$ is

$$\frac{3\sqrt{3} \cos \theta x}{3^2} + \sin \theta y = 1 \Rightarrow \frac{\sqrt{3} \cos \theta x}{3} + \sin \theta y = 1$$

Here, length of intercept on the x-axis = $\frac{9}{\sqrt{3} \cos \theta}$ and intercept on the y-axis = $\frac{1}{\sin \theta}$



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Example 5.15 The equation of the normal to the curve $y = e^{-2|x|}$ at the point where the curve cuts the line $x = \frac{1}{2}$ is

- (a) $2e(ex + 2y) = e^2 - 4$ (b) $2e(ex - 2y) = e^2 - 4$ (c) $2e(ey - 2x) = e^2 - 4$ (d) none of these

Solution At the point $x = \frac{1}{2}$, $y = e^{-1}$

Since, $x = \frac{1}{2} > 0$, $\therefore y = e^{-2x}$

On differentiation w.r.t x , we get

$$\frac{dy}{dx} = -2e^{-2x} \quad \Rightarrow \quad \left[\frac{dy}{dx} \right]_{\left(\frac{1}{2}, \frac{1}{e}\right)} = -2e^{-1} = -\frac{2}{e}$$

Thus, the equation of normal is

$$\left(y - \frac{1}{e}\right) = \frac{e}{2} \left(x - \frac{1}{2}\right) \quad \Rightarrow \quad 2e(ex - 2y) = e^2 - 4.$$

Hence (b) is the correct answer.

5.5 LENGTH OF TANGENT, SUB-TANGENT, NORMAL AND SUB-NORMAL

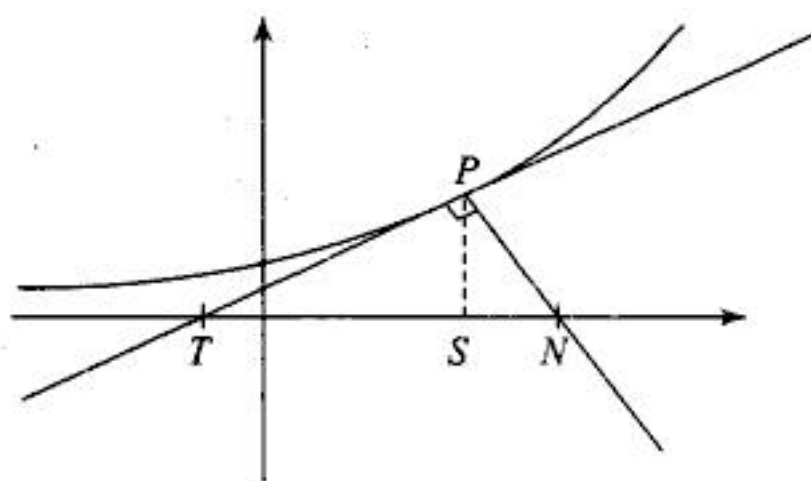


Fig. 5.14

From Figure 5.14, we observe:

- (a) **Length of Tangent:** Length of segment PT of the tangent between the point of tangent and x -axis is called *length of tangent*.

$$\text{Length of tangent } PT = \left| b \sqrt{1 + \left(\frac{dx}{dy} \right)^2_{(a,b)}} \right|$$

- (b) **Length of Normal:** The length of segment PN i.e. the portion of the normal intercepted between the point on the curve and x -axis is called *length of normal*.

$$\text{Length of normal } PN = \left| b \sqrt{1 + \left(\frac{dy}{dx} \right)^2_{(a,b)}} \right|$$



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(b) Given $y = f(x) = \frac{2-x^2}{(x-2)(x-5)}$

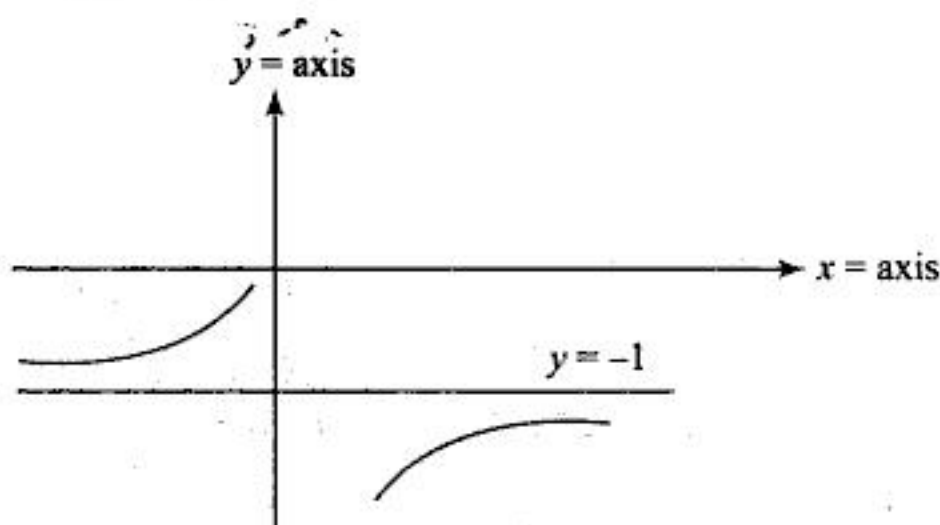


Fig 5.21

Here, If $x \rightarrow \infty$ then $y \rightarrow 1$

$$\therefore \frac{2-x^2}{(x-2)(x-5)} - (-1) = \frac{-7x+12}{(x-2)(x-5)} = \begin{cases} < 0, & \text{when } x \rightarrow +\infty \\ > 0, & \text{when } x \rightarrow -\infty \end{cases}$$

Hence, the curve lie below its asymptote $y = -1$

As, $x \rightarrow \infty$ and above its asymptote $y = -1$ as $x \rightarrow -\infty$

5.6.3 Oblique Asymptotes

Let $y = mx + c$ be an asymptote to the curve $y = f(x)$ then find

(a) $\lim_{x \rightarrow \infty} \frac{f(x)}{x} = m$

(b) $\lim_{x \rightarrow \infty} \{f(x) - mx\} = c$

Hence, $y = mx + c$ is the required oblique asymptote.

Example 5.20 Find the oblique asymptote of curve of function $f(x) = 2x + \frac{3}{x} - 1$

Solution Given function $y = f(x) = 2x + \frac{3}{x} - 1$

Let oblique asymptote of given function $y = mx + c$

Here, $\lim_{x \rightarrow \infty} \frac{f(x)}{x} = m \Rightarrow m = \lim_{x \rightarrow \infty} \frac{2x + \frac{3}{x} - 1}{x} = \lim_{x \rightarrow \infty} \frac{2x^2 - x + 3}{x^2} = 2$



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TYPE II: PROBLEMS BASED ON EQUATIONS OF TANGENT AND NORMAL

Example 5.25 The equation of the tangent to the curve $y = \sqrt{9 - 2x^2}$ at the point where the ordinate and the abscissa are equal, is

- (a) $2x + y - 3\sqrt{3} = 0$ (b) $2x + y + 3\sqrt{3} = 0$ (c) $2x - y - 3\sqrt{3} = 0$ (d) none of these

Solution On putting $y = x$ in $y = \sqrt{9 - 2x^2}$, we get

$$\begin{aligned} x = \sqrt{9 - 2x^2} &\Rightarrow x^2 = 9 - 2x^2 \\ &\Rightarrow x = \sqrt{3}, -\sqrt{3} \end{aligned}$$

Since $y > 0$, therefore, the point is the $(\sqrt{3}, \sqrt{3})$

Now, we have, $y^2 = 9 - 2x^2$

On differentiating w.r.t. x , we get

$$2y \frac{dy}{dx} - 4x \Rightarrow \frac{dy}{dx} = -\frac{2x}{y}$$

$$\therefore \left[\frac{dy}{dx} \right]_{(\sqrt{3}, \sqrt{3})} = -\frac{2\sqrt{3}}{\sqrt{3}} = -2$$

So, the equation of the tangent at $(\sqrt{3}, \sqrt{3})$ is

$$(y - \sqrt{3}) = -2(x - \sqrt{3}) \Rightarrow 2x + y - 3\sqrt{3} = 0.$$

Hence (a) is the correct answer.

Example 5.26 The tangent to the curve $5x^2 + y^2 = 1$ at $\left(\frac{1}{3}, -\frac{2}{3}\right)$ passes through the point

- (a) $(0, 0)$ (b) $(1, -1)$ (c) $(-1, 1)$ (d) none of these

Solution We have, $5x^2 + y^2 = 1$

$$\Rightarrow 10x + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{5x}{y}$$

$$\therefore \left[\frac{dy}{dx} \right]_{\left(\frac{1}{3}, -\frac{2}{3}\right)} = \frac{-5\left(\frac{1}{3}\right)}{\left(-\frac{2}{3}\right)} = \frac{5}{2}.$$

Hence, the equation of the tangent at $\left(\frac{1}{3}, -\frac{2}{3}\right)$ is



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$$\Rightarrow \frac{dy}{dx} = m_1 = \frac{x^2 - y^2}{2xy}$$

On differentiating equation (ii) w.r.t x , we get

$$6xy + 3x^2 \frac{dy}{dx} - 3y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = m_2 = -\frac{2xy}{x^2 - y^2}$$

$$\text{The product of slopes for the two curves} = \left(\frac{x^2 - y^2}{2xy} \right) \times \left(\frac{2xy}{x^2 - y^2} \right) = -1.$$

$\therefore$ The curves cut each other orthogonally.
Hence (c) is the correct answer.

Example 5.33 The angle at which the curves $y = \sin x$ and $y = \cos x$ intersecting in $[0, \pi]$ is

- (a) $\pm \tan^{-1} 2\sqrt{2}$ (b) $\pm \tan^{-1} 2\sqrt{2}$ (c) $\pm \tan^{-1} \left(\frac{1}{\sqrt{2}} \right)$ (d) none of these

Solution We have,

$$y = \sin x \quad (i)$$

and

$$y = \cos x \quad (ii)$$

The two curves intersect at $x = \frac{\pi}{4}$ in $[0, \pi]$.

On differentiating equation (i) w.r.t x , we get

$$\frac{dy}{dx} = \cos x$$

$$\therefore \left[\frac{dy}{dx} \right]_{x=\frac{\pi}{4}} = \frac{1}{\sqrt{2}} = m_1 \text{ (say).}$$

On differentiating (ii) w.r.t x , we get

$$\frac{dy}{dx} = -\sin x.$$

$$\left[\frac{dy}{dx} \right]_{x=\frac{\pi}{4}} = -\frac{1}{\sqrt{2}} = m_2 \text{ (say).}$$

Hence, the angle between the two curves is

$$\theta = \pm \tan^{-1} \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right) = \pm \tan^{-1} 2\sqrt{2}.$$

Hence (a) is the correct answer.



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16. The equation of the tangent to the curve $y = be^{-x/a}$ where it meets the y-axis is :
 (a) $x/a - y/b = -1$ (b) $x/a - y/b = 1$
 (c) $x/a + y/b = 1$ (d) $x/a + y/b = -1$
17. If the tangent at $(1, 1)$ on $y^2 = x(2 - x)^2$ meets the curve at P , then P is :
 (a) $(4, 4)$ (b) $(-1, 2)$
 (c) $(9/4, 3/8)$ (d) none
18. The equation of the tangent to the curve $y = 4 + \sin^2 x$ is $y =$
 (a) 2 (b) 3
 (c) 4 (d) 6
19. Equation of tangent to the curve $x = a \cos^3 t, y = a \sin^3 t$ at t is
 (a) $x \sec t - y \operatorname{cosec} t = a$
 (b) $x \sec t + y \operatorname{cosec} t = a$
 (c) $x \operatorname{cosec} t + y \operatorname{cosec} t = a$
 (d) none of these
20. The tangents to the function $x = a(\theta - \sin \theta) : y = a(1 + \cos \theta)$ at $\theta = (2n + 1)\pi, n \in Z$ are parallel to:
 (a) $y = x$ (b) $y = -x$
 (c) x-axis (d) y-axis
21. Equation of normal to the curve $x + y = x^y$ where it cuts x-axis, is:
 (a) $x + y - 1 = 0$ (b) $x - y - 1 = 0$
 (c) $x + y = 0$ (d) none of these
22. The chord joining the points where $x = p$ and $x = q$ on the curve $y = ax^2 + bx + c$ is parallel to the tangent at the point on the curve whose abscissa is:
 (a) $\frac{1}{2}(p + q)$ (b) $\frac{1}{2}(p - q)$
 (c) $pq/2$ (d) none
23. If the tangent to the curve $x = at^2, y = 2at$ is perpendicular to x-axis at:
 (a) (a, a) (b) $(0, a)$
 (c) $(a, 0)$ (d) $(0, 0)$
24. The tangent to the curve $y = e^{2x}$ at the point $(0, 1)$ meets the x-axis at:
 (a) $(1, 0)$ (b) $(2, 0)$
 (c) $(-1/2, 0)$ (d) none
25. If the tangent at $P(4m^2, 8m^3)$ to the curve $x^3 = y^2$ is also a normal, then $m^2 =$
 (a) 2 (b) $1/9$
 (c) $2/9$ (d) none
26. The tangent at any point on the curve $x^3 + y^3 = 2$, and cuts off length's p and q on co-ordinate axes then $p^{-3/2} + q^{-3/2} =$
 (a) $2^{3/2}$ (b) $2^{-3/2}$
 (c) $1/\sqrt{2}$ (d) none
27. If the tangent to the curve $2y^3 = ax^2 + x^3$ at (a, a) cuts off intercepts p and q on co-ordinates axes. If $p^2 + q^2 = 61$, then $a =$
 (a) 30 (b) 10
 (c) 20 (d) none
28. The distance between the origin and the normal to the curve $y = e^{2x} + x^2$ at $x = 0$ is:
 (a) $2/\sqrt{3}$ (b) $2/\sqrt{5}$
 (c) $1/\sqrt{3}$ (d) $1/\sqrt{5}$
29. The equation of one of the tangents to the curve $y = \cos(x + y), 0 \leq x \leq 2\pi$, that is parallel to the line $x + 2y = 0$ is :
 (a) $x + 2y = 3\pi/2$ (b) $x + 2y = \pi/2$
 (c) $x + 2y = \pi/4$ (d) none of these
30. The portion of the tangent to the curve $x^{2/3} + y^{2/3} = a^{2/3}$ which is intercepted between the axes is:
 (a) $2a$ (b) a
 (c) $a/2$ (d) $a\sqrt{2}$

Type IV: Problems Based on Length of Tangent, Sub-tangent, Normal and Sub-normal

- The length of the sub-normal to the curve $y^2 = x^3$ at the point $(4, 8)$ is
 (a) 24 (b) $8/3$
 (c) $3/8$ (d) None
- If any point on a curve, the sub-tangent and sub-normal are equal, then the length of tangent is:



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7. $3x + y - 10 = 0$ 8. $8x - 2y + 1 = 0$ 9. $x + 3y - 8 = 0, x + 3y + 8 = 0$
 10. $6x - 7y - 19 = 0$

Type IV: Problems Based on Length of Tangent, Sub-Tangent, Normal and Sub-Normal

(i) $a \sin \theta, 2a \sin^2 \left[\frac{\theta}{2} \right] \tan \left[\frac{\theta}{2} \right],$

$2a \sin \left[\frac{\theta}{2} \right], 2a \sin \left[\frac{\theta}{2} \right] \tan \left[\frac{\theta}{2} \right]$

(ii) $y \cot t, y \tan t,$
 $y \operatorname{cosec} t, y \sec t$

LEVEL III: MISCELLANEOUS QUESTIONS

Type I: Passage

Passage I

- (i) (a) (ii) (c) (iii) (a)

Passage II

- (i) (b) (ii) (c) (iii) (b)

Type II: Column Matching

- | | | | |
|------------|-------------|--------------|------------------|
| 1. (i) (s) | (ii) (t) | (iii) (q) | (iv) (p) (v) (r) |
| 2. (i) (p) | (ii) (r, s) | (iii) (q, t) | |
| 3. (i) (s) | (ii) (r) | (iii) (p) | (iv) (q) |
| 4. (i) (r) | (ii) (s) | (iii) (q) | (iv) (q) |
| 5. (i) (q) | (ii) (r) | (iii) (p) | (iv) (s) |

Type III: Assertion-Reason

1. (a) 2. (a) 3. (a) 4. (d) 5. (b) 6. (b)



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8.1.2 Decreasing Function

If $y = f(x)$ is any differentiable function in $[a, b]$ and satisfies $f'(x) \leq 0$ then the function must be decreasing in $[a, b]$.

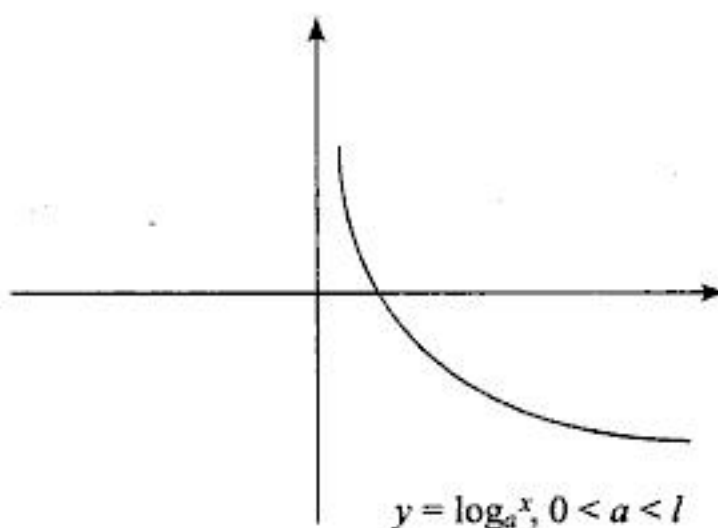


Fig. 8.3 Decreasing function

Strictly Decreasing Function

If $y = f(x)$ is any differentiable function in $[a, b]$ and satisfies $f'(x) < 0$ then the function must be strictly decreasing in $[a, b]$.

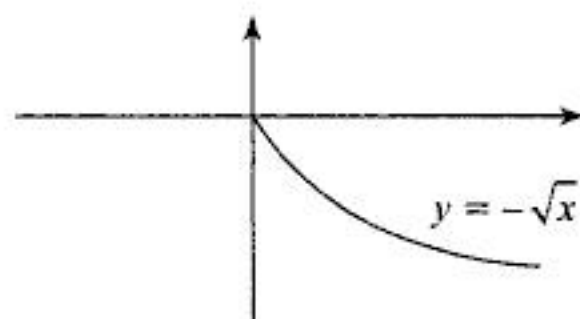


Fig. 8.4 Strictly Decreasing

8.2 SIGN OF SECOND ORDER DERIVATIVE

The sign of second derivative $f''(x)$ on an interval determines the concavity of the curve $y = f(x)$ in that interval.

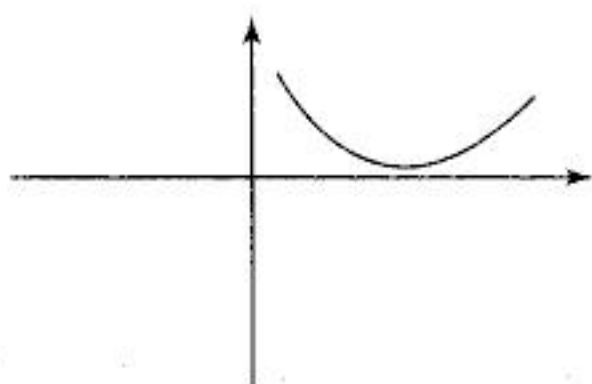


Fig. 8.5 Concave Curve

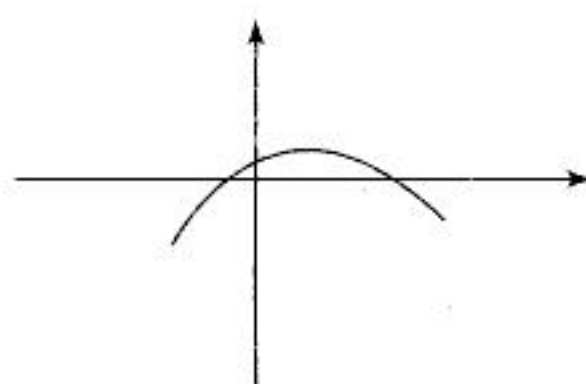


Fig. 8.6 Convex Curve

8.2.1 Concave Upward Curve

A curve is called *concave upward* in the interval $[a, b]$ if all points on the curve lie above the tangent to the curve at any point in the interval $[a, b]$.

If $f''(x) > 0$ for all x in $[a, b]$ then the curve of function $y = f(x)$ is concave upward in the interval $[a, b]$.

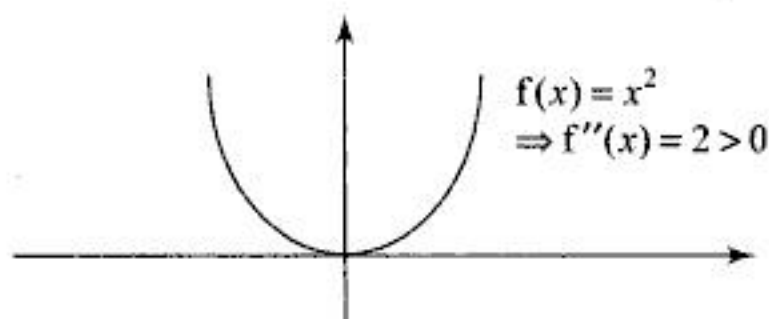


Fig. 8.7 Concave Upward



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4. If a function $f(x)$ is continuous in an interval then maximum and minimum values will occur alternatively.

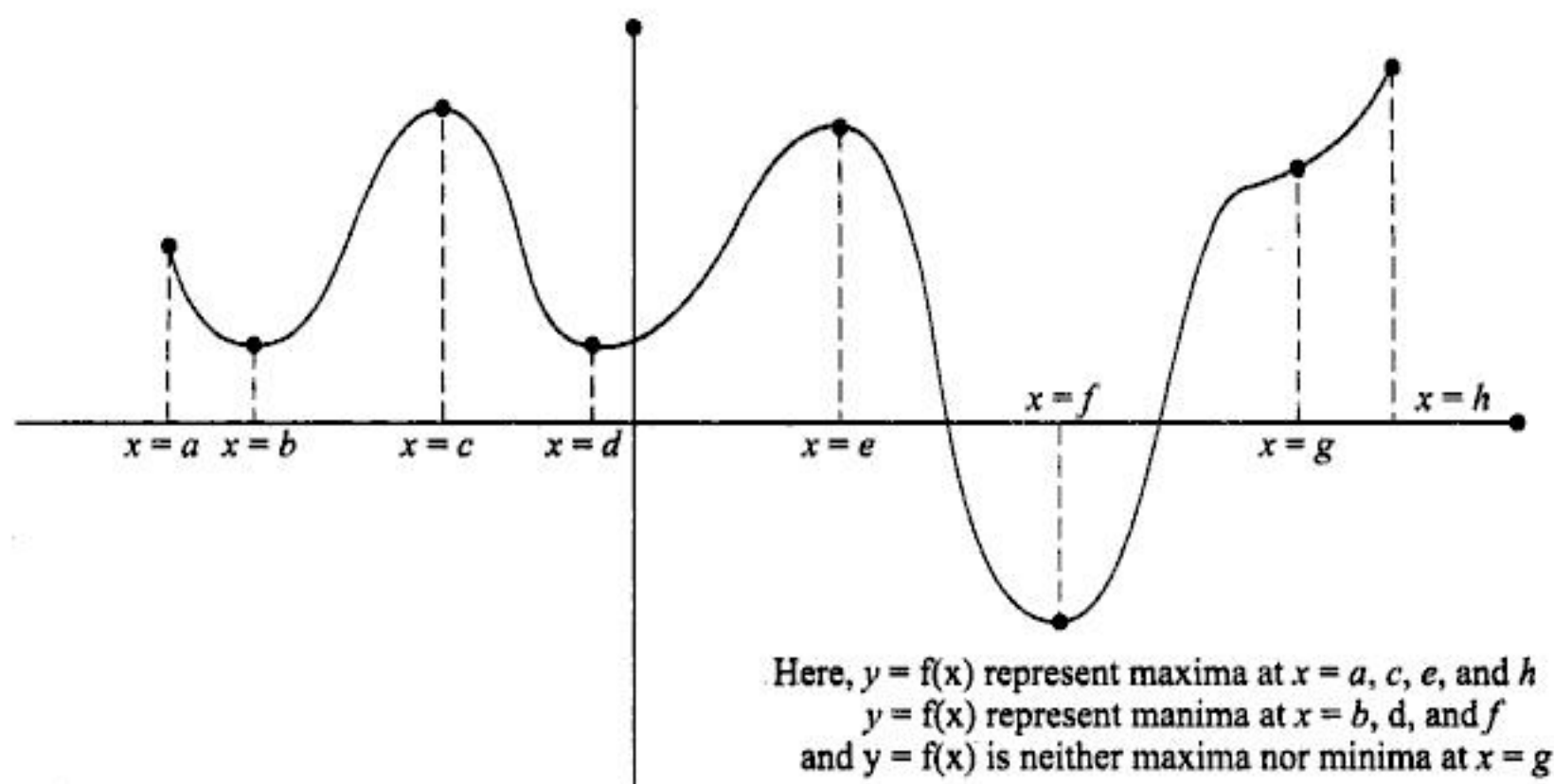


Fig. 8.19

5. Local minimum values of a function at some point may be greater than its local maximum values at some other point.

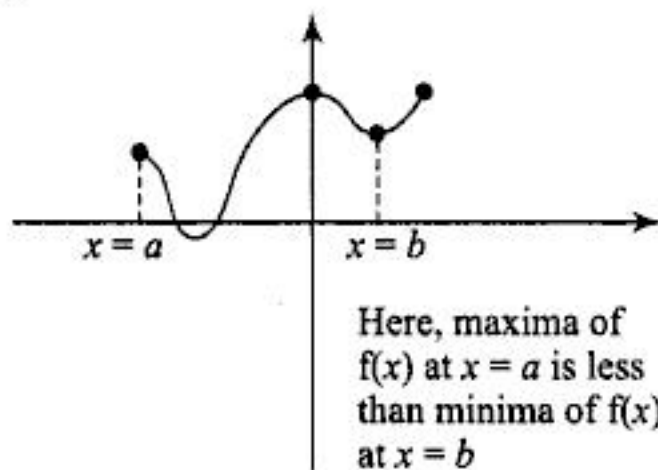


Fig. 8.20

6. **Absolute maxima** of function means that greatest value of maximum's of function and **absolute minima** of function means that least values of minimum's of function.

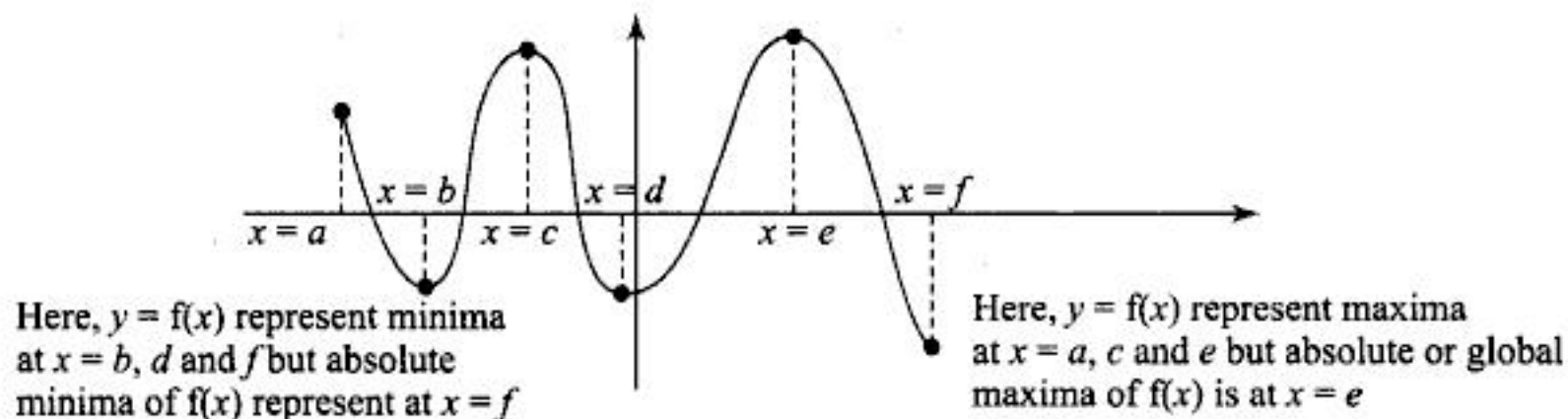


Fig. 8.21



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$\Rightarrow \frac{dy}{dx} > 0$ for $x < 0$ and $\frac{dy}{dx} < 0$ for $x > 0$. thus $x = 0$ is the point of local maxima for y . Now $y|_{x=0} = \frac{1}{2}$

(positive). thus $x = 0$ is also the point of local maximum for $y = \left| \frac{x^2 - 2}{x^2 - 4} \right|$.

Hence (d) is the correct answer.

Example 8.16 A function such that $f'(a) = f''(a) = \dots = f^{2n}(a) = 0$ and f has a local maximum value b at $x = a$. If $f(x)$ is

- (a) $(x - a)^{2n+2}$ (b) $b - 1 - (x + 1 - a)^{2n+1}$ (c) $b - (x - a)^{2n+2}$ (d) $(x - a)^{2n+2} - b$.

Solution For local maximum, or local minimum odd derivative must be equal to zero.

For local maxima, even derivative must be negative.

Since maximum value at $x = a$ is b .

$$\therefore f(x) = b - (x - a)^{2n+2} \quad (\because f^{2n+2}(a) = -ve).$$

Hence (c) is the correct answer.

8.6 FERMAT'S THEOREM

Let a function $y = f(x)$ be defined on a certain interval and have maximum or minimum values at some interior point $x = c$ of the interval. If there exists a derivative $f'(x)$ at the point $x = c$ then $f'(c) = 0$

8.6.1 Geometrical Interpretation

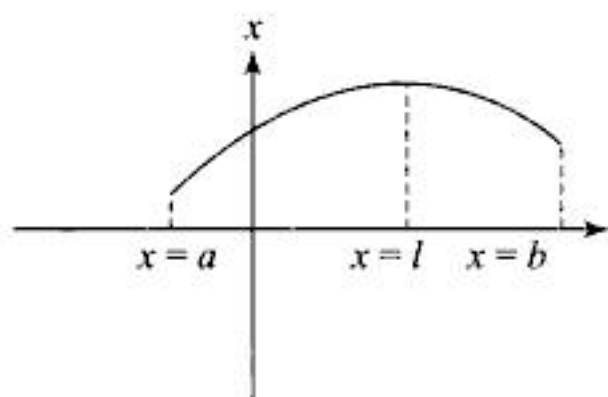


Fig. 8.27

OR

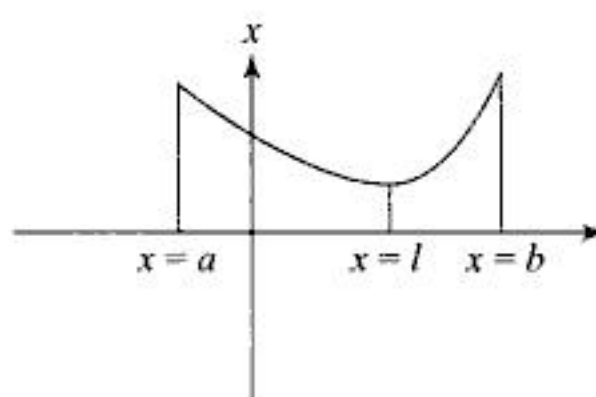


Fig. 8.28

Example 8.17 Check whether the function $f(x) = 2x^2 - 1$, $x \in [-1, 0]$ satisfies the conditions of Fermat's theorem or not.

Solution Given function $f(x) = 2x^2 - 1$, $x \in [-1, 0]$,

here, given function is strictly decreasing in $[-1, 0]$ and consequently attains maxima at $x = -1$ and minima at $x = 0$.

Since, these points are not interior points of the interval $[-1, 0]$,

hence, we cannot apply Fermat's theorem to assert that

$$f'(-1) = f'(0) = 0$$

In fact $f'(-1) = 4$.

We can see that $f'(0) = 0$ but this inference could not have been drawn by use of Fermat's theorem which is not applicable in this problem.



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6. Perimeter of a rectangle = $2(l + b)$
7. Area of a trapezium = $\frac{1}{2} \cdot (\text{sum of parallel side})$
8. Area of a circle = πr^2
9. Perimeter of a circle = $2\pi r$
10. Volume of a sphere = $\frac{4}{3} \cdot \pi r^3$
11. Surface area of a sphere = $4\pi r^2$
12. Volume of a cone = $\frac{1}{3} \cdot \pi r^2 h$
13. Curve surface area of a cone = $\pi r l$
14. Total surface area of a cone = $\pi r(l + r)$
15. Volume of a cylinder = $\pi r^2 h$
16. Curve surface area of a cylinder = $2\pi r h$
17. Total surface area of a cylinder = $2\pi r(h + r)$
18. Volume of a cuboid = $l \times b \times h$
19. Surface area of a cuboid = $2(lb + bh + lh)$
20. Area of four walls of cuboid = $2(l + b)h$
21. Volume of cube = l^3
22. Surface area of a cube = $6l^2$
23. Area of four walls of a cube = $4l^2$

8.9 TIPS AND TRICKS

1. If $f(x)$ is a monotonic increasing function in $[a, b]$ then greatest and least values of $f(x)$ in $[a, b]$ are $f(b)$ and $f(a)$ respectively.
2. If $f(x)$ is a monotonic decreasing function in $[a, b]$ then greatest and least values of $f(x)$ in $[a, b]$ are $f(a)$ and $f(b)$ respectively.
3. If $f(x)$ is monotonic increasing in $[a, b]$ and decreasing in $[b, c]$ then greatest value $f(x)$ is $f(b)$ and least value of $f(x)$ is $\min\{f(a), f(c)\}$
4. If $f(x)$ is monotonic decreasing in $[a, b]$ and increasing in $[b, c]$ then least value $f(x)$ is $f(b)$ and greatest value of $f(x)$ is $\max\{f(a), f(c)\}$
5. If $f(x) \rightarrow \infty$ as $x \rightarrow a$ or b and $f(x) = 0$ only for one value of $x = c \in (a, b)$ then $f(c)$ is necessary the minimum and least value.
6. If $f(x) \rightarrow -\infty$ as $x \rightarrow a$ or b and $f'(x) = 0$ only for one value of $x = c \in (a, b)$ then $f(c)$ is necessary the maximum and greatest value.
7. If $f(x)$ is strictly increasing function on $[a, b]$ then $f^{-1}(x)$ exists and also $f^{-1}(x)$ is strictly increasing on $[a, b]$.
8. If $f(x)$ is strictly decreasing function on $[a, b]$ then $f^{-1}(x)$ exists and also $f^{-1}(x)$ is strictly decreasing on $[a, b]$.
9. If $y = f(x)$ is any function defined on a real domain D and I is an interval contain in D then function $y = f(x)$ is said to be



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Type II: Problems Based on Maximum and Minimum Value of Function

1. Find all the points of local maxima and minima and the corresponding maximum and minimum values of the following functions, if any

(i) $f(x) = x^3 - 6x^2 + 9x + 15$

(ii) $f(x) = (x-1)(x+2)^2$

(iii) $f(x) = \frac{2}{x} - \frac{2}{x^2}, x > 0$

2. Find Maximum and minimum value of following function.

(i) $\frac{x^3}{3} - \frac{x^2}{2} - 6x + 8$

(ii) $\frac{x^2 - x - 11}{x - 5}$

3. Find the maximum value of $\frac{\log x}{x}$.

4. Find the minimum value of x^x .

5. What is the maximum slope of the curve $y = -x^3 + 3x^2 + 9x - 27$ and at what point is it?

6. Prove that $y = \sin x + \sqrt{3} \cos x$ has maximum value at $x = \frac{\pi}{6}$.

7. Find the points of local maxima and minima, if any, of the following functions. Find also the local maximum and local minimum values.

(i) $f(x) = \sin 2x - x, \frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

(ii) $f(x) = 2\sin x - x, -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

(iii) $f(x) = \sin^4 x + \cos^4 x, 0 < x < \frac{\pi}{2}$

(iv) $f(x) = \sin x(1 + \cos x); 0 \leq x \leq \frac{\pi}{2}$

8. The curve $f(x) = \frac{x^2 + ax + b}{x - 10}$ has a turning point at $(4, 1)$. Find the values of a and b and also show that $f(x)$ is maximum at this point.

9. The curve $f(x) = \frac{ax - b}{(x - 1)(x - 4)}$ has a turning point at $P(2, -1)$. Find the values of a and b and also show that $f(x)$ is maximum at point P .

10. Determine the absolute maximum and minimum value of the following functions?

(i) $f(x) = x^{50} - x^{20}$ in $[0, 1]$

(ii) $f(x) = 2 \sin x - x$ in $[0, 2\pi]$

(iii) $f(x) = x^3 + \left(\frac{1}{2} - x\right)^2$ in $[-2, 2.5]$

Type III: Problems Based on Finding Maximum and Minimum Values of a Function Determined by Given Condition

- Determine two positive real numbers whose sum is 15 and sum of whose squares is minimum.
- Prove that if product of two numbers is constant, their sum will be minimum when the two numbers are equal.
- Prove that the area of a right angle triangle drawn on a given side as hypotenuse is maximum when the triangle is isosceles.
- Show that rectangle of maximum perimeter which can be inscribed in a circle of radius a is a square of side $a\sqrt{2}$.



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Rule-IV

$$\int (f(x))^n f'(x) dx = \frac{(f(x))^{n+1}}{n+1} + c$$

Rule-V

$$\int f(x) dx = F(x) + c \text{ then } \int f(ax + b) dx = \frac{F(ax + b)}{a} + c$$

SOLVED EXAMPLES

Example 9.1 Evaluate $\int \sqrt{x}(x^{3/2} - 7x^{5/2} + 9x^{7/2}) dx$

Solution Given $\int \sqrt{x}(x^{3/2} - 7x^{5/2} + 9x^{7/2}) dx$

$$\begin{aligned}
 &= \int \sqrt{x}(x^{3/2} - 7x^{5/2} + 9x^{7/2}) dx \\
 &= \int (x^{1/2} \cdot x^{3/2} - 7x^{1/2} \cdot x^{5/2} + 9x^{1/2} \cdot x^{7/2}) dx \\
 &= \int (x^2 - 7x^3 + 9x^4) dx \\
 &= \int x^2 dx - \int 7x^3 dx + 9 \int x^4 dx \\
 &= \frac{x^{2+1}}{2+1} - 7 \cdot \frac{x^{3+1}}{3+1} + \frac{x^{4+1}}{4+1} + c \\
 &= \frac{x^3}{3} - 7 \cdot \frac{x^4}{4} + 9 \cdot \frac{x^5}{5} + c
 \end{aligned}$$

Example 9.2 Evaluate $\int \frac{(1+x)^3}{\sqrt{x}} dx$

Solution $\int \frac{(1+x)^3}{\sqrt{x}} dx$

$$\begin{aligned}
 &= \int \frac{1 + 3x + 3x^2 + x^3}{\sqrt{x}} dx \\
 &= \int \left(\frac{1}{\sqrt{x}} + \frac{3x}{\sqrt{x}} + \frac{3x^2}{\sqrt{x}} + \frac{x^3}{\sqrt{x}} \right) dx \\
 &= \int (x^{-1/2} + 3x^{-1/2} + 3x^{2-1/2} + x^{3-1/2}) dx \\
 &= \int x^{-1/2} dx + 3 \int x^{1/2} dx + 3 \int x^{3/2} dx + \int x^{5/2} dx
 \end{aligned}$$



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$$\begin{aligned}
 &= \frac{3}{4} \int \sin x dx - \frac{1}{4} \int \sin 3x dx \\
 &= \frac{3}{4} (-\cos x) - \frac{1}{4} \left(-\frac{\cos 3x}{3} \right) + c \\
 &= -\frac{3}{4} \cos x + \frac{\cos 3x}{12} + c
 \end{aligned}$$

Example 9.9 Evaluate $\int \sqrt{1 + \cos x} dx$.

Solution $\int \sqrt{1 + \cos x} dx = \int \sqrt{2 \cos^2 \frac{x}{2}} dx$

$$= \sqrt{2} \int \cos \frac{x}{2} dx = \sqrt{2} \cdot \frac{\sin \frac{x}{2}}{\frac{1}{2}} + c$$

$$= 2\sqrt{2} \sin\left(\frac{x}{2}\right) + c$$

Example 9.10 Evaluate $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx$.

Solution $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx = \int \frac{\sin^3 x}{\sin^2 x \cos^2 x} dx + \int \frac{\cos^3 x}{\sin^2 x \cos^2 x} dx$

$$= \int \frac{\sin x}{\cos^2 x} dx + \int \frac{\cos x}{\sin^2 x} dx = \int \tan x \sec x dx + \int \cot x \operatorname{cosec} x dx$$

$$= \sec x - \operatorname{cosec} x + c.$$

Example 9.11 For what value of c and a , is the following equation satisfactory?

$$\int (\sin 2x + \cos 2x) dx = \frac{1}{\sqrt{2}} \sin(2x - c) + a.$$

Solution $\int (\sin 2x + \cos 2x) dx = \int \sin 2x dx + \int \cos 2x dx.$

$$= -\frac{\cos 2x}{2} + \frac{\sin 2x}{2} + k, \text{ where } k \text{ is an arbitrary constant}$$

$$= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} \sin 2x - \frac{1}{\sqrt{2}} \cos 2x \right) + k$$



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$$\begin{aligned}
 &= a^3 \cdot \frac{e^{3x \log a}}{3 \log a} + c = \frac{a^3 \cdot a^{3x}}{3 \log a} + c \quad [\because e^{3x \log a} = a^{3x}] \\
 &= \frac{a^{3x+3}}{3 \log a} + c
 \end{aligned}$$

Example 9.17 $\int \frac{(a^x + b^x)^2}{a^x b^x} dx$ is equal to

- (a) $\frac{(a/b)^x}{\log_e(a/b)} + \frac{(a/b)^x}{\log_e(b/a)} + x + c$ (b) $\frac{(a/b)^x}{\log_e(b/a)} + \frac{(b/a)^x}{\log_e(a/b)} + x + c$
- (c) $\frac{(a/b)^x}{\log_e(b/a)} + \frac{(b/a)^x}{\log_e(a/b)} + 2x + c$ (d) $\frac{(a/b)^x}{\log_e(a/b)} + \frac{(b/a)^x}{\log_e(b/a)} + 2x + c$

Solution $\int \frac{(a^x + b^x)^2}{a^x b^x} dx$

$$\begin{aligned}
 &= \int \frac{a^{2x} + b^{2x} + 2a^x b^x}{a^x b^x} dx \\
 &= \int \left(\frac{a^x}{b^x} + \frac{b^x}{a^x} + 2 \right) dx \\
 &= \int \left\{ \left(\frac{a}{b} \right)^x + \left(\frac{b}{a} \right)^x + 2 \right\} dx \\
 &= \frac{(a/b)^x}{\log_e(a/b)} + \frac{(b/a)^x}{\log_e(b/a)} + 2x + c.
 \end{aligned}$$

Example 9.18 $\int \frac{e^{5 \log_e x} - e^{4 \log_e x}}{e^{3 \log_e x} - e^{2 \log_e x}} dx$ is equal to

- (a) $\frac{x^3}{3} + c$ (b) $\frac{x^2}{3} + c$ (c) $\frac{x^4}{3} + c$ (d) $\frac{x^3}{2} + c$

Solution $\int \frac{e^{5 \log_e x} - e^{4 \log_e x}}{e^{3 \log_e x} - e^{2 \log_e x}} dx$

$$\begin{aligned}
 &= \int \frac{x^5 - x^4}{x^3 - x^2} dx && [\because e^{a \log_e x} = x^a] \\
 &= \int \frac{x^4(x-1)}{x^2(x-1)} dx \\
 &= \int x^2 dx = \frac{x^3}{3} + c.
 \end{aligned}$$



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ANSWER KEY

1. $\frac{e^{2x}}{4}(2x-1)+c$

2. $x \tan x - \log |\sec x| - \frac{x^2}{2} + c$

3. $\frac{\sin 3x}{18} - \frac{x \cos 3x}{6} + \frac{\sin x}{2} - \frac{x \cos x}{2} + c$

4. $\frac{-x^2 \cos 2x}{2} + \frac{x \sin 2x}{2} + \frac{\cos 2x}{4} + c$

5. $\frac{x^3}{3} \left(\log x - \frac{1}{3} \right) + c$

6. $x \tan \frac{x}{2} - 2 \log \left| \sec \frac{x}{2} \right| + c$

Note I

If $f(x)$ is logarithmic or inverse trigonometrical function then $\int f(x) dx$ can be evaluated by parts using:-

$$\int 1 \cdot f(x) dx = f(x) \int 1 \cdot dx - \int \left\{ \frac{d\{f(x)\}}{dx} \int 1 \cdot dx \right\} dx$$

Example 9.24 Evaluate $\int \log x dx$ is equal to

- (a) $x(\log x) - 1 + c$ (b) $x(\log x) - x + c$ (c) $x(\log x) - \frac{1}{x} + c$ (d) $x(\log x) + c$

Solution Given $\int \log x dx$

$$= \int \log x \cdot 1 dx$$

$$= \log x \cdot (\int 1 dx) - \int \left\{ \frac{d}{dx} (\log x) \cdot \int 1 dx \right\} dx$$

$$= (\log x) - \int \frac{1}{x} \cdot x dx$$

$$= x(\log x) - \int 1 dx$$

$$= x(\log x) - x + c.$$

Example 9.25 Evaluate $\int \sin^{-1} x dx$ is equal to

- (a) $x \sin^{-1} x + \sqrt{\sin^2 x - 1} + c$ (b) $x \sin^{-1} x + \sqrt{1 - \sin^2 x} + c$
 (c) $x \sin^{-1} x + \sqrt{x^2 - 1} + c$ (d) none of these

Solution Given $\int \sin^{-1} x dx$

$$= \int \sin^{-1} x \cdot 1 dx$$

$$= \sin^{-1} x \cdot (\int 1 dx) - \int \left\{ \frac{d}{dx} (\sin^{-1} x) \cdot \int 1 dx \right\} dx$$

$$= x \cdot \sin^{-1} x - \int \frac{1}{\sqrt{1-x^2}} \cdot x dx$$

Let

$$y = 1 - x^2 \Rightarrow \frac{dy}{dx} = -2x \Rightarrow \frac{dy}{-2} = x dx$$



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$$\begin{aligned}
 &= \frac{2}{3} \sqrt{1+3x} \left[\frac{9-(1+3x)}{3} \right] + c \\
 &= \frac{2}{3} \sqrt{1+3x} \left(\frac{8-3x}{3} \right) + c = \frac{2}{9} (8-3x) \sqrt{1+3x} + c
 \end{aligned}$$

EXERCISE-I

Solve:-

1. $\int (x+2) \sqrt{3x+5} \, dx$
2. $\int x^2 \sqrt{x+1} \, dx$
3. $\int (2x^2+3) \sqrt{x+2} \, dx$
4. $\int \frac{2-3x}{\sqrt{1+3x}} \, dx$
5. $\int \frac{2x-1}{(x-1)^2} \, dx$
6. $\int \frac{x^2}{\sqrt{1+x}} \, dx$

ANSWER KEY

1. $\frac{2}{135} (9x+20)(3x+5)^{3/2} + c$
2. $2(x+1)^{3/2} \left[\frac{(x+1)^2}{7} - \frac{2}{5}(x+1) + \frac{1}{3} \right] + c$
3. $2(x+2)^{3/2} \left[\frac{2(x+2)^2}{7} - \frac{8}{5}(x+2) + \frac{11}{3} \right] + c$
4. $\frac{2}{9} (8-3x)(3x+1)^{1/2} + c$
5. $2 \log |x-1| - \frac{1}{x-1} + c$
6. $2(x+1)^{1/2} \left[\frac{(x+1)^2}{5} - \frac{2}{3}(x+1) + 1 \right] + c$

Form-II

Form of integration $\int \sin^n x \cdot \cos^m x \, dx$ can be evaluated by using:

- (a) If power on $\sin x$ is odd positive integer then put $z = \cos x$
- (b) If power on $\cos x$ is odd positive integer then put $z = \sin x$
- (c) If power on $\cos x$ and $\sin x$ are odd positive integer then put $z = \sin x$ or $\cos x$
- (d) If power on $\cos x$ or $\sin x$ is not odd positive then find sum of power on $\cos x$ and $\sin x$. If sum of power on $\cos x$ and $\sin x$ is even negative then put $z = \tan x$ otherwise use method of transformation.

Example 9.35 Evaluate $\int \sin^3 x \cos^2 x \, dx$.

Solution Given $\int \sin^3 x \cos^2 x \, dx$.

Here power of $\sin x$ is odd positive integer, therefore, put $z = \cos x$

Let $z = \cos x$, then $dz = -\sin x \, dx$

$$\begin{aligned}
 \text{Now } \int \sin^3 x \cos^2 x \, dx &= \int \sin^2 x \cos^2 x \sin x \, dx \\
 &= \int (1 - \cos^2 x) \cos^2 x \sin x \, dx
 \end{aligned}$$



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ANSWER KEY

1. $\frac{\tan^2 x}{2} + c$

2. $\frac{\tan^5 x}{5} + c$

3. $\frac{\sec^3 x}{3} + c$

4. $\frac{2 \sec^{\frac{7}{2}} x}{7} + c$

5. $\frac{\sec^4 x}{4} - \sec^2 x \log |\sec x| + c$

6. $\frac{\sec^4 (x/2)}{2} + c$

Form-IV

Form of integration $\int \operatorname{cosec}^n x \cdot \cot^m x \, dx$ can be evaluated by using:-

- If power on $\operatorname{cosec} x$ is even positive integer then put $z = \cot x$
- If power on $\cot x$ is odd positive integer then put $z = \operatorname{cosec} x$
- If power on $\cot x$ is even positive integer then first put $\operatorname{cosec}^2 x - 1$ in place of $\cot^2 x$ then put $z = \cot x$ otherwise use integration by parts.

Example 9.46 Evaluate $\int \operatorname{cosec}^4 x \, dx$.

Solution Given $\int \operatorname{cosec}^4 x \, dx$.

Here power of $\operatorname{cosec} x$ is even positive integer therefore, put $z = \cot x$

$$\Rightarrow dz = -\operatorname{cosec}^2 x \, dx$$

$$\begin{aligned} \text{Now } \int \operatorname{cosec}^4 x \, dx &= \int \operatorname{cosec}^2 x \cdot \operatorname{cosec}^2 x \, dx \\ &= \int (1 + \cot^2 x) \operatorname{cosec}^2 x \, dx \\ &= \int (1 + z^2) (-dz) = -\int (1 + z^2) dz \\ &= -\cot x - \frac{\cot^3 x}{3} + c \end{aligned}$$

EXERCISE-L

Solve:-

1. $\int \cot^4 x \, dx$

2. $\int \sqrt{\cot x} \operatorname{cosec}^2 x \, dx$

3. $\int \cot^n x \operatorname{cosec}^2 x \, dx$

4. $\int \cot^3 x \, dx$

ANSWER KEY

1. $x + \cot x - \frac{\cot^3 x}{3} + c$

2. $-\frac{2 \cot^{3/2} x}{3} + c$

3. $-\frac{\cot^{n+1} x}{n+1} + c$

4. $-\frac{\operatorname{cosec}^2 x}{2} + \log |\operatorname{cosec} x| + c$



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Here $\sqrt{e^{2x} - 4} = \sqrt{(e^x)^2 - 2^2}$ which is of the form, $\sqrt{x^2 - a^2}$

$$\text{As, } \int \frac{e^x}{\sqrt{e^{2x} - 4}} dx = \int \frac{e^x}{\sqrt{(e^x)^2 - 2^2}} dx \quad (i)$$

Let $e^x = 2 \sec \theta$, then $e^x dx = 2 \sec \theta \tan \theta d\theta$

$$\begin{aligned} \text{Now from (i), } \int \frac{e^x}{\sqrt{e^{2x} - 4}} dx &= \int \frac{2 \sec \theta \tan \theta}{\sqrt{4 \sec^2 \theta - 4}} d\theta = \int \frac{2 \sec \theta \tan \theta}{2 \tan \theta} d\theta \\ &= \int \sec \theta d\theta = \log |\sec \theta + \tan \theta| + c \quad (ii) \end{aligned}$$

$$\therefore e^x = 2 \sec \theta \quad \therefore \sec \theta = \frac{e^x}{2}$$

$$\therefore \tan \theta = \sqrt{\sec^2 \theta - 1} = \sqrt{\frac{e^{2x}}{4} - 1} = \frac{\sqrt{e^{2x} - 4}}{2}$$

$$\begin{aligned} \text{From (ii), } \int \frac{e^x}{\sqrt{e^{2x} - 4}} dx &= \log \left| \frac{e^x}{2} + \frac{\sqrt{e^{2x} - 4}}{2} \right| + c \\ &= \log \left(\frac{e^x + \sqrt{e^{2x} - 4}}{2} \right) + c \\ &= \log (e^x + \sqrt{e^{2x} - 4}) - \log 2 + c \\ &= \log (e^x + \sqrt{e^{2x} - 4}) + c' \end{aligned}$$

Form (c)

$\int \sqrt{x^2 + a^2} dx$ or $\int \frac{dx}{\sqrt{x^2 + a^2}}$ can be solved by putting $x = a \tan \theta$ or $a \cot \theta$

Particular Case

$\int \sqrt{x^2 + 1} dx$ or $\int \frac{dx}{\sqrt{x^2 + 1}}$ Can be solved by putting $x = \tan \theta$ or $\cot \theta$

Example 9.58 Evaluate $\int \frac{x^3}{1+x^8} dx$

Solution Given $\int \frac{x^3}{1+x^8} dx$

Here $1 + x^8 = 1 + (x^4)^2$, which is of the form of $a^2 + x^2$

Let $x^4 = \tan \theta$ then $4x^3 dx = \sec^2 \theta d\theta$



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$$\begin{aligned} \text{or } \int \frac{(px+q)}{\sqrt{ax^2+bx+c}} dx &= \int \frac{A(2ax+b)+B}{\sqrt{ax^2+bx+c}} dx \\ &= A \int \frac{2ax+b}{\sqrt{ax^2+bx+c}} dx + B \int \frac{dx}{\sqrt{ax^2+bx+c}} \end{aligned}$$

Then integrate.

Example 9.68 Evaluate $\int \frac{x+1}{\sqrt{2x^2+x-3}} dx$

Solution Given Let $x+1 = A$ [d.c. of $(2x^2+x-3)$] + B

$$\Rightarrow x+1 = A(4x+1) + B = 4Ax + A + B$$

Equating the coefficient of similar power of x , we get

$$4A = 1 \quad \therefore A = \frac{1}{4} \text{ and } A + B = 1$$

$$\therefore B = 1 - A = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\text{Now } I = \int \frac{x+1}{\sqrt{2x^2+x-3}} dx = \int \frac{A(4x+1)+B}{\sqrt{2x^2+x-3}} dx$$

$$= A \int \frac{4x+1}{\sqrt{2x^2+x-3}} dx + B \int \frac{dx}{\sqrt{2x^2+x-3}}$$

$$= \frac{1}{4} \int \frac{4x+1}{\sqrt{2x^2+x-3}} dx + \frac{3}{4} \int \frac{dx}{\sqrt{2x^2+x-3}} \quad \text{(ii)}$$

$$\text{Let } I_1 = \frac{1}{4} \int \frac{4x+1}{\sqrt{2x^2+x-3}} dx$$

$$\text{Let } z = 2x^2 + x - 3, \text{ then } dz = (4x+1)dx$$

$$\text{Now, } I_1 = \frac{1}{4} \int \frac{dz}{\sqrt{z}} = \frac{1}{4} \int z^{-\frac{1}{2}} dz = \frac{1}{4} \cdot \frac{z^{\frac{1}{2}}}{\frac{1}{2}} = \frac{1}{2} \sqrt{2x^2+x-3}$$

$$\text{Let } I_2 = \frac{3}{4} \int \frac{dx}{\sqrt{2x^2+x-3}} = \frac{3}{4} \int \frac{dx}{\sqrt{2\left(x^2 + \frac{1}{2}x - \frac{3}{2}\right)}}$$



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ANSWER KEY

$$1. \frac{2}{\sqrt{23}} \tan^{-1} \left(\frac{4x+3}{\sqrt{23}} \right) + c$$

$$2. \log \left| \frac{x+2}{x+3} \right| + c$$

$$3. \frac{1}{2\sqrt{2}} \log \left| \frac{\sqrt{2}+1+x}{\sqrt{2}-1-x} \right| + c$$

$$4. \frac{1}{\sqrt{11}} \tan^{-1} \left(\frac{3x+2}{\sqrt{11}} \right) + c$$

$$5. \log |x^2 + 3x - 18| - \frac{2}{3} \log \left| \frac{x-3}{x+6} \right| + c.$$

$$6. x + \log(1 + e^x) - 2 \log(1 + 2e^x) + c$$

Form-III

If $f(x)$ and $g(x)$ are any two polynomial functions then form of integration $\int \frac{f(x)}{g(x)} dx$ can be evaluated by using:

Form-(a)

$\int \frac{f(x)}{g(x)} dx$ where degree of $f(x)$ is equal to degree of $g(x)$.

To solve, first expressed numerator $f(x)$ in the form of denominator $g(x)$, then integrate.

Example 9.73 The value of $\int \frac{x^2 + 5x + 3}{x^2 + 3x + 2} dx$ is

$$(a) \frac{x}{2} + \log |x^2 + 3x + 2| - 2 \log \left| \frac{x+1}{x+2} \right| + c$$

$$(b) x + \log |x^2 + 3x + 2| - 2 \log \left| \frac{x+1}{x+2} \right| + c$$

$$(c) x + \log |x^2 + 3x + 2| - \frac{1}{2} \log \left| \frac{x+1}{x+2} \right| + c$$

(d) none of these

Solution Given, $\int \frac{x^2 + 5x + 3}{x^2 + 3x + 2} dx$

$$= \int \left(1 + \frac{2x+1}{x^2 + 3x + 2} \right) dx$$

$$= \int 1 \cdot dx + \int \frac{2x+3-2}{x^2 + 3x + 2} dx$$

$$= \int 1 \cdot dx + \int \frac{2x+3}{x^2 + 3x + 2} dx - 2 \int \frac{1}{x^2 + 3x + 2} dx$$



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$$\begin{aligned}
&= \int \frac{1}{x \log_e ex \cdot \log_e e^2 x \cdot \log_e e^3 x} dx \\
&= \int \frac{1}{x(\log_e e + \log_e x)(\log_e e^2 + \log_e x)(\log_e e^3 + \log_e x)} dx \\
&= \int \frac{1}{x(1 + \log_e x)(2 + \log_e x)(3 + \log_e x)} d(\log_e x) \\
&= \int \frac{1}{(1+t)(2+t)(3+t)} dt, \quad \text{where } t = \log_e x \\
&= \int \left(\frac{1}{2} \cdot \frac{1}{1+t} - \frac{1}{2+t} + \frac{1}{3+t} \right) dt \quad [\text{Using partial fractions}] \\
&= \frac{1}{2} \log |1+t| - \log(2+t) + \log(3+t) + c \\
&= \frac{1}{2} \log |1 + \log_e x| - \log(2 + \log_e x) + \log |3 + \log_e x| + c \\
&= \frac{1}{2} \log \{\log_e ex\} - \log \{\log_e e^2 x\} + \log \{\log_e e^3 x\} + c.
\end{aligned}$$

Calculus or Limit Method Use following steps:

Step-(i) Write the given fraction $\frac{f(x)}{(x-a_1)(x-a_2)(x-a_3)\dots(x-a_n)}$

Step-(ii) Let $\frac{f(x)}{(x-a_1)(x-a_2)(x-a_3)\dots(x-a_n)} = \frac{A}{(x-a_1)} + \frac{B}{(x-a_2)} + \frac{C}{(x-a_3)} + \dots + \frac{n}{(x-a_n)}$

Step-(iii) For value of A use

$$A = \lim_{x \rightarrow a_1} (x-a_1) \left\{ \frac{f(x)}{(x-a_1)(x-a_2)\dots(x-a_n)} \right\}$$

For value of B use

$$B = \lim_{x \rightarrow a_2} (x-a_2) \left\{ \frac{f(x)}{(x-a_1)(x-a_2)\dots(x-a_n)} \right\}$$

For value of C , use

$$C = \lim_{x \rightarrow a_3} (x-a_3) \left\{ \frac{f(x)}{(x-a_1)(x-a_2)\dots(x-a_n)} \right\}$$

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- (b) $a = \frac{1}{10}, b = -\frac{2}{5}$
- (c) $a = -\frac{1}{10}, b = \frac{2}{5}$
- (d) $a = \frac{1}{10}, b = \frac{2}{5}$
36. If $\int \frac{3 \sin x + 2 \cos x}{3 \cos x + 2 \sin x} dx = ax + b \ln |2 \sin x + 3 \cos x| + C$, then
- (a) $a = -\frac{12}{13}, b = \frac{15}{39}$
- (b) $a = -\frac{7}{13}, b = \frac{6}{13}$
- (c) $a = \frac{12}{13}, b = -\frac{15}{39}$
- (d) $a = -\frac{7}{13}, b = -\frac{6}{13}$
37. $\int \frac{3e^x - 5e^{-x}}{4e^x + 5e^{-x}} dx = ax + b \ln |4e^x + 5e^{-x}| + C$, then
- (a) $a = -\frac{1}{8}, b = \frac{7}{8}$
- (b) $a = \frac{1}{8}, b = \frac{7}{8}$
- (c) $a = -\frac{1}{8}, b = -\frac{7}{8}$
- (d) $a = \frac{1}{8}, b = -\frac{7}{8}$
38. $\int \sqrt{\frac{\cos x - \cos^3 x}{1 - \cos^3 x}} dx$ is equal to
- (a) $\frac{2}{3} \sin^{-1}(\cos^{3/2} x) + C$
- (b) $\frac{3}{2} \sin^{-1}(\cos^{3/2} x) + C$
- (c) $\frac{2}{3} \cos^{-1}(\cos^{3/2} x) + C$
- (d) None of these
39. If $I^r(x)$ means $\log \log \log \dots x$, the log being repeated r , then $\int [x I(x) I^2(x) I^3(x) \dots I^r(x)]^{-1} dx$ is equal to
- (a) $I^{r+1}(x) + C$
- (b) $\frac{I^{r+1}(x)}{r+1} + C$
- (c) $I^r(x) + C$
- (d) None of these
40. If $I = \int (\sqrt{\cot x} - \sqrt{\tan x}) dx$, then I equals
- (a) $\sqrt{2} \log(\sqrt{\tan x} - \sqrt{\cot x}) + C$
- (b) $\sqrt{2} \log |\sin x + \cos x + \sqrt{\sin 2x}| + C$
- (c) $\sqrt{2} \log |\sin x - \cos x + \sqrt{2} \sin x \cos x| + C$
- (d) $\sqrt{2} \log |\sin(x + \pi/4) + \sqrt{2} \sin x \cos x| + C$
41. If $I = \int \frac{dx}{(a^2 - b^2 x^2)^{3/2}}$, then I equals
- (a) $\frac{x}{\sqrt{a^2 - b^2 x^2}} + C$
- (b) $\frac{x}{a^2 \sqrt{a^2 - b^2 x^2}} + C$
- (c) $\frac{ax}{\sqrt{a^2 - b^2 x^2}} + C$
- (d) None of these
42. $\int e^{x^4} (x + x^3 + 2x^5) e^{x^2} dx$ is equal to
- (a) $\frac{1}{2} x e^{x^2} e^{x^4} + c$
- (b) $\frac{1}{2} x^2 e^{x^4} + c$



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Example 10.9 $\int_2^4 \frac{6x^2 - 1}{\sqrt{2x^3 - x}} dx$

Solution Given $\int_2^4 \frac{6x^2 - 1}{\sqrt{2x^3 - x}} dx$

put $z = 2x^3 - x \Rightarrow dz = (6x^2 - 1)dx$

$$\begin{aligned} \Rightarrow \int_2^4 \frac{6x^2 - 1}{\sqrt{2x^3 - x}} dx &= \int_2^4 \frac{dz}{\sqrt{z}} = \left[\frac{z^{1/2}}{1/2} \right]_2^4 \\ &= \left[2(2x^3 - x)^{1/2} \right]_2^4 = 4\sqrt{31} - 2\sqrt{14} \end{aligned}$$

Example 10.10 $\int_1^e \frac{\cos(\log x)}{x} dx$

Solution Given $\int_1^e \frac{\cos(\log x)}{x} dx$

Put $Z = \log x \Rightarrow dz = \frac{dx}{x}$

$$\begin{aligned} \therefore \int_1^e \frac{\cos(\log x)}{x} dx &= \int_1^e \cos z dz = [\sin z]_1^e \\ &= [\sin(\log x)]_1^e = \sin 1 \end{aligned}$$

Example 10.11 $\int_0^\pi x \cdot \sin^2 x dx$

Solution Given $\int_0^\pi x \cdot \sin^2 x dx$

As,

$$\begin{aligned} \int x \cdot \sin^2 x dx &= \int x \cdot \left(\frac{1 - \cos 2x}{2} \right) dx = \frac{1}{2} \int x dx - \frac{1}{2} \int x \cdot \cos 2x dx \\ &= \frac{x^2}{4} - \frac{1}{2} \left[x \cdot \frac{\sin 2x}{2} - \int 1 \cdot \frac{\sin 2x}{2} dx \right] = \frac{x^2}{4} - \frac{x \sin 2x}{4} - \frac{\cos 2x}{8} \end{aligned}$$

Hence,

$$\int_0^\pi x \cdot \sin^2 x dx = \left[\frac{x^2}{4} - \frac{x \sin 2x}{4} - \frac{\cos 2x}{8} \right]_0^\pi = \frac{\pi^2}{4}$$



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- (ii) $y(1 + xy)dx - xdy = 0$
 (iii) $y(2xy + e^x)dx - e^x dy = 0$
 (iv) $ydx - xdy = xydy$
 2. Solve the equation $(x^2 + 1)y_2 = 2xy_1$ where

$$y_1 = \frac{dy}{dx} \text{ and } y_2 = \frac{d^2y}{dx^2}.$$

3. The value of a and b are respectively
 (a) 0, 1 (b) 1, 0
 (c) -1, 0 (d) 0, -1
 4. The value of c is
 (a) 6 (b) -7
 (c) 2 (d) -1
 5. Value of d is
 (a) 6 (b) -7
 (c) 2 (d) -1

Type XI: Problems Based on Application of Differential Equation

- The surface area of a balloon being inflated changes at a constant rate. If initial, its radius is 3 units and after 2 second it is 5 units, find the radius after t seconds.
- A particle starting with velocity u is falling freely under gravity with a constant acceleration. Find the velocity v and distance s traveled by particle at time t .

LEVEL III: MISCELLANEOUS QUESTIONS

Type I: Passage

Passage I

The differential equation corresponding to $y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x}$ where c_1, c_2, c_3 are arbitrary constants and m_1, m_2, m_3 are the roots

of equation $m^3 - 7m + 6 = 0$ is $a \frac{d^3 y}{dx^3} + b \frac{d^2 y}{dx^2} +$

$c \frac{dy}{dx} + d = 0$ where a, b, c are constants.

On the Basis of above information, answer the following questions.

- The order of differential equation is
 (a) 1 (b) 2
 (c) 3 (d) None of these
- The degree of differential equation is
 (a) 1 (b) 2
 (c) 3 (d) None of these

Passage II

Consider the curve which passes through (1,1) and any tangent to which in the first quadrant cuts x-axis at A and y-axis at B. Area of ΔOAB is 2 square units.

On the Basis of above Information, Answer the Following Questions.

- The differential equation of all such curves must be

(a) $2xy + y^2 \left(\frac{dy}{dx} \right)^{-1} - x^2 \frac{dy}{dx} = 4$

(b) $2xy - y^2 \left(\frac{dy}{dx} \right)^{-1} - x^2 \frac{dy}{dx} = 4$

(c) $2xy - y^2 \left(\frac{dy}{dx} \right) - x^2 \frac{dy}{dx} = 4$

(d) None of these

- If all the differential equations in Q.1 are differentiated then we cannot get

(a) $\frac{dy}{dx} = \frac{y}{x}$ (b) $\frac{dy}{dx} = -\frac{y}{x}$

(c) $\frac{d^2 y}{dx^2} = 0$ (d) $\frac{dy}{dx} = \frac{x}{y}$

- The second degree curve satisfying the differential equation is

(a) $x^2 + y^2 = 1$ (b) $xy = 1$
 (c) $y^2 = x$ (d) None of these



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Annexure

Questions of Calculus in IIT-JEE 2010

Problem 1 Let $f(x)$ is the real valued function defined on the interval $(-1, 1)$ such that $e^{-x}f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt$

for all x in $(-1, 1)$ and let f^{-1} is inverse of $f(x)$ then $(f^{-1}(2))'$ is equal to

- (a) 1 (b) $1/3$ (c) $1/2$ (d) $1/e$

Solution Given $f(x)$ is the inverse of $f(x)$ i.e. $f(f^{-1}(x)) = x$

Differentiate both sides w.r.t x , we get $\frac{df(f^{-1}(x))}{dx} = \frac{dx}{dx}$

$$\Rightarrow f'(f^{-1}(x))(f^{-1}(x))' = 1 \Rightarrow (f^{-1}(x))' = \frac{1}{f'(f^{-1}(x))} \quad (i)$$

$$\text{Since, } e^{-x}f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt \quad (ii)$$

Put $x = 0$ we get $f(0) = 2 \Rightarrow f^{-1}(2) = 0$

And differentiate (ii) both sides to get $e^{-x}f'(x) - e^{-x}f(x) = \sqrt{x^4 + 1}$

Put $x = 0$ to get $e^{-0}f'(0) - e^{-0}f(0) = \sqrt{0^4 + 1} \Rightarrow f'(0) - f(0) = 1 \Rightarrow f'(0) = 3$

$$\text{From (i) } (f^{-1}(x))' = \frac{1}{f'(f^{-1}(x))} \Rightarrow (f^{-1}(2))' = \frac{1}{f'(f^{-1}(2))} = \frac{1}{f'(0)} = \frac{1}{3}$$

Hence correct option is (b).

Problem 2 Let $S = \{1, 2, 3, 4\}$ the total number of unordered pairs of disjoint subsets of S is equal to
(a) 25 (b) 34 (c) 42 (d) 41

Solution Given set $S = \{1, 2, 3, 4\}$

Disjoint sets means common elements is not possible

Hence, for disjoint sets of A and B

Let $x \in S$ then three cases are possible.



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